
6

POISSON PROCESS

The count has arrived ...

–Thomas Carlyle, *Count Cagliostro*

6.1 INTRODUCTION

Text messages arrive on your cell phone at irregular times throughout the day. Accidents occur on the highway in a seemingly random distribution of time and place. Babies are born at chance moments on a maternity ward.

All of these phenomena are well modeled by the Poisson process, a stochastic process used to model the occurrence, or arrival, of events over a continuous interval. Typically, the interval represents time.

A Poisson process is a special type of *counting process*. Given a stream of events that arrive at random times starting at $t = 0$, let N_t denote the number of arrivals that occur by time t , that is, the number of events in $[0, t]$. For instance, N_t might be the number of text messages received up to time t .

For each $t \geq 0$, N_t is a random variable. The collection of random variables $(N_t)_{t \geq 0}$ is a continuous-time, integer-valued stochastic process, called a counting process. Since N_t counts events in $[0, t]$, as t increases, the number of events N_t increases.

Counting Process

A *counting process* $(N_t)_{t \geq 0}$ is a collection of non-negative, integer-valued random variables such that if $0 \leq s \leq t$, then $N_s \leq N_t$.

Unlike a Markov chain, which is a *sequence* of random variables, a counting process forms an uncountable collection, since it is indexed over a continuous time interval.

Figure 6.1 shows the path of a counting process in which events occur at times t_1, t_2, t_3, t_4, t_5 . As shown, the path of a counting process is a right-continuous step function. If $0 \leq s < t$, then $N_t - N_s$ is the number of events in the interval $(s, t]$.

Note that in the discussion of the Poisson process, we use the word *event* in a loose, generic sense, and not in the rigorous sense of probability theory where an event is a subset of the sample space.

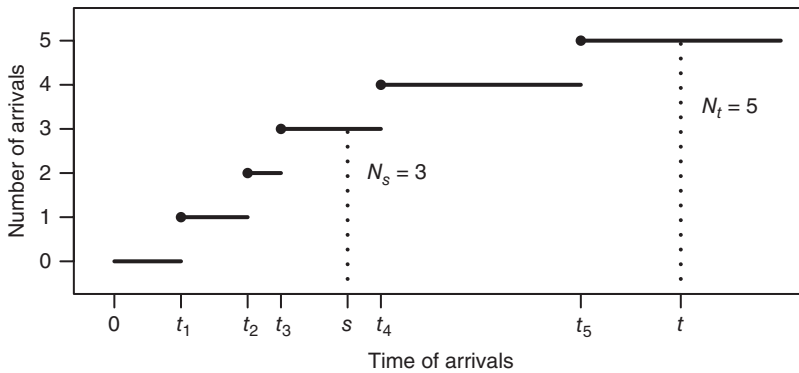


Figure 6.1 Counting process.

There are several ways to characterize the Poisson process. One can focus on (i) the *number* of events that occur in fixed intervals, (ii) *when* events occur, and the times between those events, or (iii) the probabilistic behavior of individual events on infinitesimal intervals. This leads to three equivalent definitions of a Poisson process, each of which gives special insights into the stochastic model.

Poisson Process—Definition 1

A *Poisson process with parameter λ* is a counting process $(N_t)_{t \geq 0}$ with the following properties:

1. $N_0 = 0$.
2. For all $t > 0$, N_t has a Poisson distribution with parameter λt .

3. (Stationary increments) For all $s, t > 0$, $N_{t+s} - N_s$ has the same distribution as N_t . That is,

$$P(N_{t+s} - N_s = k) = P(N_t = k) = \frac{e^{-\lambda t} (\lambda t)^k}{k!}, \text{ for } k = 0, 1, \dots$$

4. (Independent increments) For $0 \leq q < r \leq s < t$, $N_t - N_s$ and $N_r - N_q$ are independent random variables.

The stationary increments property says that the distribution of the number of arrivals in an interval only depends on the *length* of the interval.

The independent increments property says that the number of arrivals on disjoint intervals are independent random variables.

Since N_t has a Poisson distribution, $E(N_t) = \lambda t$. That is, we expect about λt arrivals in t time units. Thus, the *rate* of arrivals is $E(N_t)/t = \lambda$.

Example 6.1 Starting at 6 a.m., customers arrive at Martha's bakery according to a Poisson process at the rate of 30 customers per hour. Find the probability that more than 65 customers arrive between 9 and 11 a.m.

Solution Let $t = 0$ represent 6 a.m. Then, the desired probability is $P(N_5 - N_3 > 65)$. By stationary increments,

$$\begin{aligned} P(N_5 - N_3 > 65) &= P(N_2 > 65) = 1 - P(N_2 \leq 65) \\ &= 1 - \sum_{k=0}^{65} P(N_2 = k) \\ &= 1 - \sum_{k=0}^{65} \frac{e^{-30(2)} (30(2))^k}{k!} = 0.2355. \end{aligned}$$

In R, the result is obtained by typing

```
> 1-ppois(65, 2*30)
[1] 0.2355065
```

■

Example 6.2 Joe receives text messages starting at 10 a.m. at the rate of 10 texts per hour according to a Poisson process. Find the probability that he will receive exactly 18 texts by noon and 70 texts by 5 p.m.

Solution The desired probability is $P(N_2 = 18, N_7 = 70)$, with time as hours. If 18 texts arrive in $[0, 2]$ and 70 texts arrive in $[0, 7]$, then there are $70 - 18 = 52$ texts in

$(2, 7]$. That is,

$$\{N_2 = 18, N_7 = 70\} = \{N_2 = 18, N_7 - N_2 = 52\}.$$

The intervals $[0, 2]$ and $(2, 7]$ are disjoint, which gives

$$\begin{aligned} P(N_2 = 18, N_7 = 70) &= P(N_2 = 18, N_7 - N_2 = 52) \\ &= P(N_2 = 18) P(N_7 - N_2 = 52) \\ &= P(N_2 = 18) P(N_5 = 52) \\ &= \left(\frac{e^{-2(10)} (2(10))^{18}}{18!} \right) \left(\frac{e^{-5(10)} (5(10))^{52}}{52!} \right) \\ &= 0.0045, \end{aligned}$$

where the second equality is because of independent increments, and the third equality is because of stationary increments. The final calculation in R is

```
> dpois(18, 2*10) * dpois(52, 5*10)
[1] 0.004481021
```

Warning: It would be incorrect to write

$$P(N_2 = 18, N_7 - N_2 = 52) = P(N_2 = 18, N_5 = 52).$$

It is not true that $N_7 - N_2 = N_5$. The number of arrivals in $(2, 7]$ is not necessarily equal to the number of arrivals in $(0, 5]$. What is true is that the *distribution* of $N_7 - N_2$ is equal to the distribution of N_5 . Note that while $N_7 - N_2$ is independent of N_2 , the random variable N_5 is not independent of N_2 . Indeed, $N_5 \geq N_2$. ■

Translated Poisson Process

Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ . For fixed time $s > 0$, consider the *translated process* $(N_{t+s} - N_s)_{t \geq 0}$. The translated process is probabilistically equivalent to the original process.

Translated Poisson Process is a Poisson Process

Proposition 6.1. *Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ . For $s > 0$, let*

$$\tilde{N}_t = N_{t+s} - N_s, \text{ for } t \geq 0.$$

Then, $(\tilde{N}_t)_{t \geq 0}$ is a Poisson process with parameter λ .

We have that $(\tilde{N}_t)_{t \geq 0}$ is a counting process with $\tilde{N}_0 = N_s - N_s = 0$. By stationary increments, $\tilde{N}_t$ has the same distribution as N_t . And the new process inherits stationary and independent increments from the original. It follows that if $N_s = k$, the distribution of $N_{t+s} - k$ is equal to the distribution of N_t .

■ **Example 6.3** On election day, people arrive at a voting center according to a Poisson process. On average, 100 voters arrive every hour. If 150 people arrive during the first hour, what is the probability that at most 350 people arrive before the third hour?

Solution Let N_t denote the number of arrivals in the first t hours. Then, $(N_t)_{t \geq 0}$ is a Poisson process with parameter $\lambda = 100$. Given $N_1 = 150$, the distribution of $N_3 - N_1 = N_3 - 150$ is equal to the distribution of N_2 . This gives

$$\begin{aligned} P(N_3 \leq 350 | N_1 = 150) &= P(N_3 - 150 \leq 200 | N_1 = 150) \\ &= P(N_2 \leq 200) \\ &= \sum_{k=0}^{200} \frac{e^{-100} (100)^k}{k!} \\ &= 0.519. \end{aligned}$$

■

6.2 ARRIVAL, INTERARRIVAL TIMES

For a Poisson process with parameter λ , let X denote the time of the first arrival. Then, $X > t$ if and only if there are no arrivals in $[0, t]$. Thus,

$$P(X > t) = P(N_t = 0) = e^{-\lambda t}, \text{ for } t > 0.$$

Hence, X has an exponential distribution with parameter λ .

The exponential distribution plays a central role in the Poisson process. What is true for the time of the first arrival is also true for the time between the first and second arrival, and for all *interarrival times*. A Poisson process is a counting process for which interarrival times are independent and identically distributed exponential random variables.

Poisson Process—Definition 2

Let $X_1, X_2, \dots$ be a sequence of i.i.d. exponential random variables with parameter λ . For $t > 0$, let

$$N_t = \max\{n : X_1 + \dots + X_n \leq t\},$$

with $N_0 = 0$. Then, $(N_t)_{t \geq 0}$ defines a Poisson process with parameter λ .

Let

$$S_n = X_1 + \cdots + X_n, \text{ for } n = 1, 2, \dots$$

We call $S_1, S_2, \dots$ the *arrival times* of the process, where S_k is the time of the k th arrival. Furthermore,

$$X_k = S_k - S_{k-1}, \text{ for } k = 1, 2, \dots$$

is the *interarrival time* between the $(k - 1)$ th and k th arrival, with $S_0 = 0$.

The relationship between the interarrival and arrival times for a Poisson process is illustrated in Figure 6.2.

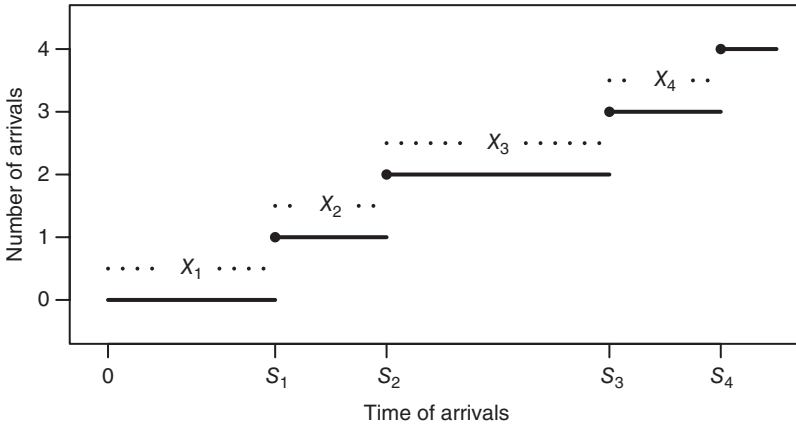


Figure 6.2 Arrival times $S_1, S_2, \dots$, and interarrival times $X_1, X_2, \dots$

In the next section, we show that Definitions 1 and 2 are equivalent. A benefit of Definition 2 is that it leads to a direct method for constructing, and simulating, a Poisson process:

1. Let $S_0 = 0$.
2. Generate i.i.d. exponential random variables $X_1, X_2, \dots$
3. Let $S_n = X_1 + \cdots + X_n$, for $n = 1, 2, \dots$
4. For each $k = 0, 1, \dots$, let $N_t = k$, for $S_k \leq t < S_{k+1}$.

Two realizations of a Poisson process with $\lambda = 0.1$ obtained by this method on the interval $[0, 100]$ are shown in Figure 6.3.

The importance of the exponential distribution to the Poisson process lies in its unique memoryless property, a topic from probability that merits review.

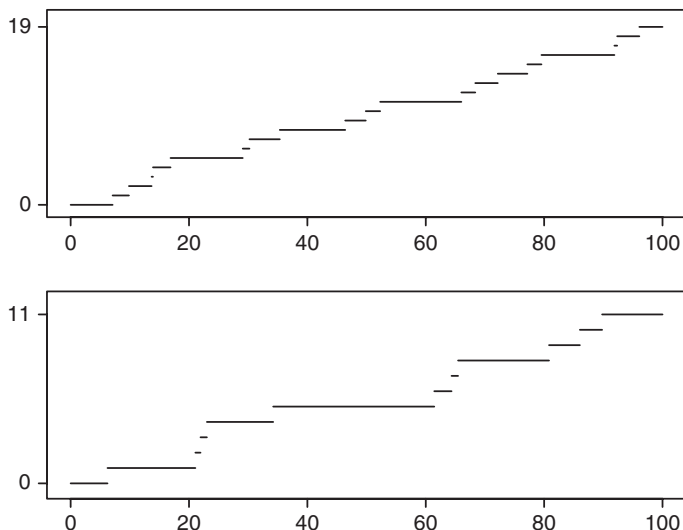


Figure 6.3 Realizations of a Poisson process with $\lambda = 0.1$.

Memorylessness

To illustrate, assume that Amy and Zach each want to take a bus. Buses arrive at a bus stop according to a Poisson process with parameter $\lambda = 1/30$. That is, the times between buses have an exponential distribution, and buses arrive, on average, once every 30 minutes. Unlucky Amy gets to the bus stop just as a bus pulls out of the station. Her waiting time for the next bus is about 30 minutes. Zach arrives at the bus stop 10 minutes after Amy. Remarkably, the time that Zach waits for a bus also has an exponential distribution with parameter $\lambda = 1/30$. Memorylessness means that their waiting time distributions are the same, and they will both wait, on average, the same amount of time!

To prove it true, observe that Zach waits more than t minutes if and only if Amy waits more than $t + 10$ minutes, given that a bus does not come in the first 10 minutes. Let A and Z denote Amy and Zach's waiting times, respectively. Amy's waiting time is exponentially distributed. Hence,

$$\begin{aligned} P(Z > t) &= P(A > t + 10 | A > 10) = \frac{P(A > t + 10)}{P(A > 10)} \\ &= \frac{e^{-(t+10)/30}}{e^{-10/30}} = e^{-t/30} = P(A > t), \end{aligned}$$

from which it follows that A and Z have the same distribution. See the R code and Figure 6.4 for the results of a simulation.

Of course, there is nothing special about $t = 10$. Memorylessness means that regardless of how long you have waited, the distribution of the time you still have to wait is the same as the original waiting time.

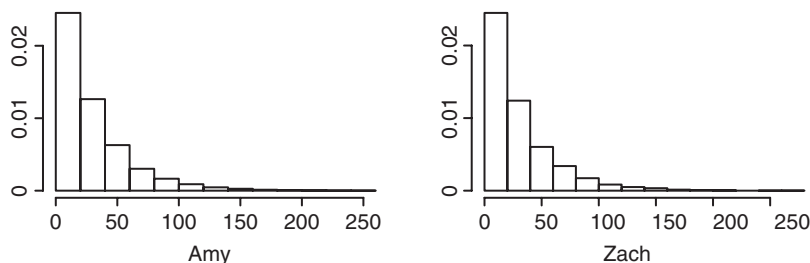


Figure 6.4 Waiting time distributions for Amy and Zach. Zach arrives 10 minutes after Amy. By memorylessness, the distributions are the same.

R : Bus Waiting Times

```
# buswaiting.R
> trials <- 5000
> amy <- numeric(trials)
> zach <- numeric(trials)
> for (i in 1:trials) {
+   bus <- rexp(1,1/30)
+   amy[i] <- bus
+   while (bus < 10) { bus <- bus + rexp(1,1/30) }
+   zach[i] <- bus-10 }
> mean(amy)
[1] 29.8043
> mean(zach)
[1] 30.39833
> hist(amy,xlab="Amy",prob=T,ylab="",main="")
> hist(zach,xlab="Zach",prob=T,ylab="",main="")
```

The exponential distribution is the only continuous distribution that is memoryless. (The geometric distribution has the honors for the discrete case.) Here is the general statement of the property.

Memorylessness

A random variable X is *memoryless* if, for all $s, t > 0$,

$$P(X > s + t | X > s) = P(X > t).$$

Results for the *minimum* of independent exponential random variables are particularly useful when working with the Poisson process. We highlight two properties that arise in many settings.

Minimum of Independent Exponential Random Variables

Let $X_1, \dots, X_n$ be independent exponential random variables with respective parameters $\lambda_1, \dots, \lambda_n$. Let $M = \min(X_1, \dots, X_n)$.

1. For $t > 0$,

$$P(M > t) = e^{-t(\lambda_1 + \dots + \lambda_n)}. \quad (6.1)$$

That is, M has an exponential distribution with parameter $\lambda_1 + \dots + \lambda_n$.

2. For $k = 1, \dots, n$,

$$P(M = X_k) = \frac{\lambda_k}{\lambda_1 + \dots + \lambda_n}. \quad (6.2)$$

Proof.

1. For $t > 0$,

$$\begin{aligned} P(M > t) &= P(X_1 > t, \dots, X_n > t) = P(X_1 > t) \cdots P(X_n > t) \\ &= e^{-\lambda_1 t} \cdots e^{-\lambda_n t} = e^{-t(\lambda_1 + \dots + \lambda_n)}. \end{aligned}$$

2. For $1 \leq k \leq n$, conditioning on X_k gives

$$\begin{aligned} P(M = X_k) &= P(\min(X_1, \dots, X_n) = X_k) \\ &= P(X_1 \geq X_k, \dots, X_n \geq X_k) \\ &= \int_0^\infty P(X_1 \geq t, \dots, X_n \geq t | X_k = t) \lambda_k e^{-\lambda_k t} dt \\ &= \int_0^\infty P(X_1 \geq t, \dots, X_{k-1} \geq t, X_{k+1} \geq t, \dots, X_n \geq t) \lambda_k e^{-\lambda_k t} dt \\ &= \int_0^\infty P(X_1 \geq t) \cdots P(X_{k-1} \geq t) P(X_{k+1} \geq t) \cdots P(X_n \geq t) \lambda_k e^{-\lambda_k t} dt \\ &= \lambda_k \int_0^\infty e^{-t(\lambda_1 + \dots + \lambda_n)} dt = \frac{\lambda_k}{\lambda_1 + \dots + \lambda_n}. \end{aligned}$$

■

Example 6.4 A Boston subway station services the red, green, and orange lines. Subways on each line arrive at the station according to three independent Poisson processes. On average, there is one red train every 10 minutes, one green train every 15 minutes, and one orange train every 20 minutes.

- (i) When you arrive at the station what is the probability that the first subway that arrives is for the green line?

- (ii) How long will you wait, on average, before some train arrives?
- (iii) You have been waiting 20 minutes for a red train and have watched three orange trains go by. What is the expected additional time you will wait for your subway?

Solution

- (i) Let X_G , X_R , and X_O denote, respectively, the times of the first green, red, and orange subways that arrive at the station. The event that the first subway is green is the event that X_G is the minimum of the three independent random variables. The desired probability is

$$P(\min(X_G, X_R, X_O) = X_G) = \frac{1/15}{1/10 + 1/15 + 1/20} = \frac{4}{13} = 0.31.$$

- (ii) The time of the first train arrival is the minimum of X_G , X_R , and X_O , which has an exponential distribution with parameter

$$\frac{1}{10} + \frac{1}{15} + \frac{1}{20} = \frac{13}{60}.$$

Thus, you will wait, on average $60/13 = 4.615$ minutes. A *quick* simulation gives

```
> sim <- replicate(10000,
+   min(rexp(1,1/10), rexp(1,1/15), rexp(1,1/20)))
> mean(sim)
[1] 4.588456
```

- (iii) Your waiting time is independent of the orange arrivals. By memorylessness of interarrival times, the additional waiting time for the red line has the same distribution as the original waiting time. You will wait, on average, 10 more minutes. ■

For a Poisson process, each arrival time S_n is a sum of n i.i.d. exponential interarrival times. A sum of i.i.d. exponential variables has a gamma distribution.

Arrival Times and Gamma Distribution

For $n = 1, 2, \dots$, let S_n be the time of the n th arrival in a Poisson process with parameter λ . Then, S_n has a gamma distribution with parameters n and λ . The density function of S_n is

$$f_{S_n}(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \text{ for } t > 0.$$

Mean and variance are

$$E(S_n) = \frac{n}{\lambda} \quad \text{and} \quad \text{Var}(S_n) = \frac{n}{\lambda^2}.$$

For a general gamma distribution, the parameter n does not have to be an integer. When it is, the distribution is sometimes called an *Erlang distribution*. Observe that if $n = 1$, the gamma distribution reduces to the exponential distribution with parameter λ .

Two ways to derive the gamma distribution for sums of i.i.d. exponentials are (i) by induction on n , using the fact that $S_n = S_{n-1} + X_n$, and (ii) by moment-generating functions. We leave both derivations to the exercises.

■ **Example 6.5** The times when goals are scored in hockey are modeled as a Poisson process in Morrison (1976). For such a process, assume that the average time between goals is 15 minutes.

- (i) In a 60-minute game, find the probability that a fourth goal occurs in the last 5 minutes of the game.
- (ii) Assume that at least three goals are scored in a game. What is the mean time of the third goal?

Solution The parameter of the hockey Poisson process is $\lambda = 1/15$.

- (i) The desired probability is

$$P(55 < S_4 \leq 60) = \frac{1}{6} \int_{55}^{60} (1/15)^4 t^3 e^{-t/15} dt = 0.068.$$

In R, the probability is found by typing

```
> pgamma(60, 4, 1/15) - pgamma(55, 4, 1/15)
[1] 0.06766216
```

- (ii) The desired expectation is

$$\begin{aligned} E(S_3 | S_3 < 60) &= \frac{1}{P(S_3 < 60)} \int_0^{60} t f_{S_3}(t) dt \\ &= \frac{1}{P(S_3 < 60)} \int_0^{60} t \frac{(1/15)^3 t^2 e^{-t/15}}{2} dt \\ &= \frac{25.4938}{0.7619} = 33.461 \text{ minutes.} \end{aligned}$$

■

6.3 INFINITESIMAL PROBABILITIES

A third way to define the Poisson process is based on an infinitesimal description of the distribution of points (e.g., events) in small intervals.

To state the new definition, we use *little-oh* notation. Write $f(h) = o(h)$ to mean that

$$\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0.$$

More generally, say that a function f is *little-oh of* g , and write $f(h) = o(g(h))$, to mean that

$$\lim_{h \rightarrow 0} \frac{f(h)}{g(h)} = 0.$$

Little-oh notation is often used when making order of magnitude statements about a function, or in referencing the remainder term of an approximation.

For example, the Taylor series expansion of e^h with remainder term is

$$e^h = 1 + h + \frac{h^2}{2} + \frac{h^3}{6} + \cdots = 1 + h + R(h),$$

where $R(h) = e^z h^2/2$, for some $z \in (-h, h)$. Since $R(h)/h = e^z h/2 \rightarrow 0$, as $h \rightarrow 0$, we can write

$$e^h = 1 + h + o(h).$$

Note that if two functions f and g are little-oh of h , then $f(h) + g(h) = o(h)$, since $(f(h) + g(h))/h \rightarrow 0$, as $h \rightarrow 0$. Similarly, if $f(h) = o(h)$, then $cf(h) = o(h)$, for any constant c . If $f(h) = o(1)$, then $f(h) \rightarrow 0$, as $h \rightarrow 0$.

Poisson Process—Definition 3

A Poisson process with parameter λ is a counting process $(N_t)_{t \geq 0}$ with the following properties:

1. $N_0 = 0$.
2. The process has stationary and independent increments.
3. $P(N_h = 0) = 1 - \lambda h + o(h)$.
4. $P(N_h = 1) = \lambda h + o(h)$.
5. $P(N_h > 1) = o(h)$.

Properties 3–5 essentially ensure that there cannot be infinitely many arrivals in a finite interval, and that in an infinitesimal interval there may occur at most one event.

It is straightforward to show that Definition 3 is a consequence of Definition 1. If N_h has a Poisson distribution with parameter λh , then

$$P(N_h = 0) = e^{-\lambda h} = 1 - \lambda h + o(h),$$

$$P(N_h = 1) = e^{-\lambda h} \lambda h = (1 - \lambda h + o(h)) \lambda h = \lambda h + o(h),$$

and

$$\begin{aligned} P(N_h > 1) &= 1 - P(N_h = 0) - P(N_h = 1) \\ &= 1 - (1 - \lambda h + o(h)) - (\lambda h + o(h)) \\ &= o(h). \end{aligned}$$

The converse, that Definition 3 implies Definition 1, is often shown by deriving and solving a differential equation. We will forego the rigorous proof, but give a heuristic explanation, which offers insight into the nature of Poisson arrivals.

Assume that Definition 3 holds. We need to show that N_t has a Poisson distribution with parameter λt .

Consider N_t , the number of points in the interval $[0, t]$. Partition $[0, t]$ into n subintervals each of length t/n . Properties 3–5 imply that for sufficiently large n , each subinterval will contain either 0 or 1 point with high probability. The chance that a small subinterval contains 2 or more points is negligible. See Figure 6.5.

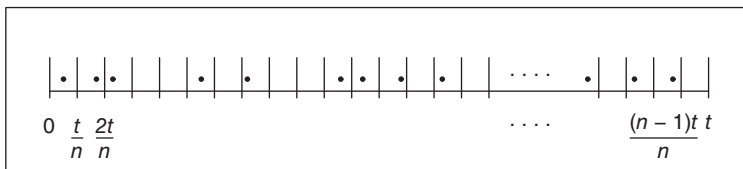


Figure 6.5 A partition of $[0, t]$ where each subinterval contains 0 or 1 point.

A subinterval has the form $((k-1)t/n, kt/n]$ for some $k = 1, \dots, n$. By stationary increments,

$$P(N_{kt/n} - N_{(k-1)t/n} = 1) = P(N_{t/n} = 1) = \frac{\lambda t}{n} + o\left(\frac{t}{n}\right), \text{ for all } k.$$

Furthermore, by independent increments, whether or not a point is contained in a particular subinterval is independent of the points in any other subinterval.

Hence, for large n , the outcomes in each subinterval can be considered a sequence of n i.i.d. Bernoulli trials, where the probability p_n that a point is contained in a subinterval is

$$p_n = \frac{\lambda t}{n} + o\left(\frac{t}{n}\right).$$

Thus, the number of points in $[0, t]$, being the sum of n i.i.d. Bernoulli trials, has a binomial distribution with parameters n and p_n .

The result that N_t has a Poisson distribution is obtained by letting n tend to infinity and appealing to the Poisson approximation of the binomial distribution. Since

$$np_n = n \left[\frac{\lambda t}{n} + o\left(\frac{t}{n}\right) \right] = \lambda t + n \left[o\left(\frac{t}{n}\right) \right] \rightarrow \lambda t > 0, \text{ as } n \rightarrow \infty,$$

the approximation holds, which gives that N_t has a Poisson distribution with parameter λt . Note that for the $o(t/n)$ term, since

$$\lim_{1/n \rightarrow 0} \frac{o(t/n)}{t/n} = 0, \text{ then equivalently } \lim_{n \rightarrow \infty} n[o(t/n)] = 0.$$

A statement and proof of the Poisson approximation of the binomial distribution is given in Appendix B, Section B.4.

Equivalence of Poisson Definitions

We have shown that Definitions 1 and 3 are equivalent. Here, we show that Definitions 1 and 2 are equivalent.

Assume Definition 2. That is, let $X_1, X_2, \dots$ be an i.i.d. sequence of exponential random variables with parameter λ . For each n , let $S_n = X_1 + \dots + X_n$, with $S_0 = 0$, and let $N_t = \max\{n : S_n \leq t\}$. We show N_t has a Poisson distribution with parameter λt .

Observe that for $k \geq 0$, $N_t = k$ if and only if the k th arrival occurs by time t and the $(k+1)$ th arrival occurs after t . That is, $S_k \leq t < S_k + X_{k+1}$. Since S_k is a function of $X_1, \dots, X_k$, S_k and X_{k+1} are independent random variables, and their joint density is the product of their marginal densities. This gives

$$f_{S_k, X_{k+1}}(s, x) = f_{S_k}(s) f_{X_{k+1}}(x) = \left(\frac{\lambda^k s^{k-1} e^{-\lambda s}}{(k-1)!} \right) \lambda e^{-\lambda x}, \text{ for } s, x > 0.$$

For $k \geq 0$,

$$\begin{aligned} P(N_t = k) &= P(S_k \leq t \leq S_k + X_{k+1}) \\ &= P(S_k \leq t, X_{k+1} \geq t - S_k) \\ &= \int_0^t \int_{t-s}^{\infty} f_{S_k, X_{k+1}}(s, x) dx ds \\ &= \int_0^t \int_{t-s}^{\infty} \left(\frac{\lambda^k s^{k-1} e^{-\lambda s}}{(k-1)!} \right) \lambda e^{-\lambda x} dx ds \\ &= \frac{\lambda^k}{(k-1)!} \int_0^t (s^{k-1} e^{-\lambda s}) e^{-\lambda(t-s)} ds \\ &= \frac{e^{-\lambda t} \lambda^k}{(k-1)!} \int_0^t s^{k-1} ds = \frac{e^{-\lambda t} (\lambda t)^k}{k!}, \end{aligned}$$

which gives the desired Poisson distribution.

The fact that the interarrival times of a Poisson process are memoryless means that the pattern of arrivals from an arbitrary time s onward behaves the same as the pattern of arrivals from time 0 onward. It follows that the number of arrivals in the interval $(s, s + t]$ has the same distribution as the number of arrivals in $(0, t]$. Stationary, as well as independent, increments are direct consequences.

Conversely, assume Definition 1. Consider the distribution of the first interarrival time X_1 . As shown at the beginning of Section 6.2 the distribution is exponential with parameter λ .

For X_2 , consider $P(X_2 > t | X_1 = s)$. If the first arrival occurs at time s , and the second arrival occurs more than t time units later, then there are no arrivals in the interval $(s, s + t]$. Conversely, if there are no arrivals in $(s, s + t]$ and $X_1 = s$, then $X_2 > t$. Thus,

$$\begin{aligned} P(X_2 > t | X_1 = s) &= P(N_{s+t} - N_s = 0 | X_1 = s) \\ &= P(N_{s+t} - N_s = 0) \\ &= P(N_t = 0) = e^{-\lambda t}, \text{ for } t > 0. \end{aligned}$$

The second equality is because of independent increments. It follows that X_1 and X_2 are independent, and the distribution of X_2 is exponential with parameter λ .

For the general case, consider $P(X_{k+1} > t | X_1 = x_1, X_2 = x_2, \dots, X_k = x_k)$. If the k th arrival occurs at time $x_1 + \dots + x_k$ and $X_{k+1} > t$, then there are no arrivals in the interval $(x_1 + \dots + x_k, x_1 + \dots + x_k + t]$. See Figure 6.6.

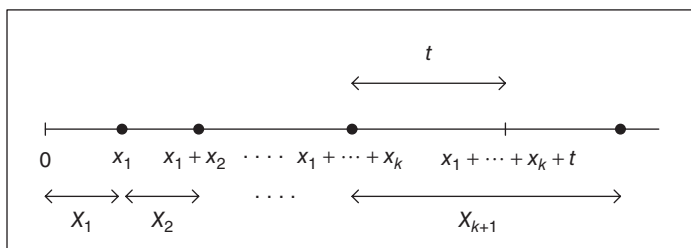


Figure 6.6 With $X_{k+1} > t$, there are no arrivals between $x_1 + \dots + x_k$ and $x_1 + \dots + x_k + t$.

We have that

$$\begin{aligned} P(X_{k+1} > t | X_1 = x_1, \dots, X_k = x_k) &= P(N_{x_1 + \dots + x_k + t} - N_{x_1 + \dots + x_k} = 0 | X_1 = x_1, \dots, X_k = x_k) \\ &= P(N_{x_1 + \dots + x_k + t} - N_{x_1 + \dots + x_k} = 0) \\ &= P(N_t = 0) = e^{-\lambda t}, \text{ for } t > 0. \end{aligned}$$

Hence, X_{k+1} is independent of $X_1, \dots, X_k$, and the distribution of X_{k+1} is exponential with parameter λ .

It follows that $X_1, X_2, \dots$ is an i.i.d. sequence of exponential random variables with parameter λ , which gives Definition 2.

The equivalence of all three definitions has been established.

6.4 THINNING, SUPERPOSITION

According to the United Nations Population Division, the worldwide sex ratio at birth is 108 boys to 100 girls. Thus, the probability that any birth is a boy is

$$p = \frac{108}{108 + 100} = 0.519.$$

That this probability is greater than one-half is said to be nature's way of balancing the fact that boys have a slightly higher risk than girls of not surviving birth.

Assume that babies are born on a maternity ward according to a Poisson process $(N_t)_{t \geq 0}$ with parameter λ . How can the number of male births and the number of female births be described?

Babies' sex is independent of each other. We can think of a male birth as the result of a coin flip whose heads probability is p . Assume that there are n births by time t . Then, the number of male births by time t is the number of heads in n i.i.d. coin flips, which has a binomial distribution with parameters n and p . Similarly the number of female births in $[0, t]$ has a binomial distribution with parameters n and $1 - p$.

Let M_t denote the number of male births by time t . Similarly define the number of female births F_t . Thus, $M_t + F_t = N_t$. The joint probability mass function of (M_t, F_t) is

$$\begin{aligned} P(M_t = m, F_t = f) &= P(M_t = m, F_t = f, N_t = m + f) \\ &= P(M_t = m, F_t = f | N_t = m + f) P(N_t = m + f) \\ &= P(M_t = m | N_t = m + f) P(N_t = m + f) \\ &= \frac{(m + f)!}{m! f!} p^m (1 - p)^f \frac{e^{-\lambda t} (\lambda t)^{m+f}}{(m + f)!} \\ &= \frac{p^m (1 - p)^f e^{-\lambda t(p + (1-p))} (\lambda t)^{m+f}}{m! f!} \\ &= \left(\frac{e^{-\lambda p t} (\lambda p t)^m}{m!} \right) \left(\frac{e^{-\lambda(1-p)t} (\lambda(1-p)t)^f}{f!} \right), \end{aligned}$$

for $m, f = 0, 1, \dots$. This shows that M_t and F_t are independent Poisson random variables with parameters $\lambda p t$ and $\lambda(1 - p)t$, respectively. In fact, each process $(M_t)_{t \geq 0}$ and $(F_t)_{t \geq 0}$ is a Poisson process, called a *thinned process*. It is not hard to show that both processes inherit stationary and independent increments from the original Poisson birth process.

The birth example with two thinned processes illustrates a general result.

Thinned Poisson Process

Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ . Assume that each arrival, independent of other arrivals, is marked as a *type k* event with probability p_k , for $k = 1, \dots, n$, where $p_1 + \dots + p_n = 1$. Let $N_t^{(k)}$ be the number of type k events in $[0, t]$. Then, $(N_t^{(k)})_{t \geq 0}$ is a Poisson process with parameter λp_k . Furthermore, the processes

$$(N_t^{(1)})_{t \geq 0}, \dots, (N_t^{(n)})_{t \geq 0}$$

are independent. Each process is called a *thinned Poisson process*.

Example 6.6 Consider the male and female birth processes. Assume that births occur on a maternity ward at the average rate of 2 births per hour.

- (i) On an 8-hour shift, what is the expectation and standard deviation of the number of female births?
- (ii) Find the probability that only girls were born between 2 and 5 p.m.
- (iii) Assume that five babies were born on the ward yesterday. Find the probability that two are boys.

Solution Let $(N_t)_{t \geq 0}$, $(M_t)_{t \geq 0}$, and $(F_t)_{t \geq 0}$ denote the overall birth, male, and female processes, respectively.

- (i) Female births form a Poisson process with parameter

$$\lambda(1 - p) = 2(0.481) = 0.962.$$

The number of female births on an 8-hour shift F_8 has a Poisson distribution with expectation

$$E(F_8) = \lambda(1 - p)8 = 2(0.481)8 = 7.696,$$

and standard deviation

$$SD(F_8) = \sqrt{7.696} = 2.774.$$

- (ii) The desired probability is $P(M_3 = 0, F_3 > 0)$. By independence,

$$\begin{aligned} P(M_3 = 0, F_3 > 0) &= P(M_3 = 0)P(F_3 > 0) \\ &= e^{-2(0.519)3} (1 - e^{-2(0.481)3}) \\ &= e^{-3.114} (1 - e^{-2.886}) = 0.042. \end{aligned}$$

- (iii) Conditional on there being five births in a given interval, the number of boys in that interval has a binomial distribution with parameters $n = 5$ and $p = 0.519$. The desired probability is

$$\frac{5!}{2!3!}(0.519)^2(0.481)^3 = 0.30.$$

■

Related to the thinned process is the *superposition* process obtained by merging, or adding, independent Poisson processes. We state the following intuitive result without proof:

Superposition Process

Assume that $(N_t^{(1)})_{t \geq 0}, \dots, (N_t^{(n)})_{t \geq 0}$ are n independent Poisson processes with respective parameters $\lambda_1, \dots, \lambda_n$. Let $N_t = N_t^{(1)} + \dots + N_t^{(n)}$, for $t \geq 0$. Then, $(N_t)_{t \geq 0}$ is a Poisson process with parameter $\lambda = \lambda_1 + \dots + \lambda_n$.

See Figure 6.7 to visualize the superposition of three independent Poisson processes.

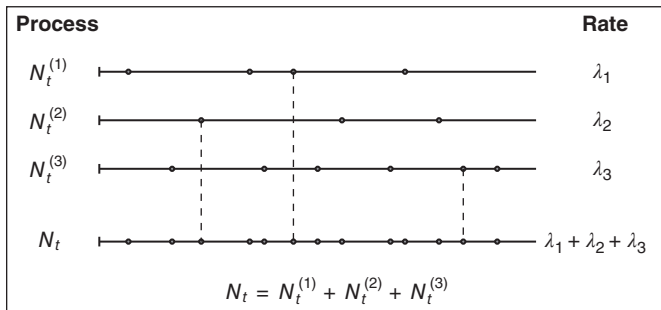


Figure 6.7 The N_t process is the superposition of $N_t^{(1)}$, $N_t^{(2)}$, and $N_t^{(3)}$.

■ **Example 6.7 (Oh my!)** In the land of Oz, sightings of lions, tigers, and bears each follow a Poisson process with respective parameters, $\lambda_L, \lambda_T, \lambda_B$, where the time unit is hours. Sightings of the three species are independent of each other.

- Find the probability that Dorothy will not see any animal in the first 24 hours from when she arrives in Oz.
- Dorothy saw three animals one day. Find the probability that each species was seen.

Solution

- (i) The process of animal sightings $(N_t)_{t \geq 0}$ is the superposition of three independent Poisson processes. Thus, it is a Poisson process with parameter $\lambda_L + \lambda_T + \lambda_B$. The desired probability is

$$P(N_{24} = 0) = e^{-24(\lambda_L + \lambda_T + \lambda_B)}.$$

- (ii) Let L_t , T_t , and B_t be the numbers of lions, tigers, and bears, respectively, seen by time t . The desired probability is

$$\begin{aligned} P(L_{24} = 1, B_{24} = 1, T_{24} = 1 | N_{24} = 3) \\ &= \frac{P(L_{24} = 1, B_{24} = 1, T_{24} = 1, N_{24} = 3)}{P(N_{24} = 3)} \\ &= \frac{P(L_{24} = 1, B_{24} = 1, T_{24} = 1)}{P(N_{24} = 3)} \\ &= \frac{P(L_{24} = 1)P(B_{24} = 1)P(T_{24} = 1)}{P(N_{24} = 3)} \\ &= \frac{(e^{-24\lambda_L} 24\lambda_L)(e^{-24\lambda_T} 24\lambda_T)(e^{-24\lambda_B} 24\lambda_B)}{e^{-24(\lambda_L + \lambda_T + \lambda_B)} (24(\lambda_L + \lambda_T + \lambda_B))^3 / 3!} \\ &= \frac{6\lambda_L \lambda_B \lambda_T}{(\lambda_L + \lambda_B + \lambda_T)^3}. \end{aligned}$$

■

Embedding and the Birthday Problem

Sometimes discrete problems can be solved by *embedding* them in continuous ones. The methods in this section were popularized in Blom and Holst (1991) for solving discrete balls-and-urn models, which involve sampling with and without replacement. We illustrate the method on the famous birthday problem. (If your probability class did not cover the birthday problem, you should ask for your money back.)

The classic birthday problem asks, “How many people must be in a room before the probability that some share a birthday, ignoring year and leap days, is at least 50%?”

The probability that two people have the same birthday is 1 minus the probability that no one shares a birthday, which is

$$p_k = 1 - \prod_{i=1}^k \frac{366 - i}{365} = 1 - \frac{365!}{(365 - k)! 365^k}.$$

One finds that $p_{22} = 0.476$ and $p_{23} = 0.507$. Thus, 23 people are needed.

Consider this variant of the birthday problem, assuming a random person’s birthday is uniformly distributed on the 365 days of the year. People enter a room one by

one. How many people are in the room the first time that two people share the same birthday? Let K be the desired number. We show how to find the mean and standard deviation of K by embedding.

Consider a continuous-time version of the previous question. People enter a room according to a Poisson process $(N_t)_{t \geq 0}$ with rate $\lambda = 1$. Each person is independently *marked* with one of 365 birthdays, where all birthdays are equally likely. The procedure creates 365 thinned Poisson processes, one for each birthday. Each of the 365 processes are independent, and their superposition gives the process of people entering the room. See Figure 6.8.

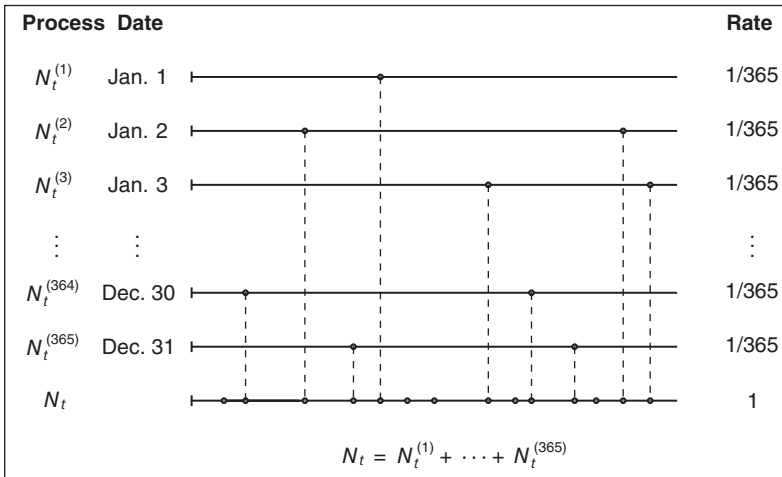


Figure 6.8 Embedding the birthday problem in a superposition of Poisson processes.

Let $X_1, X_2, \dots$, be the interarrival sequence for the process of people entering the room. The X_i are i.i.d. exponential random variables with mean 1. Let T be the first time when two people in the room share the same birthday. Then,

$$T = \sum_{i=1}^K X_i. \quad (6.3)$$

Equation (6.3) relates the interarrival times $X_1, X_2, \dots$, the continuous time T , and the discrete count K .

The X_i are independent of K . The random variable T is represented as a random sum of random variables. By results for such sums (see Example 1.27),

$$E[T] = E[K] E[X_1] = E[K].$$

For each $k = 1, \dots, 365$, let Z_k be the time when the second person marked with birthday k enters the room. Then, the first time two people in the room have the same birthday is $T = \min_{1 \leq k \leq 365} Z_k$. Each Z_k , being the arrival time of the second event of

a Poisson process, has a gamma distribution with parameters $n = 2$ and $\lambda = 1/365$, with density

$$f(t) = \frac{te^{-t/365}}{365^2}, \text{ for } t > 0,$$

and cumulative distribution function

$$P(Z_1 \leq t) = \int_0^t \frac{se^{-s/365}}{365^2} ds = 1 - \frac{e^{-t/365}(365 + t)}{365}.$$

This gives

$$\begin{aligned} P(T > t) &= P(\min(Z_1, \dots, Z_{365}) > t) \\ &= P(Z_1 > t, \dots, Z_{365} > t) \\ &= P(Z_1 > t)^{365} = \left(1 + \frac{t}{365}\right)^{365} e^{-t}, \text{ for } t > 0. \end{aligned}$$

The desired birthday expectation is

$$E(K) = E(T) = \int_0^\infty P(T > t) dt = \int_0^\infty \left(1 + \frac{t}{365}\right)^{365} e^{-t} dt. \quad (6.4)$$

The second equality makes use of a general result for the expectation of a positive, continuous random variable. (For reference, see Exercise 1.21.) The last integral of Equation (6.4) is difficult to solve exactly. It can be estimated using a Taylor series approximation. A numerical software package finds $E(K) = 24.617$ and $\text{Var}(K) = 779.23$, with standard deviation 27.91.

We invite the reader to use embedding to find the expected number of people needed for *three* people to share the same birthday. See Exercise 6.19.

6.5 UNIFORM DISTRIBUTION

It is common to think of a Poisson process as modeling a *completely random* distribution of events, or points, on the positive number line. Although it is not possible to have a uniform distribution on $[0, \infty)$, or any unbounded interval, there is nevertheless a strong connection between a Poisson process and the uniform distribution.

If a Poisson process contains exactly n events in an interval $[0, t]$, then the unordered *locations*, or times, of those events are uniformly distributed on the interval.

For the case of one event, the assertion is easily shown. Consider the distribution of the time of the first arrival, conditional on there being one arrival by time t . For $0 \leq s \leq t$,

$$\begin{aligned} P(S_1 \leq s | N_t = 1) &= \frac{P(S_1 \leq s, N_t = 1)}{P(N_t = 1)} = \frac{P(N_s = 1, N_t = 1)}{P(N_t = 1)} \\ &= \frac{P(N_s = 1, N_t - N_s = 0)}{P(N_t = 1)} \\ &= \frac{P(N_s = 1)P(N_{t-s} = 0)}{P(N_t = 1)} \\ &= \frac{e^{-\lambda s} \lambda s e^{-\lambda(t-s)}}{e^{-\lambda t} \lambda t} = \frac{s}{t}, \end{aligned}$$

which is the cumulative distribution function of the uniform distribution on $[0, t]$.

To discuss the case of more than one arrival in $[0, t]$, we introduce the topic of *order statistics*. Let $U_1, \dots, U_n$ be an i.i.d. sequence of random variables uniformly distributed on $[0, t]$. Their joint density function is

$$f_{U_1, \dots, U_n}(u_1, \dots, u_n) = \frac{1}{t^n}, \quad \text{for } 0 \leq u_1, \dots, u_n \leq t.$$

Arrange the U_i in increasing order $U_{(1)} \leq U_{(2)} \leq \dots \leq U_{(n)}$, where $U_{(k)}$ is the k th smallest of the U_i . The ordered sequence $(U_{(1)}, \dots, U_{(n)})$ is called the *order statistics* of the original sequence.

The joint density function of the order statistics is

$$f_{U_{(1)}, \dots, U_{(n)}}(u_1, \dots, u_n) = \frac{n!}{t^n}, \quad \text{for } 0 \leq u_1 < \dots < u_n \leq t.$$

We will not prove this rigorously, but give an intuitive argument. Assume that

$$U_{(1)} = u_1, \dots, U_{(n)} = u_n, \quad \text{for } 0 < u_1 < \dots < u_n < t.$$

Consider a sample of n independent uniform random variables on $[0, t]$. There are $n!$ such samples that would give rise to these order statistic values, as there are $n!$ orderings of the n distinct numbers $u_1, \dots, u_n$. The value of the joint density for each of these uniform samples is equal to $1/t^n$. Hence, the infinitesimal probability

$$\begin{aligned} f_{U_{(1)}, \dots, U_{(n)}}(u_1, \dots, u_n) du_1 \cdots du_n &= n! f_{U_1, \dots, U_n}(u_1, \dots, u_n) du_1 \cdots du_n \\ &= \frac{n!}{t^n} du_1 \cdots du_n, \end{aligned}$$

which establishes the claim.

We can now describe the joint distribution of the arrival times in a Poisson process, conditional on the number of arrivals.

Arrival Times and Uniform Distribution

Let $S_1, S_2, \dots$, be the arrival times of a Poisson process with parameter λ . Conditional on $N_t = n$, the joint distribution of $(S_1, \dots, S_n)$ is the distribution of the order statistics of n i.i.d. uniform random variables on $[0, t]$. That is, the joint density function of $S_1, \dots, S_n$ is

$$f(s_1, \dots, s_n) = \frac{n!}{t^n}, \text{ for } 0 < s_1 < \dots < s_n < t. \quad (6.5)$$

Equivalently, let $U_1, \dots, U_n$ be an i.i.d. sequence of random variables uniformly distributed on $[0, t]$. Then, conditional on $N_t = n$,

$$(S_1, \dots, S_n) \text{ and } (U_{(1)}, \dots, U_{(n)})$$

have the same distribution.

Proof. For jointly distributed random variables $S_1, \dots, S_n$ with joint density f ,

$$f(s_1, \dots, s_n) = \lim_{\epsilon_1 \rightarrow 0} \dots \lim_{\epsilon_n \rightarrow 0} \frac{P(s_1 \leq S_1 \leq s_1 + \epsilon_1, \dots, s_n \leq S_n \leq s_n + \epsilon_n)}{\epsilon_1 \dots \epsilon_n}.$$

To establish Equation (6.5), assume that $0 < s_1 < s_2 < \dots < s_n < t$ and consider the event

$$\{s_1 \leq S_1 \leq s_1 + \epsilon_1, \dots, s_n \leq S_n \leq s_n + \epsilon_n\},$$

given that there are exactly n arrivals in $[0, t]$. For $\epsilon_1, \dots, \epsilon_n$ sufficiently small, this is the event that each of the intervals $(s_i, s_i + \epsilon_i]$ contains exactly one arrival, and no arrivals occur elsewhere in $[0, t]$. By stationary and independent increments,

$$\begin{aligned} & P(s_1 \leq S_1 \leq s_1 + \epsilon_1, \dots, s_n \leq S_n \leq s_n + \epsilon_n | N_t = n) \\ &= \frac{P(N_{s_1 + \epsilon_1} - N_{s_1} = 1, \dots, N_{s_n + \epsilon_n} - N_{s_n} = 1, N_t = n)}{P(N_t = n)} \\ &= \frac{\lambda \epsilon_1 e^{-\lambda \epsilon_1} \dots \lambda \epsilon_n e^{-\lambda \epsilon_n} e^{-\lambda(t - \epsilon_1 - \dots - \epsilon_n)}}{e^{-\lambda t} (\lambda t)^n / n!} \\ &= \frac{n! \epsilon_1 \dots \epsilon_n}{t^n}. \end{aligned}$$

Dividing by $\epsilon_1 \dots \epsilon_n$, and letting each $\epsilon_i \rightarrow 0$, gives the result. ■

There is a lot of information contained in Equation (6.5). From this, one can obtain joint distributions of any subset of the arrival times, including the marginal distributions. For instance, the *last (first)* arrival time, conditional on there being n arrivals in $[0, t]$, has the same distribution as the maximum (minimum) of n independent uniform random variables on $(0, t)$.

■ **Example 6.8** Starting at time $t = 0$, patrons arrive at a restaurant according to a Poisson process with rate 20 customers per hour.

- (i) Find the probability that the 60th customer arrives in the interval $[2.9, 3]$.
- (ii) If 60 people arrive at the restaurant by time $t = 3$, find the probability that the 60th customer arrives in the interval $[2.9, 3]$.

Solution

- (i) The time of the 60th arrival S_{60} has a gamma distribution with parameters $n = 60$ and $\lambda = 20$. The desired probability is $P(2.9 < S_{60} < 3)$. In R, type

```
> pgamma(3, 60, 20) - pgamma(2.9, 60, 20)
[1] 0.1034368
```

- (ii) Given $N_3 = 60$, the arrival time of the 60th customer has the same distribution as the maximum M of 60 i.i.d. random variables uniformly distributed on $(0, 3)$. The desired probability is

$$\begin{aligned}
 P(2.9 < S_{60} < 3 | N_3 = 60) &= P(2.9 < M < 3) = 1 - P(M \leq 2.9) \\
 &= 1 - P(U_1 \leq 2.9, \dots, U_{60} \leq 2.9) \\
 &= 1 - P(U_1 \leq 2.9)^{60} \\
 &= 1 - \left(\frac{2.9}{3}\right)^{60} \\
 &= 1 - 0.131 = 0.869.
 \end{aligned}$$

■

■ **Example 6.9** Concert-goers arrive at a show according to a Poisson process with parameter λ . The band starts playing at time t . The k th person to arrive in $[0, t]$ waits $t - S_k$ time units for the start of the concert, where S_k is the k th arrival time. Find the expected total waiting time of concert-goers who arrive before the band starts.

Solution The desired expectation is $E\left(\sum_{k=1}^{N_t} (t - S_k)\right)$. Conditioning on N_t ,

$$\begin{aligned}
 E\left(\sum_{k=1}^{N_t} (t - S_k)\right) &= \sum_{n=1}^{\infty} E\left(\sum_{k=1}^n (t - S_k) | N_t = n\right) P(N_t = n) \\
 &= \sum_{n=1}^{\infty} E\left(tn - \sum_{k=1}^n S_k | N_t = n\right) P(N_t = n) \\
 &= \sum_{n=1}^{\infty} \left(tn - E\left(\sum_{k=1}^n S_k | N_t = n\right)\right) P(N_t = n) \\
 &= \lambda t^2 - \sum_{n=1}^{\infty} E\left(\sum_{k=1}^n S_k | N_t = n\right) P(N_t = n).
 \end{aligned}$$

Conditional on n arrivals in $[0, t]$, $S_1 + \cdots + S_n$ has the same distribution as the sum of the uniform order statistics. Furthermore, $\sum_{k=1}^n U_{(k)} = \sum_{k=1}^n U_k$. This gives

$$\begin{aligned} \sum_{n=1}^{\infty} E\left(\sum_{k=1}^n S_k | N_t = n\right) P(N_t = n) &= \sum_{n=1}^{\infty} E\left(\sum_{k=1}^n U_{(k)}\right) P(N_t = n) \\ &= \sum_{n=1}^{\infty} E\left(\sum_{k=1}^n U_k\right) P(N_t = n) \\ &= \sum_{n=1}^{\infty} \frac{nt}{2} P(N_t = n) \\ &= \frac{\lambda t^2}{2}. \end{aligned}$$

The desired expectation is

$$E\left(\sum_{k=1}^{N_t} (t - S_k)\right) = \lambda t^2 - \frac{\lambda t^2}{2} = \frac{\lambda t^2}{2}.$$

■

Example 6.10 Students enter a campus building according to a Poisson process $(N_t)_{t \geq 0}$ with parameter λ . The times spent by each student in the building are i.i.d. random variables with continuous cumulative distribution function $F(t)$. Find the probability mass function of the number of students in the building at time t , assuming there are no students in the building at time 0.

Solution Let B_t denote the number of students in the building at time t . Conditioning on N_t ,

$$\begin{aligned} P(B_t = k) &= \sum_{n=k}^{\infty} P(B_t = k | N_t = n) P(N_t = n) \\ &= \sum_{n=k}^{\infty} P(B_t = k | N_t = n) \frac{e^{-\lambda t} (\lambda t)^n}{n!}. \end{aligned}$$

Assume that n students enter the building by time t , with arrival times $S_1, \dots, S_n$. Let Z_k be the length of time spent in the building by the k th student, for $1 \leq k \leq n$. Then, $Z_1, \dots, Z_n$ are i.i.d. random variables with cdf F , and students leave the building at times $S_1 + Z_1, \dots, S_n + Z_n$. There are k students in the building at time t if and only

if k of the departure times $S_1 + Z_1, \dots, S_n + Z_n$ exceed t . This gives

$$\begin{aligned}
 P(B_t = k | N_t = n) &= P(k \text{ of the } S_1 + Z_1, \dots, S_n + Z_n \text{ exceed } t | N_t = n) \\
 &= P(k \text{ of the } U_{(1)} + Z_1, \dots, U_{(n)} + Z_n \text{ exceed } t) \\
 &= P(k \text{ of the } U_1 + Z_1, \dots, U_n + Z_n \text{ exceed } t) \\
 &= \binom{n}{k} p^k (1-p)^{n-k},
 \end{aligned}$$

where

$$p = P(U_1 + Z_1 > t) = \frac{1}{t} \int_0^t P(Z_1 > t - x) dx = \frac{1}{t} \int_0^t [1 - F(x)] dx.$$

This gives

$$\begin{aligned}
 P(B_t = k) &= \sum_{n=k}^{\infty} \binom{n}{k} p^k (1-p)^{n-k} e^{-\lambda t} \frac{(\lambda t)^n}{n!} \\
 &= \frac{p^k (\lambda t)^k}{k!} \sum_{n=k}^{\infty} \frac{(1-p)^{n-k} (\lambda t)^{n-k}}{(n-k)!} \\
 &= \frac{p^k (\lambda t)^k}{k!} e^{\lambda(1-p)t} \\
 &= \frac{e^{-\lambda p t} (\lambda p t)^k}{k!}, \text{ for } k = 0, 1, \dots
 \end{aligned}$$

That is, B_t has a Poisson distribution with parameter λp , where

$$p = \int_0^t [1 - F(x)] dx.$$

■

Simulation

Results for arrival times and the uniform distribution offer a new method for simulating a Poisson process with parameter λ on an interval $[0, t]$:

1. Simulate the number of arrivals N in $[0, t]$ from a Poisson distribution with parameter λt .
2. Generate N i.i.d. random variables uniformly distributed on $(0, t)$.
3. Sort the variables in increasing order to give the Poisson arrival times.

R: Simulating a Poisson Process on $[0, t]$

Following is a simulation of the arrival times of a Poisson process with parameter $\lambda = 1/2$ on $[0, 30]$.

```
# poissonsimsim.R
> t <- 30
> lambda <- 1/2
> N <- rpois(1, lambda*t)
> unifs <- runif(N, 0, t)
> arrivals <- sort(unifs)
> arrivals
[1] 8.943 9.835 11.478 12.039 16.009 17.064 17.568
[8] 17.696 18.663 22.961 24.082 24.440 28.250
```

6.6 SPATIAL POISSON PROCESS

The spatial Poisson process is a model for the distribution of events, or *points*, in two- or higher-dimensional space. Such processes have been used to model the location of trees in a forest, galaxies in the night sky, and cancer clusters across the United States. For $d \geq 1$ and $A \subseteq \mathbb{R}^d$, let N_A denote the number of points in the set A . Write $|A|$ for the *size* of A (e.g., area in $\mathbb{R}^2$, volume in $\mathbb{R}^3$).

Spatial Poisson Process

A collection of random variables $(N_A)_{A \subseteq \mathbb{R}^d}$ is a spatial Poisson process with parameter λ if

1. for each bounded set $A \subseteq \mathbb{R}^d$, N_A has a Poisson distribution with parameter $\lambda|A|$.
2. whenever A and B are disjoint sets, N_A and N_B are independent random variables.

Observe how the spatial Poisson process generalizes the regular one-dimensional Poisson process. Property 1 gives the analogue of stationary increments, where the *size* of an interval is the length of the interval. Property 2 is the counterpart of independent increments.

 **Example 6.11** A spatial Poisson process in the plane has parameter $\lambda = 1/2$. Find the probability that a disk of radius 2 centered at $(3, 4)$ contains exactly 5 points.

Solution Let C denote the disk. Then, $|C| = \pi r^2 = 4\pi$. The desired probability is

$$P(N_C = 5) = \frac{e^{-\lambda|C|}(\lambda|C|)^5}{5!} = \frac{e^{-2\pi}(2\pi)^5}{5!} = 0.152.$$

■

The uniform distribution arises for the spatial process in a similar way to how it does for the one-dimensional Poisson process. Given a bounded set $A \subseteq \mathbb{R}^d$, then conditional on there being n points in A , the locations of the points are uniformly distributed in A . For this reason, a spatial Poisson process is sometimes called a model of *complete spatial randomness*.

To simulate a spatial Poisson process with parameter λ on a bounded set A , first simulate the number of points N in A according to a Poisson distribution with parameter $\lambda|A|$. Then, generate N points uniformly distributed in A .

Four realizations of a spatial Poisson process with parameter $\lambda = 100$ on the square $[0, 1] \times [0, 1]$ are shown in Figure 6.9. The circle C inside the square is centered at $(0.7, 0.7)$ with radius $r = 0.2$. The simulation was repeated 100,000 times, counting the number of points in the circle at each iteration. Table 6.1 shows the numerical results. The distribution of counts is seen to be close to the expected counts for a Poisson distribution with mean $\lambda|C| = 100\pi(0.2)^2 = 12.567$.

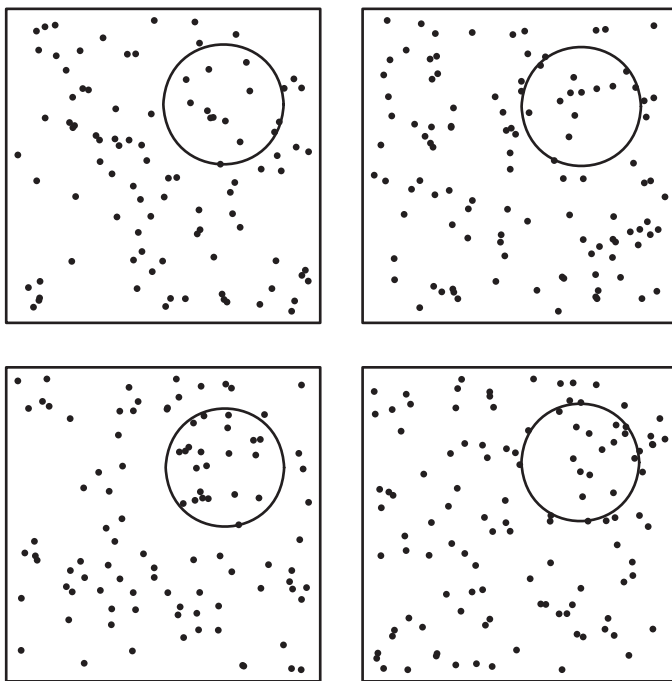


Figure 6.9 Samples of a spatial Poisson process.

TABLE 6.1 Number of Points in a Circle of Radius $r = 0.2$ for a Spatial Poisson Process with $\lambda = 100$.

Counts	0–4	5–9	10–14	15–19	20–24	25–29
Observed	522	19200	52058	24975	3135	106
Expected	510.0	19130.0	522215.6	24941.5	3075.00	125.9

Simulation is based on 100,000 trials.

R: Simulating a Spatial Poisson Process

```
# spatialPoisson.R
> lambda <- 100
> squarearea <- 1
> trials <- 100000
> simlist <- numeric(trials)
> for (i in 1:trials) {
+   N <- rpois(1,lambda*squarearea)
+   xpoints <- runif(N,0,1)
+   ypoints <- runif(N,0,1)
+   ct <- sum(((xpoints-0.7)^2+(ypoints-0.7)^2)<=0.2^2)
+   simlist[i] <- ct } # number of points in circle
> mean(simlist)
[1] 12.57771
> var(simlist)
[1] 12.57435
> # Compare with theoretical mean and variance
> lambda*pi*(0.2)^2
[1] 12.56637
```

A spatial Poisson process is a special case of a *point process*, which is a general model for the distribution of points in space. There is an abundance of applications of point processes, which include models that incorporate clustering, attraction, repulsion, time and space dependence, and so on. Often one wants to measure how close or far a given point pattern is from complete spatial randomness, that is, from a spatial Poisson process. A common measure is the *nearest-neighbor distance* defined to be the distance between an arbitrary point and the point of the process closest to it.

Consider a spatial Poisson process in $\mathbb{R}^2$ with parameter λ . Let x denote a fixed point in the plane. Let D be the distance from x to its nearest neighbor. The event $\{D > t\}$ occurs if and only if there are no points in the circle centered at x of radius t . Let C_x denote such a circle. Then,

$$P(D > t) = P\left(N_{C_x} = 0\right) = e^{-\lambda|C_x|} = e^{-\lambda\pi t^2}, \text{ for } t > 0.$$

Differentiating gives the density function for the nearest-neighbor distance

$$f_D(t) = e^{-\lambda\pi t^2} 2\lambda\pi t, \text{ for } t > 0,$$

with mean and variance

$$E(D) = \frac{1}{2\sqrt{\lambda}} \quad \text{and} \quad \text{Var}(D) = \frac{4 - \pi}{4\pi\lambda}.$$

■ **Example 6.12 (South Carolina swamp forest)** Jones et al. (1994) introduce data for the locations of 630 trees, including 91 cypress trees, in a swamp hardwood forest in South Carolina for the purpose of studying tree population dynamics. See Figure 6.10. The data are contained in a $200 \times 50 \text{ m}^2$ area. The average nearest-neighbor distance for all tree locations is 1.990 m. For the cypress trees it is 5.08 m.

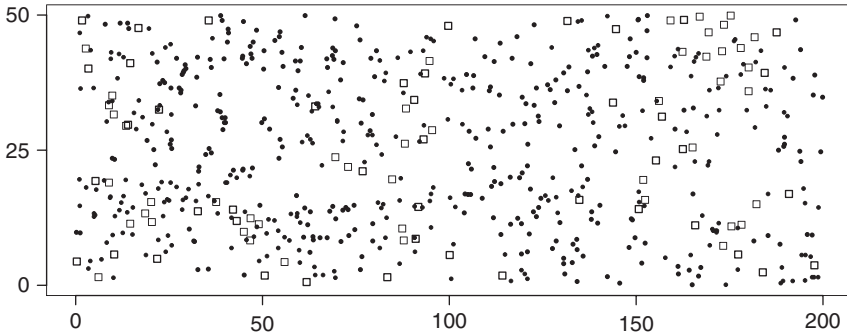


Figure 6.10 Plot of tree locations in a hardwood swamp in South Carolina. Squares are locations of cypress trees and dots locations of any other species. *Source:* Dixon (2012).

Researchers are interested in whether there is evidence of clustering. In a spatial Poisson process, with points distributed at the rate of 630 per $10,000 \text{ m}^2$, the expected nearest-neighbor distance is

$$E(D) = \frac{1}{2\sqrt{630/10000}} = 1.992 \text{ m},$$

with standard error

$$SD(D)/\sqrt{n} = \sqrt{\frac{4 - \pi}{4\pi 630/10000}} / \sqrt{630} = 0.041.$$

For the 91 cypress trees, a model of complete spatial randomness would yield

$$E(D) = \frac{1}{2\sqrt{91/100000}} = 5.241 \text{ m},$$

with standard error

$$SD(D)/\sqrt{n} = \sqrt{\frac{4 - \pi}{4\pi 630/10000}} / \sqrt{91} = 1.041 \text{ m.}$$

As measured by nearest-neighbor distance, the data do not show evidence of clustering. We note that researchers were able to detect some small evidence of clustering for these data by using more sophisticated spatial statistic tools. ■

6.7 NONHOMOGENEOUS POISSON PROCESS

In a Poisson process, arrivals occur at a constant rate, independent of time. However, for many applications this is an unrealistic assumption. Consider lunch time at a college cafeteria. The doors open at 11 a.m. Students arrive at an increasing rate until the noon peak hour. Then, the rate stays constant for 2 hours, after which it declines until 3 p.m., when the cafeteria closes.

Such activity can be modeled by a nonhomogeneous Poisson process with rate $\lambda = \lambda(t)$, which depends on t . Such a rate *function* is called the *intensity function*.

Nonhomogeneous Poisson Process

A counting process $(N_t)_{t \geq 0}$ is a *nonhomogeneous Poisson process with intensity function* $\lambda(t)$, if

1. $N_0 = 0$.
2. For all $t > 0$, N_t has a Poisson distribution with mean

$$E(N_t) = \int_0^t \lambda(x) dx.$$

3. For $0 \leq q < r \leq s < t$, $N_r - N_q$ and $N_t - N_s$ are independent random variables.

A nonhomogeneous Poisson process has independent increments, but not necessarily stationary increments. It can be shown that for $0 < s < t$, $N_t - N_s$ has a Poisson distribution with parameter $\int_s^t \lambda(x) dx$. If $\lambda(t) = \lambda$ is constant, we obtain the regular Poisson process with parameter λ .

■ **Example 6.13** Students arrive at the cafeteria for lunch according to a nonhomogeneous Poisson process. The arrival rate increases linearly from 100 to 200 students per hour between 11 a.m. and noon. The rate stays constant for the next 2 hours, and then decreases linearly down to 100 from 2 to 3 p.m. Find the probability that there are at least 400 people in the cafeteria between 11:30 a.m. and 1:30 p.m.

Solution The intensity function is

$$\lambda(t) = \begin{cases} 100 + 100t, & 0 \leq t \leq 1, \\ 200, & 1 < t \leq 3, \\ 500 - 100t, & 3 \leq t < 4, \end{cases}$$

where t represents hours past 11 a.m. See Figure 6.11.

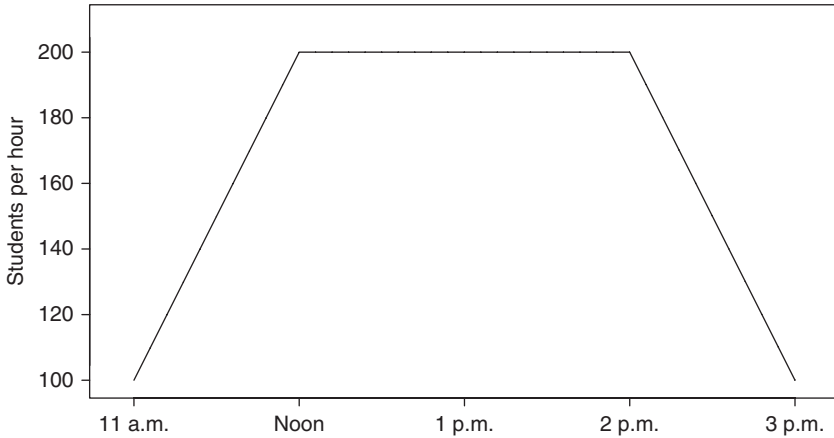


Figure 6.11 Intensity function.

The desired probability is $P(N_{2.5} - N_{0.5} \geq 400)$, where $N_{2.5} - N_{0.5}$ has a Poisson distribution with mean

$$E(N_{2.5} - N_{0.5}) = \int_{0.5}^{2.5} \lambda(t) dt = \int_{0.5}^1 (100 + 100t) dt + \int_1^{2.5} 200 dt = 387.5.$$

Then,

$$P(N_{2.5} - N_{0.5} \geq 400) = 1 - \sum_{k=0}^{399} \frac{e^{-387.5} (387.5)^k}{k!} = 0.269.$$

■

■ **Example 6.14** In reliability engineering one is concerned with the probability that a system is working during an interval of time. A common model for failure times is a nonhomogeneous Poisson process with intensity function of the form

$$\lambda(t) = \alpha \beta t^{\beta-1},$$

where $\alpha, \beta > 0$ are parameters, and t represents the age of the system. At $\beta = 1$, the model reduces to a homogeneous Poisson process with parameter α . If the system starts at $t = 0$, the expected number of failures after t time units is

$$E(N_t) = \int_0^t \lambda(x) dx = \int_0^t \alpha \beta x^{\beta-1} dx = \alpha t^\beta.$$

Because of the power law form of the mean failure time, the process is sometimes called a *power law Poisson process*.

Of interest, is the *reliability* $R(t)$, defined as the probability that a system, which starts at time t , is operational up through time $t + c$, for some constant c , that is, the probability of no failures in the interval $(t, t + c]$. This gives

$$R(t) = P(N_{t+c} - N_t = 0) = e^{-\int_t^{t+c} \lambda(x) dx} = e^{-\int_t^{t+c} \alpha \beta x^{\beta-1} dx} = e^{-\alpha((t+c)^\beta - t^\beta)}.$$

■

6.8 PARTING PARADOX

How wonderful that we have met with a paradox. Now we have some hope of making progress.

—Niels Bohr

The following classic is based on Feller (1968). Buses arrive at a bus stop according to a Poisson process. The time between buses, on average, is 10 minutes. Lisa gets to the bus stop at time t . How long can she expect to wait for a bus?

Here are two possible answers:

- (i) By memorylessness, the time until the next bus is exponentially distributed with mean 10 minutes. Lisa will wait, on average, 10 minutes.
- (ii) Lisa arrives at some time between two consecutive buses. The expected time between consecutive buses is 10 minutes. By symmetry, her expected waiting time should be half that, or 5 minutes.

Paradoxically, *both* answers have some truth to them! On the one hand, the time until the next bus will be shown to have an exponential distribution with mean 10 minutes. But the *backwards* time to the previous bus is almost exponential as well, with mean close to 10 minutes. Thus, the time when Lisa arrives at the bus stop is a point in an interval whose length is about 20 minutes. And the argument in (ii) essentially holds. By symmetry, her expected waiting time should be half that, or 10 minutes.

The surprising result is that the interarrival time of the buses before and after Lisa's arrival is about 20 minutes. And yet the expected interarrival time for buses is 10 minutes!

R : Waiting Time Paradox

A Poisson process with parameter $\lambda = 1/10$ is generated on $[0, 200]$. Lisa arrives at the bus stop at time $t = 50$. Simulation shows her average wait time is about 10 minutes.

```
# waitingparadox.R
> mytime <- 50
> lambda <- 1/10
> trials <- 10000
> simlist <- numeric(trials)
> for (i in 1:trials) {
+   N <- rpois(1, 300*lambda)
+   arrivals <- sort(runif(N, 0, 300))
+   wait <- arrivals[arrivals > mytime][1] - mytime
+   simlist[i] <- wait }
> mean(simlist)
[1] 10.04728
```

To explain the paradox, consider the process of bus arrivals. The rate of one arrival per 10 minutes is an average. The time between buses is random, and buses may arrive one right after the other, or there may be a long time between consecutive buses. When Lisa gets to the bus stop, she is more likely to get there during a longer interval between buses than a shorter interval.

To illustrate the idea, pick a random number between 1 and 200. Do it now before reading on. Now look at Figure 6.12, which gives arrival times for a Poisson process with parameter $\lambda = 1/10$ on $[0, 200]$. Find your number. Is your number in a short interval (length less than 10) or a long interval (length greater than 10)? Most readers will find their number in a long interval.

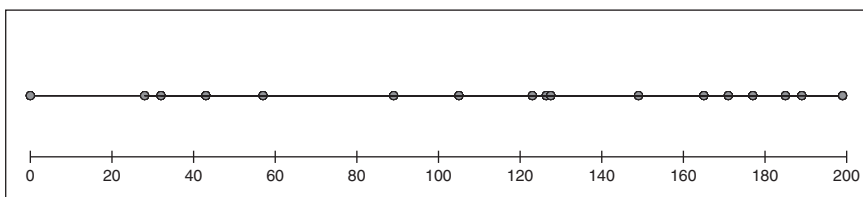


Figure 6.12 Pick a number from 0 to 200. Is your number in a short or long interval?

This example illustrates the phenomenon of *length-biased* or *size-biased* sampling. If you reach into a bag containing pieces of string of different lengths and pick a string *at random*, you tend to pick a longer rather than a shorter piece. Bus interarrival times are analogous to lengths of string.

Here is another example of size-biased sampling. Suppose you want to estimate how much time people spend exercising at the gym. If you go to the gym to survey

people at random, you are likely to get a biased estimate, as you are more likely to sample people who work out a lot. Those who rarely go to the gym are not likely to be there when you go!

For the bus waiting problem, the expected length of an interarrival time, *which contains a fixed time t* , is larger—about twice as large—than the average interval length between buses.

Here is an exact analysis. Consider a fixed $t > 0$. The time of the last bus before t is S_{N_t} . The time of the next bus after t is S_{N_t+1} . The expected length of the interval containing t is

$$E(S_{N_t+1} - S_{N_t}) = E(S_{N_t+1}) - E(S_{N_t}).$$

For $E(S_{N_t+1})$, condition on N_t . Consider

$$E(S_{N_t+1} | N_t = k) = E(S_{k+1} | N_t = k) = E(S_{k+1}) = \frac{k+1}{\lambda}.$$

The second equality is because the $(k+1)$ th arrival occurs after time t , and is thus independent of N_t . It follows that $E(S_{N_t+1} | N_t) = (N_t + 1)/\lambda$. By the law of total expectation,

$$E(S_{N_t+1}) = E(E(S_{N_t+1} | N_t)) = E\left(\frac{N_t + 1}{\lambda}\right) = \frac{\lambda t + 1}{\lambda} = t + \frac{1}{\lambda}. \quad (6.6)$$

For $E(S_{N_t})$, we have that $E(S_{N_t} | N_t = k) = E(S_k | N_t = k)$. Conditional on $N_t = k$, the k th arrival time has the same distribution as the maximum of k i.i.d. uniform random variables distributed on $(0, t)$. We leave it to the reader to show that this expectation is equal to $tk/(k+1)$. That is, $E(S_{N_t} | N_t = k) = tk/(k+1)$ and thus

$$E(S_{N_t} | N_t) = tN_t/(N_t + 1) = t - t/(N_t + 1).$$

This gives

$$E(S_{N_t}) = E(E(S_{N_t} | N_t)) = E\left(t - \frac{t}{N_t + 1}\right) = t - tE\left(\frac{1}{N_t + 1}\right). \quad (6.7)$$

We find

$$\begin{aligned} E\left(\frac{1}{N_t + 1}\right) &= \sum_{k=0}^{\infty} \left(\frac{1}{k+1}\right) \frac{e^{-\lambda t} (\lambda t)^k}{k!} \\ &= \frac{e^{-\lambda t}}{\lambda t} \sum_{k=0}^{\infty} \frac{(\lambda t)^{k+1}}{(k+1)!} = \frac{e^{-\lambda t}}{\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda t)^k}{k!} \\ &= \frac{e^{-\lambda t}}{\lambda t} (e^{\lambda t} - 1) = \frac{1 - e^{-\lambda t}}{\lambda t}. \end{aligned}$$

Together with Equation (6.7),

$$E(S_{N_t}) = t - \frac{1}{\lambda} + \frac{e^{-\lambda t}}{\lambda}.$$

Finally with Equation (6.6), the expected length of the interval that contains t is

$$E(S_{N_{t+1}} - S_{N_t}) = \left(t + \frac{1}{\lambda}\right) - \left(t - \frac{1}{\lambda} + \frac{e^{-\lambda t}}{\lambda}\right) = \frac{2 - e^{-\lambda t}}{\lambda} \approx \frac{2}{\lambda},$$

for large (or even moderate) t .

EXERCISES

- 6.1** Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter $\lambda = 1.5$. Find the following:
- (a) $P(N_1 = 2, N_4 = 6)$
 - (b) $P(N_4 = 6 | N_1 = 2)$
 - (c) $P(N_1 = 2 | N_4 = 6)$
- 6.2** Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter $\lambda = 2$. Find the following:
- (a) $E(N_3 N_4)$
 - (b) $E(X_3 X_4)$
 - (c) $E(S_3 S_4)$
- 6.3** Calls are received at a company call center according to a Poisson process at the rate of five calls per minute.
- (a) Find the probability that no call occurs over a 30-second period.
 - (b) Find the probability that exactly four calls occur in the first minute, and six calls occur in the second minute.
 - (c) Find the probability that 25 calls are received in the first 5 minutes and six of those calls occur in the first minute.
- 6.4** Starting at 9 a.m., patients arrive at a doctor's office according to a Poisson process. On average, three patients arrive every hour.
- (a) Find the probability that at least two patients arrive by 9:30 a.m.
 - (b) Find the probability that 10 patients arrive by noon and eight of them come to the office before 11 a.m.
 - (c) If six patients arrive by 10 a.m., find the probability that only one patient arrives by 9:15 a.m.

- 6.5** Let $(N_t)_{t \geq 0}$ be a Poisson process. Explain what is wrong with the following *proof* that N_3 is a constant.

$$\begin{aligned} E((N_3)^2) &= E(N_3 N_3) = E(N_3(N_3 - N_3)) \\ &= E(N_3)E(N_3 - N_3) = E(N_3)E(N_3) \\ &= E(N_3)^2. \end{aligned}$$

Thus, $\text{Var}(N_3) = E((N_3)^2) - E(N_3)^2 = 0$, which gives that N_3 is a constant with probability 1.

- 6.6** Occurrences of landfalling hurricanes during an El Niño event are modeled as a Poisson process in Bove et al. (1998). The authors assert that “During an El Niño year, the probability of two or more hurricanes making landfall in the United States is 28%.” Find the rate of the Poisson process.
- 6.7** Ben, Max, and Yolanda are at the front of three separate lines in the cafeteria waiting to be served. The serving times for the three lines follow independent Poisson processes with respective parameters 1, 2, and 3.
- Find the probability that Yolanda is served first.
 - Find the probability that Ben is served before Yolanda.
 - Find the expected waiting time for the first person served.
- 6.8** Starting at 6 a.m., cars, buses, and motorcycles arrive at a highway toll booth according to independent Poisson processes. Cars arrive about once every 5 minutes. Buses arrive about once every 10 minutes. Motorcycles arrive about once every 30 minutes.
- Find the probability that in the first 20 minutes, exactly three vehicles—two cars and one motorcycle—arrive at the booth.
 - At the toll booth, the chance that a driver has exact change is $1/4$, independent of vehicle. Find the probability that no vehicle has exact change in the first 10 minutes.
 - Find the probability that the seventh motorcycle arrives within 45 minutes of the third motorcycle.
 - Find the probability that at least one other vehicle arrives at the toll booth between the third and fourth car arrival.
- 6.9** Show that the geometric distribution is memoryless.
- 6.10** Assume that $X_1, X_2, \dots$ is an i.i.d. sequence of exponential random variables with parameter λ . Let $S_n = X_1 + \dots + X_n$. Show that S_n has a gamma distribution with parameters n and λ
- by moment-generating functions,
 - by induction on n .

- 6.11** Show that a continuous probability distribution that is memoryless must be exponential. Hint: For $g(t) = P(X > t)$, show that $g(t) = (g(1))^t$ for all positive, rational t .
- 6.12** Starting at noon, diners arrive at a restaurant according to a Poisson process at the rate of five customers per minute. The time each customer spends eating at the restaurant has an exponential distribution with mean 40 minutes, independent of other customers and independent of arrival times. Find the distribution, as well as the mean and variance, of the number of diners in the restaurant at 2 p.m.
- 6.13** Assume that $(N_t)_{t \geq 0}$ is a Poisson process with parameter λ . Find the conditional distribution of N_s given $N_t = n$, for
- $s < t$,
 - $s > t$.
- 6.14** Red cars arrive at an intersection according to a Poisson process with parameter r . Blue cars arrive, independently of red cars, according to a Poisson process with parameter b . Let X be the number of blue cars which arrive between two successive red cars. Show that X has a geometric distribution.
- 6.15** Failures occur for a mechanical process according to a Poisson process. Failures are classified as either major or minor. Major failures occur at the rate of 1.5 failures per hour. Minor failures occur at the rate of 3.0 failures per hour.
- Find the probability that two failures occur in 1 hour.
 - Find the probability that in half an hour, no major failures occur.
 - Find the probability that in 2 hours, at least two major failures occur or at least two minor failures occur.
- 6.16** Accidents occur at a busy intersection according to a Poisson process at the rate of two accidents per week. Three out of four accidents involve the use of alcohol.
- What is the probability that three accidents involving alcohol will occur next week?
 - What is the probability that at least one accident occurs tomorrow?
 - If six accidents occur in February (four weeks), what is the probability that less than half of them involve alcohol?
- 6.17** Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ and arrival times $S_1, S_2, \dots$. Evaluate the expected sum of squares of the arrival times before time t ,

$$E \left(\sum_{n=1}^{N_t} S_n^2 \right).$$

- 6.18** The planets of the Galactic Empire are distributed in space according to a spatial Poisson process at an approximate density of one planet per cubic parsec. From the Death Star, let X be the distance to the nearest planet.

- (a) Find the probability density function of X .
- (b) Find the mean distance from the Death Star to the nearest planet.
- 6.19** Members of a large audience are asked to state their birthdays, one at a time. How many people will be asked before three persons are found to have the same birthday? Use embedding to estimate the expected number. You will need to use numerical methods to evaluate the resulting integral.
- 6.20** Consider a spatial point process in $\mathbb{R}^2$ with parameter λ . Assume that A is a bounded set in $\mathbb{R}^2$ which contains exactly one point of the process. Given $B \subseteq A$, find the probability that B contains one point.
- 6.21** For a Poisson process with parameter λ show that for $s < t$, the correlation between N_s and N_t is
- $$\text{Corr}(N_s, N_t) = \sqrt{\frac{s}{t}}.$$
- 6.22** At the Poisson Casino, two dice are rolled at random times according to a Poisson process with parameter λ . Find the probability that in $[0, t]$ every pair of dice rolled comes up 7.
- 6.23** Oak and maple trees are each located in the arboretum according to independent spatial Poisson processes with parameters λ_O and λ_M , respectively.
- (a) In a region of x square meters, find the probability that both species of trees are present.
- (b) In the arboretum, there is a circular pond of radius 100 m. Find the probability that within 20 m of the pond there are only oaks.
- 6.24** Tom is bird watching in the arboretum. The times when he sights a meadowlark occur in accordance with a Poisson process with parameter λ . The times when he sights a sparrow occur as a Poisson process with parameter μ . Assume that meadowlark and sparrow sightings are independent.
- (a) Find the probability that a meadowlark is seen first.
- (b) Find the probability the one bird is seen by time $t = 1$.
- (c) Find the probability that one sparrow and two meadowlarks are seen by time $t = 2$.
- 6.25** Computers in the lab fail, on average, twice a day, according to a Poisson process. Last week, 10 computers failed. Find the expected time of the last failure, and give an approximate time of day when the last failure occurred.
- 6.26** If $(N_t)_{t \geq 0}$ is a Poisson process with parameter λ , find the probability generating function of N_t .
- 6.27** In a small parliamentary election, votes are counted according to a Poisson process at the rate of 60 votes per minute. There are six political parties, whose popularity among the electorate is shown by this distribution.

A	B	C	D	E	F
0.05	0.30	0.10	0.10	0.25	0.20

- (a) In the first 2 minutes of the vote tally, 40 people had voted for parties E and F. Find the probability that more than 100 votes were counted in the first 2 minutes.
- (b) If vote counting starts at 3 p.m., find the probability that the first vote is counted by the first second after 3 p.m.
- (c) Find the probability that the first vote for party C is counted before a vote for B or D.
- 6.28** Tornadoes hit a region according to a Poisson process with $\lambda = 2$. The number of insurance claims filed after any tornado has a Poisson distribution with mean 30. The number of tornadoes is independent of the number of insurance claims. Find the expectation and standard deviation of the total number of claims filed by time t .
- 6.29** Job offers for a recent college graduate arrive according to a Poisson process with mean two per month. A job offer is acceptable if the salary offered is at least \$35,000. Salary offers follow an exponential distribution with mean \$25,000. Find the probability that an acceptable job offer will be received within 3 months.
- 6.30** See Example 6.9. Find the variance of total waiting time of concert-goers who arrive before the band starts.
- 6.31** See Example 6.9. Find the expected *average* waiting time of concert-goers who arrive before the band starts.
- 6.32** Investors purchase \$1,000 bonds at the random times of a Poisson process with parameter λ . If the interest rate is r , then the *present value* of an investment purchased at time t is $1000e^{-rt}$. Show that the expected total present value of the bonds purchased by time t is $1000\lambda(1 - e^{-rt})/r$.
- 6.33** Let $S_1, S_2, \dots$ be the arrival times of a Poisson process with parameter λ . Given the time of the n th arrival, find the expected time $E(S_1 | S_n)$ of the first arrival.
- 6.34** Describe in words the random variable S_{N_t} . Find the distribution of S_{N_t} by giving the cumulative distribution function.
- 6.35** See the definitions for the spatial and nonhomogeneous Poisson processes. Define a nonhomogeneous, spatial Poisson process in $\mathbb{R}^2$. Consider such a process $(N_A)_{A \subseteq \mathbb{R}^2}$ with intensity function

$$\lambda(x, y) = e^{-(x^2+y^2)}, \text{ for } -\infty < x, y < \infty.$$

Let C denote the unit circle, that is, the circle of radius 1 centered at the origin. Find $P(N_C = 0)$.

- 6.36** Starting at 9 a.m., customers arrive at a store according to a nonhomogeneous Poisson process with intensity function $\lambda(t) = t^2$, for $t > 0$, where the time unit is hours. Find the probability mass function of the number of customers who enter the store by noon.
- 6.37** A *compound Poisson process* $(C_t)_{t \geq 0}$ is defined as

$$C_t = \sum_{k=1}^{N_t} X_k,$$

where $(N_t)_{t \geq 0}$ is a Poisson process, and $X_1, X_2, \dots$ is an i.i.d. sequence of random variables that are independent of $(N_t)_{t \geq 0}$.

Assume that automobile accidents at a dangerous intersection occur according to a Poisson process at the rate of 3 accidents per week. Furthermore, the number of people seriously injured in an accident has a Poisson distribution with mean 2. Show that the process of serious injuries is a compound Poisson process, and find the mean and standard deviation of the number of serious injuries over 1 year's time.

- 6.38** A *mixed Poisson process*, also called a *Cox process* or *doubly stochastic process*, arises from a Poisson process where the parameter Λ is itself a random variable. If $(N_t)_{t \geq 0}$ is a mixed Poisson process, then the conditional distribution of N_t given $\Lambda = \lambda$ is Poisson with parameter λt . Assume that for such a process Λ has an exponential distribution with parameter μ . Find the probability mass function of N_t .
- 6.39** See Exercise 6.38. Assume that $(N_t)_{t \geq 0}$ is a mixed Poisson process with rate parameter uniformly distributed on $(0, 1)$. Find $P(N_1 = 1)$.
- 6.40** Assume that $(N_t)_{t \geq 0}$ is a mixed Poisson process whose rate Λ has a gamma distribution with parameters n and λ . Show that

$$P(N_t = k) = \binom{n+k-1}{k} \left(\frac{\lambda}{\lambda+t} \right)^n \left(\frac{t}{\lambda+t} \right)^k, \text{ for } k = 0, 1, \dots$$

- 6.41** R : Goals occur in a soccer game according to a Poisson process. The average total number of goals scored for a 90-minute match is 2.68. Assume that two teams are evenly matched. Use simulation to estimate the probability both teams will score the same number of goals. Compare with the theoretical result.
- 6.42** R : Simulate the restaurant results of Exercise 6.12.
- 6.43** R : Simulate a spatial Poisson process with $\lambda = 10$ on the box of volume 8 with vertices at the 8 points $(\pm 1, \pm 1, \pm 1)$. Estimate the mean and variance of the number of points in the ball centered at the origin of radius 1. Compare with the exact values.

- 6.44** R : See Exercise 6.32. Simulate the expected total present value of the bonds if the interest rate is 4%, the Poisson parameter is $\lambda = 50$, and $t = 10$. Compare with the exact value.
- 6.45** R : Simulate the birthday problem of Exercise 6.19.

7

CONTINUOUS-TIME MARKOV CHAINS

Life defies our phrases, it is infinitely continuous and subtle and shaded, whilst our verbal terms are discrete, rude, and few.

—William James

7.1 INTRODUCTION

In this chapter, we extend the Markov chain model to continuous time. A continuous-time process allows one to model not only the transitions between states, but also the duration of time in each state. The central Markov property continues to hold—given the present, past and future are independent.

The Markov property is a form of memorylessness. This leads to the exponential distribution. In a continuous-time Markov chain, when a state is visited, the process stays in that state for an exponentially distributed length of time before moving to a new state. If one just watches the sequence of states that are visited, ignoring the length of time spent in each state, the process looks like a discrete-time Markov chain.

One of the Markov chains introduced in this book was the three-state weather chain of Example 2.3, with state space {rain, snow, clear}. Consider a continuous-time extension. Assume that rainfall lasts, on average, 3 hours at a time. When it snows, the duration, on average, is 6 hours. And the weather stays clear, on average, for

12 hours. Furthermore, changes in weather states are described by the stochastic transition matrix

$$\tilde{\mathbf{P}} = \begin{matrix} & \begin{matrix} \text{Rain} & \text{Snow} & \text{Clear} \end{matrix} \\ \begin{matrix} \text{Rain} \\ \text{Snow} \\ \text{Clear} \end{matrix} & \begin{pmatrix} 0 & 1/2 & 1/2 \\ 3/4 & 0 & 1/4 \\ 1/4 & 3/4 & 0 \end{pmatrix} \end{matrix}. \quad (7.1)$$

To elaborate, assume that it is currently snowing. Under this model, it snows for an exponential length of time with parameter $\lambda_s = 1/6$. (Remember that the parameter of an exponential distribution is the reciprocal of the mean.) Then, the weather changes to either rain or clear, with respective probabilities $3/4$ and $1/4$. If it switches to rain, it rains for an exponential length of time with parameter $\lambda_r = 1/3$. Then, the weather changes to either snow or clear with equal probability, and so on. Figure 7.1 gives an example of how the process unfolds over 50 hours.

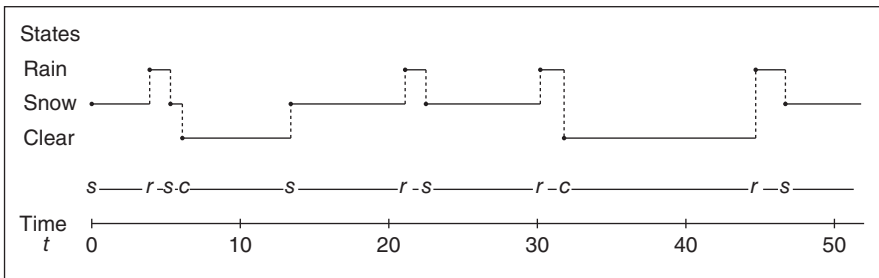


Figure 7.1 Realization of a continuous-time weather chain.

Let X_t denote the weather at time t . Then, $(X_t)_{t \geq 0}$ is a continuous-time Markov chain. The $\tilde{\mathbf{P}}$ matrix, exponential time parameters $(\lambda_r, \lambda_s, \lambda_c) = (1/3, 1/6, 1/12)$, and initial distribution completely specify the process. That is, they are sufficient for computing all probabilities of the form $P(X_{t_1} = i_1, \dots, X_{t_n} = i_n)$, for $n \geq 1$, states $i_1, \dots, i_n$, and times $t_1, \dots, t_n \geq 0$.

Markov Transition Function

The formal treatment of continuous-time Markov chains begins with the defining Markov property.

Markov Property

A continuous-time stochastic process $(X_t)_{t \geq 0}$ with discrete state space S is a *continuous-time Markov chain* if

$$P(X_{t+s} = j | X_s = i, X_u = x_u, 0 \leq u < s) = P(X_{t+s} = j | X_s = i),$$

for all $s, t, \geq 0$, $i, j, x_u \in S$, and $0 \leq u < s$.

The process is said to be *time-homogeneous* if this probability does not depend on s . That is,

$$P(X_{t+s} = j | X_s = i) = P(X_t = j | X_0 = i), \text{ for } s \geq 0. \quad (7.2)$$

The Markov chains we treat in this book are all time-homogeneous. For each $t \geq 0$, the transition probabilities in Equation (7.2) can be arranged in a matrix function $\mathbf{P}(t)$ called the *transition function*, with

$$P_{ij}(t) = P(X_t = j | X_0 = i).$$

Note that for the weather chain, the $\tilde{\mathbf{P}}$ matrix in Equation (7.1) is *not* the transition function. The $\tilde{\mathbf{P}}$ matrix gives the probabilities of moving from state to state in a discretized process in which time has been ignored.

The transition function $\mathbf{P}(t)$ has similar properties as that of the transition matrix for a discrete-time Markov chain. For instance, the Chapman–Kolmogorov equations hold.

Chapman–Kolmogorov Equations

For a continuous-time Markov chain $(X_t)_{t \geq 0}$ with transition function $\mathbf{P}(t)$,

$$\mathbf{P}(s+t) = \mathbf{P}(s)\mathbf{P}(t),$$

for $s, t \geq 0$. That is,

$$P_{ij}(s+t) = [P(s)P(t)]_{ij} = \sum_k P_{ik}(s)P_{kj}(t), \text{ for states } i, j, \text{ and } s, t \geq 0.$$

Proof. By conditioning on X_s ,

$$\begin{aligned} P_{ij}(s+t) &= P(X_{s+t} = j | X_0 = i) \\ &= \sum_k P(X_{s+t} = j | X_s = k, X_0 = i) P(X_s = k | X_0 = i) \\ &= \sum_k P(X_{s+t} = j | X_s = k) P(X_s = k | X_0 = i) \\ &= \sum_k P(X_t = j | X_0 = k) P(X_s = k | X_0 = i) \\ &= \sum_k P_{ik}(s) P_{kj}(t) = [P(s)P(t)]_{ij}, \end{aligned}$$

where the third equality is because of the Markov property, and the fourth equality is by time-homogeneity. ■

■ **Example 7.1 (Poisson process)** A Poisson process $(N_t)_{t \geq 0}$ with parameter λ is a continuous-time Markov chain. The Markov property holds as a consequence of stationary and independent increments. For $0 \leq i \leq j$,

$$\begin{aligned} P_{ij}(t) &= P(N_{t+s} = j | N_s = i) = \frac{P(N_{t+s} = j, N_s = i)}{P(N_s = i)} \\ &= \frac{P(N_{t+s} - N_s = j - i, N_s = i)}{P(N_s = i)} \\ &= P(N_{t+s} - N_s = j - i) \\ &= P(N_t = j - i) = \frac{e^{-\lambda t} (\lambda t)^{j-i}}{(j-i)!}. \end{aligned}$$

The transition function is

$$P(t) = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{pmatrix} e^{-\lambda t} & (\lambda t)e^{-\lambda t} & (\lambda t)^2 e^{-\lambda t}/2 & (\lambda t)^3 e^{-\lambda t}/6 & \dots \\ 0 & e^{-\lambda t} & (\lambda t)e^{-\lambda t} & (\lambda t)^2 e^{-\lambda t}/2 & \dots \\ 0 & 0 & e^{-\lambda t} & (\lambda t)e^{-\lambda t} & \dots \\ 0 & 0 & 0 & e^{-\lambda t} & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

Holding Times and Embedded Chains

By homogeneity, when a Markov chain visits state i its forward evolution from that time onward behaves the same as the process started in i at time $t = 0$. Time-homogeneity and the Markov property characterizes the distribution of the length of time that a continuous-time chain stays in state i before transitioning to a new state.

Holding Times are Exponentially Distributed

Let T_i be the *holding time* at state i , that is, the length of time that a continuous-time Markov chain started in i stays in i before transitioning to a new state. Then, T_i has an exponential distribution.

Proof. We show that T_i is memoryless. Let $s, t \geq 0$. For the chain started in i , the event $\{T_i > s\}$ is equal to the event that $\{X_u = i, \text{ for } 0 \leq u \leq s\}$. Since $\{T_i > s + t\}$ implies $\{T_i > s\}$,

$$P(T_i > s + t | X_0 = i) = P(T_i > s + t, T_i > s | X_0 = i)$$

$$\begin{aligned}
&= P(T_i > s + t | X_0 = i, T_i > s) P(T_i > s | X_0 = i) \\
&= P(T_i > s + t | X_u = i, \text{ for } 0 \leq u \leq s) P(T_i > s | X_0 = i) \\
&= P(T_i > s + t | X_s = i) P(T_i > s | X_0 = i) \\
&= P(T_i > t | X_0 = i) P(T_i > s | X_0 = i),
\end{aligned}$$

where the next to last equality is because of the Markov property, and the last equality is because of homogeneity. This gives that T_i is memoryless. The result follows since the exponential distribution is the only continuous distribution that is memoryless. ■

For each i , let q_i be the parameter of the exponential distribution for the holding time T_i . We assume that $0 < q_i < \infty$. Technically, a continuous-time process can be defined where $q_i = 0$ or $+\infty$. In the former case, when i is visited the process never leaves, and i is called an *absorbing state*. In the latter case, the process leaves i immediately upon entering i . This would allow for infinitely many transitions in a finite interval. Such a process is called *explosive*.

The evolution of a continuous-time Markov chain which is neither absorbing nor explosive can be described as follows. Starting from i , the process stays in i for an exponentially distributed length of time, on average $1/q_i$ time units. Then, it hits a new state $j \neq i$, with some probability p_{ij} . The process stays in j for an exponentially distributed length of time, on average $1/q_j$ time units. It then hits a new state $k \neq j$, with probability p_{jk} , and so on.

The transition probabilities p_{ij} describe the discrete transitions from state to state. If we ignore time, and just watch state to state transitions, we see a sequence $Y_0, Y_1, \dots$, where Y_n is the n th state visited by the continuous process. The sequence $Y_0, Y_1, \dots$ is a discrete-time Markov chain called the *embedded chain*.

Let $\tilde{P}$ be the transition matrix for the embedded chain. That is, $\tilde{P}_{ij} = p_{ij}$. Then, $\tilde{P}$ is a stochastic matrix whose diagonal entries are 0.

■ **Example 7.2 (Poisson process)** For a Poisson process with parameter λ , the holding time parameters are constant. That is, $q_i = \lambda$, for $i = 0, 1, 2, \dots$. The process moves from 0 to 1 to 2, and so on. The transition matrix of the embedded chain is

$$\tilde{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ \vdots \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & \dots \\ 0 & 0 & 1 & 0 & 0 & \dots \\ 0 & 0 & 0 & 1 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & \dots \\ 0 & 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

■ **Example 7.3 (Two-state chain)** A two-state continuous-time Markov chain is specified by two holding time parameters as depicted in the transition graph in Figure 7.2. Edges of the graph are labeled with holding time rates, not probabilities.

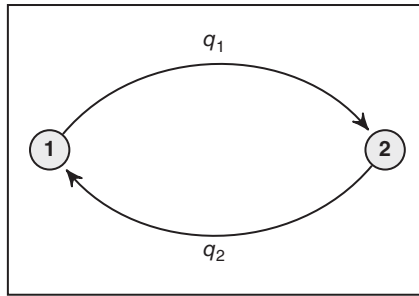


Figure 7.2 Two-state process.

The embedded chain transition matrix is

$$\tilde{P} = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{matrix}.$$

■

The process stays in state 1 an exponential length of time with parameter q_1 before moving to 2. It stays in 2 an exponential length of time with parameter q_2 before moving to 1, and so on.

7.2 ALARM CLOCKS AND TRANSITION RATES

A continuous-time Markov chain can also be described by specifying *transition rates* between pairs of states. Central to this approach is the notion of the *exponential alarm clock*.

Imagine that for each state i , there are independent alarm clocks associated with each of the states that the process can visit after i . If j can be hit from i , then the alarm clock associated with (i, j) will ring after an exponentially distributed length of time with parameter q_{ij} . When the process first hits i , the clocks are started simultaneously. The first alarm that rings determines the next state to visit. If the (i, j) clock rings first and the process moves to j , a new set of exponential alarm clocks are started, with transition rates $q_{j1}, q_{j2}, \dots$. Again, the first alarm that rings determines the next state hit, and so on.

The q_{ij} are called the *transition rates* of the continuous-time process. From the transition rates, we can obtain the holding time parameters and the embedded chain transition probabilities.

Consider the process started in i . The clocks are started, and the first one that rings determines the next transition. The time of the first alarm is the minimum of independent exponential random variables with parameters $q_{i1}, q_{i2}, \dots$. Recall results for the exponential distribution and the minimum of independent exponentials, as given in Equations (6.1) and (6.2). The minimum has an exponential distribution with parameter $\sum_k q_{ik}$. That is, the chain stays at i for an exponentially distributed amount

of time with parameter $\sum_k q_{ik}$. From the discussion of holding times, the exponential length of time that the process stays in i has parameter q_i . That is,

$$q_i = \sum_k q_{ik}.$$

The interpretation is that the rate that the process leaves state i is equal to the sum of the rates from i to each of the next states.

From i , the chain moves to j if the (i, j) clock rings first, which occurs with probability $q_{ij}/\sum_k q_{ik} = q_{ij}/q_i$. Thus, for the embedded chain transition probabilities

$$p_{ij} = \frac{q_{ij}}{\sum_k q_{ik}} = \frac{q_{ij}}{q_i}.$$

■ **Example 7.4** The general three-state continuous-time Markov chain is described by the transition graph in Figure 7.3. In terms of the transition rates, holding time parameters are

$$(q_1, q_2, q_3) = (q_{12} + q_{13}, q_{21} + q_{23}, q_{31} + q_{32}),$$

with embedded chain transition matrix

$$\tilde{\mathbf{P}} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & q_{12}/q_1 & q_{13}/q_1 \\ q_{21}/q_2 & 0 & q_{23}/q_2 \\ q_{31}/q_3 & q_{32}/q_3 & 0 \end{pmatrix} \end{matrix}.$$

■

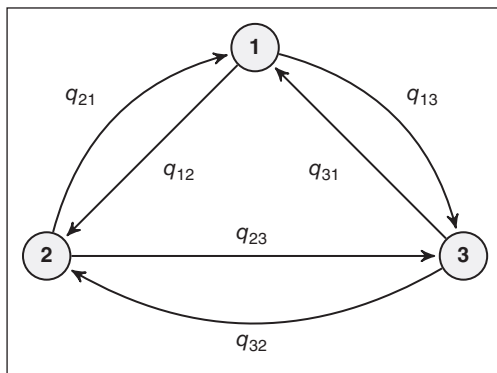


Figure 7.3 Transition rates for three-state chain.

■ **Example 7.5 (Registration line)** It is time for students to register for classes, and a line is forming at the registrar's office for those who need assistance. It takes

the registrar an exponentially distributed amount of time to service each student, at the rate of one student every 5 minutes. Students arrive at the office and get in line according to a Poisson process at the rate of one student every 4 minutes. Line size is capped at 4 people. If an arriving student finds that there are already 4 people in line, then they try again later. As soon as there is at least one person in line, the registrar starts assisting the first available student. The arrival times of the students are independent of the registrar's service time.

Let X_t be the number of students in line at time t . Then, $(X_t)_{t \geq 0}$ is a continuous-time Markov chain on $\{0, 1, 2, 3, 4\}$.

If there is no one in line, then the size of the line increases to 1 when a student arrives. If there are 4 people in line, then the number decreases to 3 when the registrar finishes assisting the student they are meeting with. If there are 1, 2, or 3 students in line, then the line size can either decrease or increase by 1. If a student arrives at the registrar's office before the registrar has finished serving the student being helped, then the line increases by 1. If the registrar finishes serving the student being helped before another student arrives, the line decreases by 1.

Imagine that when there is 1 person in line two exponential alarm clocks are started—one for student arrivals, with rate $1/4$, the other for the registrar's service time, with rate $1/5$. If the arrival time clock rings first, the line increases by one. If the service clock rings first, the line decreases by one. The same dynamics hold if there are 2 or 3 people in line. The process is described by the transition graph in Figure 7.4.

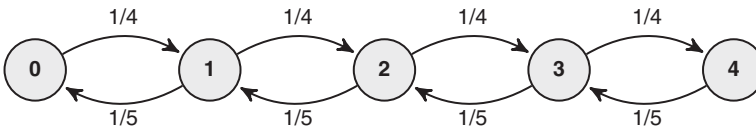


Figure 7.4 Transition rate graph for registration line Markov chain.

The holding time parameters are

$$(q_0, q_1, q_2, q_3, q_4) = \left(\frac{1}{4}, \frac{9}{20}, \frac{9}{20}, \frac{9}{20}, \frac{1}{5} \right),$$

with embedded chain transition matrix

$$\tilde{P} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 4/9 & 0 & 5/9 & 0 & 0 \\ 0 & 4/9 & 0 & 5/9 & 0 \\ 0 & 0 & 4/9 & 0 & 5/9 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

Since students arrive in line at a faster rate than the registrar's service time, the line tends to grow over time. See a realization of the process on $[0, 60]$ in Figure 7.5. ■

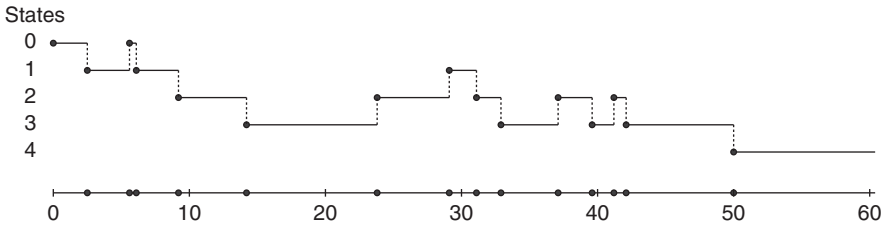


Figure 7.5 Realization of the size of the line at the registrar's office.

7.3 INFINITESIMAL GENERATOR

In continuous time, transition rates play a fundamental role when working with Markov chains. Since the derivative of a function describes its rate of change, it is not surprising that the derivative of the transition function $\mathbf{P}'(t)$ is most important.

Assume that $(X_t)_{t \geq 0}$ is a continuous-time Markov chain with transition function $\mathbf{P}(t)$. Assume the transition function is differentiable. Note that

$$P_{ij}(0) = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

If $X_t = i$, then the instantaneous transition rate of hitting $j \neq i$ is

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{E(\text{Number of transitions to } j \text{ in } (t, t+h])}{h} &= \lim_{h \rightarrow 0^+} \frac{P(X_{t+h} = j | X_t = i)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{P(X_h = j | X_0 = i)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{P_{ij}(h)}{h} \\ &= \lim_{h \rightarrow 0^+} \frac{P_{ij}(h) - P_{ij}(0)}{h} \\ &= P'_{ij}(0). \end{aligned}$$

The first equality is because for h sufficiently small, the number of transitions to j in $(t, t+h]$ is either 0 or 1. Let $\mathbf{Q} = \mathbf{P}'(0)$. The off-diagonal entries of $\mathbf{Q}$ are the instantaneous transition rates, which are the transition rates q_{ij} introduced in the last section. That is, $Q_{ij} = q_{ij}$, for $i \neq j$. In the language of infinitesimals, if $X_t = i$, then the chance that $X_{t+dt} = j$ is $q_{ij} dt$.

The diagonal entries of $\mathbf{Q}$ are

$$Q_{ii} = P'_{ii}(0) = \lim_{h \rightarrow 0^+} \frac{P_{ii}(h) - P_{ii}(0)}{h} = \lim_{h \rightarrow 0^+} \frac{P_{ii}(h) - 1}{h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0^+} - \frac{\sum_{j \neq i} P_{ij}(h)}{h} = - \sum_{j \neq i} \lim_{h \rightarrow 0^+} \frac{P_{ij}(h)}{h} \\
&= - \sum_{j \neq i} Q_{ij} = - \sum_{j \neq i} q_{ij} = -q_i.
\end{aligned}$$

The $\mathbf{Q}$ matrix is called the *generator* or *infinitesimal generator*. It is the most important matrix for continuous-time Markov chains. Here, we derived $\mathbf{Q}$ from the transition function $\mathbf{P}(t)$. However, in a modeling context one typically starts with $\mathbf{Q}$, identifying the transition rates q_{ij} based on the qualitative and quantitative dynamics of the process. The transition function and related quantities are derived from $\mathbf{Q}$.

Clearly, the generator is not a stochastic matrix. Diagonal entries are negative, entries can be greater than 1, and rows sum to 0.

■ **Example 7.6** The infinitesimal generator matrix for the registration line chain of Example 7.5 is

$$\mathbf{Q} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} -1/4 & 1/4 & 0 & 0 & 0 \\ 1/5 & -9/20 & 1/4 & 0 & 0 \\ 0 & 1/5 & -9/20 & 1/4 & 0 \\ 0 & 0 & 1/5 & -9/20 & 1/4 \\ 0 & 0 & 0 & 1/5 & -1/5 \end{pmatrix} \end{matrix}.$$

■ **Example 7.7** The generator for a Poisson process with parameter λ is

$$\mathbf{Q} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ \vdots \end{matrix} & \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & \dots \\ 0 & -\lambda & \lambda & 0 & 0 & \dots \\ 0 & 0 & -\lambda & \lambda & 0 & \dots \\ 0 & 0 & 0 & -\lambda & \lambda & \dots \\ 0 & 0 & 0 & 0 & -\lambda & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

For a continuous-time Markov chain, the transition probabilities of the embedded chain can be derived from first principles from the transition function and generator matrix. From i , the probability that if a transition occurs at time t the process moves to a different state $j \neq i$ is

$$\begin{aligned}
\lim_{h \rightarrow 0^+} P(X_{t+h} = j | X_t = i, X_{t+h} \neq i) &= \lim_{h \rightarrow 0^+} \frac{P(X_h = j | X_0 = i, X_h \neq i)}{P(X_h \neq i | X_0 = i)} \\
&= \lim_{h \rightarrow 0^+} \frac{P(X_h = j, X_h \neq i, X_0 = i)}{P(X_h \neq i, X_0 = i)}
\end{aligned}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0^+} \frac{P(X_h = j | X_0 = i)}{P(X_h \neq i | X_0 = i)} \\
&= \lim_{h \rightarrow 0^+} \frac{P_{ij}(h)/h}{1 - P_{ii}(h)/h} \\
&= \frac{q_{ij}}{q_i},
\end{aligned}$$

which is independent of t . This gives the transition probability p_{ij} of the embedded chain. It also establishes the relationship between instantaneous transition rates, holding time parameters, and the embedded chain transition probabilities.

Instantaneous Rates, Holding Times, Transition Probabilities

For a continuous-time Markov chain, let q_{ij} , q_i , and p_{ij} be defined as above. For $i \neq j$,

$$q_{ij} = q_i p_{ij}.$$

For discrete-time Markov chains, there is no generator matrix and the probabilistic properties of the stochastic process are captured by the transition matrix $\mathbf{P}$. For continuous-time Markov chains the generator matrix $\mathbf{Q}$ gives a complete description of the dynamics of the process. The distribution of any finite subset of the X_t , and all probabilistic quantities of the stochastic process, can, in principle, be obtained from the infinitesimal generator and the initial distribution.

Forward, Backward Equations

The transition function $\mathbf{P}(t)$ can be computed from the generator $\mathbf{Q}$ by solving a system of differential equations.

Kolmogorov Forward, Backward Equations

A continuous-time Markov chain with transition function $\mathbf{P}(t)$ and infinitesimal generator $\mathbf{Q}$ satisfies the *forward equation*

$$\mathbf{P}'(t) = \mathbf{P}(t)\mathbf{Q} \quad (7.3)$$

and the *backward equation*

$$\mathbf{P}'(t) = \mathbf{Q}\mathbf{P}(t). \quad (7.4)$$

Equivalently, for all states i and j ,

$$P'_{ij}(t) = \sum_k P_{ik}(t)q_{kj} = -P_{ij}(t)q_j + \sum_{k \neq j} P_{ik}(t)q_{kj}$$

and

$$P'_{ij}(t) = \sum_k q_{ik} P_{kj}(t) = -q_i P_{ij}(t) + \sum_{k \neq i} q_{ik} P_{kj}(t).$$

Proof. The equations are a consequence of the Chapman–Kolmogorov property. For the forward equation, for $h, t \geq 0$,

$$\begin{aligned} \frac{P(t+h) - P(t)}{h} &= \frac{P(t)P(h) - P(t)}{h} \\ &= P(t) \left(\frac{P(h) - I}{h} \right) \\ &= P(t) \left(\frac{P(h) - P(0)}{h} \right). \end{aligned}$$

Taking limits as $h \rightarrow 0^+$ gives $P'(t) = P(t)Q$. The backward equation is derived similarly, starting with $P(t+h) = P(h)P(t)$. ■

■ **Example 7.8 (Poisson process)** The transition probabilities for the Poisson process with parameter λ ,

$$P_{ij}(t) = \frac{e^{-\lambda t} (\lambda t)^{j-i}}{(j-i)!}, \text{ for } j \geq i,$$

were derived in Example 7.1. They satisfy the Kolmogorov forward equations

$$\begin{aligned} P'_{ii}(t) &= -\lambda P_{ii}(t), \\ P'_{ij}(t) &= -\lambda P_{ij}(t) + \lambda P_{i,j-1}, \text{ for } j = i+1, i+2, \dots, \end{aligned}$$

and the backward equations

$$\begin{aligned} P'_{ii}(t) &= -\lambda P_{ii}(t), \\ P'_{ij}(t) &= -\lambda P_{ij}(t) + \lambda P_{j+1,i}, \text{ for } j = i+1, i+2, \dots \end{aligned}$$

■ **Example 7.9 (Two-state process)** For a continuous-time process with generator

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix} \end{matrix}$$

the forward equations give

$$\begin{aligned} P'_{11}(t) &= -P_{11}(t)q_1 + P_{12}(t)q_{21} \\ &= -\lambda P_{11}(t) + (1 - P_{11}(t))\mu \\ &= \mu - (\lambda + \mu)P_{11}(t), \end{aligned}$$

using the fact that the first row of the matrix $\mathbf{P}(t)$ sums to 1. The solution to the linear differential equation is

$$P_{11}(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t}.$$

Also,

$$P'_{22}(t) = -P_{22}(t)q_2 + P_{21}(t)q_{12} = \lambda - (\lambda + \mu)P_{22}(t),$$

with solution

$$P_{22}(t) = \frac{\lambda}{\lambda + \mu} + \frac{\mu}{\lambda + \mu} e^{-(\lambda + \mu)t}.$$

The transition function is

$$\mathbf{P}(t) = \frac{1}{\lambda + \mu} \begin{pmatrix} \overset{1}{\mu + \lambda e^{-(\lambda + \mu)t}} & \overset{2}{\lambda - \lambda e^{-(\lambda + \mu)t}} \\ \mu - \mu e^{-(\lambda + \mu)t} & \lambda + \mu e^{-(\lambda + \mu)t} \end{pmatrix}. \quad (7.5)$$

■

Matrix Exponential

The Kolmogorov backward equation $\mathbf{P}'(t) = \mathbf{Q}\mathbf{P}(t)$ is a matrix equation, which bears a striking resemblance to the nonmatrix differential equation $p'(t) = qp(t)$, where p is a differentiable function and q is a constant. If $p(0) = 1$, the latter has the unique solution

$$p(t) = e^{tq}, \text{ for } t \geq 0.$$

If you are not familiar with solutions to matrix differential equations it might be tempting to try to solve the backward equation by analogy, and write

$$\mathbf{P}(t) = e^{t\mathbf{Q}}, \text{ for } t \geq 0,$$

since $\mathbf{P}(0) = \mathbf{I}$. Remarkably, this is exactly correct, as long as $e^{t\mathbf{Q}}$ is defined properly.

Matrix Exponential

Let $\mathbf{A}$ be a $k \times k$ matrix. The *matrix exponential* $e^{\mathbf{A}}$ is the $k \times k$ matrix

$$e^{\mathbf{A}} = \sum_{n=0}^{\infty} \frac{1}{n!} \mathbf{A}^n = \mathbf{I} + \mathbf{A} + \frac{1}{2} \mathbf{A}^2 + \frac{1}{6} \mathbf{A}^3 + \cdots$$

The matrix exponential is the matrix version of the exponential function and reduces to the ordinary exponential function e^x when A is a 1×1 matrix. The matrix e^A is well-defined as its defining series converges for all square matrices A .

The matrix exponential satisfies many familiar properties of the exponential function. These include

1. $e^0 = I$.
2. $e^A e^{-A} = I$.
3. $e^{(s+t)A} = e^{sA} e^{tA}$.
4. If $AB = BA$, then $e^{A+B} = e^A e^B = e^B e^A$.
5. $\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A$.

For a continuous-time Markov chain with generator Q , the matrix exponential e^{tQ} is the unique solution to the forward and backward equations. Letting $P(t) = e^{tQ}$ gives

$$P'(t) = \frac{d}{dt} e^{tQ} = Q e^{tQ} = e^{tQ} Q = P(t) Q = Q P(t).$$

Transition Function and Generator

For a continuous-time Markov chain with transition function $P(t)$ and infinitesimal generator Q ,

$$P(t) = e^{tQ} = \sum_{n=0}^{\infty} \frac{1}{n!} (tQ)^n = I + tQ + \frac{t^2}{2} Q^2 + \frac{t^3}{6} Q^3 + \cdots \quad (7.6)$$

Computing the matrix exponential is often numerically challenging. Finding accurate and efficient algorithms is still a topic of current research. Furthermore, the transition function is difficult to obtain in closed form for all but the most specialized models. For applied problems, numerical approximation methods are often needed. The R package `expm` contains the function `expm(mat)` for computing the matrix exponential of a numerical matrix `mat`.

R: Computing the Transition Function

For the registration line Markov chain of Example 7.5, we find the transition function $P(t)$ for $t = 2.5$.

```
# matrixexp.R
> install.packages("expm")
> library(expm)
> Q <- matrix(c(-1/4, 1/4, 0, 0, 0, 1/5, -9/20, 1/4, 0, 0, 0,
```



```

+ 1/5, -9/20, 1/4, 0, 0, 0, 1/5, -9/20, 1/4, 0, 0, 0, 1/5, -1/5) ,
+ nrow=5, byrow=T)
> Q
      0      1      2      3      4
0 -0.25  0.25  0.00  0.00  0.00
1  0.20 -0.45  0.25  0.00  0.00
2  0.00  0.20 -0.45  0.25  0.00
3  0.00  0.00  0.20 -0.45  0.25
4  0.00  0.00  0.00  0.20 -0.20
> P <- function(t) expm(t*Q)
> P(2.5)
      0      1      2      3      4
0 0.610 0.290 0.081 0.016 0.003
1 0.232 0.443 0.238 0.071 0.017
2 0.052 0.190 0.435 0.238 0.085
3 0.008 0.045 0.191 0.446 0.310
4 0.001 0.008 0.054 0.248 0.688

```

Diagonalization*

If the $\mathbf{Q}$ matrix is diagonalizable, then so is $e^{t\mathbf{Q}}$, and the transition function can be expressed in terms of the eigenvalues and eigenvectors of $\mathbf{Q}$. Write $\mathbf{Q} = \mathbf{S}\mathbf{D}\mathbf{S}^{-1}$, where

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_k \end{pmatrix}$$

is a diagonal matrix whose diagonal entries are the eigenvalues of $\mathbf{Q}$, and $\mathbf{S}$ is an invertible matrix whose columns are the corresponding eigenvectors. This gives

$$\begin{aligned} e^{t\mathbf{Q}} &= \sum_{n=0}^{\infty} \frac{1}{n!} (t\mathbf{Q})^n = \sum_{n=0}^{\infty} \frac{t^n}{n!} (\mathbf{S}\mathbf{D}\mathbf{S}^{-1})^n \\ &= \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{S}\mathbf{D}^n \mathbf{S}^{-1} = \mathbf{S} \left(\sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{D}^n \right) \mathbf{S}^{-1} \\ &= \mathbf{S} e^{t\mathbf{D}} \mathbf{S}^{-1}, \end{aligned}$$

where

$$e^{t\mathbf{D}} = \sum_{n=0}^{\infty} \frac{t^n}{n!} \mathbf{D}^n = \sum_{n=0}^{\infty} \frac{t^n}{n!} \begin{pmatrix} \lambda_1^n & 0 & \cdots & 0 \\ 0 & \lambda_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_k^n \end{pmatrix}$$

$$\begin{aligned}
&= \begin{pmatrix} \sum_{n=0}^{\infty} (t\lambda_1)^n/n! & 0 & \cdots & 0 \\ 0 & \sum_{n=0}^{\infty} (t\lambda_2)^n/n! & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sum_{n=0}^{\infty} (t\lambda_k)^n/n! \end{pmatrix} \\
&= \begin{pmatrix} e^{t\lambda_1} & 0 & \cdots & 0 \\ 0 & e^{t\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{t\lambda_k} \end{pmatrix}.
\end{aligned}$$

■ **Example 7.10** For the two-state chain, the generator

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

is diagonalizable with eigenvalues 0 and $-(\lambda + \mu)$, and corresponding eigenvectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -\lambda \\ \mu \end{pmatrix}$. This gives

$$Q = SDS^{-1} = \begin{pmatrix} 1 & -\lambda \\ 1 & \mu \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & -(\lambda + \mu) \end{pmatrix} \begin{pmatrix} \mu/(\lambda + \mu) & \lambda/(\lambda + \mu) \\ -1/(\lambda + \mu) & 1/(\lambda + \mu) \end{pmatrix}.$$

The transition function is

$$\begin{aligned}
P(t) &= e^{tQ} = S e^{tD} S^{-1} \\
&= \begin{pmatrix} 1 & -\lambda \\ 1 & \mu \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & e^{-t(\lambda + \mu)} \end{pmatrix} \begin{pmatrix} \mu/(\lambda + \mu) & \lambda/(\lambda + \mu) \\ -1/(\lambda + \mu) & 1/(\lambda + \mu) \end{pmatrix} \\
&= \frac{1}{\lambda + \mu} \begin{pmatrix} \mu + \lambda e^{-t(\lambda + \mu)} & \lambda - \lambda e^{-t(\lambda + \mu)} \\ \mu - \mu e^{-t(\lambda + \mu)} & \lambda + \mu e^{-t(\lambda + \mu)} \end{pmatrix}.
\end{aligned}$$

This result was also shown in Example 7.9 as the solution to the Kolmogorov forward equations. ■

■ **Example 7.11 (DNA evolution)** Continuous-time Markov chains are used to study the evolution of DNA sequences. Numerous models have been proposed for the evolutionary changes on the genome as a result of mutation. Such models are often specified in terms of transition rates between base nucleotides adenine, guanine, cytosine, and thymine at a fixed chromosome location.

The *Jukes–Cantor* model assumes that all transition rates are the same. The infinitesimal generator, with parameter $r > 0$, is

$$Q = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} -3r & r & r & r \\ r & -3r & r & r \\ r & r & -3r & r \\ r & r & r & -3r \end{pmatrix} \end{matrix}.$$

The generator is diagonalizable with linearly independent eigenvectors

$$\begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix},$$

corresponding to eigenvalues $-4r, -4r, -4r$, and 0 . This gives

$$\begin{aligned} P(t) &= e^{tQ} = S e^{tD} S^{-1} \\ &= \begin{pmatrix} -1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{-4rt} & 0 & 0 & 0 \\ 0 & e^{-4rt} & 0 & 0 \\ 0 & 0 & e^{-4rt} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -\frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \\ -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{pmatrix} \\ &= \frac{1}{4} \begin{pmatrix} 1 + 3e^{-4rt} & 1 - e^{-4rt} & 1 - e^{-4rt} & 1 - e^{-4rt} \\ 1 - e^{-4rt} & 1 + 3e^{-4rt} & 1 - e^{-4rt} & 1 - e^{-4rt} \\ 1 - e^{-4rt} & 1 - e^{-4rt} & 1 + 3e^{-4rt} & 1 - e^{-4rt} \\ 1 - e^{-4rt} & 1 - e^{-4rt} & 1 - e^{-4rt} & 1 + 3e^{-4rt} \end{pmatrix}. \end{aligned}$$

The Jukes–Cantor model does not distinguish between base types. Nucleotides a and g are purines, c and t are pyrimidines. Changes from purine to purine or pyrimidine to pyrimidine are called *transitions*. Changes from purine to pyrimidine or vice versa are called *transversions*.

The Kimura model, which includes two parameters r and s , distinguishes between transitions and transversions. The generator is

$$Q = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} -(r+2s) & r & s & s \\ r & -(r+2s) & s & s \\ s & s & -(r+2s) & r \\ s & s & r & -(r+2s) \end{pmatrix} \end{matrix}.$$

The matrix is diagonalizable. The respective matrices of eigenvalues and eigenvectors are

$$D = \begin{pmatrix} -2(r+s) & 0 & 0 & 0 \\ 0 & -2(r+s) & 0 & 0 \\ 0 & 0 & -4s & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \text{and} \quad S = \begin{pmatrix} -1 & 0 & -1 & 1 \\ 0 & -1 & 1 & 1 \\ 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}.$$

The transition function $P(t) = e^{tQ} = S e^{tD} S^{-1}$ is

$$P_{xy}(t) = \begin{cases} (1 + e^{-4st} - 2e^{-2(r+s)t}) / 4, & \text{if } xy \in \{ag, ga, ct, tc\}, \\ (1 - e^{-4st}) / 4, & \text{if } xy \in \{ac, at, gc, gt, ca, cg, ta, tg\}, \\ (1 + e^{-4st} + 2e^{-2(r+s)t}) / 4, & \text{if } xy \in \{aa, gg, cc, tt\}. \end{cases}$$

Estimating Mutation Rate and Evolutionary Distance

Continuous-time Markov models are used by biologists and geneticists to estimate the evolutionary distance between species, as well as the related mutation rate. The following application is based on Durrett (2002).

A statistic for estimating evolutionary distance from DNA sequences is the number of locations in the sequences that differ. Given two DNA strands and a common nucleotide site, consider the probability q that the nucleotides in the two strands are identical given that the most recent common ancestor occurred t time units ago. Assume at that time that the nucleotide at the given site was a . Then, the probability that the two sequences are identical, by the Jukes–Cantor model, is

$$\begin{aligned} q &= (P_{aa}(t))^2 + (P_{ag}(t))^2 + (P_{ac}(t))^2 + (P_{at}(t))^2 \\ &= \left(\frac{1}{4} + \frac{3}{4}e^{-4rt}\right)^2 + 3\left(\frac{1}{4} - \frac{1}{4}e^{-4rt}\right)^2 \\ &= \frac{1}{4} + \frac{3}{4}e^{-8rt}, \end{aligned}$$

which is independent of the starting nucleotide. Thus the probability p that the two sites are different is

$$p = 1 - q = \frac{3}{4} (1 - e^{-8rt}).$$

Solving for rt gives

$$rt = -\frac{1}{8} \ln \left(1 - \frac{4p}{3}\right).$$

In the Jukes–Cantor model, nucleotides change at the rate of $3r$ transitions per time unit. In two DNA strands we expect $2(3rt) = 6rt$ substitutions over t years. Let K denote the number of substitutions that occur over t years. Then,

$$E(K) = 6rt = -\frac{3}{4} \ln \left(1 - \frac{4p}{3}\right).$$

To estimate the actual number of substitutions that occurred since the most recent common ancestor t time units ago, take

$$\hat{K} = -\frac{3}{4} \ln \left(1 - \frac{4\hat{p}}{3}\right),$$

where $\hat{p}$ is the observed fraction of differences between two sequences.

The number of nucleotide substitutions per site between two sequences since their separation from the last common ancestor is called the *evolutionary distance* between the sequences. It is an important quantity for estimating the rate of evolution and the divergence time between species.

Durrett compares the sequence of mitochondrial RNA for the somatotropin gene (a growth hormone) for rats and humans. The observed proportion of differences between the two sequences at a fixed site in the gene is $\hat{p} = 0.366$. Hence, the estimate of the number of substitutions at that site that have occurred since divergence of the two species is

$$\hat{K} = -\frac{3}{4} \ln \left(1 - \frac{4(0.366)}{3} \right) = 0.502.$$

Using the fact that rats and humans diverged about 80 million years ago, and choosing 0.502 as the number of substitutions per nucleotide, the mutation rate at that position is estimated as $0.502/(8 \times 10^7) = 6.275 \times 10^{-9}$ per year. ■

■ **Example 7.12 (Using symbolic software)** Symbolic software systems, such as *Mathematica* and *Maple*, work in exact integer arithmetic, and can be used to find the matrix exponential when the generator matrix contains symbolic parameters. *Wolfram Alpha*, which is freely available on the web, has the command **MatrixExp** for computing the matrix exponential.

To find the transition function for the Jukes–Cantor model from the previous example, we type the following command from our web browser

> **MatrixExp**[t{{ - 3r, r, r, r}, {r, -3r, r, r}, {r, r, -3r, r}, {r, r, r, -3r}}],

which gives the output shown in Figure 7.6. ■

7.4 LONG-TERM BEHAVIOR

For continuous-time Markov chains, limiting and stationary distributions are defined similarly as for discrete time.

Limiting Distribution

A probability distribution π is the *limiting distribution* of a continuous-time Markov chain if for all states i and j ,

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j.$$

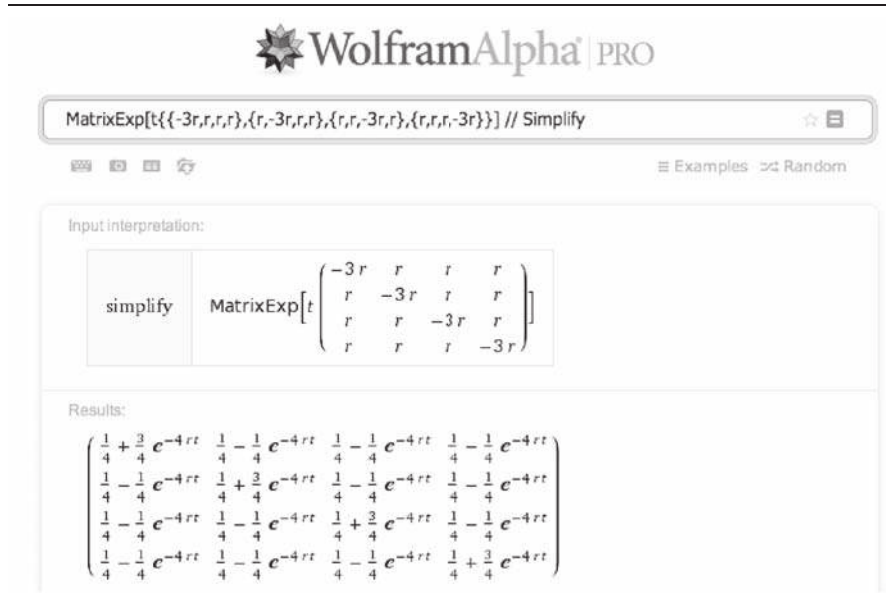


Figure 7.6 Computing the matrix exponential with *WolframAlpha*. Source: See Wolfram Alpha LLC (2015).

Stationary Distribution

A probability distribution π is a *stationary distribution* if

$$\pi = \pi P(t), \text{ for } t \geq 0.$$

That is, for all states j ,

$$\pi_j = \sum_i \pi_i P_{ij}(t), \text{ for } t \geq 0.$$

As in the discrete case, the limiting distribution, if it exists, is a stationary distribution. However, the converse is not necessarily true and depends on the class structure of the chain. For characterizing the states of a continuous-time Markov chain, notions of accessibility, communication, and irreducibility are defined as in the discrete case. A continuous-time Markov chain is *irreducible* if for all i and j , $P_{ij}(t) > 0$ for some $t > 0$.

In one regard, the classification of states is easier in continuous time since periodicity is not an issue. All states are essentially aperiodic, a consequence of the following lemma.

Lemma 7.1. *If $P_{ij}(t) > 0$, for some $t > 0$, then $P_{ij}(t) > 0$, for all $t > 0$.*

The result is intuitive since if $P_{ij}(t) > 0$ for some t , then there exists a path from i to j in the embedded chain, and for any time s there is positive probability of reaching j from i in s time units. We forego the complete proof, but show the result for forward time. Assume that $P_{ij}(t) > 0$ for some t . Then, for $s \geq 0$,

$$P_{ij}(t+s) = \sum_k P_{ik}(t)P_{kj}(s) \geq P_{ij}(t)P_{jj}(s) \geq P_{ij}(t)e^{-q_j s} > 0.$$

For the penultimate inequality, $e^{-q_j s}$ is the probability that there is no transition from j by time s . The latter event implies that the process started at j is at j at time s , whose probability is $P_{jj}(s)$.

A finite-state continuous-time Markov chain is irreducible if all the holding time parameters are positive. On the contrary, if $q_i = 0$ for some i , then i is an *absorbing state*. If we assume that all the holding time parameters are finite, then there are two possibilities: (i) the process is irreducible, all states communicate, and $P_{ij}(t) > 0$, for $t > 0$ and all i, j or (ii) the process contains one or more absorbing states.

The following fundamental limit theorem is given without proof. Note the analogies with the discrete-time results, for example, Theorems 3.6 and 3.8.

Fundamental Limit Theorem

Theorem 7.2. *Let $(X_t)_{t \geq 0}$ be a finite, irreducible, continuous-time Markov chain with transition function $P(t)$. Then, there exists a unique stationary distribution π , which is the limiting distribution. That is, for all j ,*

$$\lim_{t \rightarrow \infty} P_{ij}(t) = \pi_j, \text{ for all initial } i.$$

Equivalently,

$$\lim_{t \rightarrow \infty} P(t) = \Pi,$$

where Π is a matrix all of whose rows are equal to π .

 **Example 7.13 (Two-state process)** For the continuous-time Markov chain on two states,

$$\lim_{t \rightarrow \infty} P(t) = \lim_{t \rightarrow \infty} \frac{1}{\lambda + \mu} \begin{pmatrix} \mu + \lambda e^{-(\lambda+\mu)t} & \lambda - \lambda e^{-(\lambda+\mu)t} \\ \mu - \mu e^{-(\lambda+\mu)t} & \lambda + \mu e^{-(\lambda+\mu)t} \end{pmatrix} = \frac{1}{\lambda + \mu} \begin{pmatrix} \mu & \lambda \\ \mu & \lambda \end{pmatrix}.$$

The stationary distribution is

$$\pi = \left(\frac{\mu}{\lambda + \mu}, \frac{\lambda}{\lambda + \mu} \right).$$

■

The following result links the stationary distribution with the generator.

Stationary Distribution and Generator Matrix

A probability distribution π is a stationary distribution of a continuous-time Markov chain with generator Q if and only if

$$\pi Q = \mathbf{0}.$$

That is,

$$\sum_i \pi_i Q_{ij} = 0, \text{ for all } j.$$

Proof. Assume that $\pi = \pi P(t)$, for all $t \geq 0$. Take the derivative of both sides of the equation at $t = 0$ to get $\mathbf{0} = \pi P'(0) = \pi Q$. Conversely, assume that $\pi Q = \mathbf{0}$. Then,

$$\mathbf{0} = \pi Q P(t) = \pi P'(t), \text{ for } t \geq 0,$$

by the Kolmogorov backward equation. Since the derivative is equal to $\mathbf{0}$, $\pi P(t)$ is a constant. In particular, $P(0) = I$, and thus $\pi P(t) = \pi P(0) = \pi$, for $t \geq 0$. ■

The stationary probability π_j can be interpreted as the long-term proportion of time that the chain spends in state j . This is analogous to the discrete-time case in which the stationary probability represents the long-term fraction of transitions that the chain visits a given state.

■ **Example 7.14 (Eat, play, sleep)** Jesse is a newborn baby who is always in one of three states: eat, play, and sleep. He eats on average for 30 minutes at a time; plays on average for 1 hour; and sleeps for about 3 hours. After eating, there is a 50–50 chance he will sleep or play. After playing, there is a 50–50 chance he will eat or sleep. And after sleeping, he always plays. Jesse's life is governed by a continuous-time Markov chain. What proportion of the day does Jesse sleep?

Solution The holding time parameters for the three-state chain (in hour units) are $(q_e, q_p, q_s) = (2, 1, 1/3)$. The embedded chain transition probabilities are

$$\tilde{P} = \begin{matrix} & \begin{matrix} \text{Eat} & \text{Play} & \text{Sleep} \end{matrix} \\ \begin{matrix} \text{Eat} \\ \text{Play} \\ \text{Sleep} \end{matrix} & \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

With $q_{ij} = q_i p_{ij}$, the generator matrix is

$$Q = \begin{matrix} & \begin{matrix} \text{Eat} & \text{Play} & \text{Sleep} \end{matrix} \\ \begin{matrix} \text{Eat} \\ \text{Play} \\ \text{Sleep} \end{matrix} & \begin{pmatrix} -2 & 1 & 1 \\ 1/2 & -1 & 1/2 \\ 0 & 1/3 & -1/3 \end{pmatrix} \end{matrix}.$$

The linear system $\pi Q = \mathbf{0}$ gives

$$\begin{aligned} -2\pi_e + (1/2)\pi_p &= 0, \\ \pi_e - \pi_p + (1/3)\pi_s &= 0, \\ \pi_e + (1/2)\pi_p - (1/3)\pi_s &= 0, \end{aligned}$$

with solution

$$\pi = (\pi_e, \pi_p, \pi_s) = \left(\frac{1}{14}, \frac{4}{14}, \frac{9}{14} \right) = (0.071, 0.286, 0.643).$$

Jesses spends almost two-thirds of his day sleeping.

Using R, we compute the transition function, and then take $P(t)$ for large t (in this case $t = 100$) to find the approximate limiting distribution. ■

R: Eat, Play, Sleep

```
> install.packages("expm")
> library(expm)
> Q = matrix(c(-2, 1, 1, 1/2, -1, 1/2, 0, 1/3, -1/3),
+   nrow=3, byrow=T)
> P <- function(t) {expm(t*Q)}
> P(100)
```

	Eat	Play	Sleep
Eat	0.07143	0.28571	0.64286
Play	0.07143	0.28571	0.64286
Sleep	0.07143	0.28571	0.64286

■ **Example 7.15 (DNA evolution)** The Jukes–Cantor and Kimura models, introduced in Example 7.11, have uniform limiting distribution $\pi = (1/4, 1/4, 1/4, 1/4)$. An objection to these models is that nucleotides are not distributed uniformly on the genome. For instance, frequencies for human DNA are approximately

<i>a</i>	<i>g</i>	<i>c</i>	<i>t</i>
0.292	0.207	0.207	0.292

The four-parameter *Felsenstein* model, introduced in 1981, has generator matrix

$$Q = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} -\alpha(1-p_a) & \alpha p_g & \alpha p_c & \alpha p_t \\ \alpha p_a & \alpha(1-p_g) & \alpha p_c & \alpha p_t \\ \alpha p_a & \alpha p_g & -\alpha(1-p_c) & \alpha p_t \\ \alpha p_a & \alpha p_g & \alpha p_c & -\alpha(1-p_t) \end{pmatrix} \end{matrix},$$

where $p_a + p_g + p_c + p_t = 1$ and $\alpha > 0$. One checks that $\pi = (p_a, p_g, p_c, p_t)$ satisfies $\pi Q = \mathbf{0}$ and is the stationary distribution. The transition function, which we do not derive, is

$$P_{ij}(t) = \begin{cases} (1 - e^{-\alpha t})p_j, & \text{if } i \neq j, \\ e^{-\alpha t} + (1 - e^{-\alpha t})p_j, & \text{if } i = j. \end{cases}$$

■

Absorbing States

An absorbing Markov chain is one in which there is at least one absorbing state. Let $(X_t)_{t \geq 0}$ be an absorbing continuous-time Markov chain on $\{1, \dots, k\}$. For simplicity, assume the chain has one absorbing state a . As in the discrete case, the nonabsorbing states are *transient*. There is positive probability that the chain, started in a transient state, gets absorbed and never returns to that state. For transient state i , we derive an expression for a_i , the expected time until absorption.

Let T denote the set of transient states. Write the generator in canonical block matrix form

$$Q = \begin{matrix} & \begin{matrix} a & T \end{matrix} \\ \begin{matrix} a \\ T \end{matrix} & \begin{pmatrix} 0 & \mathbf{0} \\ * & V \end{pmatrix} \end{matrix},$$

where V is a $(k-1) \times (k-1)$ matrix.

Mean Time Until Absorption

Theorem 7.3. *For an absorbing continuous-time Markov chain, define a square matrix F on the set of transient states, where F_{ij} is the expected time, for the chain started in i , that the process is in j until absorption. Then,*

$$F = -V^{-1}.$$

For the chain started in i , the mean time until absorption is,

$$a_i = \sum_j F_{ij}.$$

The matrix F is called the fundamental matrix.

Proof. The proof is based on conditioning on the first transition. For the chain started in i , consider the mean time F_{ij} spent in j before absorption.

Assume that $j \neq i$ and that the process first moves to $k \neq i$. The probability of moving from i to k is q_{ik}/q_i . If $k = a$ is the absorbing state, the time spent in j is 0. Otherwise, the mean time in j until absorption is F_{kj} . This gives

$$\begin{aligned} F_{ij} &= \sum_{\substack{k \in T \\ k \neq i}} \frac{q_{ik}}{q_i} F_{kj} = \frac{1}{q_i} \sum_{\substack{k \in T \\ k \neq i}} V_{ik} F_{kj} \\ &= \frac{1}{q_i} [(VF)_{ij} - V_{ii} F_{ij}] = \frac{1}{q_i} [(VF)_{ij} + q_i F_{ij}] \\ &= \frac{1}{q_i} (VF)_{ij} + F_{ij}. \end{aligned}$$

Thus $(VF)_{ij} = 0$. We have used that $V_{ij} = Q_{ij}$, for $i, j \in T$.

Assume that $j = i$. The mean time spent in i before transitioning to a new state is $1/q_i$. Conditioning on the next state,

$$\begin{aligned} F_{ii} &= \frac{q_{ia}}{q_i} \left(\frac{1}{q_i} \right) + \sum_{\substack{k \in T \\ k \neq i}} \frac{q_{ik}}{q_i} \left(\frac{1}{q_i} + F_{ki} \right) \\ &= \frac{q_{ia}}{q_i^2} + \frac{1}{q_i^2} \sum_{\substack{k \in T \\ k \neq i}} q_{ik} + \frac{1}{q_i} \sum_{\substack{k \in T \\ k \neq i}} q_{ik} F_{ki} \\ &= \frac{-1}{q_i^2} Q_{ii} + \frac{1}{q_i} [(VF)_{ii} - V_{ii} F_{ii}] \\ &= \frac{1}{q_i} + \frac{1}{q_i} (VF)_{ii} + F_{ii}. \end{aligned}$$

Hence, $(VF)_{ii} = -1$.

In summary,

$$(VF)_{ij} = \begin{cases} -1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

That is, $VF = -I$, and $F = -V^{-1}$.

For the chain started in i , the mean time until absorption is the sum of the mean times in each transient state j until absorption, which gives the final result. ■

■ **Example 7.16 (Multistate models)** Multistate Markov models are used in medicine to model the course of diseases. A patient may advance into, or recover from, successively more severe stages of a disease until some terminal state. Each stage represents a state of an absorbing continuous-time Markov chain.

Bartolomeo et al. (2011) develops such a model to study the progression of liver disease among patients diagnosed with cirrhosis of the liver. The general form of the infinitesimal generator matrix for their three-parameter model is

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} -(q_{12} + q_{13}) & q_{12} & q_{13} \\ 0 & -q_{23} & q_{23} \\ 0 & 0 & 0 \end{pmatrix} \end{matrix},$$

where state 1 represents cirrhosis, state 2 denotes liver cancer (hepatocellular carcinoma), and state 3 is death. The fundamental matrix is

$$F = - \begin{pmatrix} -(q_{12} + q_{13}) & q_{12} \\ 0 & -q_{23} \end{pmatrix}^{-1} = \begin{pmatrix} 1/(q_{12} + q_{13}) & q_{12}/(q_{23}(q_{12} + q_{13})) \\ 0 & 1/q_{23} \end{pmatrix},$$

with mean absorption times

$$a_1 = \frac{1}{q_{12} + q_{13}} + \frac{q_{12}}{q_{23}(q_{12} + q_{13})} \quad \text{and} \quad a_2 = \frac{1}{q_{23}}.$$

From a sample of 1,925 patients diagnosed with cirrhosis, and data on the number of months at each stage of the disease, the authors estimate the parameters of the model for subgroups depending on age, gender, and other variables. Their mean parameter estimates are $\widehat{q}_{12} = 0.0151$, $\widehat{q}_{13} = 0.0071$, and $\widehat{q}_{23} = 0.0284$.

Plots of the transition probabilities between states are shown in Figure 7.7. The fundamental matrix is estimated to be

$$\widehat{F} = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 45.05 & 23.95 \\ 0.00 & 35.21 \end{pmatrix} \end{matrix}.$$

From the fundamental matrix, the estimated mean time to death for patients with liver cirrhosis is $45.05 + 23.95 = 69$ months. See the foregoing R code for a simulation of the mean time to death.

R: Simulation of Time to Absorption

```
# absorption.R
> trials <- 100000
> simlist <- numeric(trials)
> init <- 1 # initial state of liver cirrhosis
> for (i in 1:trials) {
+   state <- init
+   t <- 0
```

```

+ while (TRUE) {
+   if (state == 1) { q12 <- rexp(1,0.0151)
+     q13 <- rexp(1,0.0071) }
+   if (q12 < q13) {t <-t + q12
+     state <- 2}
+   else {t <- t + q13
+     break}
+   if (state == 2) {q23 <- rexp(1,0.0284)
+     t <- t + q23
+     break}
+ }
+ simlist[i] <- t }
> mean(simlist)
[1] 69.01561

```

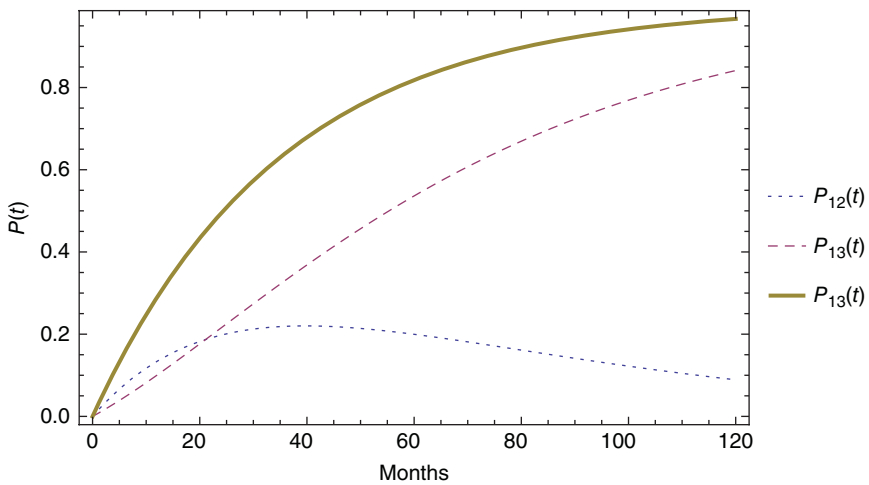


Figure 7.7 Estimated transition probabilities for stages of liver cirrhosis. *Source:* Bartolomeo et al. (2011).

Stationary Distribution of Embedded Chain

For a continuous-time Markov chain, the stationary distribution π is not the same as the stationary distribution of the embedded chain, which we denote as ψ . Consider a three-state process with generator

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} -2 & 1 & 1 \\ 2 & -4 & 2 \\ 4 & 4 & -8 \end{pmatrix} \end{matrix}.$$

The unique solution of $\pi Q = \mathbf{0}$ is $\pi = (4/7, 2/7, 1/7)$, which is the stationary distribution of the continuous-time chain. However, the embedded chain has transition matrix

$$\tilde{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} \end{matrix},$$

with uniform stationary distribution $\psi = (1/3, 1/3, 1/3)$.

Note the difference in interpretation of the two distributions. The stationary probability π_j is the long-term proportion of *time* that the process spends in state j . On the other hand, the embedded chain stationary probability ψ_j is the long-term proportion of transitions that the process makes into state j .

Each stationary distribution can be derived from the other. From $\pi Q = \mathbf{0}$, we have that

$$\pi_j q_j = \sum_{i \neq j} \pi_i q_{ij}, \text{ for all } j. \quad (7.7)$$

Define $\tilde{\psi}_j = \pi_j q_j$. Then, Equation (7.7) gives

$$\tilde{\psi}_j = \sum_{i \neq j} \tilde{\psi}_i p_{ij}.$$

Thus $\tilde{\psi}$ satisfies $\tilde{\psi} \tilde{P} = \tilde{\psi}$. To obtain a stationary distribution for the embedded chain, normalize $\tilde{\psi}$. Let

$$\psi_j = \frac{\tilde{\psi}_j}{\sum_k \tilde{\psi}_k} = \frac{\pi_j q_j}{\sum_k \pi_k q_k}.$$

Then, $\psi = \psi \tilde{P}$, and ψ is the stationary distribution of the embedded Markov chain.

Having derived ψ from π , we can also derive π from ψ . Since $\psi_j = C \pi_j q_j$, where C is an appropriate normalizing constant, we have that $\pi_j = \psi_j / (C q_j)$. This gives,

$$\pi_j = \frac{\psi_j / q_j}{\sum_k \psi_k / q_k}, \text{ for all } j.$$

Example 7.17 The embedded chain of a continuous-time process has transition matrix

$$\tilde{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

Assume that the process stays at state 1 on average 5 minutes before moving to 2. From state 2, it stays on average 2 minutes before moving to a new state. From 3, it stays on average 4 minutes before transitioning to 2. Find the stationary distribution of the continuous-time chain.

Solution Solve $\boldsymbol{\psi}\tilde{\mathbf{P}} = \boldsymbol{\psi}$ to get $\boldsymbol{\psi} = (1/6, 1/2, 1/3)$. Holding time parameters are $(q_1, q_2, q_3) = (1/5, 1/2, 1/4)$. The stationary distribution $\boldsymbol{\pi}$ is proportional to

$$(\psi_1/q_1, \psi_2/q_2, \psi_3/q_3) = (5/6, 2/2, 4/3),$$

which gives $\boldsymbol{\pi} = (5/19, 6/19, 8/19)$. ■

Global Balance

Assume that $\boldsymbol{\pi}$ is the stationary distribution of a continuous-time Markov chain. From $\boldsymbol{\pi}\mathbf{Q} = \mathbf{0}$, we have that

$$\sum_{i \neq j} \pi_i q_{ij} = \pi_j q_j, \text{ for all } j. \quad (7.8)$$

The holding time parameter q_j is the transition rate from j . Since π_j is the long-term proportion of time the process visits j , the right-hand side of Equation (7.8) is the long-term rate that the process leaves j . Also, $\pi_i q_{ij}$ is the long-term rate of transitioning from i to j . Thus, the left-hand side of Equation (7.8) is the long-term rate that the process enters j .

Equations (7.8) are known as the *global balance equations*. They say that in stationarity the rates in and out of any state are the same.

■ **Example 7.18** In Example 7.17, we have that

$$(\pi_1 q_1, \pi_2 q_2, \pi_3 q_3) = (1/19, 3/19, 2/19).$$

By global balance, in the long-term, every 19 minutes the process sees one transition to and from state 1, three transitions to and from state 2, and two transitions to and from state 3.

We illustrate one of these facts by simulating the number of transitions to and from state 2 for the first 19 minutes of the stationary continuous-time chain.

R: Simulating Global Balance

```
# globalbalance.R
> trials <- 100000
> simin2 <- simout2 <- numeric(trials)
> for (i in 1:trials) {
+   state <- sample(1:3,1, prob=c(5/19,6/19,8/19))
+   t <- 0 # time
+   in2 <- 0 # counter for transitions to 2
+   out2 <- 0 # counter for transitions from 2
+   while (t < 19) {
```

```

+ if (state == 1) {t <- t+rexp(1,1/5)
+   if (t > 19) {break}
+   state <- 2
+   in2 <- in2 + 1 }
+ if (state == 3) {t <- t + rexp(1,1/4)
+   if (t > 19) {break}
+   state <- 2
+   in2 <- in2 + 1}
+ if (state ==2) { r1 <- rexp(1, (1/2)*(1/3))
+   r3 <- rexp(1, (1/2)*(2/3))
+   if (r1 < r3) { t <- t + r1
+     if (t > 19) {break}
+     out2 <- out2 + 1
+     state <- 1} else { t <- t + r3
+     if (t > 19) {break}
+     out2 <- out2 + 1
+     state <- 3}}
+ }
+ simin2[i] <- in2
+ simout2[i] <- out2}
> mean(simin2)    # mean transitions to 2
[1] 2.99994
> mean(simout2)   # mean transitions from 2
[1] 3.00264

```

■

7.5 TIME REVERSIBILITY

Intuitively, a continuous-time Markov chain is *time reversible* if the process in forward time is indistinguishable from the process in reversed time. A consequence is that for all states i and j , the long-term *forward* transition rate from i to j is equal to the long-term *backward* rate from j to i . This is a stronger condition than global balance, which says that the long-term rate from a given state is equal to the long-term rate into that state.

Time Reversibility

A continuous-time Markov chain with generator Q and unique stationary distribution π is said to be *time reversible* if

$$\pi_i q_{ij} = \pi_j q_{ji}, \text{ for all } i, j. \quad (7.9)$$

Equations (7.9) give the *local balance*, or *detailed balance*, equations. They say that the long-term transition rate from i to j is equal to the long-term transition rate from j to i .

The local balance equations are a property of reversible chains, which can be used to find the stationary distribution. If a probability distribution λ satisfies

$$\lambda_i q_{ij} = \lambda_j q_{ji}, \text{ for all } i, j,$$

then the continuous-time chain is time reversible and $\pi = \lambda$ is the unique stationary distribution of the chain. Summing over i gives

$$\sum_i \lambda_i q_{ij} = \lambda_j \sum_i q_{ji} = 0.$$

That is, $\lambda Q = 0$.

■ **Example 7.19** The general, nine-parameter, time reversible model for DNA substitutions has generator matrix

$$Q = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} - & \alpha p_g & \beta p_c & \gamma p_t \\ \alpha p_a & - & \delta p_c & \epsilon p_t \\ \beta p_a & \delta p_g & - & \eta p_t \\ \gamma p_a & \epsilon p_g & \eta p_c & - \end{pmatrix} \end{matrix},$$

where $p_a + p_g + p_c + p_t = 1$, and $\alpha, \beta, \gamma, \delta, \epsilon, \eta > 0$. Diagonal entries are chosen so that rows sum to 0. One checks that $p_x q_{xy} = q_{yx} p_y$ for $x, y \in \{a, g, c, t\}$. It follows that the unique stationary distribution is $\pi = (p_a, p_g, p_c, p_t)$. ■

A continuous-time Markov chain is time reversible if and only if its embedded discrete-time chain is time reversible. From Equation (7.9), we obtain that

$$\pi_i q_i p_{ij} = \pi_j q_j p_{ji}, \text{ for all } i, j.$$

Dividing both sides by the normalizing constant $\sum_k \pi_k q_k$ gives

$$\psi_i p_{ij} = \psi_j p_{ji}, \text{ for all } i, j.$$

That is, ψ satisfies the detailed balance equations for the embedded chain. In stationarity, the frequency of transitions from i to j is the same as that from j to i .

Tree Theorem

A *tree* is an undirected graph in which any pair of vertices is connected by exactly one path. Equivalently, a tree is a connected graph that has no cycles. Figure 7.8 shows three examples of trees.

Consider an undirected transition graph for a continuous-time Markov chain where vertices i and j are joined by an edge if $q_{ij} > 0$ or $q_{ji} > 0$. The next theorem, from

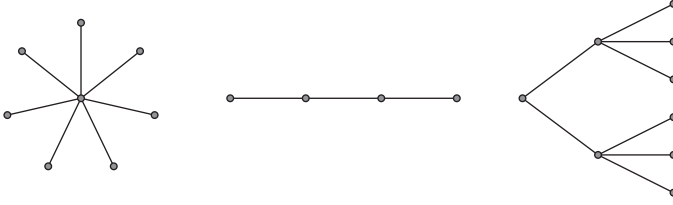


Figure 7.8 Trees.

Kelly (1994), gives a sufficient condition for time reversibility based on the structure of the graph.

Markov Processes on Trees are Time Reversible

Theorem 7.4. *Assume that the transition graph of an irreducible continuous-time Markov chain is a tree. Then, the process is time reversible.*

The result is shown by means of the following lemma, which is of independent interest. It states that the long-term transition rate out of any nonempty subset of states is equal to the long-term transition rate into that set.

Lemma 7.5. *Let S denote the state space of a continuous-time Markov chain with stationary distribution π and generator Q . For any nonempty subset of the state space $A \subseteq S$,*

$$\sum_{i \in A} \sum_{j \notin A} \pi_i q_{ij} = \sum_{i \in A} \sum_{j \notin A} \pi_j q_{ji}. \quad (7.10)$$

Proof of Lemma 7.5. The left-hand side of Equation (7.10) is equal to

$$\sum_{i \in A} \sum_{j \in S} \pi_i q_{ij} - \sum_{i \in A} \sum_{j \in A} \pi_i q_{ij} = - \sum_{i \in A} \sum_{j \in A} \pi_i q_{ij},$$

since the rows of Q sum to 0. The right-hand side of Equation (7.10) is equal to

$$\sum_{i \in A} \sum_{j \in S} \pi_j q_{ji} - \sum_{i \in A} \sum_{j \in A} \pi_j q_{ji} = - \sum_{i \in A} \sum_{j \in A} \pi_j q_{ji},$$

since π is the stationary distribution and $\pi Q = 0$. ■

Proof of Theorem 7.4. Assume that the transition graph is a tree. We show that

$$\pi_i q_{ij} = \pi_j q_{ji}, \text{ for all } i, j.$$

If there is no edge between i and j in the transition graph, then $q_{ij} = q_{ji} = 0$. Assume that there is an edge between i and j . A tree has the property, since it contains no cycles, that removal of any edge cuts the graph into two disconnected components. Let A denote the set of states connected to i , after removing the edge between i and j . Since the only vertex connected to i in the original graph, which is not in S , is j ,

$$\sum_{k \in A} \sum_{l \notin A} \pi_k q_{kl} = \pi_i q_{ij}$$

and

$$\sum_{k \in A} \sum_{l \notin A} \pi_l q_{lk} = \pi_j q_{ji}.$$

By Lemma 7.5, the result is proven. $\blacksquare$

Theorem 7.4 gives a sufficient condition for a continuous-time Markov chain to be time reversible. However, the condition is not necessary. Following is an example of a reversible Markov chain whose undirected transition graph is not a tree.

$\blacksquare$ **Example 7.20** The continuous-time Markov chain with symmetric generator matrix

$$Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

has stationary distribution $\pi = (1/3, 1/3, 1/3)$. One checks that the local balance equations are satisfied and the process is time reversible. However, the undirected transition graph is a triangle. $\blacksquare$

Birth-and-Death Process

Birth-and-death processes form a large class of time reversible, continuous-time Markov chains, which arise in many applications. For these processes, transitions only occur to neighboring states. *Births* occur from i to $i + 1$ at the rate λ_i . *Deaths* occur from i to $i - 1$ at the rate μ_i .

The generator matrix for the general birth-and-death process on $\{0, 1, \dots\}$ is

$$Q = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & 0 & \dots \\ \mu_1 & -(\lambda_1 + \mu_1) & \lambda_1 & 0 & \dots \\ 0 & \mu_2 & -(\lambda_2 + \mu_2) & \lambda_2 & \dots \\ 0 & 0 & \mu_3 & -(\lambda_3 + \mu_3) & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

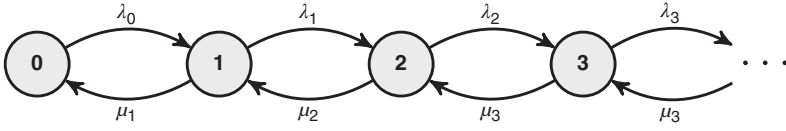


Figure 7.9 Birth-and-death process.

See Figure 7.9 for the directed transition graph. The undirected graph is a path, which is a tree. By Theorem 7.4, the chain is reversible.

The local balance equations for a birth-and-death process are

$$\pi_i \lambda_i = \pi_{i+1} \mu_{i+1}, \text{ for } i = 0, 1, \dots$$

Solving for the stationary distribution,

$$\begin{aligned} \pi_1 &= \pi_0 \frac{\lambda_0}{\mu_1}, \\ \pi_2 &= \pi_1 \frac{\lambda_1}{\mu_2} = \pi_0 \frac{\lambda_0 \lambda_1}{\mu_1 \mu_2}, \end{aligned}$$

and so on, giving

$$\pi_k = \pi_0 \frac{\lambda_0 \cdots \lambda_{k-1}}{\mu_1 \cdots \mu_k} = \pi_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i}, \text{ for } k = 0, 1, \dots, \quad (7.11)$$

where we use the convention, for $k = 0$, that an empty product is equal to 1. For the components of π to sum to 1, we need

$$1 = \sum_{k=0}^{\infty} \pi_k = \pi_0 \sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i}. \quad (7.12)$$

A necessary and sufficient condition for the stationary distribution to exist is that

$$\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} < \infty,$$

in which case we can solve for π_0 in Equation (7.12).

Stationary Distribution for Birth-and-Death Process

For a birth-and-death process with birth rates λ_i and death rates μ_i , for $i = 1, 2, \dots$, assume that

$$\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} < \infty.$$

Then, the unique stationary distribution π is

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i}, \text{ for } k = 1, 2, \dots,$$

where

$$\pi_0 = \left(\sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_{i-1}}{\mu_i} \right)^{-1}.$$

Example 7.21 (Continuous-time random walk) Consider a continuous-time version of simple random walk on $\{0, 1, 2, \dots\}$ with reflecting boundaries. From 0, the walk moves to 1 after an exponentially distributed length of time with rate λ . From $i > 0$, transitions to the left occur at the rate μ , and transitions to the right occur at the rate λ . Find the stationary distribution.

Solution The walk is a birth-and-death process with constant birth rate $\lambda_i = \lambda$ and death rate $\mu_i = \mu$. For the stationary distribution,

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda}{\mu} = \pi_0 \left(\frac{\lambda}{\mu} \right)^k, \text{ for } k = 0, 1, \dots,$$

with

$$\pi_0 = \left(\sum_{k=0}^{\infty} \left(\frac{\lambda}{\mu} \right)^k \right)^{-1} = 1 - \frac{\lambda}{\mu},$$

provided that $\lambda < \mu$. In that case,

$$\pi_k = (1 - \lambda/\mu)(\lambda/\mu)^k, \text{ for } k = 0, 1, \dots$$

For $\lambda < \mu$, the stationary distribution is the geometric distribution on $\{0, 1, \dots\}$ with parameter $1 - \lambda/\mu$. ■

Example 7.22 (Yule process) The Yule process arises in biology to describe the growth of a population where each individual gives birth to an offspring at a constant rate λ independently of other individuals. Let X_t denote the size of the population at time t . If $X_t = i$, then a new individual is born when one of the i members of the population gives birth, which occurs at rate $i\lambda$. A Yule process is a birth-and-death process with birth rate $\lambda_i = i\lambda$ and death rate $\mu_i = 0$. In a Yule process, all states are transient and no limiting distribution exists.

Assume that the initial size of the population is 1. The Yule process satisfies the Kolmogorov forward equations

$$P'_{1,j}(t) = -\lambda j P_{1,j}(t) + \lambda(j-1)P_{1,j-1}(t), \text{ for } j = 1, 2, \dots,$$

with

$$P_{1,1}(0) = 1 \quad \text{and} \quad P_{1,j} = 0, \text{ for } j \geq 2.$$

The solution to the system of differential equations, which may be verified directly, is

$$P_{1,j}(t) = e^{-\lambda t}(1 - e^{-\lambda t})^{j-1}, \text{ for } j = 1, 2, \dots,$$

which is a geometric distribution with parameter $e^{-\lambda t}$.

For the process started with i individuals, the transition function is

$$P_{ij}(t) = \binom{j-1}{i-1} e^{-\lambda it} (1 - e^{-\lambda t})^{j-i}, \text{ for } j \geq i, \quad (7.13)$$

which is a negative binomial distribution. See Exercise 7.16. ■

Example 7.23 An academic support center has N tutors who help students with their homework. Students arrive at the center according to a Poisson process at rate λ . Each tutor takes an exponential length of time to work with students. Tutors' service times and student arrival times are independent. If all the tutors are busy when a student arrives at the center, the student will leave. Let X_t denote the number of tutors who are busy at time t . Find the stationary distribution.

Solution Assume that $i < N$ tutors are busy at time t . The number of busy tutors will increase by one if a student arrives, which occurs at rate λ . That is, $q_{i,i+1} = \lambda$, for $i = 0, \dots, N-1$.

On the other hand, if $i > 0$ tutors are busy, the number of busy tutors decreases by one if one of the i busy tutors finishes with their student. Each tutor's service time is exponentially distributed with parameter μ . Thus, the first time one of the i tutors is free is the minimum of i exponential random variables with parameter μ , which has an exponential distribution with parameter $i\mu$. This gives $q_{i,i-1} = i\mu$, for $i = 1, \dots, N$.

The process is a finite birth-and-death process with constant birth rate and linear death rate. The generator for $N = 4$ tutors is

$$Q = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 \\ \mu & -(\lambda + \mu) & \lambda & 0 & 0 \\ 0 & 2\mu & -(\lambda + 2\mu) & \lambda & 0 \\ 0 & 0 & 3\mu & -(\lambda + 3\mu) & \lambda \\ 0 & 0 & 0 & 4\mu & -4\mu \end{pmatrix} \end{matrix}.$$

For general N , the stationary distribution is

$$\pi_k = \pi_0 \prod_{i=1}^k \frac{\lambda}{i\mu} = \pi_0 \frac{(\lambda/\mu)^k}{k!}, \text{ for } k = 0, 1, \dots, N,$$

with

$$\pi_0 = \left(\sum_{k=0}^N \prod_{i=1}^k \frac{\lambda}{i\mu} \right)^{-1} = \left(\sum_{k=0}^N \frac{(\lambda/\mu)^k}{k!} \right)^{-1}.$$

The distribution is a truncated Poisson distribution on $\{0, 1, \dots, N\}$ with parameter λ/μ . For large N , the distribution is an approximate Poisson distribution, and the long-term expected number of busy tutors is λ/μ . ■

Common birth-and-death processes are listed in Table 7.1.

TABLE 7.1 Types of Birth-and-Death Processes

Type	Birth Rate	Death Rate
Pure birth	λ_i	$\mu_i = 0$
Poisson process	$\lambda_i = \lambda$	$\mu_i = 0$
Pure death	$\lambda_i = 0$	μ_i
Linear process	$\lambda_i = i\lambda, i > 0$	$\mu_i = i\mu$
Yule process	$\lambda_i = \lambda i, i, \lambda > 0$	$\mu_i = 0$
Linear with immigration	$\lambda_i = i\lambda + \alpha, i, \alpha > 0$	$\mu_i = i\mu$

7.6 QUEUEING THEORY

Queueing theory is the study of waiting lines, or queues. In the terminology of queueing theory, *customers* arrive at a facility for *service*. If the service is not immediate, they wait for service, and leave the system when the service is complete. The framework is very general and could describe a diner waiting to be seated at a restaurant, a computer program waiting to be run, and a machine waiting to be repaired.

The general queueing model can be quite broad with many parameters, which describe things such as the distribution of arrival times, the distribution of service times, the number of servers, the capacity of the system, the stages of service, and how customers line up to be served.

A standard notation of the form $A/B/n$ is used to describe a queueing model, where A denotes the arrival time distribution, B the service time distribution, and n the number of servers.

The $M/M/1$ queue is a basic model. The M stands for Markov or memoryless. In this model, both arrival and service times have exponential distributions, and there is one server. The $M/M/1$ queue is a birth-and-death process with constant birth and death rates.

The queueing models we explore in this section are continuous-time Markov chains where X_t denotes the number of customers in the system at time t .

A central result in queueing theory is Little's formula, which is deceptively simple and remarkably diverse.

Little's Formula

In a queueing system, let L denote the long-term average number of customers in the system, λ the rate of arrivals, and W the long-term average time that a customer is in the system. Then,

$$L = \lambda W.$$

The power of Little's formula is that it applies to a very broadly defined queueing system. We will not prove the formula but justify it intuitively with a *proof by picture*. The argument is based on Gross et al. (2008).

Consider a realization of a queueing system between the time when a customer first enters the system and when the system is next empty. Assume that four customers enter the system in the time interval $[0, t]$ with arrival and departure times

Customer	Arrival Time	Departure Time
1	t_1	t_5
2	t_2	t_4
3	t_3	t_7
4	t_6	t

See Figure 7.10. Let A be the area of the shaded region. Little's formula is obtained by computing A in two ways.

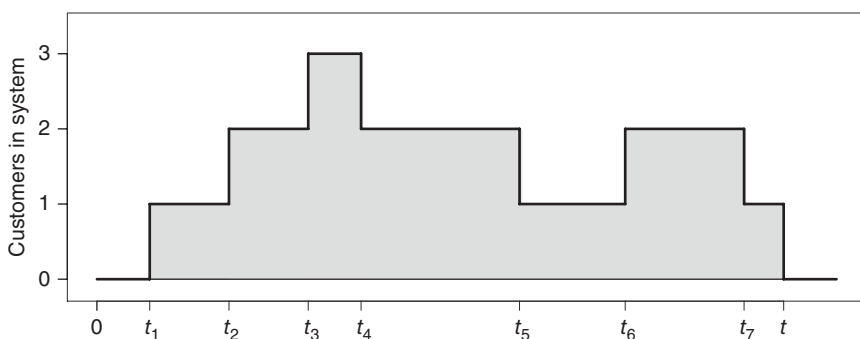


Figure 7.10 Little's formula is obtained by finding the area of the shaded region two ways.

The average length of time W that a customer spends in the system is

$$\begin{aligned} W &= \frac{(t_5 - t_1) + (t_4 - t_2) + (t_7 - t_3) + (t - t_6)}{4} \\ &= \frac{(t - t_1) + (t_5 - t_2) + (t_7 - t_6) + (t_4 - t_3)}{\text{Number of customers in } [0, t]} \\ &= \frac{A}{\text{Number of customers in } [0, t]}. \end{aligned} \quad (7.14)$$

In addition, the average number of customers L in the system is

$$\begin{aligned} L &= \frac{1}{t} [1(t_2 - t_1) + 2(t_3 - t_2) + 3(t_4 - t_3) + 2(t_5 - t_4) \\ &\quad + 1(t_6 - t_5) + 2(t_7 - t_6) + 1(t - t_7)] \\ &= \frac{A}{t}. \end{aligned} \quad (7.15)$$

Equations (7.14) and (7.15) give

$$L = \frac{A}{t} = \frac{W \times \text{Number of customers in } [0, t]}{t}.$$

For large t , this gives $L = \lambda W$.

Example 7.24 (At the carwash) Cars arrive at a drive-through carwash according to a Poisson process at the rate of nine customers per hour. The time to wash a car has an exponential distribution with mean 5 minutes. Many questions can be asked.

1. How many cars, on average, are at the carwash?
2. How long, on average, is a customer at the carwash?
3. How long, on average, does a customer wait to be served?
4. What is the expected number of cars waiting to be served?

Solution Let X_t denote the number of cars in the system at hour t . The process is an M/M/1 queueing system, which is a birth-and-death process with constant birth and death rates. The arrival rate is $\lambda = 9$. Since the average time to wash a car is one-twelfth of an hour, the service rate is $\mu = 12$. The limiting distribution probabilities (see Example 7.21) are

$$\pi_k = \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^k = \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^k, \text{ for } k = 0, 1, \dots,$$

which is a geometric distribution with parameter $p = 1 - \lambda/\mu = 1/4$.

1. The long-term expected number of cars at the carwash is the mean of the geometric distribution. A geometric distribution with parameter p , which takes values on $\{0, 1, \dots\}$, has expectation $(1 - p)/p$. The desired expectation is

$$\frac{\lambda/\mu}{1 - \lambda/\mu} = \frac{\lambda}{\mu - \lambda} = 3.$$

On average, there will be three cars at the carwash. In the notation of Little's formula, $L = 3$.

2. By Little's formula, the long-term average time W that a customer is at the carwash is

$$W = \frac{L}{\lambda} = \frac{\lambda/(\mu - \lambda)}{\lambda} = \frac{1}{\mu - \lambda} = \frac{1}{3}.$$

A customer will be at the carwash, on average, for 20 minutes.

3. Let W_q denote the long-term average time a customer spends in the queue waiting to be served. Let W_s be the average time it takes for a car to be washed. Then, $W = W_q + W_s$. The service time for a car to be washed is the mean of an exponential distribution with parameter μ . Thus, $W_s = 1/\mu$. This gives

$$W_q = W - W_s = \frac{1}{\mu - \lambda} - \frac{1}{\mu} = \frac{\lambda}{\mu(\lambda - \mu)} = \frac{1}{4}.$$

A customer waits, on average, 15 minutes to be served.

4. We can consider the process restricted to just the queue as its own queueing system. Little's formula applies with $L_q = \lambda W_q$, where L_q denotes the long-term average number of cars waiting to be served. This gives

$$L_q = \lambda W_q = \frac{\lambda^2}{\mu(\lambda - \mu)} = \frac{9}{4}.$$

On average, there are 2.25 cars in the queue. ■

M/M/c Queue

An M/M/c queue has c servers. Consider the dynamics of the process. If there are $0 < k \leq c$ customers then k servers are busy. The number of customers will decrease by one, the first time one of the servers completes their service. The time until that happens is the minimum of k independent exponential random variables, which has an exponential distribution with parameter $k\mu$. If there are more than c customers in the system, then the time until the first service time is complete is the minimum of c independent exponential random variables, and thus is exponentially distributed with parameter $c\mu$.

The M/M/c queue is a birth-and-death process with parameters $\lambda_i = \lambda$, for all i , and

$$\mu_i = \begin{cases} i\mu, & \text{for } i = 1, \dots, c, \\ c\mu, & \text{for } i = c + 1, c + 2, \dots \end{cases}$$

We have that

$$\begin{aligned}
 \sum_{k=0}^{\infty} \prod_{i=1}^k \frac{\lambda_i}{\mu_i} &= \sum_{k=0}^{c-1} \prod_{i=1}^k \frac{\lambda}{i\mu} + \sum_{k=c}^{\infty} \left(\prod_{i=1}^c \frac{\lambda}{i\mu} \right) \left(\prod_{i=c+1}^k \frac{\lambda}{c\mu} \right) \\
 &= \sum_{k=0}^{c-1} \left(\frac{\lambda}{\mu} \right)^k \frac{1}{k!} + \frac{1}{c!} \sum_{k=c}^{\infty} \left(\frac{\lambda}{\mu} \right)^k \left(\frac{1}{c} \right)^{k-c} \\
 &= \sum_{k=0}^{c-1} \left(\frac{\lambda}{\mu} \right)^k \frac{1}{k!} + \frac{(\lambda/\mu)^c}{c!} \sum_{k=c}^{\infty} \left(\frac{\lambda}{c\mu} \right)^{k-c}.
 \end{aligned}$$

The infinite sum converges for $0 < \lambda < c\mu$, in which case the stationary distribution π exists, with

$$\pi_0 = \left(\sum_{k=0}^{c-1} \left(\frac{\lambda}{\mu} \right)^k \frac{1}{k!} + \frac{(\lambda/\mu)^c}{c!} \left(\frac{1}{1 - \lambda/c\mu} \right) \right)^{-1}.$$

The stationary probabilities are

$$\pi_k = \begin{cases} \frac{\pi_0}{k!} \left(\frac{\lambda}{\mu} \right)^k, & \text{for } 0 \leq k < c, \\ \frac{\pi_0}{c^{k-c}c!} \left(\frac{\lambda}{\mu} \right)^k, & \text{for } k \geq c. \end{cases}$$

■ **Example 7.25 (At the hair salon)** A hair salon has five chairs. Customers arrive at the salon at the rate of 6 per hour. The hair stylists each take, on average, half an hour to service a customer, independent of arrival times.

1. Jill, the owner, wants to know the long-term probability that no customers are in the salon.
2. Danny, a potential customer, wants to know the average waiting time for a haircut.
3. Leslie, a hair stylist, wants to know the long-term expected number of customers in the salon.

Solution The system is an M/M/5 queue with $\lambda = 6$, $c = 5$, and $\mu = 2$.

1. The long-term probability that no customers are in the salon is

$$\pi_0 = \left(\sum_{k=0}^4 \frac{3^k}{k!} + \frac{3^5}{5!} \left(\frac{1}{1 - 6/10} \right) \right)^{-1} = \frac{16}{343} = 0.0466.$$

2. The long-time average waiting time in the queue, in the notation of Little's formula, is $W_q = L_q/\lambda$. To find L_q , the expected number of customers in the queue, observe that there are k people in the queue if and only if there are $k + c$ customers in the system. This gives

$$\begin{aligned}
 L_q &= \sum_{k=c}^{\infty} (k - c) \pi_k = \sum_{k=c}^{\infty} (k - c) \frac{\pi_0}{c^{k-c} c!} \left(\frac{\lambda}{\mu} \right)^k \\
 &= \frac{\pi_0}{c!} \left(\frac{\lambda}{\mu} \right)^c \sum_{k=c}^{\infty} (k - c) \frac{1}{c^{k-c}} \left(\frac{\lambda}{\mu} \right)^{k-c} \\
 &= \frac{\pi_0}{c!} \left(\frac{\lambda}{\mu} \right)^c \sum_{k=0}^{\infty} k \left(\frac{\lambda}{c\mu} \right)^k \\
 &= \frac{\pi_0}{c!} \left(\frac{\lambda}{\mu} \right)^c \frac{\lambda}{c\mu} \left(\frac{1}{1 - \lambda/c\mu} \right)^2 \\
 &= \frac{\pi_0 3^5}{5!} \left(\frac{6}{10} \right) \left(\frac{1}{1 - 6/10} \right)^2 \\
 &= 0.35423.
 \end{aligned}$$

By Little's formula, the expected waiting time in the queue is

$$W_q = \frac{L_q}{\lambda} = \frac{0.35423}{6} = 0.059,$$

or about 3.6 minutes.

3. The long-term expected waiting time in the system is

$$W = W_q + W_s = W_q + \frac{1}{\mu} = 0.059 + 0.5 = 0.559,$$

or about 33.54 minutes. The expected number of customers in the system, by Little's formula, is

$$L = \lambda W = 6(0.559) = 3.354.$$

Note that the last result can also be obtained by finding the expectation with respect to the stationary distribution

$$L = \sum_{k=0}^{\infty} k \pi_k.$$

■

7.7 POISSON SUBORDINATION

The times when transitions occur for a continuous-time Markov chain are exponentially distributed with holding time rates $q_1, q_2, \dots$. In this section, we present an interesting representation of a continuous-time Markov chain in which all holding

time rates are the same, but where transitions from a state to itself are allowed. The representation is remarkably useful for simulation and numerical computation.

Consider a finite-state, irreducible, discrete-time Markov chain $Y_0, Y_1, \dots$ with transition matrix $\mathbf{R}$. Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ , which is independent of the Markov chain. Define a continuous-time process $(X_t)_{t \geq 0}$ by $X_t = Y_{N_t}$. That is, transitions for the X_t process occur at the arrival times of the Poisson process. From state i , the process holds an exponentially distributed amount of time with parameter λ and then transitions to j with probability R_{ij} .

The process $(X_t)_{t \geq 0}$ is a continuous-time Markov chain whose transition function $P(t)$ has a surprisingly simple form. By conditioning on N_t ,

$$\begin{aligned} P_{ij}(t) &= P(X_t = j | X_0 = i) \\ &= \sum_{k=0}^{\infty} P(X_t = j | N_t = k, X_0 = i) P(N_t = k | X_0 = i) \\ &= \sum_{k=0}^{\infty} P(Y_k = j | N_t = k, X_0 = i) P(N_t = k) \\ &= \sum_{k=0}^{\infty} P(Y_k = j | Y_0 = i) P(N_t = k) \\ &= \sum_{k=0}^{\infty} R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!}. \end{aligned}$$

We say that the $(X_t)_{t \geq 0}$ Markov chain is *subordinated to a Poisson process*.

Not only can we construct a continuous-time Markov chain from a discrete-time chain and a Poisson process, but conversely many continuous-time Markov chains can be represented as a chain subordinated to a Poisson process.

Consider a continuous-time Markov chain with generator $\mathbf{Q}$ and holding time parameters $q_1, q_2, \dots$. Assume the parameters are uniformly bounded. That is, there exists a constant λ such that $q_i \leq \lambda$, for all i . This will always be the case if the chain is finite, and we can take $\lambda = \max_i q_i$. Let

$$\mathbf{R} = \frac{1}{\lambda} \mathbf{Q} + \mathbf{I}.$$

The matrix $\mathbf{R}$ is a stochastic matrix. Entries are non-negative and rows sum to 1. The transition function can be given in terms of $\mathbf{R}$, as

$$\begin{aligned} P(t) &= e^{t\mathbf{Q}} = e^{-\lambda t} e^{t\mathbf{Q}} e^{\lambda t} = e^{-\lambda t} e^{t(\mathbf{Q} + \lambda \mathbf{I})} \\ &= e^{-\lambda t} \sum_{k=0}^{\infty} \frac{1}{k!} t^k (\mathbf{Q} + \lambda \mathbf{I})^k \\ &= \sum_{k=0}^{\infty} \left(\frac{1}{\lambda} \mathbf{Q} + \mathbf{I} \right)^k \frac{e^{-\lambda t} (\lambda t)^k}{k!} \end{aligned}$$

$$= \sum_{k=0}^{\infty} \mathbf{R}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!}.$$

The continuous-time chain is represented as a Markov chain subordinated to a Poisson process. Other names for Poisson subordination are *randomization* and *uniformization*.

Note that the $\mathbf{R}$ matrix is *not* the matrix of the embedded Markov chain. The entries of the embedded Markov matrix are

$$\tilde{P}_{ij} = \begin{cases} q_{ij}/q_i, & \text{for } i \neq j, \\ 0, & \text{for } i = j, \end{cases}$$

while the entries of the $\mathbf{R}$ matrix are

$$R_{ij} = \begin{cases} q_{ij}/\lambda, & \text{for } i \neq j, \\ 1 - q_i/\lambda, & \text{for } i = j. \end{cases}$$

Poisson subordination can be described as follows. From a given state i , wait an exponential length of time with rate λ . Then, flip a coin whose heads probability is q_i/λ . If heads, transition to a new state according to the $\mathbf{R}$ matrix. If tails, stay at i and repeat. Thus, holding time parameters are constant, and transitions, or pseudo-transitions, from a state to itself are allowed.

To illustrate, consider a continuous-time Markov chain on $\{a, b, c\}$ with generator

$$\mathbf{Q} = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} -2 & 1 & 1 \\ 1 & -3 & 2 \\ 5 & 1 & -6 \end{pmatrix} \end{matrix}.$$

Choose $\lambda = \max\{2, 3, 6\} = 6$. Then,

$$\mathbf{R} = \frac{1}{\lambda} \mathbf{Q} + \mathbf{I} = \frac{1}{6} \begin{pmatrix} -2 & 1 & 1 \\ 1 & -3 & 2 \\ 5 & 1 & -6 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2/3 & 1/6 & 1/6 \\ 1/6 & 1/2 & 1/3 \\ 5/6 & 1/6 & 0 \end{pmatrix}.$$

See Figure 7.11 for a comparison of the dynamics of the original chain with the subordinated process.

Long-Term Behavior, Simulation, Computation

For a Markov chain subordinated to a Poisson process, the discrete $\mathbf{R}$ -chain has the same stationary distribution as the original chain. We have that

$$\pi \mathbf{Q} = \pi \lambda (\mathbf{R} - \mathbf{I}) = \lambda \pi \mathbf{R} - \lambda \pi.$$

Thus, $\pi \mathbf{Q} = \mathbf{0}$ if and only if $\pi \mathbf{R} = \pi$.

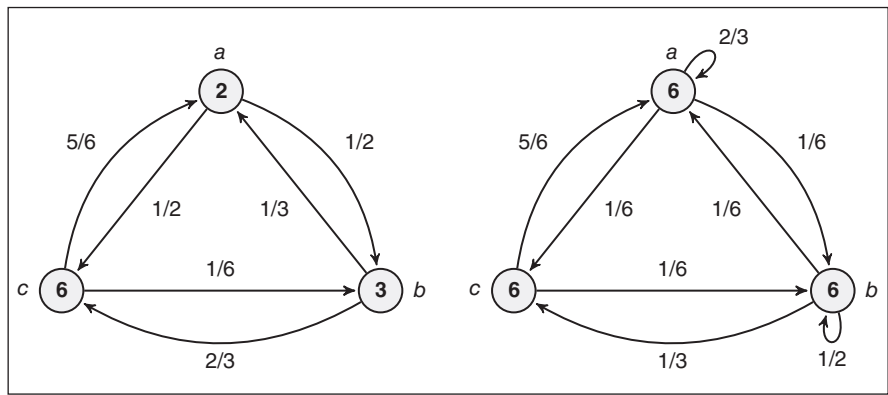


Figure 7.11 Both graphs describe the same Markov chain. Nodes are labeled with holding time parameters, edges are labeled with transition probabilities. The graph on the right shows the chain subordinated to a Poisson process. Holding time parameters are constant, and transitions to the same state are allowed.

■ **Example 7.26** Consider a Markov chain $(X_t)_{t \geq 0}$ with generator

$$Q = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -3 & 2 \\ 0 & 1 & -1 \end{pmatrix}.$$

Letting $\lambda = \max\{q_1, q_2, q_3\} = 3$ gives

$$R = \frac{1}{3}Q + I = \begin{pmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 0 & 2/3 \\ 0 & 1/3 & 2/3 \end{pmatrix}.$$

Observe that $\pi = (1/8, 2/8, 5/8)$ is the stationary distribution for both the original chain and the discrete-time R -chain. ■

Poisson subordination leads to an efficient method to simulate a continuous-time Markov chain, since transitions rates are constant and thus not dependent on the current state. Here, we simulate the distribution of $X_{1.5}$ for the Markov chain in Example 7.26.

R: Simulation of the Distribution of $X_{1.5}$

```
# Psubordination.R
> Q
  1  2  3
1 -2  1  1
2  1 -3  2
3  0  1 -1
```

```

> lambda <- 3
> R <- (1/lambda)*Q+diag(3)
> R
      1      2      3
1 0.3333333 0.3333333 0.3333333
2 0.3333333 0.0000000 0.6666667
3 0.0000000 0.3333333 0.6666667
> trials <- 100000
> simlist <- numeric(trials)
> for (i in 1:trials) {
+   s <- 0 # time
+   state <- 1
+   newstate <- 1
+   while(s < 1.5) {
+     state <- newstate
+     s <- s+rexp(1,lambda)
+     newstate <- sample(1:3,1,prob=r[state,]) }
+   simlist[i] <- state
+ }
> table(simlist)/trials
simlist
      1      2      3
0.16274 0.24958 0.58768
> expm(1.5*Q)[1,] # Compare with exact values
      1      2      3
1 0.163 0.249 0.588

```

Another benefit of Poisson subordination is that it gives a numerically stable method for computing a Markov transition function, as compared with the matrix exponential. For large matrices, the matrix exponential is notoriously difficult to compute. The infinite sum $\sum_{k=0}^{\infty} (tQ)^k/k!$ converges slowly. The matrix Q contains positive and negative numbers, some of which may be greater than 1, and thus Q^k may contain large positive and negative numbers, a source of numerical instability.

On the other hand, with Poisson subordination the transition function

$$P_{ij}(t) = \sum_{k=0}^{\infty} R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!} \quad (7.16)$$

can be computed numerically by truncating the infinite sum to a desired level of accuracy. Consider the approximation

$$\widehat{P_{ij}(t)} = \sum_{k=0}^N R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!}, \quad (7.17)$$

for some N . The absolute error of the approximation is

$$\begin{aligned}
 |P_{ij}(t) - \widehat{P_{ij}(t)}| &= \left| \sum_{k=0}^{\infty} R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!} - \sum_{k=0}^N R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!} \right| \\
 &= \sum_{k=N+1}^{\infty} R_{ij}^k \frac{e^{-\lambda t} (\lambda t)^k}{k!} \\
 &\leq \sum_{k=N+1}^{\infty} \frac{e^{-\lambda t} (\lambda t)^k}{k!} \\
 &= P(Y > N),
 \end{aligned}$$

where Y is a Poisson random variable with parameter λt . The inequality holds since R^k is a stochastic matrix, all of whose entries are between 0 and 1.

To obtain an absolute error in the approximation of at most ϵ , choose N such that $P(Y > N) \leq \epsilon$.

The following example is small enough so that exact calculations are possible and numerical approximations are not necessary. However, it illustrates the general method.

■ **Example 7.27** Consider a birth-and-death process on $\{0, \dots, 5\}$ with constant death rate $\mu_i = 2$ and linear birth rates $\lambda_i = i$, for $i = 1, \dots, 4$. The generator is

$$Q = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ 2 & -4 & 2 & 0 & 0 & 0 \\ 0 & 2 & -5 & 3 & 0 & 0 \\ 0 & 0 & 2 & -6 & 4 & 0 \\ 0 & 0 & 0 & 2 & -7 & 5 \\ 0 & 0 & 0 & 0 & 2 & -2 \end{pmatrix} \end{pmatrix}.$$

Find $P(1.5)$ to within four-digit accuracy.

Solution Let $\lambda = 7$, and set

$$R = \frac{1}{\lambda} Q + I = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 6/7 & 1/7 & 0 & 0 & 0 & 0 \\ 2/7 & 3/7 & 2/7 & 0 & 0 & 0 \\ 0 & 2/7 & 2/7 & 3/7 & 0 & 0 \\ 0 & 0 & 2/7 & 1/7 & 4/7 & 0 \\ 0 & 0 & 0 & 2/7 & 0 & 5/7 \\ 0 & 0 & 0 & 0 & 2/7 & 5/7 \end{pmatrix} \end{pmatrix}.$$

Choose N such that $P(Y > N) < 0.5 \times 10^{-4}$, where Y is a Poisson random variable with parameter $7 \times 1.5 = 10.5$. In R, we find that

```
> 1-ppois(24, 10.5)
[1] 9.933169e-05
> 1-ppois(25, 10.5)
[1] 3.921504e-05
```

Thus, truncate the infinite series at $N = 25$. This gives

$$\begin{aligned} \widehat{P(1.5)} &= \sum_{k=0}^{25} R^k \frac{e^{-\lambda(1.5)} ((1.5)t)^k}{k!} \\ &= \begin{pmatrix} 0.50706 & 0.18466 & 0.09985 & 0.06535 & 0.05600 & 0.08703 \\ 0.36932 & 0.15278 & 0.10063 & 0.08480 & 0.09944 & 0.19299 \\ 0.19970 & 0.10063 & 0.08741 & 0.10025 & 0.15741 & 0.35457 \\ 0.08713 & 0.05653 & 0.06683 & 0.10241 & 0.19880 & 0.48826 \\ 0.03734 & 0.03315 & 0.05247 & 0.09940 & 0.21813 & 0.55948 \\ 0.02321 & 0.02573 & 0.04728 & 0.09765 & 0.22379 & 0.58230 \end{pmatrix}. \end{aligned}$$

For this example, we can compare the numerical approximation with the matrix exponential computation $P(1.5) = e^{1.5Q}$, which can be found with software. The transition function, to five decimal places, is

$$P(1.5) = \begin{pmatrix} 0.50708 & 0.18467 & 0.09985 & 0.06535 & 0.05601 & 0.08704 \\ 0.36933 & 0.15278 & 0.10064 & 0.08480 & 0.09945 & 0.19300 \\ 0.19970 & 0.10064 & 0.08741 & 0.10025 & 0.15742 & 0.35458 \\ 0.08714 & 0.05653 & 0.06683 & 0.10241 & 0.19881 & 0.48828 \\ 0.03734 & 0.03315 & 0.05247 & 0.09940 & 0.21814 & 0.55950 \\ 0.02321 & 0.02573 & 0.04728 & 0.09766 & 0.2238 & 0.58232 \end{pmatrix}.$$

By inspection, we see that our approximation is accurate to the desired level of accuracy.

By contrast, if we truncate the infinite sum in the matrix exponential to 25 terms we get

$$\begin{aligned} e^{1.5Q} &\approx \sum_{k=0}^{25} (1.5Q)^k / k! \\ &= \begin{pmatrix} -4.06 & 22.01 & -69.10 & 170.51 & -284.73 & 166.39 \\ 44.01 & -208.29 & 661.43 & -1629.30 & 2723.19 & -1590.05 \\ -138.20 & 661.43 & -2099.20 & 5174.13 & -8648.70 & 5051.54 \\ 227.34 & -1086.20 & 3449.42 & -8503.51 & 14216.90 & -8302.93 \\ -189.82 & 907.73 & -2882.90 & 7108.44 & -11884.80 & 6942.40 \\ 44.37 & -212.01 & 673.54 & -1660.59 & 2776.96 & -1621.28 \end{pmatrix}, \end{aligned}$$

which is not close to converging. In fact, it takes almost twice as many terms in this case before reaching the desired level of accuracy. ■

EXERCISES

7.1 A continuous-time Markov chain has generator matrix

$$Q = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} -1 & 1 & 0 \\ 1 & -2 & 1 \\ 2 & 2 & -4 \end{pmatrix} \end{matrix}.$$

Exhibit (i) the transition matrix of the embedded Markov chain and (ii) the holding time parameter for each state.

7.2 A Markov chain on $\{1, 2, 3, 4\}$ has nonzero transition rates

$$q_{12} = q_{23} = q_{31} = q_{41} = 1 \quad \text{and} \quad q_{14} = q_{32} = q_{34} = q_{43} = 2.$$

- (a) Exhibit the (i) generator, (ii) holding time parameters, and (iii) transition matrix for the embedded Markov chain.
- (b) If the chain is at state 1, how long on average will it take before moving to a new state?
- (c) If the chain is at state 3, how long on average will it take before moving to state 4?
- (d) Over the long term, what proportion of visits will be to state 2?

7.3 A three-state Markov chain has distinct holding time parameters a , b , and c . From each state, the process is equally likely to transition to the other two states. Exhibit the generator matrix and find the stationary distribution.

7.4 During lunch hour, customers arrive at a fast-food restaurant at the rate of 120 customers per hour. The restaurant has one line, with three workers taking food orders at independent service stations. Each worker takes an exponentially distributed amount of time—on average 1 minute—to service a customer. Let X_t denote the number of customers in the restaurant (in line and being serviced) at time t . The process $(X_t)_{t \geq 0}$ is a continuous-time Markov chain. Exhibit the generator matrix.

7.5 For the fast-food restaurant chain of the previous exercise, assume that customers turn away from the store if all three service stations are busy. Let Y_t denote the number of service stations busy at time t . Then, $(Y_t)_{t \geq 0}$ is a continuous-time Markov chain. Exhibit the generator matrix.

7.6 A Markov chain $(X_t)_{t \geq 0}$ on $\{1, 2, 3, 4\}$ has generator matrix

$$Q = \begin{pmatrix} -2 & 1 & 1 & 0 \\ 1 & -3 & 1 & 1 \\ 2 & 2 & -4 & 0 \\ 1 & 2 & 3 & 6 \end{pmatrix}.$$

Use technology as needed for the following:

- (a) Find the long-term proportion of time that the chain visits state 1.
- (b) For the chain started in state 2, find the long-term probability that the chain visits state 3.
- (c) Find $P(X_1 = 3 | X_0 = 1)$.
- (d) Find $P(X_5 = 1, X_2 = 4 | X_1 = 3)$.

7.7 A Markov chain has generator matrix

$$Q = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -2 & 2 \\ 3 & 0 & -3 \end{pmatrix}.$$

- (a) Exhibit the Kolmogorov backward equations.
- (b) Find the transition function by diagonalizing the generator and finding the matrix exponential.
- (c) Find the transition function using a symbolic software system such as *Wolfram Alpha*.

7.8 Consider the Jukes–Cantor model for DNA nucleotide substitution in Example 7.11. Find the transition function $P(t)$ by solving the backward, or forward, equations.

7.9 Consider the Felsenstein DNA model of Example 7.15.

- (a) Show that $\pi = (p_a, p_c, p_c, p_t)$ is the stationary distribution.
- (b) For $\alpha = 2$ and the parameter values given in the example, find the probability that an a nucleotide mutates to a c nucleotide in 1.5 time units.

7.10 Unlike the Felsenstein DNA model, introduced in Example 7.15, the Hasegawa, Kishino, Yao model distinguishes between transitions and transversions. The generator matrix is

$$Q = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} -\alpha p_g - \beta p_r & \alpha p_g & \beta p_c & \beta p_t \\ \alpha p_a & -\alpha p_a - \beta p_r & \beta p_c & \beta p_t \\ \beta p_a & \beta p_g & -\alpha p_t - \beta p_s & \alpha p_t \\ \beta p_a & \beta p_g & \alpha p_c & -\alpha p_c - \beta p_s \end{pmatrix} \end{matrix},$$

where $p_r = p_c + p_t$, $p_s = p_a + p_g$, $p_a + p_g + p_c + p_t = 1$, and $\alpha, \beta > 0$. Show that $\pi = (p_a, p_g, p_c, p_t)$ is the stationary distribution of the chain.

7.11 Let A be a square matrix. Show that

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A.$$

7.12 The following result from linear algebra relates the determinant and trace of a matrix A :

$$\det e^A = e^{\text{tr } A}.$$

Prove this for the case that A is diagonalizable.

7.13 Assume that π is the limiting distribution of a continuous-time chain. Show that π is a stationary distribution. (*Hint*: start with the forward equation.)

7.14 For the Markov chain with transition rate graph shown in Figure 7.12, find

- the generator matrix,
- the stationary distribution of the continuous-time Markov chain,
- the transition matrix of the embedded chain,
- the stationary distribution of the embedded chain.

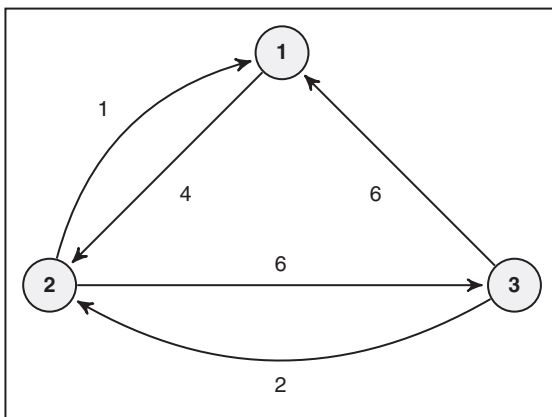


Figure 7.12

7.15 Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter $\lambda = 1$. Define the process $X_t = N_t \bmod 4$, for $t \geq 0$. Then, $(X_t)_{t \geq 0}$ is a continuous-time Markov chain on $\{0, 1, 2, 3\}$.

- Exhibit the generator matrix.
- Use a symbolic software system such as *Wolfram Alpha* to find the transition function $P(t)$.

7.16 For a Yule process started with i individuals, derive the transition probabilities

$$P_{ij}(t) = \binom{j-1}{i-1} e^{-\lambda i t} (1 - e^{-\lambda t})^{j-i}, \text{ for } j \geq i.$$

Hint: Relate the process to a Yule process started with one individual.

- 7.17** Each individual in a population gives birth to an offspring at the rate of 1.5 per unit time independently of other individuals. If the population starts with 4 individuals, find the mean and variance of the size of the population at time $t = 8$.
- 7.18** For a general birth-and-death process with birth rates λ_i and death rates μ_i , let T_i denote the time, from state i , for the process to hit state $i + 1$.
- (a) Show that

$$E[T_i] = \frac{1}{\lambda_i} + \frac{\mu_i}{\lambda_i} E[T_{i-1}], \text{ for } i = 1, 2, \dots$$

Hint: Condition on whether the first transition is a birth or a death.

- (b) Solve $E[T_i]$ for the case $\lambda_i = \lambda$ and $\mu_i = \mu$, for all i .
- 7.19** Taxis arrive at a taxi stand according to a Poisson process with parameter λ . Customers arrive, independently of taxis, at the rate μ . If there are no taxis when a customer arrives at the stand they will leave. Assume that $\lambda < \mu$. What is the long-term probability that an arriving customer gets a taxi?
- 7.20** The M/M/ ∞ queue has infinitely many servers. Show that the limiting distribution is Poisson and find the mean number of customers in the system.
- 7.21** For an M/M/ ∞ queue with $\lambda = \mu = 1$, find the mean time until state 4 is hit for the process started in state 1.
- 7.22** Tom is taking an online exam in which there are four questions of increasing difficulty. Tom needs to complete each question before moving on to the next question. It takes on average 10, 20, 30, and 40 minutes, respectively, to answer each question. The times spent for each question are independent and exponentially distributed.
- (a) Find the probability that after 45 minutes Tom has completed the exam.
- (b) Find the probability that after 45 minutes Tom is still on the third question.
- 7.23** A facility has four machines, with two repair workers to maintain them. Individual machines fail on average every 10 hours. It takes an individual repair worker on average 4 hours to fix a machine. Repair and failure times are independent and exponentially distributed.
- (a) Find the generator matrix.
- (b) In the long term, how many machines are typically operational?
- (c) If all four machines are initially working, find the probability that only two machines are working after 5 hours.
- 7.24** Customers arrive at a busy food truck according to a Poisson process with parameter λ . If there are i people already in line, the customer will join the line with probability $1/(i + 1)$. Assume that the chef at the truck takes, on average, α minutes to process an order.
- (a) Find the long-term average number of people in line.
- (b) Find the long-term probability that there are at least two people in line.

- 7.25** Over the long term, a continuous-time Markov chain on $\{1, 2, 3, 4\}$ makes 10% of its transitions to 1 and 30% each to 2, 3, and 4, respectively. From state i , it stays on average i minutes before moving to a new state, for $i = 1, 2, 3, 4$.
- Find the stationary distribution of the embedded discrete-time chain.
 - Find the stationary distribution of the continuous-time chain.
- 7.26** A facility has three machines and three mechanics. Machines break down at the rate of one per 24 hours. Breakdown times are exponentially distributed. The time it takes a mechanic to fix a machine is exponentially distributed with mean 6 hours. Only one mechanic can work on a failed machine at any given time. Let X_t be the number of machines working at time t . Find the long-term probability that all machines are working.
- 7.27** Recall the discrete-time Ehrenfest dog–flea model of Example 3.7. In the continuous-time version, there are N fleas distributed between two dogs. Fleas jump from one dog to another independently at rate λ . Let X_t denote the number of fleas on the first dog.
- Show that the process is a birth-and-death process. Give the birth and death rates.
 - Find the stationary distribution.
 - Assume that fleas jump at the rate of 2 per minute. If there are 10 fleas on Cooper and no fleas on Lisa, how long, on average, will it take for Lisa to get 4 fleas?
- 7.28** A linear birth-and-death process with immigration (see Table 7.1) has parameters $\lambda = 3$, $\mu = 4$, and $\alpha = 2$. Find the stationary distribution.
- 7.29** Consider an absorbing, continuous-time Markov chain with possibly more than one absorbing states.
- Argue that the continuous-time chain is absorbed in state a if and only if the embedded discrete-time chain is absorbed in state a .
 - Let

$$Q = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & -3 & 2 & 0 & 0 \\ 0 & 2 & -4 & 2 & 0 \\ 0 & 0 & 2 & -5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

be the generator matrix for a continuous-time Markov chain. For the chain started in state 2, find the probability that the chain is absorbed in state 5.

- 7.30** Cars arrive at a toll booth according to a Poisson process at the rate of two cars per minute. The time taken by the attendant to collect the toll is exponentially distributed with mean 20 seconds.
- Find the long-term mean number of cars in line at the toll booth.

- (b) Find the long-term probability that there are more than three cars at the toll booth.
- 7.31** See the last exercise. Assume that the toll booth has two attendants on duty. Find the long-term probability there are no cars at the toll booth.
- 7.32** Calls come in to a computer help center at the rate of 15 calls per hour. There are three tech support workers on duty, and the times they take to provide assistance are exponentially distributed with a mean of 10 minutes.
- (a) Find the average number of callers waiting to be helped.
- (b) Find the average amount of time that a caller spends waiting.
- 7.33** Consider an M/M/1 queue. Assume that the arrival and service time rates are both increased by a factor of k . What effect does this have on
- (a) the long-term expected number of customers in the system?
- (b) the long-term expected time that a customer is in the system?
- 7.34** You are a frequent customer at a coffee shop, where you typically wait 3 minutes to be served. Furthermore, on average you spend \$4 per visit. Over many months you estimate that on a typical day there are 20 customers in the shop, which is open from 6 a.m. to 10 p.m. Estimate the shop's total revenue per day.
- 7.35** Consider a continuous-time Markov chain with generator

$$Q = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -2 & 1 \\ 1 & 3 & -4 \end{pmatrix}.$$

Represent the process as a Markov chain subordinated to a Poisson process. Exhibit the transition function $P(t)$ in terms of R .

- 7.36** A discrete-time Markov chain has transition matrix

$$P = \begin{pmatrix} p & q \\ p & q \end{pmatrix},$$

for $p + q = 1$. Extend the process to continuous time by allowing transitions to occur at the points of a Poisson process with parameter λ . Find the transition function $P(t)$.

- 7.37** R: Consider a continuous-time Markov chain with generator

$$Q = \begin{pmatrix} -4 & 1 & 2 & 1 \\ 2 & -3 & 0 & 1 \\ 3 & 3 & -9 & 3 \\ 4 & 2 & 0 & -6 \end{pmatrix}.$$

- (a) To use Poisson subordination to estimate $P(0.8)$ to three significant digits, how many terms of the series are needed?

(b) Use (a) and compute $P(0.8)$.

(c) Use R's matrix exponential command to find $P(0.8)$ and check that your result in (a) is accurate to three digits.

7.38 R: Simulate Tom's online exam in Exercise 7.22 and estimate the probabilities in that problem.

7.39 R: Simulate an M/M/ ∞ queue and verify the result of Exercise 7.20, choosing your own values for λ and μ .

7.40 R: A multistate Markov model for the progression of HIV infection is developed in Hendricks et al. (1996). A sample of 467 HIV-positive men were monitored between 1984 and 1993. A 7-state absorbing chain is developed where stages 1–6 represent a range of values of an immunological marker for HIV, and state 7 corresponds to AIDS. The researchers give the following estimates for the monthly transition rates of the model.

λ_{12}	λ_{21}	λ_{23}	λ_{32}	λ_{34}	λ_{43}	λ_{45}	λ_{47}	λ_{54}	λ_{56}	λ_{57}	λ_{65}	λ_{67}
0.055	0.008	0.060	0.008	0.039	0.008	0.033	0.006	0.009	0.029	0.007	0.002	0.042

(a) Find the mean time to develop AIDS from each HIV state.

(b) Starting from the first stage of HIV infection, estimate the probability of developing AIDS within k years, for $k = 5, 10, 15, 20$.

8

BROWNIAN MOTION

Observe what happens when sunbeams are admitted into a building and shed light on its shadowy places. You will see a multitude of tiny particles mingling in a multitude of ways ... their dancing is an actual indication of underlying movements of matter that are hidden from our sight. ... It originates with the atoms which move of themselves [i.e., spontaneously]. Then those small compound bodies that are least removed from the impetus of the atoms are set in motion by the impact of their invisible blows and in turn cannon against slightly larger bodies. So the movement mounts up from the atoms and gradually emerges to the level of our senses, so that those bodies are in motion that we see in sunbeams, moved by blows that remain invisible.

—Lucretius, *On the Nature of Things*, 60 B.C.

8.1 INTRODUCTION

Brownian motion is a stochastic process, which is rooted in a physical phenomenon discovered almost 200 years ago. In 1827, the botanist Robert Brown, observing pollen grains suspended in water, noted the erratic and continuous movement of tiny particles ejected from the grains. He studied the phenomenon for many years, ruled out the belief that it emanated from some “life force” within the pollen, but could not explain the motion. Neither could any other scientist of the 19th century.

In 1905, Albert Einstein solved the riddle in his paper, *On the movement of small particles suspended in a stationary liquid demanded by the molecular-kinetic theory of heat*. Einstein explained the movement by the continual bombardment of the

immersed particles by the molecules in the liquid, resulting in “motions of such magnitude that these motions can easily be detected by a microscope.” Einstein’s theoretical explanation was confirmed 3 years later by empirical experiment, which led to the acceptance of the atomic nature of matter.

The description of the motion of dust particles in the classic poem *On the Nature of Things*, written by the Roman philosopher Lucretius over 2,000 years ago as an ancient proof of the existence of atoms, could have been a summary of Einstein’s work!

Einstein showed that the position x of a particle at time t was described by the partial differential heat equation

$$\frac{\partial}{\partial t}f(x, t) = \frac{1}{2} \frac{\partial^2}{\partial x^2}f(x, t),$$

where $f(x, t)$ represents the density (number of particles per unit volume) at position x and time t . The solution to that equation is

$$f(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t},$$

which is the probability density function of the normal distribution with mean 0 and variance t .

The mathematical object we call Brownian motion is a continuous-time, continuous-state stochastic process, also called the *Wiener process*, named after the American mathematician Norbert Wiener. The British mathematician Bertrand Russell influenced Wiener to take up the theory of Brownian motion as had been studied by Einstein. In his 1956 autobiography, Wiener writes,

The Brownian motion was nothing new as an object of study by physicists. There were fundamental papers by Einstein and Smoluchowski that covered it. But whereas these papers concerned what was happening to any given particle at a specific time, or the long-time statistics of many particles, they did not concern themselves with the mathematical properties of the curve followed by a single particle.

Here the literature was very scant, but it did include a telling comment by the French physicist Perrin in his book *Les Atomes*, where he said in effect that the very irregular curves followed by particles in the Brownian motion led one to think of the supposed continuous non-differentiable curves of the mathematicians.

Standard Brownian Motion

A continuous-time stochastic process $(B_t)_{t \geq 0}$ is a *standard Brownian motion* if it satisfies the following properties:

1. $B_0 = 0$.

2. (*Normal distribution*) For $t > 0$, B_t has a normal distribution with mean 0 and variance t .
3. (*Stationary increments*) For $s, t > 0$, $B_{t+s} - B_s$ has the same distribution as B_t . That is,

$$P(B_{t+s} - B_s \leq z) = P(B_t \leq z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx,$$

for $-\infty < z < \infty$.

4. (*Independent increments*) If $0 \leq q < r \leq s < t$, then $B_t - B_s$ and $B_r - B_q$ are independent random variables.
5. (*Continuous paths*) The function $t \mapsto B_t$ is continuous, with probability 1.

The normal distribution plays a central role in Brownian motion. The reader may find it helpful to review properties of the univariate and bivariate normal distributions in Appendix B, Section B.4. We write $X \sim \text{Normal}(\mu, \sigma^2)$ to mean that the random variable X is normally distributed with mean μ and variance σ^2 .

Brownian motion can be thought of as the motion of a particle that diffuses randomly along a line. At each point t , the particle's position is normally distributed about the line with variance t . As t increases, the particle's position is more *diffuse*; see Figure 8.1.

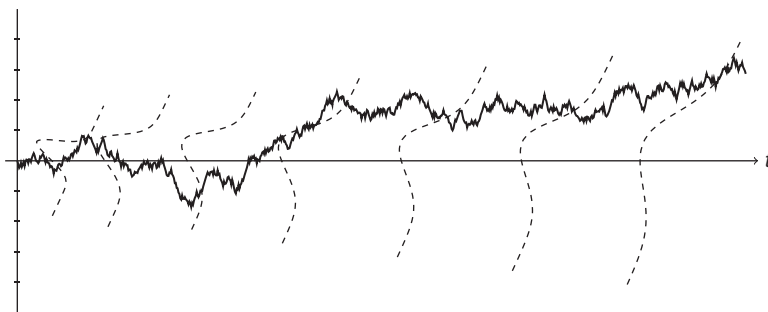


Figure 8.1 Brownian motion path. Superimposed on the graph are normal density curves with mean 0 and variance t .

It is not at all obvious that a stochastic process with the properties of Brownian motion actually exists. And Wiener's fundamental contribution was proving this existence. A rigorous derivation of Brownian motion is beyond the scope of this book, and requires measure theory and advanced analysis. The difficult part is showing the existence of a process that has stationary and independent increments together with continuous paths. The issue is touched upon at the end of the next section. First, however, we get our hands dirty with some calculations.

Computations involving Brownian motion are often tackled by exploiting stationary and independent increments. In the following examples, the reader may recognize similarities with the Poisson process, another stochastic process with stationary and independent increments. Unless stated otherwise, B_t denotes standard Brownian motion.

■ **Example 8.1** For $0 < s < t$, find the distribution of $B_s + B_t$.

Solution Write $B_s + B_t = 2B_s + (B_t - B_s)$. By independent increments, B_s and $B_t - B_s$ are independent random variables, and thus $2B_s$ and $B_t - B_s$ are independent. The sum of independent normal variables is normal. Thus, $B_s + B_t$ is normally distributed with mean $E(B_s + B_t) = E(B_s) + E(B_t) = 0$, and variance

$$\begin{aligned} \text{Var}(B_s + B_t) &= \text{Var}(2B_s + (B_t - B_s)) = \text{Var}(2B_s) + \text{Var}(B_t - B_s) \\ &= 4\text{Var}(B_s) + \text{Var}(B_{t-s}) = 4s + (t - s) \\ &= 3s + t. \end{aligned}$$

The second equality is because the variance of a sum of independent random variables is the sum of their variances. The third equality uses stationary increments. We have that $B_s + B_t \sim \text{Normal}(0, 3s + t)$. ■

■ **Example 8.2** A particle's position is modeled with a standard Brownian motion. If the particle is at position 1 at time $t = 2$, find the probability that its position is at most 3 at time $t = 5$.

Solution The desired probability is

$$\begin{aligned} P(B_5 \leq 3 | B_2 = 1) &= P(B_5 - B_2 \leq 3 - B_2 | B_2 = 1) \\ &= P(B_5 - B_2 \leq 2 | B_2 = 1) \\ &= P(B_5 - B_2 \leq 2) \\ &= P(B_3 \leq 2) = 0.876. \end{aligned}$$

The third equality is because $B_5 - B_2$ and B_2 are independent. The penultimate equality is by stationary increments. The desired probability in R is

```
> pnorm(2, 0, sqrt(3))
[1] 0.8758935
```

Note that in R commands involving the normal distribution are parameterized by standard deviation, not variance. ■

■ **Example 8.3** Find the covariance of B_s and B_t .

Solution For the covariance,

$$\text{Cov}(B_s, B_t) = E(B_s B_t) - E(B_s)E(B_t) = E(B_s B_t).$$

For $s < t$, write $B_t = (B_t - B_s) + B_s$, which gives

$$\begin{aligned} E(B_s B_t) &= E(B_s(B_t - B_s + B_s)) \\ &= E(B_s(B_t - B_s)) + E(B_s^2) \\ &= E(B_s)E(B_t - B_s) + E(B_s^2) \\ &= 0 + \text{Var}(B_s) = s. \end{aligned}$$

Thus, $\text{Cov}(B_s, B_t) = s$. For $t < s$, by symmetry $\text{Cov}(B_s, B_t) = t$. In either case,

$$\text{Cov}(B_s, B_t) = \min\{s, t\}. \quad \blacksquare$$

Simulating Brownian Motion

Consider simulating Brownian motion on $[0, t]$. Assume that we want to generate n variables at equally spaced time points, that is $B_{t_1}, B_{t_2}, \dots, B_{t_n}$, where $t_i = it/n$, for $i = 1, 2, \dots, n$. By stationary and independent increments, with $B_{t_0} = B_0 = 0$,

$$B_{t_i} = B_{t_{i-1}} + \left(B_{t_i} - B_{t_{i-1}}\right) \stackrel{d}{=} B_{t_{i-1}} + X_i,$$

where X_i is normally distributed with mean 0 and variance $t_i - t_{i-1} = t/n$, and is independent of $B_{t_{i-1}}$. The notation $X \stackrel{d}{=} Y$ means that random variables X and Y have the same distribution.

This leads to a recursive simulation method. Let $Z_1, Z_2, \dots, Z_n$ be independent and identically distributed standard normal random variables. Set

$$B_{t_i} = B_{t_{i-1}} + \sqrt{t/n} Z_i, \text{ for } i = 1, 2, \dots, n.$$

This gives

$$B_{t_i} = \sqrt{\frac{t}{n}}(Z_1 + \dots + Z_n).$$

In R, the cumulative sum command

```
> cumsum(rnorm(n, 0, sqrt(t/n)))
```

generates the Brownian motion variables $B_{t/n}, B_{2t/n}, \dots, B_t$.

Simulations of Brownian motion on $[0, 1]$ are shown in Figure 8.2. The paths were drawn by simulating $n = 1,000$ points in $[0, 1]$ and then connecting the dots.

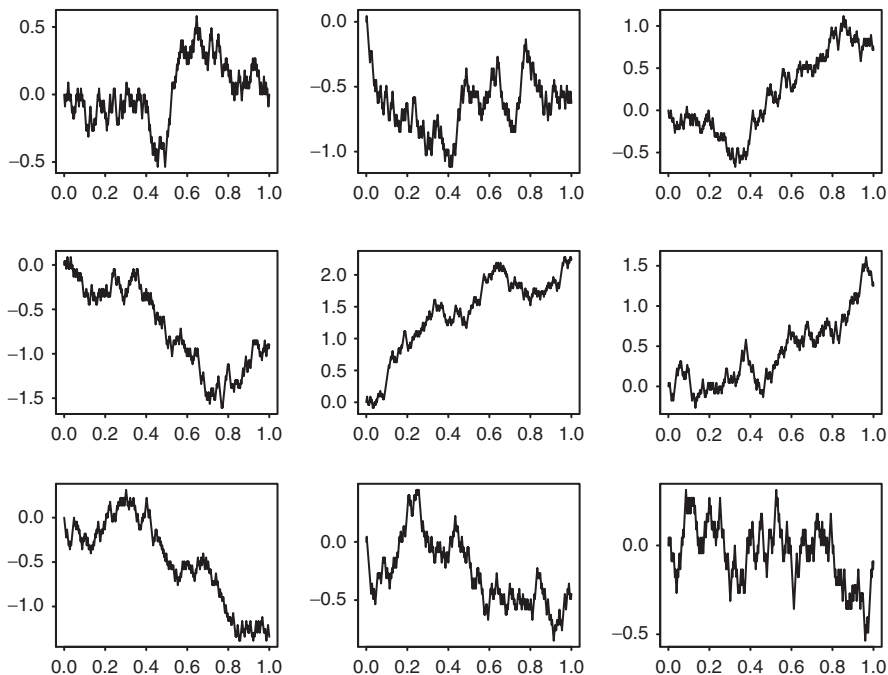


Figure 8.2 Sample paths of Brownian motion on $[0, 1]$.

R: Simulating Brownian Motion

```
# bm.R
> n <- 1000
> t <- 1
> bm <- c(0, cumsum(rnorm(n,0,sqrt(t/n))))
> steps <- seq(0,t,length=n+1)
> plot(steps,bm,type="l")
```

More generally, to simulate $B_{t_1}, B_{t_2}, \dots, B_{t_n}$, for time points $t_1 < t_2 < \dots < t_n$, set

$$B_{t_i} = B_{t_{i-1}} + \sqrt{t_i - t_{i-1}} Z_i, \text{ for } i = 1, 2, \dots, n,$$

with $t_0 = 0$.

Sample Space for Brownian Motion and Continuous Paths*

Consider the fifth defining property of Brownian motion: the function $t \mapsto B_t$ is continuous.

A continuous-time stochastic process $(X_t)_{-\infty < t < \infty}$ is a collection of random variables defined on a common sample space, or probability space, Ω . A random variable is really a function on a probability space. If $X : \Omega \rightarrow \mathbb{R}$ is a random variable, then X takes values depending on the outcome $\omega \in \Omega$. Thus, we can write $X = X(\omega)$ to emphasize the dependence on the outcome ω , although we usually suppress ω for simplicity.

In the context of stochastic processes, $X_t = X_t(\omega)$ is a function of two variables: t and ω . For fixed $t \in \mathbb{R}$, X_t is a random variable. Letting $X_t(\omega)$ vary as $\omega \in \Omega$ generates the different values of the process at the fixed time t . On the other hand, for fixed $\omega \in \Omega$, $X_t(\omega)$ is a function of t . Letting t vary generates a *sample path* or *realization*. One can think of these realizations as *random functions*.

For instance, each of the graphs in Figure 8.2, is one realization of such a random function. Using function notation, we could write $f(t) = B_t(\omega)$, for $-\infty < t < \infty$. For fixed ω , it makes sense to ask whether f is continuous, differentiable, etc.

A more precise statement of Property 5 is that

$$P(\omega \in \Omega : B_t(\omega) \text{ is a continuous function of } t) = 1.$$

Implicit in this statement is the existence of (i) the probability space Ω and (ii) a suitable probability function, or *probability measure*, P , which is consistent with the other defining properties of Brownian motion.

A probability space Ω is easy enough to identify. Let Ω be the set of all continuous functions on $[0, \infty)$. Each $\omega \in \Omega$ is a continuous function. Then, $B_t(\omega) = \omega_t$, the value of ω evaluated at t . This is the easiest way to insure that B_t has continuous sample paths. The hard part is to construct a probability function P on the set of continuous functions, which is consistent with the properties of Brownian motion. This was precisely Norbert Wiener's contribution. That probability function, introduced by Wiener in 1923, is called *Wiener measure*.

8.2 BROWNIAN MOTION AND RANDOM WALK

Continuous-time, continuous-state Brownian motion is intimately related to discrete-time, discrete-state random walk. Brownian motion can be constructed from simple symmetric random walk by suitably scaling the values of the walk while simultaneously speeding up the steps of the walk.

Let $X_1, X_2, \dots$ be an i.i.d. sequence with each X_i taking values ± 1 with probability $1/2$ each. Set $S_0 = 0$ and for integer $t > 0$, let $S_t = X_1 + \dots + X_t$. Then, $S_0, S_1, S_2, \dots$ is a simple symmetric random walk with $E(S_t) = 0$ and $\text{Var}(S_t) = t$ for $t = 0, 1, \dots$. As a sum of i.i.d. random variables, for large t , S_t is approximately normally distributed by the central limit theorem.

The random walk has independent increments. For integers $0 < q < r < s < t$, $S_t - S_s = X_{s+1} + \dots + X_t$, and $S_r - S_q = X_{q+1} + \dots + X_r$. Since $X_{s+1}, \dots, X_t$ is independent of $X_{q+1}, \dots, X_r$, it follows that $S_t - S_s$ and $S_r - S_q$ are independent random variables. Furthermore, for all integers $0 < s < t$, the distribution of $S_t - S_s$ and S_{t-s}

is the same since they are both a function of $t - s$ distinct X_t . Thus, the walk has stationary increments.

The simple random walk is a discrete-state process. To obtain a continuous-time process with continuous sample paths, values are connected by linear interpolation. Recall that $\lfloor x \rfloor$ is the *floor* of x , or integer part of x , which is the largest integer not greater than x . Extending the definition of S_t to real $t \geq 0$, let

$$S_t = \begin{cases} X_1 + \cdots + X_t, & \text{if } t \text{ is an integer,} \\ S_{\lfloor t \rfloor} + X_{\lfloor t \rfloor + 1} (t - \lfloor t \rfloor), & \text{otherwise.} \end{cases}$$

Observe that if k is a positive integer, then for $k \leq t \leq k + 1$, S_t is the linear interpolation of the points (k, S_k) and $(k + 1, S_{k+1})$. See Figure 8.3 to visualize the construction. We have $E(S_t) = 0$ and

$$\begin{aligned} \text{Var}(S_t) &= \text{Var}(S_{\lfloor t \rfloor} + X_{\lfloor t \rfloor + 1} (t - \lfloor t \rfloor)) \\ &= \text{Var}(S_{\lfloor t \rfloor}) + (t - \lfloor t \rfloor)^2 \text{Var}(X_{\lfloor t \rfloor + 1}) \\ &= \lfloor t \rfloor + (t - \lfloor t \rfloor)^2 \approx t, \end{aligned}$$

as $0 \leq t - \lfloor t \rfloor < 1$.

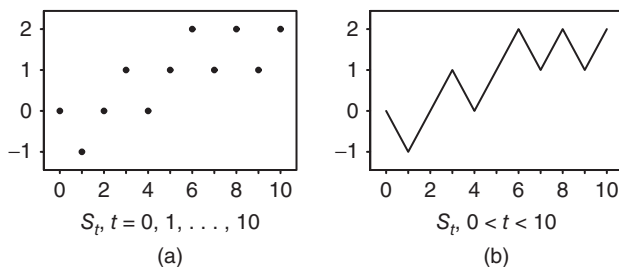


Figure 8.3 (a) Realization of a simple random walk. (b) Walk is extended to a continuous path by linear interpolation.

The process is now scaled both vertically and horizontally. Let $S_t^{(n)} = S_{nt}/\sqrt{n}$, for $n = 1, 2, \dots$. On any interval, the new process has n times as many steps as the original walk. And the height at each step is shrunk by a factor of $1/\sqrt{n}$. The construction is illustrated in Figure 8.4.

The scaling preserves mean and variance, as $E(S_t^{(n)}) = 0$ and $\text{Var}(S_t^{(n)}) = \text{Var}(S_{nt})/n \approx t$. Sample paths are continuous and for each n , the process retains independent and stationary increments. Considering the central limit theorem, it is reasonable to think that as $n \rightarrow \infty$, the process converges to Brownian motion. This, in fact, is the case, a result proven by Monroe Donsker in 1951.

We have not precisely said what *convergence* actually means since we are not talking about convergence of a sequence of numbers, nor even convergence of random variables, but rather convergence of stochastic processes. We do not

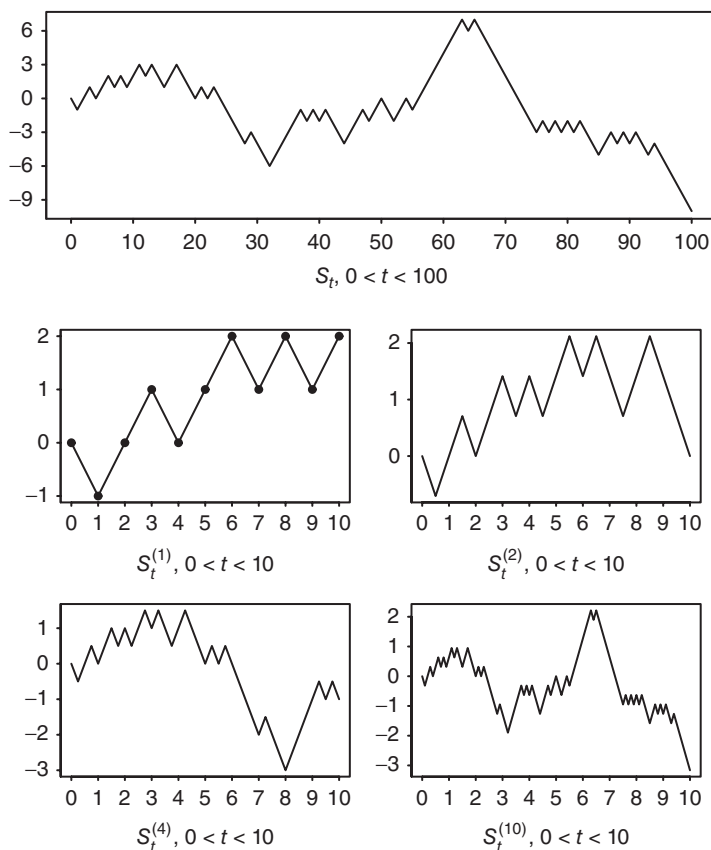


Figure 8.4 Top graph is a simple symmetric random walk for 100 steps. Bottom graphs show the scaled process $S_t^{(n)} = S_{nt}/\sqrt{n}$, for $n = 1, 2, 4, 10$.

give a rigorous statement. Nevertheless, the reader can trust their intuition that for large n , $\left(S_{nt}/\sqrt{n}\right)_{t \geq 0}$ behaves like a standard Brownian motion process $(B_t)_{t \geq 0}$.

Invariance Principle*

The construction of Brownian motion from simple symmetric random walk can be generalized so that we start with any i.i.d. sequence $X_1, \dots, X_n$ with mean 0 and variance 1. Let $S_n = X_1 + \dots + X_n$. Then, $S_{nt}/\sqrt{n}$ converges to B_t , as $n \rightarrow \infty$. This is known as *Donsker's invariance principle*. The word *invariance* is used because all random walks with increments that have mean 0 and variance 1, regardless of distribution, give the same Brownian motion limit.

A consequence of the invariance principle and continuity is that *functions* of the discrete process $S_{nt}/\sqrt{n}$ converge to the corresponding function of

Brownian motion, as $n \rightarrow \infty$. If g is a bounded, continuous function, whose domain is the set of continuous functions on $[0, 1]$, then $g(S_{nt}/\sqrt{n}) \approx g(B_t)$, for large n .

Functions whose domain is a set of functions are called *functionals*. The invariance principle means that properties of random walk and of functionals of random walk can be derived by considering analogous properties of Brownian motion, and vice versa.

For example, assume that f is a continuous function on $[0, 1]$. Let $g(f) = f(1)$, the evaluation of $f(t)$ at $t = 1$. Then, $g(S_{nt}/\sqrt{n}) = S_n/\sqrt{n}$ and $g(B_t) = B_1$. By the invariance principle, $S_n/\sqrt{n} \rightarrow B_1$, as $n \rightarrow \infty$. The random variable B_1 is normally distributed with mean 0 and variance 1, and we have thus recaptured the central limit theorem from Donsker's invariance principle.

■ **Example 8.4** For a simple symmetric random walk, consider the maximum value of the walk in the first n steps. Let $g(f) = \max_{0 \leq t \leq 1} f(t)$. By the invariance principle,

$$\lim_{n \rightarrow \infty} g\left(\frac{S_m}{\sqrt{n}}\right) = \lim_{n \rightarrow \infty} \max_{0 \leq t \leq 1} \frac{S_m}{\sqrt{n}} = \lim_{n \rightarrow \infty} \max_{0 \leq k \leq n} \frac{S_k}{\sqrt{n}} = g(B_t) = \max_{0 \leq t \leq 1} B_t.$$

This gives $\max_{0 \leq k \leq n} S_k \approx \sqrt{n} \max_{0 \leq t \leq 1} B_t$, for large n .

In Section 8.4, it is shown that the random variable $M = \max_{0 \leq t \leq 1} B_t$ has density function

$$f_M(x) = \sqrt{\frac{2}{\pi}} e^{-x^2/2}, \text{ for } x > 0.$$

Mean and standard deviation are

$$E(M) = \sqrt{\frac{2}{\pi}} \approx 0.80 \quad \text{and} \quad SD(M) = \frac{\pi - 2}{\pi} \approx 0.60.$$

With these results, we see that in the first n steps of simple symmetric random walk, the maximum value is about $(0.80)\sqrt{n}$ give or take $(0.60)\sqrt{n}$. In $n = 10,000$ steps, the probability that a value greater than 200 is reached is

$$\begin{aligned} P\left(\max_{0 \leq k \leq n} S_k > 200\right) &= P\left(\max_{0 \leq k \leq n} \frac{S_k}{100} > 2\right) \\ &= P\left(\max_{0 \leq k \leq n} \frac{S_k}{\sqrt{n}} > 2\right) \\ &\approx P(M > 2) \\ &= \int_2^\infty \sqrt{\frac{2}{\pi}} e^{-x^2/2} dx = 0.0455. \end{aligned}$$

■

R: Maximum for Simple Symmetric Random Walk

```
# maxssrw.R
> n <- 10000
> sim <- replicate(10000,
+   max(cumsum(sample(c(-1,1),n,replace=T))))
> mean(sim)
[1] 79.7128
> sd(sim)
[1] 60.02429
> sim <- replicate(10000,
+   if(max(cumsum(sample(c(-1,1),n,rep=T)))>200)
+   1 else 0)
> mean(sim) # P(max > 200)
[1] 0.0461
```

8.3 GAUSSIAN PROCESS

The normal distribution is also called the *Gaussian distribution*, in honor of Carl Friedrich Gauss, who introduced the distribution more than 200 years ago. The bivariate and multivariate normal distributions extend the univariate distribution to higher finite-dimensional spaces. A *Gaussian process* is a continuous-time stochastic process, which extends the Gaussian distribution to infinite dimensions. In this section, we show that Brownian motion is a Gaussian process and identify the conditions for when a Gaussian process is Brownian motion.

(Gauss, considered by historians to have been one of the greatest mathematicians of all time, first used the normal distribution as a model for measurement error in celestial observations, which led to computing the orbit of the planetoid Ceres. The story of the discovery of Ceres, and the contest to compute its orbit, is a mathematical page-turner. See *The Discovery of Ceres: How Gauss Became Famous* by Teets and Whitehead (1999).)

We first define the multivariate normal distribution and establish some of its key properties.

Multivariate Normal Distribution

Random variables $X_1, \dots, X_k$ have a *multivariate normal distribution* if for all real numbers $a_1, \dots, a_k$, the linear combination

$$a_1 X_1 + \dots + a_k X_k$$

has a univariate normal distribution. A multivariate normal distribution is completely determined by its *mean vector*

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_k) = (E(X_1), \dots, E(X_k))$$

and *covariance matrix* $\mathbf{V}$, where

$$V_{ij} = \text{Cov}(X_i, X_j), \text{ for } 1 \leq i, j \leq k.$$

The joint density function of the multivariate normal distribution is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{k/2} |\mathbf{V}|^{1/2}} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{V}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right),$$

where $\mathbf{x} = (x_1, \dots, x_k)$ and $|\mathbf{V}|$ is the determinant of $\mathbf{V}$.

The multivariate normal distribution has the remarkable property that all marginal and conditional distributions are normal. If $X_1, \dots, X_k$ have a multivariate normal distribution, then the X_i are normally distributed, joint distributions of subsets of the X_i have multivariate normal distributions, and conditional distributions given subsets of the X_i are normal.

If $X_1, \dots, X_k$ are independent normal random variables, then their joint distribution is multivariate normal. For jointly distributed normal random variables, independence is equivalent to being uncorrelated. That is, if X and Y are jointly distributed normal random variables, then X and Y are independent if and only if $E(XY) = E(X)E(Y)$.

A *Gaussian process* extends the multivariate normal distribution to stochastic processes.

Gaussian Process

A *Gaussian process* $(X_t)_{t \geq 0}$ is a continuous-time stochastic process with the property that for all $n = 1, 2, \dots$ and $0 \leq t_1 < \dots < t_n$, the random variables $X_{t_1}, \dots, X_{t_n}$ have a multivariate normal distribution.

A Gaussian process is completely determined by its *mean function* $E(X_t)$, for $t \geq 0$, and *covariance function* $\text{Cov}(X_s, X_t)$, for $s, t \geq 0$.

Standard Brownian motion is a Gaussian process. The following characterization gives conditions for when a Gaussian process is a standard Brownian motion.

Gaussian Process and Brownian Motion

A stochastic process $(B_t)_{t \geq 0}$ is a standard Brownian motion if and only if it is a Gaussian process with the following properties:

1. $B_0 = 0$.
2. (Mean function) $E(B_t) = 0$, for all t .
3. (Covariance function) $\text{Cov}(B_s, B_t) = \min\{s, t\}$, for all s, t .
4. (Continuous paths) The function $t \mapsto B_t$ is continuous, with probability 1.

Proof. Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. Consider $(B_{t_1}, B_{t_2}, \dots, B_{t_k})$, for $0 < t_1 < t_2 < \dots < t_k$. For constants $a_1, a_2, \dots, a_k$, we need to show that $a_1 B_{t_1} + a_2 B_{t_2} + \dots + a_k B_{t_k}$ has a univariate normal distribution. By independent increments, $B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}}$ are independent normal random variables, whose joint distribution is multivariate normal. Write

$$\begin{aligned} & a_1 B_{t_1} + a_2 B_{t_2} + \dots + a_k B_{t_k} \\ &= (a_1 + \dots + a_k) B_{t_1} + (a_2 + \dots + a_k)(B_{t_2} - B_{t_1}) \\ & \quad + \dots + (a_{k-1} + a_k)(B_{t_{k-1}} - B_{t_{k-2}}) + a_k(B_{t_k} - B_{t_{k-1}}), \end{aligned}$$

which is a linear combination of $B_{t_1}, B_{t_2} - B_{t_1}, \dots, B_{t_k} - B_{t_{k-1}}$, and thus has a univariate normal distribution.

The mean and covariance functions for Brownian motion are

$$E(B_t) = 0 \quad \text{and} \quad \text{Cov}(B_s, B_t) = \min\{s, t\}.$$

It follows that standard Brownian motion is the unique Gaussian process with these mean and covariance functions.

Conversely, assume that $(B_t)_{t \geq 0}$ is a Gaussian process that satisfies the stated properties. We need to show the process has stationary and independent increments.

Since the process is Gaussian, for $s, t \geq 0$, $B_{t+s} - B_s$ is normally distributed with mean $E(B_{t+s} - B_s) = E(B_{t+s}) - E(B_s) = 0$, and variance

$$\text{Var}(B_{t+s} - B_s) = \text{Var}(B_{t+s}) + \text{Var}(B_s) - 2\text{Cov}(B_{t+s}, B_s) = (t+s) + s - 2s = t.$$

Thus, $B_{t+s} - B_s$ has the same distribution as B_t , which gives stationary increments.

For $0 \leq q < r \leq s < t$,

$$\begin{aligned} E((B_r - B_q)(B_t - B_s)) &= E(B_r B_t) - E(B_r B_s) - E(B_q B_t) + E(B_q B_s) \\ &= \text{Cov}(B_r, B_t) - \text{Cov}(B_r, B_s) - \text{Cov}(B_q, B_t) + \text{Cov}(B_q, B_s) \\ &= r - r - q + q = 0. \end{aligned}$$

Thus, $B_r - B_q$ and $B_t - B_s$ are uncorrelated. Since $B_r - B_q$ and $B_t - B_s$ are normally distributed, it follows that they are independent.

We have shown that $(B_t)_{t \geq 0}$ is a standard Brownian motion. ■

■ **Example 8.5** For $a > 0$, let $X_t = B_{at}/\sqrt{a}$, for $t \geq 0$. Show that $(X_t)_{t \geq 0}$ is a standard Brownian motion.

Solution For $0 \leq t_1 < \dots < t_k$ and real numbers $a_1, \dots, a_k$,

$$\sum_{i=1}^k a_i X_{t_i} = \sum_{i=1}^k \frac{a_i}{\sqrt{a}} B_{at_i},$$

which has a univariate normal distribution, since $(B_t)_{t \geq 0}$ is a Gaussian process. Thus, $(X_t)_{t \geq 0}$ is a Gaussian process. Clearly, $X_0 = 0$. The mean function is

$$E(X_t) = E\left(\frac{1}{\sqrt{a}} B_{at}\right) = \frac{1}{\sqrt{a}} E(B_{at}) = 0.$$

The covariance function is

$$\begin{aligned} \text{Cov}(X_s, X_t) &= \text{Cov}\left(\frac{1}{\sqrt{a}} B_{as}, \frac{1}{\sqrt{a}} B_{at}\right) = \frac{1}{a} \text{Cov}(B_{as}, B_{at}) \\ &= \frac{1}{a} \min\{as, at\} = \min\{s, t\}. \end{aligned}$$

Finally, path continuity of $(X_t)_{t \geq 0}$ follows from the path continuity of standard Brownian motion, as the function $t \mapsto B_{at}/\sqrt{a}$ is continuous for all $a > 0$, with probability 1. ■

Nowhere Differentiable Paths

The property illustrated in Example 8.5 shows that Brownian motion preserves its character after rescaling. For instance, given a standard Brownian motion on $[0, 1]$, if we look at the process on, say, an interval of length one-trillionth ($= 10^{-12}$) then after resizing by a factor of $1/\sqrt{10^{-12}} = 1,000,000$, what we see is indistinguishable from the original Brownian motion!

This highlights the invariance, or *fractal*, structure of Brownian motion sample paths. It means that the jagged character of these paths remains jagged at all time scales. This leads to the remarkable fact that Brownian motion sample paths are *nowhere differentiable*. It is hard to even contemplate a function that is continuous at every point on its domain, but not differentiable at any point. Indeed, for many years, mathematicians believed that such a function was impossible, until Karl Weierstrass, considered the founder of modern analysis, demonstrated their existence in 1872.

The proof that Brownian motion is nowhere differentiable requires advanced analysis. Here is a heuristic argument. Consider the formal derivative

$$\frac{d}{dt}B_t = \lim_{h \rightarrow 0} \frac{B_{t+h} - B_t}{h}.$$

By stationary increments, $B_{t+h} - B_t$ has the same distribution as B_h , which is normal with mean 0 and variance h . Thus, the difference quotient $(B_{t+h} - B_t)/h$ is normally distributed with mean 0 and variance $1/h$. As h tends to 0, the variance tends to infinity. Since the difference quotient takes arbitrarily large values, the limit, and hence the derivative, does not exist.

8.4 TRANSFORMATIONS AND PROPERTIES

In addition to invariance under rescaling, Brownian motion satisfies numerous reflection, translation, and symmetry properties.

Transformations of Brownian Motion

Let $(B_t)_{t \geq 0}$ be a standard Brownian motion. Then, each of the following transformations is a standard Brownian motion.

1. (*Reflection*) $(-B_t)_{t \geq 0}$.
2. (*Translation*) $(B_{t+s} - B_s)_{t \geq 0}$, for all $s \geq 0$.
3. (*Rescaling*) $(a^{-1/2}B_{at})_{t \geq 0}$, for all $a > 0$.
4. (*Inversion*) The process $(X_t)_{t \geq 0}$ defined by $X_0 = 0$ and $X_t = tB_{1/t}$, for $t > 0$.

We leave the proofs that reflection and translation are standard Brownian motions as exercises. Rescaling was shown in Example 8.5. For inversion, let $t_1 < \dots < t_k$. For constants $a_1, \dots, a_k$,

$$a_1 X_{t_1} + \dots + a_k X_{t_k} = a_1 t_1 B_{1/t_1} + \dots + a_k t_k B_{1/t_k}$$

is normally distributed and thus $(X_t)_{t \geq 0}$ is a Gaussian process. The mean function is $E(X_t) = E(tB_{1/t}) = tE(B_{1/t}) = 0$, for $t > 0$. Covariance is

$$\begin{aligned} \text{Cov}(X_s, X_t) &= \text{Cov}(sB_{1/s}, tB_{1/t}) = E(sB_{1/s}tB_{1/t}) \\ &= stE(B_{1/s}B_{1/t}) = st \text{Cov}(B_{1/s}, B_{1/t}) \\ &= st \min\{1/s, 1/t\} = \min\{t, s\}. \end{aligned}$$

Continuity, for all $t > 0$, is inherited from $(B_t)_{t \geq 0}$. What remains is to show the process is continuous at $t = 0$. We do not prove this rigorously. Suffice it to show that

$$0 = \lim_{t \rightarrow 0} X_t = \lim_{t \rightarrow 0} tB_{1/t}, \text{ with probability 1}$$

is equivalent to

$$0 = \lim_{s \rightarrow \infty} \frac{B_s}{s}, \text{ with probability 1,}$$

and the latter is intuitive by the strong law of large numbers.

For real x , the process defined by $X_t = x + B_t$, for $t \geq 0$, is called *Brownian motion started at x* . For such a process, $X_0 = x$ and X_t is normally distributed with mean function $E(X_t) = x$, for all t . The process retains all other defining properties of standard Brownian motion: stationary and independent increments, and continuous sample paths.

See Figure 8.5 for examples of transformations of Brownian motion.

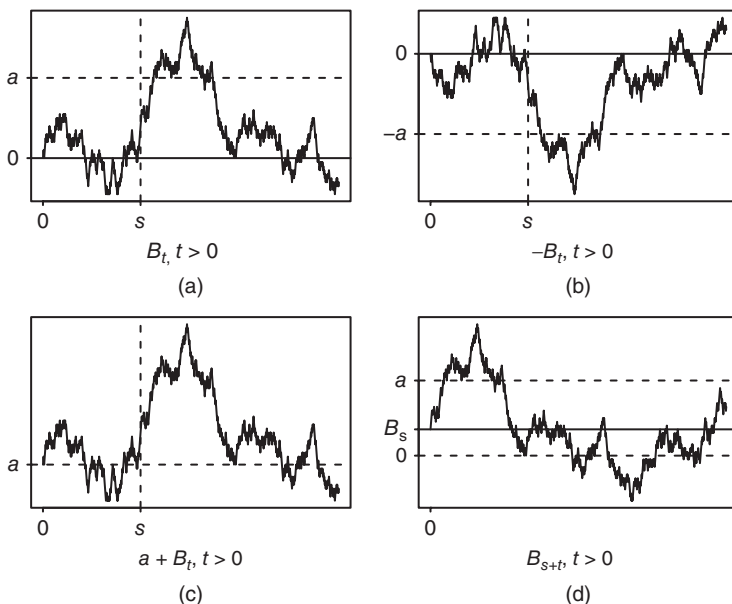


Figure 8.5 Transformations of Brownian motion. (a) Standard Brownian motion. (b) Reflection across t -axis. (c) Brownian motion started at a . (d) Translation.

Example 8.6 Let $(X_t)_{t \geq 0}$ be a Brownian motion process started at $x = 3$. Find $P(X_2 > 0)$.

Solution Write $X_t = B_t + 3$. Then,

$$P(X_2 > 0) = P(B_2 + 3 > 0) = P(B_2 > -3) = \int_{-3}^{\infty} \frac{1}{\sqrt{4\pi}} e^{-x^2/4} dx = 0.983.$$

In R, type

```
> 1-pnorm(-3, 0, sqrt(2))
[1] 0.9830526
```

■

Markov Process

Brownian motion satisfies the Markov property that conditional on the present, past and future are independent. This is a consequence of independent increments. Brownian motion is an example of a *Markov process*. A continuous-state stochastic process $(X_t)_{t \geq 0}$ is a Markov process if

$$P(X_{t+s} \leq y | X_u, 0 \leq u \leq s) = P(X_{t+s} \leq y | X_s), \quad (8.1)$$

for all $s, t \geq 0$ and real y . The process is *time-homogeneous* if the probability in Equation (8.1) does not depend on s . That is,

$$P(X_{t+s} \leq y | X_s) = P(X_t \leq y | X_0).$$

For a continuous-state Markov process, the *transition function*, or *transition kernel*, $K_t(x, y)$ plays the role that the transition matrix plays for a discrete-state Markov chain. The function $K_t(x, \cdot)$ is the conditional density of X_t given $X_0 = x$.

If $(X_t)_{t \geq 0}$ is Brownian motion started at x , then X_t is normally distributed with mean x and variance t . The transition kernel is

$$K_t(x, y) = \frac{1}{\sqrt{2\pi t}} e^{-(y-x)^2/2t}.$$

The transition kernel of a Markov process satisfies the Chapman–Kolmogorov equations. For continuous-state processes, integrals replace sums. The equations are

$$K_{s+t}(x, y) = \int_{-\infty}^{\infty} K_s(x, z) K_t(z, y) dz, \text{ for all } s, t, \quad (8.2)$$

as

$$\begin{aligned} \int_{-\infty}^y K_{s+t}(x, w) dw &= P(X_{s+t} \leq y | X_0 = x) \\ &= \int_{-\infty}^{\infty} P(X_{s+t} \leq y | X_s = z, X_0 = x) K_s(x, z) dz \\ &= \int_{-\infty}^{\infty} P(X_t \leq y | X_0 = z) K_s(x, z) dz \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^y K_t(z, w) dw \right) K_s(x, z) dz \\ &= \int_{-\infty}^y \left(\int_{-\infty}^{\infty} K_s(x, z) K_t(z, w) dz \right) dw. \end{aligned}$$

Taking derivatives with respect to y gives Equation (8.2).

First Hitting Time and Strong Markov Property

Brownian motion satisfies the strong Markov property. Recall that Brownian motion translated left or right by a constant is also a Brownian motion. For $s > 0$, with $B_s = x$, the process $(B_{t+s})_{t \geq 0}$ is a Brownian motion started at x .

By the strong Markov property this also holds for some random times as well. If S is a *stopping time*, $(B_{t+S})_{t \geq 0}$ is a Brownian motion process. See Section 3.9 to reference the strong Markov property for discrete-time Markov chains.

A common setting is when Brownian motion first hits a particular state or set of states. Let $T_a = \min\{t : B_t = a\}$ be the *first hitting time* that Brownian motion hits level a . See Figure 8.6. The random variable T_a is a stopping time. Moving forward from T_a , the translated process is a Brownian motion started at a .

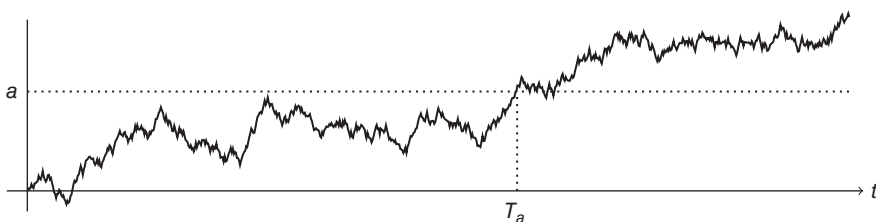


Figure 8.6 T_a is the first time Brownian motion hits level a .

The strong Markov property is used to find the distribution of T_a . Consider standard Brownian motion. At any time t , B_t is equally likely to be above or below the line $y = 0$. Assume that $a > 0$. For Brownian motion started at a , the process is equally likely to be above or below the line $y = a$. This gives,

$$P(B_t > a | T_a < t) = P(B_t > 0) = \frac{1}{2},$$

and thus,

$$\frac{1}{2} = P(B_t > a | T_a < t) = \frac{P(B_t > a, T_a < t)}{P(T_a < t)} = \frac{P(B_t > a)}{P(T_a < t)}.$$

The last equality is because the event $\{B_t > a\}$ implies $\{T_a < t\}$ by continuity of sample paths. We have that

$$P(T_a < t) = 2P(B_t > a) \tag{8.3}$$

$$\begin{aligned} &= 2 \int_a^\infty \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx \\ &= 2 \int_{a/\sqrt{t}}^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx. \end{aligned} \tag{8.4}$$

If $a < 0$, the argument is similar with $1/2 = P(B_t < a | T_a < t)$. In either case,

$$P(T_a < t) = 2 \int_{|a|/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx.$$

Differentiating with respect to t gives the probability density function of the first hitting time.

First Hitting Time Distribution

For a standard Brownian motion, let T_a be the first time the process hits level a . The density function of T_a is

$$f_{T_a}(t) = \frac{|a|}{\sqrt{2\pi t^3}} e^{-a^2/2t}, \text{ for } t > 0. \quad (8.5)$$

■ **Example 8.7** A particle moves according to Brownian motion started at $x = 1$. After $t = 3$ hours, the particle is at level 1.5. Find the probability that the particle reaches level 2 sometime in the next hour.

Solution For $t \geq 3$, the translated process is a Brownian motion started at $x = 1.5$. The event that the translated process reaches level 2 in the next hour, is equal to the event that a standard Brownian motion first hits level $a = 2 - 1.5 = 0.5$ in $[0, 1]$. The desired probability is

$$P(T_{0.5} \leq 1) = \int_0^1 \frac{0.5}{\sqrt{2\pi t^3}} e^{-(0.5)^2/2t} dt = 0.617. \quad \blacksquare$$

The first hitting time distribution has some surprising properties. Consider

$$\begin{aligned} P(T_a < \infty) &= \lim_{t \rightarrow \infty} P(T_a < t) = \lim_{t \rightarrow \infty} 2 \int_{|a|/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= 2 \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 2 \left(\frac{1}{2} \right) = 1. \end{aligned} \quad (8.6)$$

Brownian motion hits level a , with probability 1, for all a , no matter how large.

On the contrary,

$$E(T_a) = \int_0^{\infty} \frac{t|a|}{\sqrt{2\pi t^3}} e^{-a^2/2t} dt = +\infty.$$

The expected time to first reach level a is infinite. This is true for all a , no matter how small!

Reflection Principle and the Maximum of Brownian Motion

Brownian motion reflected at a first hitting time is a Brownian motion. This property is known as the *reflection principle* and is a consequence of the strong Markov property.

For a standard Brownian motion, and first hitting time T_a , consider the process $(\tilde{B}_t)_{t \geq 0}$ defined by

$$\tilde{B}_t = \begin{cases} B_t, & \text{if } 0 \leq t \leq T_a, \\ 2a - B_t, & \text{if } t \geq T_a. \end{cases}$$

This is called *Brownian motion reflected at T_a* . The construction is shown in Figure 8.7. The reflected portion $a - (B_t - a) = 2a - B_t$ is a Brownian motion process started at a by the strong Markov property and the fact that $-B_t$ is a standard Brownian motion. Concatenating the front of the original process $(B_t)_{0 \leq t \leq T_a}$ to the reflected piece preserves continuity.

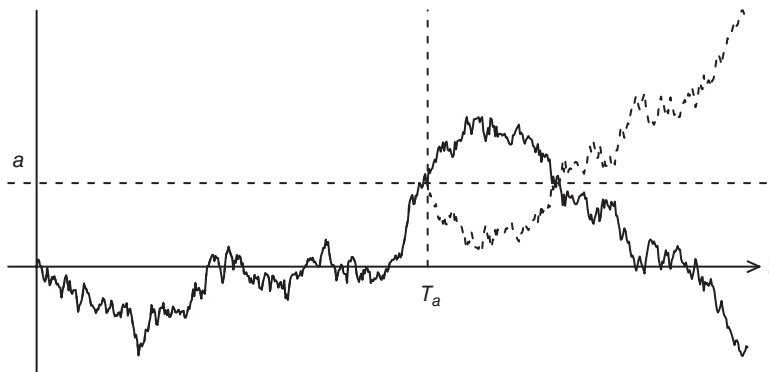


Figure 8.7 Reflection principle.

Another way of thinking of the reflection principle is that it establishes a one-to-one correspondence between paths that exceed level a at time t and paths that are below a at time t and have hit a by time t .

The reflection principle is applied in the following derivation of the distribution of $M_t = \max_{0 \leq s \leq t} B_s$, the maximum value of Brownian motion on $[0, t]$.

Let $a > 0$. If at time t , B_t exceeds a , then the maximum on $[0, t]$ is greater than a . That is, $\{B_t > a\}$ implies $\{M_t > a\}$. This gives

$$\begin{aligned} \{M_t > a\} &= \{M_t > a, B_t > a\} \cup \{M_t > a, B_t \leq a\} \\ &= \{B_t > a\} \cup \{M_t > a, B_t \leq a\}. \end{aligned}$$

As the union is disjoint, $P(M_t > a) = P(B_t > a) + P(M_t > a, B_t \leq a)$.

Consider a sample path that has crossed a by time t and is at most a at time t . By the reflection principle, the path corresponds to a reflected path that is at least a at

time t . This gives that $P(M_t > a, B_t \leq a) = P(\tilde{B}_t \geq a) = P(B_t > a)$. Thus,

$$P(M_t > a) = 2P(B_t > a) = \int_a^\infty \sqrt{\frac{2}{\pi t}} e^{-x^2/2t} dx, \text{ for } a > 0.$$

The distribution of M_t is equal to the distribution of $|B_t|$, the absolute value of a normally distributed random variable with mean 0 and variance t .

Since we have already found the distribution of T_a , a different derivation of the distribution of M_t is possible, using the fact that $M_t > a$ if and only if the process hits a by time t , that is $T_a < t$. This gives

$$P(M_t > a) = P(T_a < t) = \int_0^t \frac{a}{\sqrt{2\pi s^3}} e^{-a^2/2s} ds \quad (8.7)$$

$$= \int_a^\infty \sqrt{\frac{2}{\pi t}} e^{-x^2/2t} dx, \text{ for } a > 0. \quad (8.8)$$

The last equality is achieved by the change of variables $a^2/s = x^2/t$.

■ **Example 8.8** A laboratory instrument takes annual temperature measurements. Measurement errors are assumed to be independent and normally distributed. As precision decreases over time, errors are modeled as standard Brownian motion. For how many years can the lab be guaranteed that there is at least 90% probability that all errors are less than 4 degrees?

Solution The problem asks for the largest t such that $P(M_t \leq 4) \geq 0.90$. We have

$$0.90 \leq P(M_t \leq 4) = 1 - P(M_t > 4) = 1 - 2P(B_t > 4) = 2P(B_t \leq 4) - 1.$$

This gives

$$0.95 \leq P(B_t \leq 4) = P\left(Z \leq \frac{4}{\sqrt{t}}\right),$$

where Z is a standard normal random variable. The 95th percentile of the standard normal distribution is 1.645. Solving $4/\sqrt{t} = 1.645$ gives

$$t = \left(\frac{4}{1.645}\right)^2 = 5.91 \text{ years.}$$

■

Zeros of Brownian Motion and Arcsine Distribution

Brownian motion reaches level x , no matter how large x , with probability 1. Brownian motion also returns to the origin infinitely often. In fact, on any interval $(0, \epsilon)$, no matter how small ϵ , the process crosses the t -axis infinitely many times.

The times when the process crosses the t -axis are the *zeros* of Brownian motion. That Brownian motion has infinitely many zeros in any interval of the form $(0, \epsilon)$ is a consequence of the following.

Zeros of Brownian Motion

Theorem 8.1. For $0 \leq r < t$, let $z_{r,t}$ be the probability that standard Brownian motion has at least one zero in (r, t) . Then,

$$z_{r,t} = \frac{2}{\pi} \arccos \left(\sqrt{\frac{r}{t}} \right).$$

With $r = 0$, the result gives that standard Brownian motion has at least one zero in $(0, \epsilon)$ with probability

$$z_{0,\epsilon} = (2/\pi) \arccos(0) = (2/\pi)(\pi/2) = 1.$$

That is, $B_t = 0$, for some $0 < t < \epsilon$. By the strong Markov property, for Brownian motion restarted at t , there is at least one zero in (t, ϵ) , with probability 1. Continuing in this way, there are infinitely many zeros in $(0, \epsilon)$.

The *arcsine distribution* arises in the proof of Theorem 8.1 and other results related to the zeros of Brownian motion. The distribution is a special case of the beta distribution, and has cumulative distribution function

$$F(t) = \frac{2}{\pi} \arcsin \left(\sqrt{t} \right), \text{ for } 0 \leq t \leq 1. \quad (8.9)$$

The arcsine density function is

$$f(t) = F'(t) = \frac{1}{\pi \sqrt{t(1-t)}}, \text{ for } 0 \leq t \leq 1.$$

The density is bimodal and symmetric, as shown in Figure 8.8.

Proof of Theorem 8.1. Conditioning on B_r ,

$$\begin{aligned} z_{r,t} &= P(B_s = 0 \text{ for some } s \in (r, t)) \\ &= \int_{-\infty}^{\infty} P(B_s = 0 \text{ for some } s \in (r, t) | B_r = x) \frac{1}{\sqrt{2\pi r}} e^{-x^2/2r} dx. \end{aligned} \quad (8.10)$$

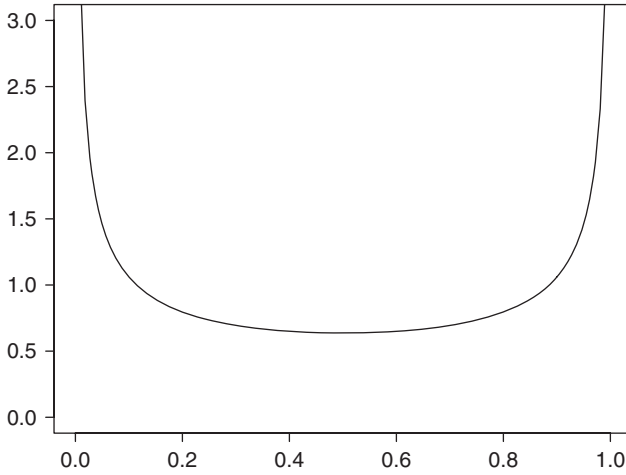


Figure 8.8 Arcsine distribution.

Assume that $B_r = x < 0$. The probability that $B_s = 0$ for some $s \in (r, t)$ is equal to the probability that for the process started in x , the maximum on $(0, t - r)$ is greater than 0. By translation, the latter is equal to the probability that for the process started in 0, the maximum on $(0, t - r)$ is greater than x . That is,

$$P(B_s = 0 \text{ for some } s \in (r, t) | B_r = x) = P(M_{t-r} > x).$$

For $x > 0$, consider the reflected process $-B_s$ started in $-x$. In either case, with Equations (8.7) and (8.10),

$$\begin{aligned} z_{r,t} &= \int_{-\infty}^{\infty} P(M_{t-r} > |x|) \frac{1}{\sqrt{2\pi r}} e^{-x^2/2r} dx \\ &= \int_{-\infty}^{\infty} \int_0^{t-r} \frac{1}{\sqrt{2\pi s^3}} |x| e^{-x^2/2s} ds \frac{1}{\sqrt{2\pi r}} e^{-x^2/2r} dx \\ &= \frac{1}{\pi} \int_0^{t-r} \frac{1}{\sqrt{rs^3}} \int_0^{\infty} x e^{-x^2(r+s)/2rs} dx ds \\ &= \frac{1}{\pi} \int_0^{t-r} \frac{1}{\sqrt{rs^3}} \int_0^{\infty} e^{-z(r+s)/rs} dz ds \\ &= \frac{1}{\pi} \int_0^{t-r} \frac{1}{\sqrt{rs^3}} \left(\frac{rs}{r+s} \right) ds \\ &= \frac{1}{\pi} \int_{r/t}^1 \frac{1}{\sqrt{x(1-x)}} dx. \end{aligned}$$

The last equality is by the change of variables $r/(r+s) = x$. The last expression is an arcsine probability, which, by Equation (8.9), gives

$$\begin{aligned} z_{r,t} &= \frac{2}{\pi} \left(\arcsin(\sqrt{1}) - \arcsin\left(\sqrt{\frac{r}{t}}\right) \right) \\ &= 1 - \frac{2}{\pi} \arcsin\left(\sqrt{\frac{r}{t}}\right) = \frac{2}{\pi} \arccos\left(\sqrt{\frac{r}{t}}\right). \end{aligned}$$

■

Last Zero Standing

Corollary 8.2. *Let L_t be the last zero in $(0, t)$. Then,*

$$P(L_t \leq x) = \frac{2}{\pi} \arcsin\left(\sqrt{\frac{x}{t}}\right), \text{ for } 0 < x < t.$$

Proof of corollary. The last zero occurs by time $x < t$ if and only if there are no zeros in (x, t) . This occurs with probability

$$1 - z_{x,t} = 1 - \frac{2}{\pi} \arccos\left(\sqrt{\frac{x}{t}}\right) = \frac{2}{\pi} \arcsin\left(\sqrt{\frac{x}{t}}\right).$$

■

Example 8.9 (Fluctuations in Coin Tossing)

We shall encounter theoretical conclusions which not only are unexpected but actually come as a shock to intuition and common sense. They will reveal that commonly accepted notions concerning chance fluctuations are without foundation and that the implications of the law of large numbers are widely misconstrued.

—William Feller

Feller, the author of the quotation and of the classic and ground-breaking probability textbook *An Introduction to Probability Theory and Its Applications*, was discussing fluctuations in coin tossing and random walk. As a discrete process random walk is often studied with counting and combinatorial tools. Because of the connection between random walk and Brownian motion many discrete results can be obtained by taking suitable limits and invoking the invariance principle.

Consider a fair coin-tossing game between two players, A and B. If the coin lands heads, A pays B one dollar. If tails, B pays A. The coin is flipped $n = 10,000$ times. To test your “intuition and common sense,” when would you guess is the *last* time the players are even? That is, at what point will the remaining duration of the game see one player always ahead?

Perhaps you think that in an evenly matched game there will be frequent changes in the lead and thus the last time the players are even is likely to be close to n , near the end of the game?

Let $\tilde{L}_n$ be the last time, in n plays, that the players are tied. This is the last zero for simple symmetric random walk on $\{0, 1, \dots, n\}$. Before continuing, we invite the reader to sketch their guesstimate of the distribution of $\tilde{L}_n$.

We simulated the coin-flipping game 5,000 times, generating the histogram of $\tilde{L}_n$ in Figure 8.9. The distribution is symmetric. It is equally likely that the last zero of the random walk is either k or $n - k$. Furthermore, the probabilities near the ends are the greatest. There is a surprisingly large probability of one player gaining the lead early in the game, and keeping the lead throughout. After just 20% of the game, there is almost 30% probability that one player will be in the lead for the remainder.

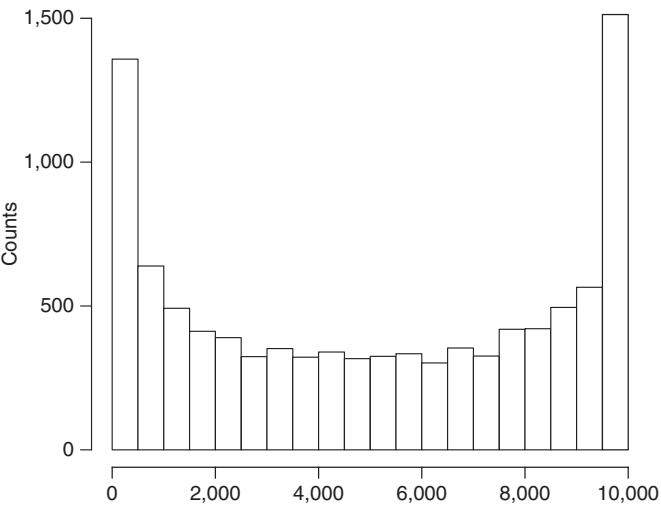


Figure 8.9 Last time players are tied in 10,000 coin flips.

It is no accident, of course, that the histogram bears a striking resemblance to the arcsine density curve in Figure 8.8. Let $0 < t < 1$. The probability that the last zero of the random walk occurs by step tn , that is, after 100*t* percent of the walk, is approximately the probability that the last zero of Brownian motion on $[0, 1]$ occurs by time t . For large n ,

$$P\left(\tilde{L}_n \leq tn\right) \approx P(L_1 \leq t) = \frac{2}{\pi} \arcsin \left(\sqrt{t}\right).$$

Simulated probabilities for the random walk and theoretical values for Brownian motion are compared in Table 8.1.

TABLE 8.1 Random Walk and Brownian Motion Probabilities for the Last Zero ($n = 10,000$)

t	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
$P\left(\tilde{L}_n \leq tn\right)$	0.207	0.297	0.367	0.434	0.499	0.565	0.632	0.704	0.795
$P(L_1 \leq t)$	0.205	0.295	0.369	0.436	0.500	0.564	0.631	0.705	0.795

R: Random Walk and Coin Tossing

```
# coinflips.R
trials <- 10000
simlist <- numeric(trials)
for (i in 1:trials) {
  rw <- c(0, cumsum(sample(c(-1,1), (trials-1),
    replace=T)))
  simlist[i] <- tail(which(rw==0), 1)
}
mean(simlist)
hist(simlist, xlab="", ylab="Counts", main="")
```

■

8.5 VARIATIONS AND APPLICATIONS

Standard Brownian motion is often too simple a model for real-life applications. Many variations arise in practice. Brownian motion started in x has a constant mean function. A common variant of Brownian motion has linear mean function as well as an additional variance parameter.

Brownian Motion with Drift

For real μ and $\sigma > 0$, the process defined by

$$X_t = \mu t + \sigma B_t, \text{ for } t \geq 0,$$

is called *Brownian motion with drift parameter μ and variance parameter σ^2* .

Brownian motion with drift is a Gaussian process with continuous sample paths and independent and stationary increments. For $s, t > 0$, $X_{t+s} - X_t$ is normally distributed with mean μs and variance $\sigma^2 s$.

■ **Example 8.10** Find the probability that Brownian motion with drift parameter $\mu = 0.6$ and variance $\sigma^2 = 0.25$ takes values between 1 and 3 at time $t = 4$.

Solution Write $X_t = (0.6)t + (0.5)B_t$. The desired probability is

$$\begin{aligned} P(1 \leq X_4 \leq 3) &= P(1 \leq (0.6)4 + (0.5)B_4 \leq 3) = P(-2.8 \leq B_4 \leq 1.2) \\ &= \int_{-2.8}^{1.2} \frac{1}{\sqrt{8\pi}} e^{-x^2/8} dx = 0.645. \end{aligned}$$

■

■ **Example 8.11 (Home team advantage)** A novel application of Brownian motion to sports scores is given in Stern (1994). The goal is to quantify the home team advantage by finding the probability in a sports match that the home team wins the game given that they lead by l points after a fraction $0 \leq t \leq 1$ of the game is completed. The model is applied to basketball where scores can be reasonably approximated by a continuous distribution.

For $0 \leq t \leq 1$, let X_t denote the difference in scores between the home and visiting teams after $100t$ percent of the game has been completed. The process is modeled as a Brownian motion with drift, where the mean parameter μ is a measure of home team advantage. The probability that the home team wins the game, given that they have an l point lead at time $t < 1$, is

$$\begin{aligned} p(l,t) &= P(X_1 > 0 | X_t = l) = P(X_1 - X_t > -l) \\ &= P(X_{1-t} > -l) = P(\mu(1-t) + \sigma B_{1-t} > -l) \\ &= P\left(B_{1-t} < \frac{l + \mu(1-t)}{\sigma}\right) \\ &= P\left(B_t < \frac{\sqrt{t}[l + \mu(1-t)]}{\sigma\sqrt{1-t}}\right). \end{aligned}$$

The last equality is because B_t has the same distribution as $\sqrt{t/(1-t)}B_{1-t}$.

The model is applied to the results of 493 National Basketball Association games in 1992. Drift and variance parameters are fit from the available data with estimates $\hat{\mu} = 4.87$ and $\hat{\sigma} = 15.82$.

Table 8.2 gives the probability of a home team win for several values of l and t . Due to the home court advantage, the home team has a greater than 50% chance of winning even if it is behind by two points at halftime ($t = 0.50$). Even in the last

TABLE 8.2 Table for Basketball Data Probabilities $p(l,t)$ that the Home Team Wins the Game Given that they are in the Lead by l Points After a Fraction t of the Game is Completed

Time t	Lead						
	$l = -10$	$l = -5$	$l = -2$	$l = 0$	$l = 2$	$l = 5$	$l = 10$
0.00				0.62			
0.25	0.32	0.46	0.55	0.61	0.66	0.74	0.84
0.50	0.25	0.41	0.52	0.59	0.65	0.75	0.87
0.75	0.13	0.32	0.46	0.56	0.66	0.78	0.92
0.90	0.03	0.18	0.38	0.54	0.69	0.86	0.98
1.00	0.00	0.00	0.00		1.00	1.0	1.0

Source: Stern (1994).

five minutes of play ($t = 0.90$), home team comebacks from five points are not that unusual, according to the model, with probability 0.18.

We recommend this paper to the mathematically inclined sports fan. It is both accessible and readable. The author discusses model assumptions and limitations, the extent to which theoretical predictions follow empirical results, and an interesting extension from basketball to baseball. ■

Brownian Bridge

The two ends of a bridge are both secured to level ground. A Brownian bridge is a Brownian motion process conditioned so that the process ends at the same level as where it begins.

Brownian Bridge

From standard Brownian motion, the conditional process $(B_t)_{0 \leq t \leq 1}$ given that $B_1 = 0$ is called a *Brownian bridge*. The Brownian bridge is *tied down* to 0 at the endpoints of $[0, 1]$.

Examples of Brownian bridge are shown in Figure 8.10. Let $(X_t)_{t \geq 0}$ denote a Brownian bridge. For $0 \leq t \leq 1$, the distribution of X_t is equal to the conditional distribution of B_t given $B_1 = 0$. Since the conditional distributions of a Gaussian process are Gaussian, it follows that Brownian bridge is a Gaussian process. Continuity of sample paths, and independent and stationary increments are inherited from standard Brownian motion.

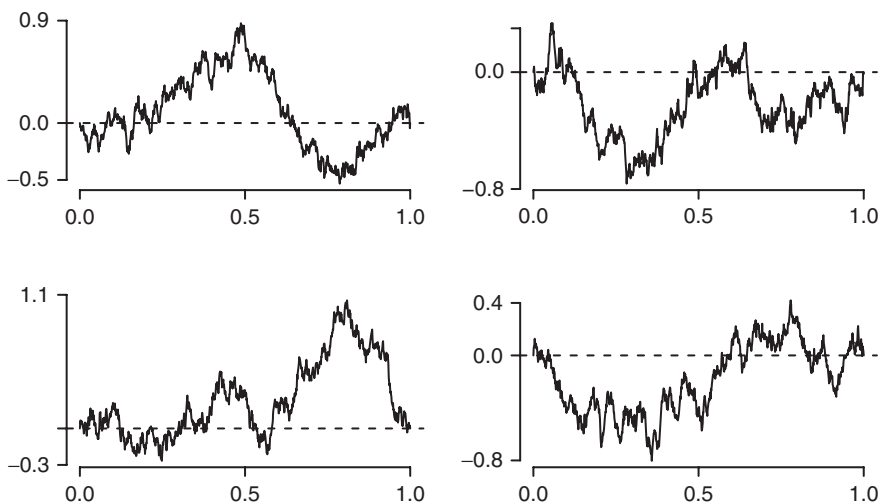


Figure 8.10 Brownian bridge sample paths.

To find the mean and covariance functions results are needed for bivariate normal distributions. We encourage the reader to work through Exercise 8.5(b) and show that for $0 < s < t$,

$$E(B_s | B_t = y) = \frac{sy}{t} \quad \text{and} \quad \text{Var}(B_s | B_t = y) = \frac{s(t-s)}{t}.$$

This gives the mean function of Brownian bridge

$$E(X_t) = E(B_t | B_1 = 0) = 0, \quad \text{for } 0 \leq t \leq 1.$$

For the covariance, $\text{Cov}(X_s, X_t) = E(X_s X_t)$. By the law of total expectation,

$$\begin{aligned} E(X_s X_t) &= E(E(X_s X_t) | X_t) = E(X_t E(X_s | X_t)) \\ &= E\left(X_t \frac{sX_t}{t}\right) = \frac{s}{t} E(X_t^2) = \frac{s}{t} E(B_t^2 | B_1 = 0) \\ &= \frac{s}{t} \text{Var}(B_t | B_1 = 0) = \left(\frac{s}{t}\right) \frac{t(1-t)}{1} = s - st. \end{aligned}$$

By symmetry, for $t < s$, $E(X_s X_t) = t - st$. In either case, the covariance function is

$$\text{Cov}(X_s, X_t) = \min\{s, t\} - st.$$

■ **Example 8.12** Let $X_t = B_t - tB_1$, for $0 \leq t \leq 1$. Show that $(X_t)_{0 \leq t \leq 1}$ is a Brownian bridge.

Solution The process is a Gaussian process since $(B_t)_{t \geq 0}$ is a Gaussian process. Sample paths are continuous, with probability 1. It is suffice to show that the process has the same mean and covariance functions as a Brownian bridge.

The mean function is $E(X_t) = E(B_t - tB_1) = E(B_t) - tE(B_1) = 0$. The covariance function is

$$\begin{aligned} \text{Cov}(X_s, X_t) &= E(X_s X_t) = E((B_s - sB_1)(B_t - tB_1)) \\ &= E(B_s B_t) - tE(B_s B_1) - sE(B_t B_1) + stE(B_1^2) \\ &= \min\{s, t\} - ts - st + st = \min\{s, t\} - st, \end{aligned}$$

which is the covariance function of Brownian bridge.

The construction described in this example gives a direct method for simulating a Brownian bridge used to draw the graphs in Figure 8.10.

R: Simulating Brownian Bridge

```
# bbridge.R
> n <- 1000
> t <- seq(0, 1, length=n)
```

```

> bm <- c(0, cumsum(rnorm(n-1, 0, 1))) / sqrt(n)
> bb <- bm - t * bm[n]
> plot(t, bb, type="l")

```

■

■ **Example 8.13 (Animal tracking)** In *Analyzing Animal Movements Using Brownian Bridges*, ecologists Horne et al. (2007) develop a Brownian bridge model for estimating the expected movement path of an animal. The model is based on a two-dimensional Brownian motion, where $Z_t^{a,b} = Z_t$ is defined to be the position in $\mathbb{R}^2$ of an animal at time $t \in [0, T]$, which starts at $\mathbf{a}$ and ends at $\mathbf{b}$. Each Z_t is normally distributed with mean vector $E(Z_t) = \mathbf{a} + \frac{t}{T}(\mathbf{b} - \mathbf{a})$ and covariance matrix

$$\sigma_t^2 = \frac{t(T-t)}{T} \sigma_m^2 \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix and σ_m^2 is a parameter related to the mobility of the animal. The probability that the animal is in region A at time t is $P(Z_t \in A)$. The model is applied to animal location data, often obtained through global positioning system telemetry, which allows for monitoring animal movements over great distances.

An objective of the researchers is to estimate the frequency of use of a region over the time of observation. Let $I_A(x)$ be the usual indicator function, which takes the value 1, if $x \in A$, and 0, otherwise. The *occupation time* for region A is defined as the random variable

$$\int_0^T I_A(Z_t) dt.$$

The expected fraction of time an animal occupies A is then

$$E\left(\frac{1}{T} \int_0^T I_A(Z_t) dt\right) = \frac{1}{T} \int_0^T P(Z_t \in A) dt,$$

a quantity used to estimate the home range of a male black bear in northern Idaho and the fall migration route of 11 caribou in Alaska.

The authors argue that the Brownian bridge movement model (BBMM) has the advantage over other methods in that BBMM assumes successive animal locations are not independent and explicitly incorporates the time between locations into the model. ■

■ **Example 8.14 (Kolmogorov–Smirnov statistic)** Following are 40 measurements, which take values between 0 and 1. We would like to test the claim that they are an i.i.d. sample from the uniform distribution on $(0, 1)$. The Brownian bridge arises in the analysis of the *Kolmogorov–Smirnov test*, a common statistical method to test such claims.

0.100	0.296	0.212	0.385	0.993	0.870	0.070	0.815	0.123	0.588
0.332	0.035	0.758	0.362	0.453	0.047	0.134	0.389	0.147	0.424
0.060	0.003	0.800	0.011	0.085	0.674	0.196	0.715	0.342	0.519
0.675	0.799	0.768	0.721	0.315	0.009	0.109	0.835	0.044	0.152

Given a sample $X_1, \dots, X_n$, define the *empirical distribution function*

$$F_n(t) = \frac{1}{n} \sum_{i=1}^n I_{\{X_i \leq t\}},$$

where $I_{\{X_i \leq t\}}$ is the indicator function equal to 1, if $X_i \leq t$, and 0, otherwise. The empirical distribution function gives the proportion of values in the sample that are at most t . If the data are a sample from a population with cumulative distribution function F , then $F_n(t)$ is an estimate of $P(X_i \leq t) = F(t)$.

If the data in our example is an i.i.d. sample from the uniform distribution on $(0, 1)$, then $F(t) = t$, for $0 \leq t \leq 1$, and we would expect $F_n(t) \approx t$. Figure 8.11 shows the empirical distribution function for these data plotted alongside the line $y = t$.

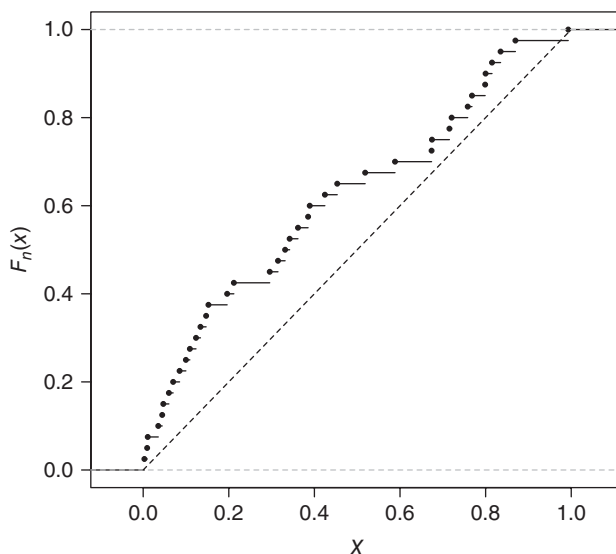


Figure 8.11 Empirical distribution function for sample data.

For a given cdf F , the *Kolmogorov–Smirnov statistic* is

$$D_n = \max_{0 \leq t \leq 1} |F_n(t) - F(t)|,$$

the maximum absolute distance between the empirical cdf and F . If the data are a sample from F , we expect $F_n(t) \approx F(t)$ and the value of D_n to be close to 0. Furthermore, large values of D_n are evidence against the hypothesis that the data are a sample

from F . For the test of uniformity, $F(t) = t$, and

$$D_n = \max_{0 \leq t \leq 1} |F_n(t) - t|.$$

For our data, the Kolmogorov–Smirnov test statistic is $D_{40} = 0.223$. Is this large or small? Does it support or contradict the uniformity hypothesis? A statistician would ask: if the data do in fact come from a uniform distribution, what is the probability that D_{40} would be as large as 0.223? The probability $P(D_{40} > 0.223)$ is the P -value of the test. A small P -value means that it is unusual for D_{40} to be as large as 0.223, which would be evidence against uniformity. The distribution of D_n is difficult to obtain, which leads one to look for a good approximation.

If $X_1, \dots, X_n$ is an i.i.d. sample from the uniform distribution on $(0, 1)$, then for $0 < t < 1$,

$$F_n(t) - t = \sum_{i=1}^n \frac{I_{\{X_i \leq t\}} - t}{n}$$

is a sum of i.i.d. random variables with common mean

$$E\left(\frac{I_{\{X_i \leq t\}} - t}{n}\right) = \frac{P(X_i \leq t) - t}{n} = \frac{t - t}{n} = 0$$

and variance

$$\text{Var}\left(\frac{I_{\{X_i \leq t\}} - t}{n}\right) = \frac{1}{n^2} \text{Var}(I_{\{X_i \leq t\}}) = \frac{P(X_i \leq t)(1 - P(X_i \leq t))}{n^2} = \frac{t(1 - t)}{n^2}.$$

Thus, $F_n(t) - t$ has mean 0 and variance $t(1 - t)/n$. For fixed $0 < t < 1$, the central limit theorem gives that $\sqrt{n}(F_n(t) - t)/\sqrt{t(1 - t)}$ converges to a standard normal random variable, as $n \rightarrow \infty$. That is, for all real x ,

$$\lim_{n \rightarrow \infty} P\left(\sqrt{n}(F_n(t) - t) \leq x\right) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi t(1 - t)}} e^{-z^2/(2t(1 - t))} dz = P(Y \leq x),$$

where $Y \sim \text{Normal}(0, t(1 - t))$.

By Donsker's invariance principle, it can be shown that the process $\sqrt{n}(F_n(t) - t)$, for $0 \leq t \leq 1$, converges to $B_t - tB_1$. The limiting process $(B_t - tB_1)_{0 \leq t \leq 1}$ is a Brownian bridge. The invariance principle further gives that

$$\sqrt{n}D_n = \sqrt{n} \max_{0 \leq t \leq 1} |F_n(t) - t|$$

converges to the maximum of a Brownian bridge. For large n , the distribution of D_n is approximately that of $M/\sqrt{n}$, where M is the maximum of a Brownian bridge.

We simulated the maximum of a Brownian bridge to find the P -value for our data, which is found to be

$$P(D_{40} > 0.223) \approx P(M/\sqrt{40} > 0.223) = P(M > 1.41) \approx 0.018.$$

The exact distribution of the maximum of a Brownian bridge is known. By using that distribution, the P -value is 0.0157. The P -value for the Kolmogorov–Smirnov test is obtained in R with the command `ks.test`.

The interpretation is that if the data were in fact uniformly distributed then the probability that D_{40} would be as large as 0.223 is less than 2%. Since the P -value is so small, we reject the claim and conclude that the data do not originate from a uniform distribution.

R: Test for Uniformity: Finding the P -value

```
# kstest.R
> trials <- 10000
> n <- 1000
> simlist <- numeric(trials)
> for (i in 1:trials) {
+   t <- seq(0,1,length=n)
+   # Brownian motion
+   bm <- c(0,cumsum(rnorm(n-1,0,1)))/sqrt(n)
+   bb <- bm-t*bm[n] # Brownian bridge
+   z <- max(bb) # maximum of Brownian bridge on [0,1]
+   simlist[i] <- if (z > 0.223*sqrt(40)) 1 else 0
+ }
> mean(simlist) # P-value = P(Z>1.41)
[1] 0.018
> ks.test(data,"punif",0,1)$p.value # exact P-value
[1] 0.015743
```



Geometric Brownian Motion

Geometric Brownian motion is a nonnegative process, which can be thought of as a stochastic model for exponential growth or decay. It is a favorite tool in mathematical finance, where it is used extensively to model stock prices.

Geometric Brownian Motion

Let $(X_t)_{t \geq 0}$ be a Brownian motion with drift parameter μ and variance parameter σ^2 . The process $(G_t)_{t \geq 0}$ defined by

$$G_t = G_0 e^{X_t}, \text{ for } t \geq 0,$$

where $G_0 > 0$, is called *geometric Brownian motion*.

Taking logarithms, we see that $\ln G_t = \ln G_0 + X_t$ is normally distributed with mean

$$E(\ln G_t) = E(\ln G_0 + X_t) = \ln G_0 + \mu t$$

and variance

$$\text{Var}(\ln G_t) = \text{Var}(\ln G_0 + X_t) = \text{Var}(X_t) = \sigma^2 t.$$

A random variable whose logarithm is normally distributed is said to have a *lognormal distribution*. For each $t > 0$, G_t has a lognormal distribution.

We leave as an exercise the derivation of mean and variance for geometric Brownian motion

$$E(G_t) = G_0 e^{t(\mu + \sigma^2/2)} \text{ and } \text{Var}(G_t) = G_0^2 e^{2t(\mu + \sigma^2/2)} (e^{t\sigma^2} - 1). \quad (8.11)$$

The exponential mean function shows that, on average, geometric Brownian motion exhibits exponential growth with growth rate $\mu + \sigma^2/2$.

Geometric Brownian motion arises as a model for quantities which can be expressed as the *product* of independent random multipliers. For $s, t \geq 0$, consider the ratio

$$\frac{G_{t+s}}{G_t} = \frac{G_0 e^{\mu(t+s) + \sigma X_{t+s}}}{G_0 e^{\mu t + \sigma X_t}} = e^{\mu s + \sigma(X_{t+s} - X_t)},$$

which has the same distribution as $e^{\mu s + \sigma X_s} = G_s/G_0$, because of stationary increments for the $(X_t)_{t \geq 0}$ process. For $0 \leq q < r \leq s < t$,

$$\frac{G_t}{G_s} = e^{\mu(t-s) + \sigma(X_t - X_s)} \text{ and } \frac{G_r}{G_q} = e^{\mu(r-q) + \sigma(X_r - X_q)}$$

are independent random variables, because of independent increments for $(X_t)_{t \geq 0}$.

Let $Y_k = G_k/G_{k-1}$, for $k = 1, 2, \dots$. Then, $Y_1, Y_2, \dots$ is an i.i.d. sequence, and

$$G_n = \left(\frac{G_n}{G_{n-1}} \right) \left(\frac{G_{n-1}}{G_{n-2}} \right) \cdots \left(\frac{G_2}{G_1} \right) \left(\frac{G_1}{G_0} \right) G_0 = G_0 Y_1 Y_2 \cdots Y_{n-1} Y_n.$$

Example 8.15 Stock prices are commonly modeled with geometric Brownian motion. The process is attractive to economists because of several assumptions.

Historical data for many stocks indicate long-term exponential growth or decline. Prices cannot be negative and geometric Brownian motion takes only positive values. Let Y_t denote the price of a stock after t days. Since the price on a given day is probably close to the price on the next day (assuming normal market conditions), stock prices are not independent. However, the *percent changes in price* from day to day Y_t/Y_{t-1} , for $t = 1, 2, \dots$ might be reasonably modeled as independent and identically distributed. This leads to geometric Brownian motion. In the context of stock prices, the standard deviation σ is called the *volatility*.

A criticism of the geometric Brownian motion model is that it does not account for extreme events like the stock market crash on October 19, 1987, when the world's stock markets lost more than 20% of their value within a few hours.

Assume that XYZ stock currently sells for \$80 a share and follows a geometric Brownian motion with drift parameter 0.10 and volatility 0.50. Find the probability that in 90 days the price of XYZ will rise to at least \$100.

Solution Let Y_t denote the price of XYZ after t years. Round 90 days as 1/4 of a year. Then,

$$\begin{aligned} P(Y_{0.25} \geq 100) &= P\left(80e^{\mu(0.25) + \sigma B_{0.25}} \geq 100\right) \\ &= P((0.1)(0.25) + (0.5)B_{0.25} \geq \ln 1.25) \\ &= P(B_{0.25} \geq 0.396) = 0.214. \end{aligned}$$

In R, type

```
> x <- (log(100/80) - (0.1)/4) / 0.5
> 1-pnorm(x, 0, sqrt(1/4))
[1] 0.214013
```

Simulations of the stock price are shown in Figure 8.12. ■

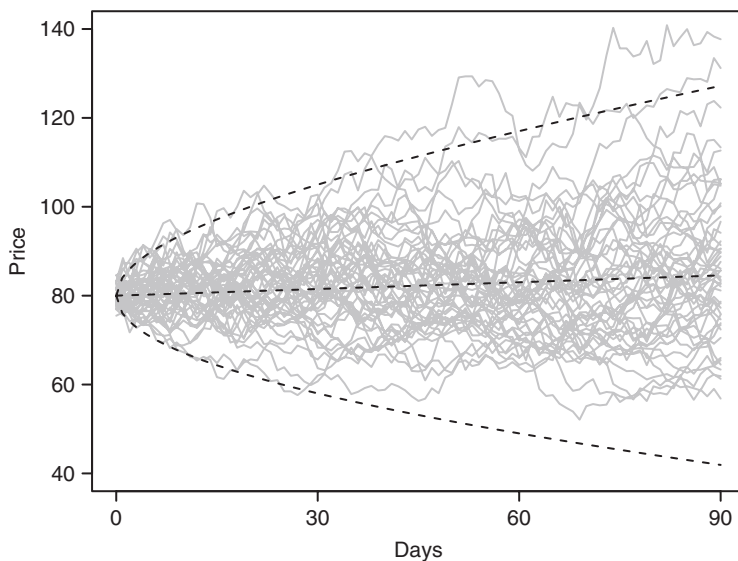


Figure 8.12 Fifty sample paths of a stock's price over 90 days modeled as geometric Brownian motion. Dotted lines are drawn at the mean function, and the mean plus or minus two standard deviations.

■ **Example 8.16 (Financial options)** An *option* is a contract that gives the buyer the right to buy shares of stock sometime in the future at a fixed price. In Example 8.13, we assumed that XYZ stock currently sells for \$80 a share. Assume that an XYZ option is selling for \$10. Under the terms of the option, in 90 days you may buy a share of XYZ stock for \$100.

If you decide to purchase the option, consider the payoff. Assume that in 90 days the price of XYZ is greater than \$100. Then, you can *exercise* the option, buy the stock for \$100, and turn around and sell XYZ for its current price. Your payoff would be $G_{90/365} - 100$, where $G_{90/365}$ is the price of XYZ in 90 days.

On the other hand, if XYZ sells for less than \$100 in 90 days, you would not exercise the option, and your payoff is nil. In either case, the payoff in 90 days is $\max\{G_{90/365} - 100, 0\}$. Your final profit would be the payoff minus the initial \$10 cost of the option.

Find the future expected payoff of the option, assuming the price of XYZ follows a geometric Brownian motion.

Solution Let G_0 denote the current stock price. Let t be the *expiration date*, which is the time until the option is exercised. Let K be the *strike price*, which is how much you can buy the stock for if you exercise the option. For XYZ, $G_0 = 80$, $t = 90/365$ (measuring time in years), and $K = 100$.

The goal is to find the expected payoff $E(\max\{G_t - K, 0\})$, assuming $(G_t)_{t \geq 0}$ is a geometric Brownian motion. Let $f(x)$ be the density function of a normal distribution with mean 0 and variance t . Then,

$$\begin{aligned} E(\max\{G_t - K, 0\}) &= E(\max\{G_0 e^{\mu t + \sigma B_t} - K, 0\}) \\ &= \int_{-\infty}^{\infty} \max\{G_0 e^{\mu t + \sigma x} - K, 0\} f(x) dx \\ &= \int_{\beta}^{\infty} (G_0 e^{\mu t + \sigma x} - K) f(x) dx \\ &= G_0 e^{\mu t} \int_{\beta}^{\infty} e^{\sigma x} f(x) dx - KP \left(Z > \frac{\beta}{\sqrt{t}} \right), \end{aligned}$$

where $\beta = (\ln(K/G_0) - \mu t) / \sigma$, and Z is a standard normal random variable.

By completing the square, the integral in the last expression is

$$\begin{aligned} \int_{\beta}^{\infty} e^{\sigma x} f(x) dx &= \int_{\beta}^{\infty} e^{\sigma x} \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx \\ &= e^{\sigma^2 t/2} \int_{\beta}^{\infty} \frac{1}{\sqrt{2\pi t}} e^{-(x - \sigma t)^2/2t} dx \end{aligned}$$

$$\begin{aligned}
&= e^{\sigma^2 t/2} \int_{(\beta - \sigma t)/\sqrt{t}}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\
&= e^{\sigma^2 t/2} P\left(Z > \frac{\beta - \sigma t}{\sqrt{t}}\right).
\end{aligned}$$

This gives

$$\begin{aligned}
&E(\max\{G_t - K, 0\}) \\
&= G_0 e^{t(\mu + \sigma^2/2)} P\left(Z > \frac{\beta - \sigma t}{\sqrt{t}}\right) - KP\left(Z > \frac{\beta}{\sqrt{t}}\right). \quad (8.12)
\end{aligned}$$

Assume that XYZ stock follows a geometric Brownian motion with drift parameter $\mu = 0.10$ and variance $\sigma^2 = 0.25$. Then,

$$\beta = \frac{\ln(K/G_0) - \mu t}{\sigma} = \frac{\ln(100/80) - (0.10)(90/365)}{0.5} = 0.3970,$$

which gives

$$\begin{aligned}
&E(\max\{G_{90/365} - 100, 0\}) \\
&= 80e^{(90/365)(0.10 + 0.25/2)} P\left(Z > \frac{0.397 - 0.5(90/365)}{\sqrt{90/365}}\right) \\
&\quad - 100P\left(Z > \frac{0.397}{\sqrt{90/365}}\right) \\
&= 1.788.
\end{aligned}$$

Given that the initial cost of the option is \$10, your expected profit is $1.788 - 10 < 0$. So you can expect to lose money.

For this example, we set an arbitrary initial cost of the option. A fundamental question in finance is how such an option should be priced. This leads to the Black–Scholes model for option pricing, which is introduced in the next section. ■

8.6 MARTINGALES

A martingale is a stochastic process that generalizes the notion of a *fair game*. Assume that after n plays of a gambling game your winnings are x . Then, by *fair*, we mean that your expected future winnings should be x regardless of past history.

Martingale

A stochastic process $(Y_t)_{t \geq 0}$ is a *martingale*, if for all $t \geq 0$,

1. $E(Y_t | Y_r, 0 \leq r \leq s) = Y_s$, for all $0 \leq s \leq t$.
2. $E(|Y_t|) < \infty$.

A discrete-time martingale $Y_0, Y_1, \dots$ satisfies

1. $E(Y_{n+1} | Y_0, \dots, Y_n) = Y_n$, for all $n \geq 0$.
2. $E(|Y_n|) < \infty$.

A most important property of martingales is that they have constant expectation. By the law of total expectation,

$$E(Y_t) = E(E(Y_t | Y_r, 0 \leq r \leq s)) = E(Y_s),$$

for all $0 \leq s \leq t$. That is,

$$E(Y_t) = E(Y_0), \text{ for all } t.$$

Example 8.17 (Random walk) Show that simple symmetric random walk is a martingale.

Solution Let

$$X_i = \begin{cases} +1, & \text{with probability } 1/2, \\ -1, & \text{with probability } 1/2, \end{cases}$$

for $i = 1, 2, \dots$. For $n \geq 1$, let $S_n = X_1 + \dots + X_n$, with $S_0 = 0$. Then,

$$\begin{aligned} E(S_{n+1} | S_0, \dots, S_n) &= E(X_{n+1} + S_n | S_0, \dots, S_n) \\ &= E(X_{n+1} | S_0, \dots, S_n) + E(S_n | S_0, \dots, S_n) \\ &= E(X_{n+1}) + S_n = S_n. \end{aligned}$$

The third equality is because X_{n+1} is independent of $X_1, \dots, X_n$, and thus independent of $S_0, S_1, \dots, S_n$. The fact that $E(S_n | S_0, \dots, S_n) = S_n$ is a consequence of a general property of conditional expectation, which states that if X is a random variable and g is a function, then $E(g(X) | X) = g(X)$.

The second part of the martingale definition is satisfied as

$$E(|S_n|) = E\left(\left|\sum_{i=1}^n X_i\right|\right) \leq E\left(\sum_{i=1}^n |X_i|\right) = \sum_{i=1}^n E(|X_i|) = n < \infty.$$

■

Since simple symmetric random walk is a martingale, the next example should not be surprising.

■ **Example 8.18 (Brownian motion)** Show that standard Brownian motion $(B_t)_{t \geq 0}$ is a martingale.

Solution We have that

$$\begin{aligned} E(B_t | B_r, 0 \leq r \leq s) &= E(B_t - B_s + B_s | B_r, 0 \leq r \leq s) \\ &= E(B_t - B_s | B_r, 0 \leq r \leq s) + E(B_s | B_r, 0 \leq r \leq s) \\ &= E(B_t - B_s) + B_s = B_s, \end{aligned}$$

where the second equality is because of independent increments. Also,

$$E(|B_t|) = \int_{-\infty}^{\infty} |x| \frac{1}{\sqrt{2\pi t}} e^{-x^2/2t} dx = \int_0^{\infty} x \sqrt{\frac{2}{\pi t}} e^{-x^2/2t} dx = \sqrt{\frac{2t}{\pi}} < \infty.$$

■

The following extension of the martingale definition is used frequently.

Martingale with Respect to Another Process

Let $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ be stochastic processes. Then, $(Y_t)_{t \geq 0}$ is a *martingale with respect to* $(X_t)_{t \geq 0}$, if for all $t \geq 0$,

1. $E(Y_t | X_r, 0 \leq r \leq s) = Y_s$, for all $0 \leq s \leq t$.
2. $E(|Y_t|) < \infty$.

The most common application of this is when Y_t is a function of X_t . That is, $Y_t = g(X_t)$ for some function g . It is useful to think of the conditioning random variables $(X_r)_{0 \leq r \leq s}$ as representing past information, or history, of the process up to time s .

Following are several examples of martingales that are functions of Brownian motion.

■ **Example 8.19 (Quadratic martingale)** Let $Y_t = B_t^2 - t$, for $t \geq 0$. Show that $(Y_t)_{t \geq 0}$ is a martingale with respect to Brownian motion. This is called the *quadratic martingale*.

Solution For $0 \leq s < t$,

$$\begin{aligned}
 E(Y_t | B_r, 0 \leq r \leq s) &= E\left(B_t^2 - t | B_r, 0 \leq r \leq s\right) \\
 &= E\left((B_t - B_s + B_s)^2 | B_r, 0 \leq r \leq s\right) - t \\
 &= E\left((B_t - B_s)^2 + 2(B_t - B_s)B_s + B_s^2 | B_r, 0 \leq r \leq s\right) - t \\
 &= E\left((B_t - B_s)^2\right) + 2B_s E(B_t - B_s) + B_s^2 - t \\
 &= (t - s) + B_s^2 - t = B_s^2 - s = Y_s.
 \end{aligned}$$

Furthermore,

$$E(|Y_t|) = E(|B_t^2 - t|) \leq E(B_t^2 + t) = E(B_t^2) + t = 2t < \infty. \quad \blacksquare$$

Example 8.20 Let $G_t = G_0 e^{X_t}$ be a geometric Brownian motion, where $(X_t)_{t \geq 0}$ is Brownian motion with drift μ and variance σ^2 . Let $r = \mu + \sigma^2/2$. Show that $e^{-rt} G_t$ is a martingale with respect to standard Brownian motion.

Solution For $0 \leq s < t$,

$$\begin{aligned}
 E(e^{-rt} G_t | B_r, 0 \leq r \leq s) &= e^{-rt} E(G_0 e^{\mu t + \sigma B_t} | B_r, 0 \leq r \leq s) \\
 &= e^{-rt} E(G_0 e^{\mu(t-s) + \sigma(B_t - B_s)} e^{\mu s + \sigma B_s} | B_r, 0 \leq r \leq s) \\
 &= e^{-rt} e^{\mu s + \sigma B_s} E(G_0 e^{\mu(t-s) + \sigma(B_t - B_s)}) \\
 &= e^{-rt} e^{\mu s + \sigma B_s} E(G_{t-s}) \\
 &= e^{-t(\mu + \sigma^2/2)} e^{\mu s + \sigma B_s} G_0 e^{(t-s)(\mu + \sigma^2/2)} \\
 &= e^{-s(\mu + \sigma^2/2)} G_0 e^{\mu s + \sigma B_s} \\
 &= e^{-rs} G_s.
 \end{aligned}$$

Also,

$$E(|e^{-rt} G_t|) = e^{-rt} E(G_t) = e^{-t(\mu + \sigma^2/2)} G_0 e^{t(\mu + \sigma^2/2)} = G_0 < \infty, \text{ for all } t. \quad \blacksquare$$

Example 8.21 (Black–Scholes) In Example 8.16, the expected payoff for a financial option was derived. This leads to the *Black–Scholes formula* for pricing options, a fundamental formula in mathematical finance.

The formula was first published by Fisher Black and Myron Scholes in 1973 and then further developed by Robert Merton. Merton and Scholes received the 1997 Nobel Prize in Economics for their work. The ability to price options and other financial *derivatives* opened up a massive global market for trading ever-more complicated

financial instruments. Black–Scholes has been described both as a formula which “changed the world” and as “the mathematical equation that caused the banks to crash.” See Stewart (2012).

There are several critical assumptions underlying the Black–Scholes formula. One is that stock prices follow a geometric Brownian motion. Another is that an investment should be *risk neutral*. What this means is that the expected return on an investment should be equal to the so-called risk-free rate of return, such as what is obtained by a short-term U.S. government bond.

Let r denote the risk-free interest rate. Starting with P dollars, because of compound interest, after t years of investing risk free your money will grow to $P(1 + r)^t$ dollars. Under continuous compounding, the future value is $F = Pe^{rt}$. This gives the future value of your present dollars. On the other hand, assume that t years from now you will be given F dollars. To find its *present value* requires *discounting* the future amount by a factor of e^{-rt} . That is, $P = e^{-rt}F$.

Let G_t denote the price of a stock t years from today. Then, the present value of the stock price is $e^{-rt}G_t$. The Black–Scholes risk neutral assumption means that the discounted stock price process is a *fair game*, that is, a martingale. For any time $0 < s < t$, the expected present value of the stock t years from now given knowledge of the stock price up until time s should be equal to the present value of the stock price s years from now. In other words,

$$E(e^{-rt}G_t | G_s, 0 \leq x \leq s) = e^{-rs}G_s. \quad (8.13)$$

In Example 8.20, it was shown that Equation (8.13) holds for geometric Brownian motion if $r = \mu + \sigma^2/2$, or $\mu = r - \sigma^2/2$. The probability calculations for the Black–Scholes formula are obtained with this choice of μ . In the language of Black–Scholes, this gives the *risk-neutral probability* for computing the options price formula. The Black–Scholes formula for the price of an option is then the present value of the expected payoff of the option under the risk-neutral probability.

See Equation (8.12) for the future expected payoff of a financial option. The present value of the expected payoff is obtained by multiplying by the discount factor e^{-rt} . Replace μ with $r - \sigma^2$ to obtain the Black–Scholes option price formula

$$e^{-rt}E(\max\{G_t - K, 0\}) = G_0P\left(Z > \frac{\alpha - \sigma t}{\sqrt{t}}\right) - e^{-rt}KP\left(Z > \frac{\alpha}{\sqrt{t}}\right),$$

where

$$\alpha = \frac{\ln(K/G_0) - (r - \sigma^2/2)t}{\sigma}.$$

For the XYZ stock example, $G_0 = 80$, $K = 100$, $\sigma^2 = 0.25$, and $t = 90/365$. Furthermore, assume $r = 0.02$ is the risk-free interest rate. Then,

$$\alpha = \frac{\ln(100/80) - (0.02 - 0.25/2)(90/365)}{0.5} = 0.498068,$$

and the Black–Scholes option price is

$$80P\left(Z > \frac{\alpha - 0.50(90/365)}{0.5}\right) - e^{-0.02(90/365)}(100)P\left(Z > \frac{\alpha}{0.5}\right) = \$2.426.$$

Remarks:

1. For a given expiration date and strike price, the Black–Scholes formula depends only on volatility σ and the risk-free interest rate r , and not on the drift parameter μ . In practice, volatility is often estimated from historical price data. The model assumes that σ and r are constant.
2. The original derivation of the Black–Scholes formula in the paper by Black and Scholes was based on deriving and solving a partial differential equation

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} = rV,$$

where $V = V(S, t)$ is the value of the option, a function of stock price S and time t . This is known as the *Black–Scholes equation*, and can be reduced to the simpler heat equation.

3. The Black–Scholes formula in the example is for a European call option, in which the option can only be exercised on the expiration date. In an American call option, one can exercise the option at any time before the expiration date. There are many types of options and financial derivatives whose pricing structure is based on Black–Scholes. ■

Optional Stopping Theorem

A martingale $(Y_t)_{t \geq 0}$ has constant expectation. For all $t \geq 0$, $E(Y_t) = E(Y_0)$. The property holds for all fixed, deterministic times, but not necessarily for *random times*. If T is a random variable, which takes values in the index set of a martingale, it is not necessarily true that $E(Y_T) = E(Y_0)$. For instance, let T be the first time that a standard Brownian motion hits level a . Then, $B_T = a = E(B_T)$. However, $E(B_t) = 0$, for all $t \geq 0$.

The optional stopping theorem gives conditions for when a random time T satisfies $E(Y_T) = E(Y_0)$. While that might not sound like a big deal, the theorem is remarkably powerful. As mathematicians David Aldous and Jim Fill write in *Reversible Markov chains and random walks on graphs*, “Modern probabilists regard the martingale optional stopping theorem as one of the most important results in their subject.” (Aldous and Fill, 2002.)

In Section 3.9, we introduced the notion of a stopping time in the context of discrete-time Markov chains. For a stochastic process $(Y_t)_{t \geq 0}$, a nonnegative random variable T is a stopping time if for each t , the event $\{T \leq t\}$ can be determined from $\{Y_s, 0 \leq s \leq t\}$. That is, if the outcomes of Y_s are known for $0 \leq s \leq t$, then it can be determined whether or not $\{T \leq t\}$ occurs.

On the interval $[0, 1]$, the first time Brownian motion hits level a is a stopping time. Whether or not a is first hit by time t can be determined from $\{B_s, 0 \leq s \leq t\}$. On the other hand, the *last time* a is hit on $[0, 1]$ is not a stopping time. To determine whether or not a was last hit by time $t < 1$ requires full knowledge of B_s , for all $s \in [0, 1]$.

Optional Stopping Theorem

Let $(Y_t)_{t \geq 0}$ be a martingale with respect to a stochastic process $(X_t)_{t \geq 0}$. Assume that T is a stopping time for $(X_t)_{t \geq 0}$. Then, $E(Y_T) = E(Y_0)$ if one of the following is satisfied.

1. T is bounded. That is, $T \leq c$, for some constant c .
2. $P(T < \infty) = 1$ and $E(|Y_t|) \leq c$, for some constant c , whenever $T > t$.

The proof of the optional stopping theorem is beyond the scope of this book. There are several versions of the theorem with alternate sets of conditions.

In the context of a fair gambling game, one can interpret the optional stopping theorem to mean that a gambler has no reasonable strategy for increasing their initial stake. Let Y_t denote the gambler's winnings at time t . If a gambler strategizes to stop play at time T , their expected winnings will be $E(Y_T) = E(Y_0) = Y_0$, the gambler's initial stake.

The word *martingale* has its origins in an 18th century French gambling strategy. Mansuy (2009) quotes the dictionary of the Académie Française, that "To play the martingale is to always bet all that was lost."

Consider such a betting strategy for a game where at each round a gambler can win or lose one franc with equal probability. After one turn, if you win you stop. Otherwise, bet 2 francs on the next round. If you win, stop. If you lose, bet 4 francs on the next round. And so on. Let T be the number of bets needed until you win. The random variable T is a stopping time. If the gambler wins their game after k plays, they gain

$$2^k - (1 + 2 + \cdots + 2^{k-1}) = 2^k - (2^k - 1) = 1 \text{ franc,}$$

and, with probability 1, the gambler will eventually win some bet. Thus, T seems to be a winning strategy.

However, if Y_n denotes the gambler's winnings after n plays, then $E(Y_T) = 1 \neq 0 = E(Y_0)$. The random variable T does not satisfy the conditions of the optional sampling theorem. The reason this martingale gambling strategy is not *reasonable* is that it assumes the gambler has infinite capital.

The optional stopping theorem gives elegant and sometimes remarkably simple solutions to seemingly difficult problems.

■ **Example 8.22** Let $a, b > 0$. For a standard Brownian motion, find the probability that the process hits level a before hitting level $-b$.

Solution Let p be the desired probability. Consider the time T that Brownian motion first hits either a or $-b$. That is, $T = \min\{t : B_t = a \text{ or } B_t = -b\}$. See Figure 8.13.

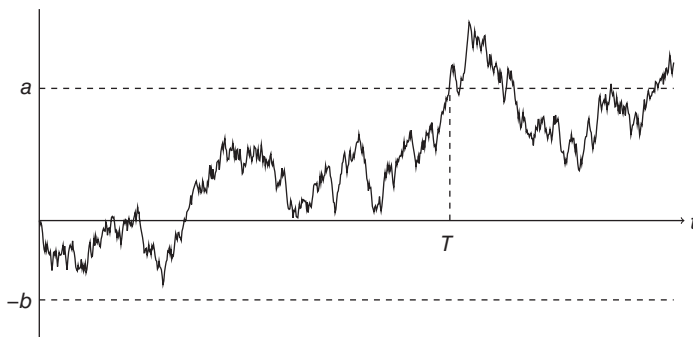


Figure 8.13 First hitting time T that Brownian motion hits level a or $-b$.

The random variable T is a stopping time. Furthermore, it satisfies the conditions of the optional stopping theorem. From Equation (8.6), the first hitting time T_a is finite with probability 1. By the strong Markov property, from a , the first time to hit $-b$ is also finite with probability 1. Thus, the first part of condition 2 is satisfied. Furthermore, for $t < T$, $B_t \in (-b, a)$. Thus, $|B_t| < \max\{a, b\}$, and the second part of condition 2 is satisfied.

Observe that $B_T = a$, with probability p , and $B_T = -b$, with probability $1 - p$. By the optional stopping theorem

$$0 = E(B_0) = E(B_T) = pa + (1 - p)(-b).$$

Solving for p gives $p = b/(a + b)$. ■

■ **Example 8.23 (Expected hitting time)** Apply the optional stopping theorem with the same stopping time as in Example 8.22, but with the quadratic martingale $B_t^2 - t$. This gives

$$E(B_T^2 - T) = E(B_0^2 - 0) = 0,$$

from which it follows that

$$E(T) = E(B_T^2) = a^2 \left(\frac{b}{a + b} \right) + b^2 \left(\frac{a}{a + b} \right) = ab.$$

We have thus discovered that the expected time that standard Brownian motion first hits the boundary of the region defined by the lines $y = a$ and $y = -b$ is ab . ■

■ **Example 8.24 (Gambler's ruin)** It was shown that discrete-time simple symmetric random walk $S_0, S_1, \dots$ is a martingale. As with the quadratic martingale, the process $S_n^2 - n$ is a martingale. The results from the last two examples for Brownian motion can be restated for gambler's ruin on $\{-b, -b+1, \dots, 0, \dots, a-1, a\}$ starting at the origin. This gives the following:

1. The probability that the gambler gains a before losing b is $b/(a+b)$.
2. The expected duration of the game is ab . ■

■ **Example 8.25 (Time to first hit the line $y = a - bt$)** For $a, b > 0$, let $T = \min\{t : B_t = a - bt\}$ be the first time a standard Brownian motion hits the line $y = a - bt$. The random variable T is a stopping time, which satisfies the optional stopping theorem. This gives

$$0 = E(B_0) = E(B_T) = E(a - bT) = a - bE(T).$$

Hence, $E(T) = a/b$. For the line $y = 4 - (0.5)t$ in Figure 8.14, the mean time to first hit the line is $4/(0.5) = 8$. ■

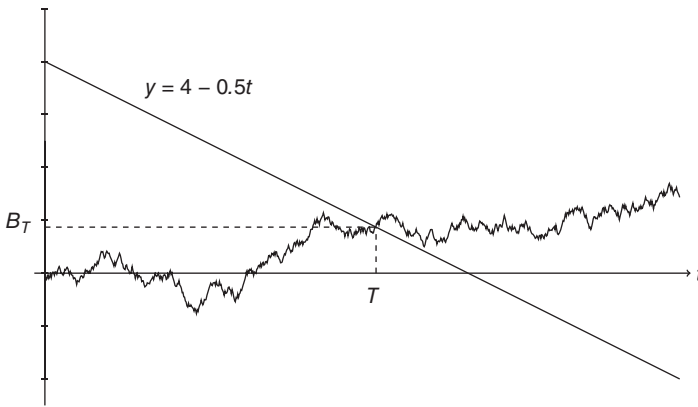


Figure 8.14 First time T Brownian motion hits the line $y = 4 - 0.5t$.

■ **Example 8.26 (Time to reach a for Brownian motion with drift)** Assume that $(X_t)_{t \geq 0}$ is a Brownian motion process with drift parameter μ and variance parameter σ^2 , where $\mu > 0$. For $a > 0$, find the expected time that the process first hits level a .

Solution Let $T = \min\{t : X_t = a\}$ be the first time that the process hits level a . Write $X_t = \mu t + \sigma B_t$. Then, $X_t = a$ if and only if

$$B_t = \frac{a - \mu t}{\sigma} = \frac{a}{\sigma} - \left(\frac{\mu}{\sigma}\right)t.$$

Applying the result of Example 8.25, $E(T) = (a/\sigma)/(\mu/\sigma) = a/\mu$. ■

■ **Example 8.27 (Variance of first hitting time)** Assume that $(X_t)_{t \geq 0}$ is a Brownian motion process with drift μ and variance σ^2 . Let $T = \min\{t : X_t = a\}$ be the first hitting time to reach level a . In the last example, the expectation of T was derived. Here, the variance of T is obtained using the quadratic martingale $Y_t = B_t^2 - t$.

Solution Since T is a stopping time with respect to B_t ,

$$0 = E(Y_0) = E(Y_T) = E(B_T^2 - T) = E(B_T^2) - E(T).$$

Thus, $E(B_T^2) = E(T) = a/\mu$. Write $X_t = \mu t + \sigma B_t$. Then, $X_T = \mu T + \sigma B_T$, giving

$$B_T = \frac{X_T - \mu T}{\sigma} = \frac{a - \mu T}{\sigma}.$$

Thus,

$$\begin{aligned} \text{Var}(T) &= E((T - E(T))^2) = E\left(\left(T - \frac{a}{\mu}\right)^2\right) \\ &= \frac{1}{\mu^2} E((\mu T - a)^2) = \frac{\sigma^2}{\mu^2} E\left(\left(\frac{a - \mu T}{\sigma}\right)^2\right) \\ &= \frac{\sigma^2}{\mu^2} E(B_T^2) = \frac{\sigma^2}{\mu^2} \left(\frac{a}{\mu}\right) = \frac{a\sigma^2}{\mu^3}. \end{aligned}$$

■

R: Hitting Time Simulation for Brownian Motion with Drift

Let $(X_t)_{t \geq 0}$ be a Brownian motion with drift $\mu = 0.5$ and variance $\sigma^2 = 1$. Consider the first hitting time T of level $a = 10$. Exact results are

$$E(T) = a/\mu = 20 \quad \text{and} \quad \text{Var}(T) = a\sigma^2/\mu^3 = 80.$$

Simulated results are based on 1,000 trials of a Brownian motion process on $[0, 80]$. See Figure 8.15 for several realizations.

```
# bmhitting.R
> mu <- 1/2
> sig <- 1
> a <- 10
> simlist <- numeric(1000)
> for (i in 1:1000) {
  t <- 80
  n <- 50000
  bm <- c(0, cumsum(rnorm(n, 0, sqrt(t/n))))
```

```

xproc <- mu*seq(0,t,t/n) + sig*bm
simlist[i] <- which(xproc >= a)[1] * (t/n) }
> mean(simlist)
[1] 20.07139
> var(simlist)
[1] 81.31142

```

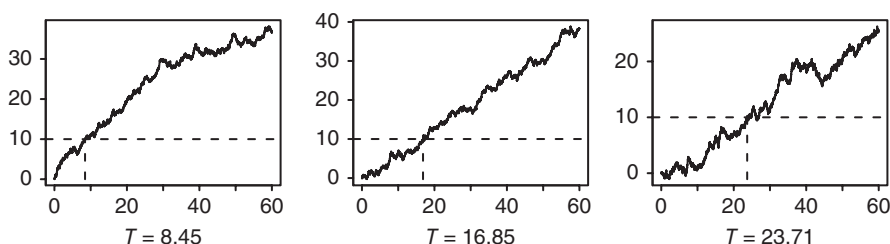


Figure 8.15 First time T to hit $a = 10$ for Brownian motion with drift $\mu = 0.5$ and variance $\sigma^2 = 1$.

EXERCISES

8.1 Show that

$$f(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-x^2/(2t)},$$

satisfies the partial differential heat equation

$$\frac{\partial f}{\partial t} = \frac{1}{2} \frac{\partial^2 f}{\partial x^2}.$$

8.2 For standard Brownian motion, find

- (a) $P(B_2 \leq 1)$
- (b) $E(B_4 | B_1 = x)$
- (c) $\text{Corr}(B_{t+s}, B_s)$
- (d) $\text{Var}(B_4 | B_1)$
- (e) $P(B_3 \leq 5 | B_1 = 2)$.

8.3 For standard Brownian motion started at $x = -3$, find

- (a) $P(X_1 + X_2 > -1)$
- (b) The conditional density of X_2 given $X_1 = 0$.
- (c) $\text{Cov}(X_3, X_4)$
- (d) $E(X_4 | X_1)$.

8.4 In a race between Lisa and Cooper, let X_t denote the amount of time (in seconds) by which Lisa is ahead when 100t percent of the race has been completed. Assume that $(X_t)_{0 \leq t \leq 1}$ can be modeled by a Brownian motion with drift parameter 0 and variance parameter σ^2 . If Lisa is leading by $\sigma/2$ seconds after three-fourths of the race is complete, what is the probability that she is the winner?

8.5 Consider standard Brownian motion. Let $0 < s < t$.

(a) Find the joint density of (B_s, B_t) .

(b) Show that the conditional distribution of B_s given $B_t = y$ is normal, with mean and variance

$$E(B_s | B_t = y) = \frac{sy}{t} \quad \text{and} \quad \text{Var}(B_s | B_t = y) = \frac{s(t-s)}{t}.$$

8.6 For $s > 0$, show that the translation $(B_{t+s} - B_s)_{t \geq 0}$ is a standard Brownian motion.

8.7 Show that the reflection $(-B_t)_{t \geq 0}$ is a standard Brownian motion.

8.8 Find the covariance function for Brownian motion with drift.

8.9 Let $W_t = B_{2t} - B_t$, for $t \geq 0$.

(a) Is $(W_t)_{t \geq 0}$ a Gaussian process?

(b) Is $(W_t)_{t \geq 0}$ a Brownian motion process?

8.10 Let $(B_t)_{t \geq 0}$ be a Brownian motion started in x . Let

$$X_t = B_t - t(B_1 - y), \quad \text{for } 0 \leq t \leq 1.$$

The process $(X_t)_{t \geq 0}$ is a Brownian bridge with start in x and end in y . Find the mean and covariance functions.

8.11 A standard Brownian motion crosses the t -axis at times $t = 2$ and $t = 5$. Find the probability that the process exceeds level $x = 1$ at time $t = 4$.

8.12 Show that Brownian motion with drift has independent and stationary increments.

8.13 Let $(X_t)_{t \geq 0}$ denote a Brownian motion with drift μ and variance σ^2 . For $0 < s < t$, find $E(X_s X_t)$.

8.14 A Brownian motion with drift parameter $\mu = -1$ and variance $\sigma^2 = 4$ starts at $x = 1.5$. Find the probability that the process is positive at $t = 3$.

8.15 See Example 8.11 on using Brownian motion to model the home team advantage in basketball. In Stern (1994), Table 8.3 is given based on the outcomes of 493 basketball games played in 1992.

Here, X_t is the difference between the home team's score and the visiting team's score after $t(100)$ percent of the game is played. The data show the mean

TABLE 8.3 Results by Quarter of 493 NBA Games

Quarter	Variable	Mean	Standard Deviation
1	$X_{0.25}$	1.41	7.58
2	$X_{0.50} - X_{0.25}$	1.57	7.40
3	$X_{0.75} - X_{0.50}$	1.51	7.30
4	$X_1 - X_{0.75}$	0.22	6.99

and standard deviation of these differences at the end of each quarter. Why might the data support the use of a Brownian motion model? What aspects of the data give reason to doubt the Brownian motion model?

- 8.16** Let $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$ be independent, standard Brownian motions. Show that $Z_t = a(X_t - Y_t)$ defines a standard Brownian motion for some a . Find a .
- 8.17** For $a > 0$, show that for standard Brownian motion the first hitting time T_a has the same distribution as $1/X^2$, where X is a normal random variable with mean 0 and variance $1/a^2$.
- 8.18** Show that the first hitting time T_a has the same distribution as $a^2 T_1$.
- 8.19** Find the mean and variance of the maximum value of standard Brownian motion on $[0, t]$.
- 8.20** Use the reflection principle to show

$$P(M_t \geq a, B_t \leq a - b) = P(B_t \geq a + b), \text{ for } a, b > 0. \quad (8.14)$$

- 8.21** From standard Brownian motion, let X_t be the process defined by

$$X_t = \begin{cases} B_t, & \text{if } t < T_a, \\ a, & \text{if } t \geq T_a, \end{cases}$$

where T_a is the first hitting time of $a > 0$. The process $(X_t)_{t \geq 0}$ is called *Brownian motion absorbed at a* . The distribution of X_t has discrete and continuous parts.

- (a) Show

$$P(X_t = a) = \frac{2}{\sqrt{2\pi t}} \int_a^\infty e^{-x^2/2t} dx.$$

- (b) For $x < a$, show

$$P(X_t \leq x) = P(B_t \leq x) - P(B_t \leq x - 2a) = \frac{1}{\sqrt{2\pi t}} \int_{x-2a}^x e^{-z^2/2t} dz.$$

Hint: Use the result of Exercise 8.20.

8.22 Let Z be the smallest zero of standard Brownian motion past t . Show that

$$P(Z \leq z) = \frac{2}{\pi} \arccos \sqrt{\frac{t}{z}}, \text{ for } z > 0.$$

8.23 Let $0 < r < s < t$.

- (a) Assume that standard Brownian motion is not zero in (r, s) . Find the probability that standard Brownian motion is not zero in (r, t) .
- (b) Assume that standard Brownian motion is not zero in $(0, s)$. Find the probability that standard Brownian motion is not zero in $(0, t)$.

8.24 Derive the mean and variance of geometric Brownian motion.

8.25 The price of a stock is modeled with a geometric Brownian motion with drift $\mu = -0.25$ and volatility $\sigma = 0.4$. The stock currently sells for \$35. What is the probability that the price will be at least \$40 in 1 year?

8.26 For the stock price model in Exercise 8.25, assume that an option is available to purchase the stock in six months for \$40. Find the expected payoff of the option.

8.27 Assume that $Z_0, Z_1, \dots$ is a branching process whose offspring distribution has mean μ . Show that Z_n/μ^n is a martingale.

8.28 An urn contains two balls—one red and one blue. At each discrete step, a ball is chosen at random from the urn. It is returned to the urn along with another ball of the same color. Let X_n denote the number of red balls in the urn after n draws. (Thus, $X_0 = 1$.) Let $R_n = X_n/(n+2)$ be the proportion of red balls in the urn after n draws. Show that $R_0, R_1, \dots$ is a martingale with respect to $X_0, X_1, \dots$. The process is called *Polya's Urn*.

8.29 Show that $(B_t^3 - 3tB_t)_{t \geq 0}$ is a martingale with respect to Brownian motion.

8.30 A Brownian motion with drift $\mu = 2$ and variance $\sigma^2 = 4$ is run until level $a = 3$ is first hit. The process is repeated 25 times. Find the approximate probability that the average first hitting time of 25 runs is between 1 and 2.

8.31 Consider standard Brownian motion started at $x = -3$.

- (a) Find the probability of reaching level 2 before -7 .
- (b) When, on average, will the process leave the region between the lines $y = 2$ and $y = -7$?

8.32 Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ . Let $X_t = N_t - \lambda t$, for $t \geq 0$. Show that X_t is a martingale with respect to N_t .

8.33 Let $X_1, X_2, \dots$ be i.i.d. random variables with mean $\mu < \infty$. Let $Z_n = \sum_{i=1}^n (X_i - \mu)$, for $n = 0, 1, 2, \dots$

- (a) Show that $Z_0, Z_1, \dots$ is a martingale with respect to $X_0, X_1, \dots$

- (b) Assume that N is a stopping time that satisfies the conditions of the optional stopping theorem. Show that

$$E\left(\sum_{i=1}^N X_i\right) = E(N)\mu.$$

This result is known as *Wald's equation*.

- 8.34** Let $(N_t)_{t \geq 0}$ be a Poisson process with parameter λ .
- (a) Find the quantity $m(t)$ such that $M_t = (N_t - \lambda t)^2 - m(t)$ is a martingale.
 - (b) For fixed integer $k > 0$, let $T = \min\{t : N_t = k\}$ be the first time k arrivals occur for a Poisson process. Show that T is a stopping time that satisfies the conditions of the optional stopping theorem.
 - (c) Use the optional stopping theorem to find the standard deviation of T .
- 8.35** For $a > 0$, let T be the first time that standard Brownian motion exits the interval $(-a, a)$.
- (a) Show that T is a stopping time that satisfies the optional stopping theorem.
 - (b) Find the expected time $E(T)$ to exit $(-a, a)$.
 - (c) Let $M_t = B_t^4 - 6tB_t^2 + 3t^2$. Then, $(M_t)_{t \geq 0}$ is a martingale, a fact that you do not need to prove. Use this to find the standard deviation of T .
- 8.36** Let $(X_t)_{t \geq 0}$ be a Brownian motion with drift $\mu \neq 0$ and variance σ^2 . The goal of this exercise is to find the probability p that X_t hits a before $-b$, for $a, b > 0$.
- (a) Let $Y_t = e^{-t\mu^2/2 + cB_t}$, where B_t denotes standard Brownian motion. Show that $(Y_t)_{t \geq 0}$ is a martingale for constant $c \neq 0$.
 - (b) Let $T = \min\{t : X_t = a \text{ or } -b\}$. Then, T is a stopping time that satisfies the optional stopping theorem. Use (a) and appropriate choice of c to show

$$E\left(e^{-\frac{2\mu X_T}{\sigma^2}}\right) = 1.$$

- (c) Show that

$$p = \frac{1 - e^{2\mu b/\sigma^2}}{e^{-2\mu a/\sigma^2} - e^{2\mu b/\sigma^2}}. \quad (8.15)$$

- 8.37** Consider a Brownian motion with drift μ and variance σ^2 . Assume that $\mu < 0$. The process tends to $-\infty$, with probability 1. Let M be the maximum value reached.
- (a) See Exercise 8.36. By letting $b \rightarrow \infty$ in Equation (8.15), show

$$P(M > a) = e^{2\mu a/\sigma^2}, \text{ for } a > 0.$$

Conclude that M has an exponential distribution.

- (b) A particle moves according to a Brownian motion with drift $\mu = -1.6$ and variance $\sigma^2 = 0.4$. Find the mean and standard deviation of the largest level reached.

8.38 Let $T = \min\{t : B_t \notin (-a, a)\}$. Show that

$$E(e^{-\lambda T}) = \frac{2}{e^{a\sqrt{2\lambda}} + e^{-a\sqrt{2\lambda}}} = \frac{1}{\cosh(a\sqrt{2\lambda})}.$$

- (a) Apply the optional stopping theorem to the exponential martingale in Exercise 8.36(a) to show that

$$E\left(e^{\sqrt{2\lambda}B_T}e^{-\lambda T}\right) = 1.$$

- (b) Show that

$$P(B_T = a, T < x) = P(B_T = -a, T < x) = \frac{1}{2}P(T < x)$$

and conclude that B_T and T are independent, and thus establish the result.

- 8.39** R: Simulate a Brownian motion $(X_t)_{t \geq 0}$ with drift $\mu = 1.5$ and variance $\sigma^2 = 4$. Simulate the probability $P(X_3 > 4)$ and compare with the exact result.
- 8.40** R: Use the script file **bbridge.R** to simulate a Brownian bridge $(X_t)_{t \geq 0}$. Estimate the probability $P(X_{3/4} \leq 1/3)$. Compare with the exact result.
- 8.41** R: The price of a stock is modeled as a geometric Brownian motion with drift $\mu = -0.85$ and variance $\sigma^2 = 2.4$. If the current price is \$50, find the probability that the price is under \$40 in 2 years. Simulate the stock price, and compare with the exact value.
- 8.42** R: Simulate the mean and standard deviation of the maximum of standard Brownian motion on $[0, 1]$. Compare with the exact values.
- 8.43** R: Write a function for pricing an option using the Black–Scholes formula. Option price is a function of initial stock price, strike price, expiration date, interest rate, and volatility.
- (a) Price an option for a stock that currently sells for \$400 and has volatility 40%. Assume that the option strike price is \$420 and the expiration date is 90 days. Assume a risk-free interest rate of 3%.
- (b) For each of the five parameters in (a), how does varying the parameter, holding the other four fixed, effect the price of the option?
- (c) For the option in (a), assume that volatility is not known. However, based on market experience, the option sells for \$30. Estimate the volatility.

9

A GENTLE INTRODUCTION TO STOCHASTIC CALCULUS*

The study of mathematics, like the Nile, begins in minuteness but ends in magnificence.

—Charles Caleb Colton

9.1 INTRODUCTION

Brownian motion paths are continuous functions. Continuous functions are integrable. Integration of Brownian motion opens the door to powerful calculus-based modeling tools, such as stochastic differential equations (SDEs). Stochastic calculus is an advanced topic, which requires measure theory, and often several graduate-level probability courses. Our goal in this section is to introduce the subject by emphasizing intuition, and whet your appetite for what is possible in this fascinating field.

We will make sense of integrals such as

$$\int_0^t B_s \, ds \quad \text{and} \quad \int_0^t B_s \, dB_s.$$

In the first integral, Brownian motion is integrated over the interval $[0, t]$. Think of the integral as representing the area under the Brownian motion curve on $[0, t]$. The fact that the integrand is random means that the integral is random, hence a random variable. As a function of t , it is a *random function*, that is, a stochastic process.

If that is not strange enough, in the second integral Brownian motion appears in both the integrand and the *integrator*, where dB_s replaces the usual ds . Here, Brownian motion is integrated *with respect to* Brownian motion. To make sense of this will require new ideas, and even new rules, of calculus.

To start off, for $0 \leq a < b$, consider the integral

$$\int_a^b B_s ds.$$

For each outcome ω , $B_s(\omega)$ is a continuous function, and thus the integral

$$\int_a^b B_s(\omega) ds$$

is well defined in the usual sense as the limit of a Riemann sum. For a partition $a = t_0 < t_1 < \cdots < t_{n-1} < t_n = b$ of $[a, b]$, define the Riemann sum

$$I^{(n)}(\omega) = \sum_{k=1}^n B_{t_k^*}(\omega)(t_k - t_{k-1}),$$

where $t_k^* \in [t_{k-1}, t_k]$ is an arbitrary point in the subinterval $[t_{k-1}, t_k]$. The integral $\int_a^b B_s(\omega) ds$ is defined as the limit of the Riemann sum as n tends to infinity and the length of the longest subinterval of the partition converges to 0.

For each $n \geq 1$, the Riemann sum $I^{(n)}$ is a random variable, which is a linear combination of normal random variables. Since Brownian motion is a Gaussian process, $I^{(n)}$ is normally distributed. As this is true for all n , it is reasonable to expect that $\lim_{n \rightarrow \infty} I^{(n)}$ is normally distributed.

Let $I_t = \int_0^t B_s ds$, for $t \geq 0$. It can be shown that $(I_t)_{t \geq 0}$ is a Gaussian process with continuous sample paths. The mean function is

$$E(I_t) = E\left(\int_0^t B_s ds\right) = \int_0^t E(B_s) ds = 0,$$

where the interchange of expectation and integral can be justified.

For $s \leq t$, the covariance function is

$$\begin{aligned} \text{Cov}(I_s, I_t) &= E(I_s I_t) = E\left(\int_0^s B_x dx \int_0^t B_y dy\right) \\ &= \int_0^s \int_0^t E(B_x B_y) dy dx = \int_0^s \int_0^t \min\{x, y\} dy dx \\ &= \int_0^s \int_0^x y dy dx + \int_0^s \int_x^t x dy dx \\ &= \frac{s^3}{6} + \left(\frac{ts^2}{2} - \frac{s^3}{3}\right) = \frac{3ts^2 - s^3}{6}. \end{aligned}$$

Letting $s = t$ gives $\text{Var}(I_t) = t^3/3$.

Thus, the stochastic integral $\int_0^t B_s ds$ is a normally distributed random variable with mean 0 and variance $t^3/3$. The integral $\int_0^t B_s ds$ is called *integrated Brownian motion*. See Figure 9.1 for realizations when $t = 1$.

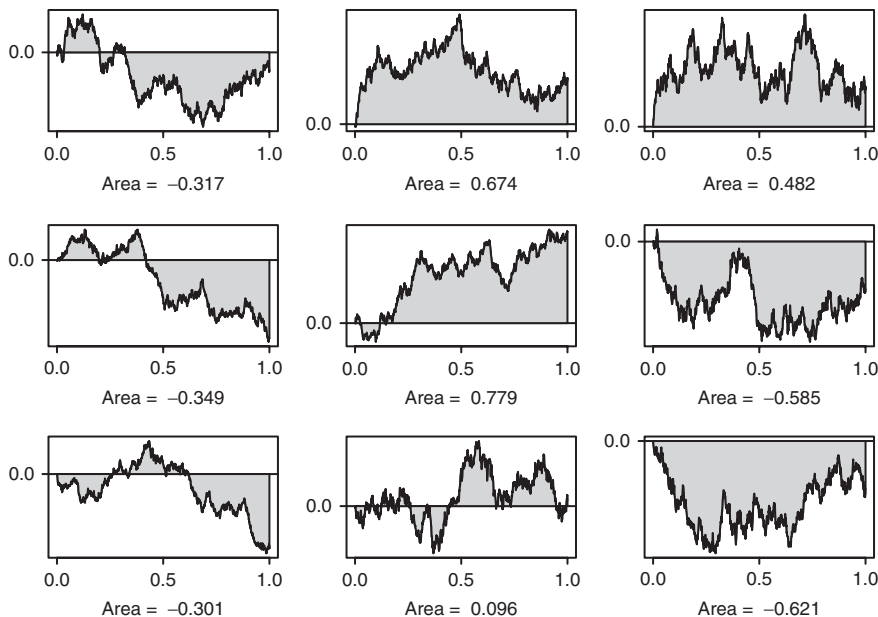


Figure 9.1 Realizations of the stochastic integral $\int_0^1 B_s ds$. The integral is normally distributed with mean 0 and variance $1/3$.

We next introduce the *Riemann–Stieltjes integral* of g with respect to f

$$\int_0^t g(x) df(x),$$

where f and g are continuous, and nonrandom, functions. The integral is defined as the limit, as n tends to infinity, of the approximating sum

$$\sum_{k=1}^n g(t_k^*) (f(t_k) - f(t_{k-1})),$$

where $0 = t_0 < t_1 < \cdots < t_{n-1} < t_n = t$ is a partition of $[0, t]$, and $t_k^* \in [t_{k-1}, t_k]$. The definition generalizes the usual Riemann integral by letting $f(x) = x$. The integral can be interpreted as a *weighted* summation, or weighted average, of g , with weights determined by f .

If f is differentiable, with continuous derivative, then

$$\int_0^t g(x) df(x) = \int_0^t g(x) f'(x) dx,$$

which gives the usual Riemann integral. In probability, if X is a continuous random variable with density function f and cumulative distribution function F , and g is a function, the expectation $E(g(X))$ can be written as

$$E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x) dx = \int_{-\infty}^{\infty} g(x)F'(x) dx = \int_{-\infty}^{\infty} g(x)dF(x),$$

that is, as a Riemann–Stieltjes integral of g with respect to the cumulative distribution function F .

Based on the Riemann–Stieltjes integral, we can define the integral of a function g with respect to Brownian motion

$$I_t = \int_0^t g(s) dB_s. \quad (9.1)$$

Technical conditions require that g be a bounded and continuous function, and satisfy $\int_0^\infty g^2(s) ds < \infty$. By analogy with the Riemann–Stieltjes integral, for the partition

$$0 = t_0 < t_1 < \cdots < t_{n-1} < t_n = t,$$

let

$$I_t^{(n)} = \sum_{k=1}^n g(t_k^*) (B_{t_k} - B_{t_{k-1}}),$$

where $t_k^* \in [t_{k-1}, t_k]$. Since $B_{t_k} - B_{t_{k-1}}$ is normally distributed with mean 0 and variance $t_k - t_{k-1}$, the approximating sum $I_t^{(n)}$ is normally distributed for all n . It can be shown that in the limit, as $n \rightarrow \infty$, the approximating sum converges to a normally distributed random variable, which we take to be the stochastic integral of Equation (9.1). Furthermore,

$$\begin{aligned} E(I_t) &= E\left(\lim_{n \rightarrow \infty} I_t^{(n)}\right) = \lim_{n \rightarrow \infty} E\left(\sum_{k=1}^n g(t_k^*) (B_{t_k} - B_{t_{k-1}})\right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n g(t_k^*) E(B_{t_k} - B_{t_{k-1}}) = 0. \end{aligned}$$

By independent increments,

$$\text{Var}(I_t^{(n)}) = \sum_{k=1}^n g^2(t_k^*) \text{Var}(B_{t_k} - B_{t_{k-1}}) = \sum_{k=1}^n g^2(t_k^*) (t_k - t_{k-1}).$$

The last expression is a Riemann sum whose limit, as n tends to infinity, is $\int_0^t g^2(s) ds$. In summary,

$$\int_0^t g(s) dB_s \sim \text{Normal} \left(0, \int_0^t g^2(s) ds \right). \quad (9.2)$$

In fact, it can be shown that $(I_t)_{t \geq 0}$ is a Gaussian process with continuous sample paths, independent increments, mean function 0, and covariance function

$$\text{Cov}(I_s, I_t) = E \left(\int_0^s g(x) dB_x \int_0^t g(y) dB_y \right) = \int_0^{\min\{s,t\}} g^2(x) dx.$$

■ **Example 9.1** Evaluate

$$\int_0^t dB_s.$$

Solution With $g(x) = 1$, the integral is normally distributed with mean 0 and variance $\int_0^t ds = t$. That is, the stochastic integral has the same distribution as B_t . Furthermore, the integral defines a continuous Gaussian process with mean 0 and covariance function

$$\int_0^{\min\{s,t\}} dx = \min\{s, t\}.$$

That is, $\left(\int_0^t dB_s\right)_{t \geq 0}$ is a standard Brownian motion, and

$$\int_0^t dB_s = B_t.$$

■

■ **Example 9.2** Evaluate

$$\int_0^t e^s dB_s.$$

Solution The integral is normally distributed with mean 0 and variance

$$\int_0^t (e^s)^2 ds = \int_0^t e^{2s} ds = \frac{1}{2} e^{2t}.$$

■

The stochastic integral

$$\int_a^b g(s) dB_s$$

has many familiar properties. Linearity holds. For functions g and h , for which the integral is defined, and constants α, β ,

$$\int_a^b [\alpha g(s) + \beta h(s)] dB_s = \alpha \int_a^b g(s) dB_s + \beta \int_a^b h(s) dB_s.$$

For $a < c < b$,

$$\int_a^b g(s) dB_s = \int_a^c g(s) dB_s + \int_c^b g(s) dB_s.$$

The integral also satisfies an *integration-by-parts* formula. If g is differentiable,

$$\int_0^t g(s) dB_s = g(t)B_t - \int_0^t B_s g'(s) ds.$$

By letting $g(t) = 1$, we capture the identity

$$\int_0^t dB_s = B_t. \quad (9.3)$$

■ **Example 9.3** Evaluate

$$\int_0^t s dB_s$$

in terms of integrated Brownian motion.

Solution Integration by parts gives

$$\int_0^t s dB_s = tB_t - \int_0^t B_s ds.$$

See Figure 9.2 for simulations of the process. ■

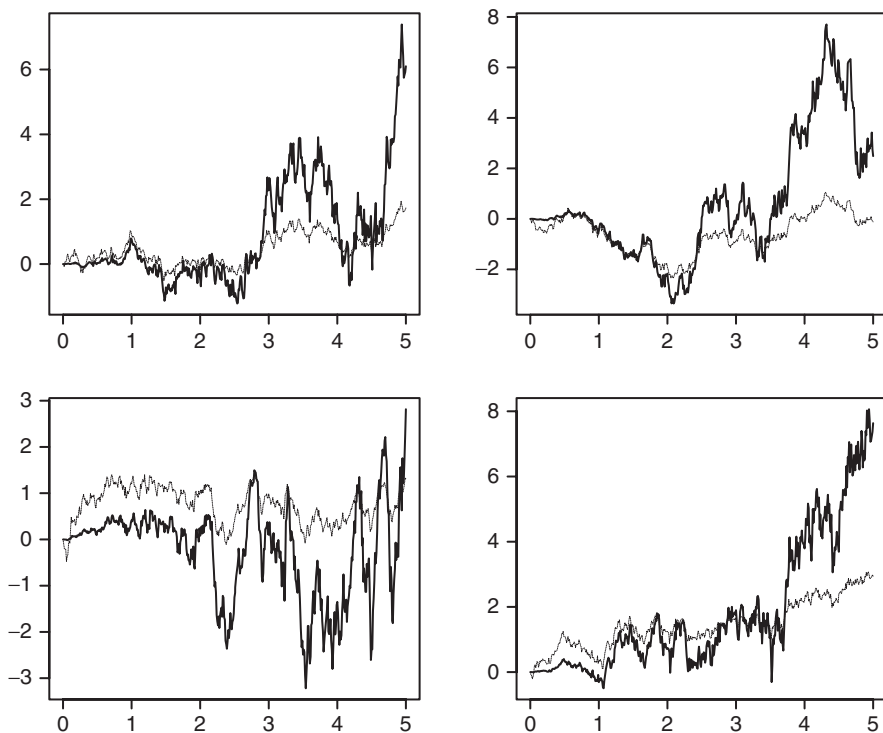


Figure 9.2 Simulations of $(\int_0^t s dB_s)_{0 \leq t \leq 5}$. The light gray curve is the underlying standard Brownian motion.

White Noise

If Brownian motion paths were differentiable, Equation (9.3) could be written as

$$B_t = \int_0^t dB_s = \int_0^t \frac{dB_s}{ds} ds.$$

The “process” $W_t = dB_t/dt$ is called *white noise*. The reason for the quotation marks is because W_t is not a stochastic process in the usual sense, as Brownian motion derivatives do not exist. Nevertheless, Brownian motion is sometimes described as *integrated white noise*.

Consider the following formal treatment of the distribution of white noise. Letting Δ_t represent a small incremental change in t ,

$$W_t \approx \frac{B_{t+\Delta_t} - B_t}{\Delta_t}.$$

The random variable W_t is approximately normally distributed with mean 0 and variance $1/\Delta_t$. We can think of W_t as the result of letting $\Delta_t \rightarrow 0$. White noise can be thought of as an *idealized* continuous-time Gaussian process, where W_t is normally distributed with mean 0 and variance $1/dt$. Furthermore, for $s \neq t$,

$$E(W_s W_t) = E\left(\frac{dB_s}{ds} \frac{dB_t}{dt}\right) = \frac{1}{ds dt} E((B_{s+ds} - B_s)(B_{t+dt} - B_t)) = 0,$$

by independent increments. Hence, W_s and W_t are independent, for all $s \neq t$.

It is hard to conceive of a real-world, time-indexed process in which all variables, no matter how close in time, are independent. Yet white noise is an extremely useful concept for real-world modeling, particularly in engineering, biology, physics, communication, and economics. In a physical context, white noise refers to sound that contains all frequencies in equal amounts, the analog of white light. See Figure 9.3. Applied to a time-varying signal g , the stochastic integral $\int_0^t g(s) dB_s$ can be interpreted as the output after the signal is transformed by white noise. For the case $g(s) = s$, see again Figure 9.2.

9.2 ITO INTEGRAL

We are now ready to consider a more general stochastic integral of the form

$$I_t = \int_0^t X_s dB_s,$$

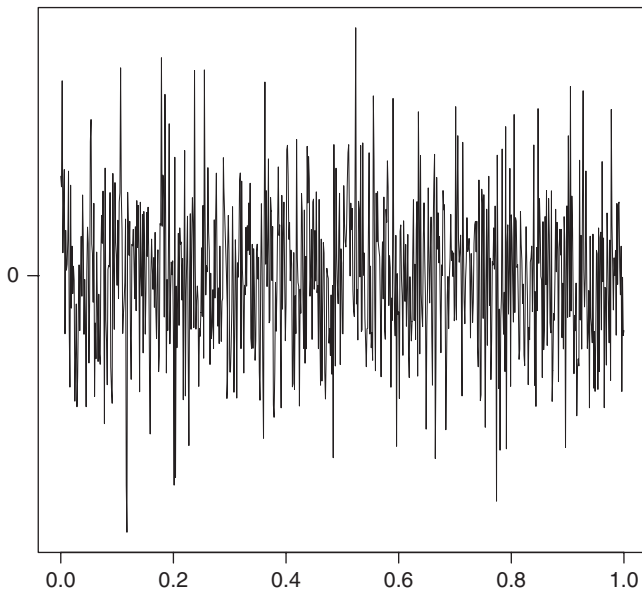


Figure 9.3 Simulation of white noise signal.

where $(X_t)_{t \geq 0}$ is a stochastic process, and $(B_t)_{t \geq 0}$ is standard Brownian motion. By analogy with what has come before, it would seem that a reasonable definition for the integral would be

$$\int_0^t X_s dB_s = \lim_{n \rightarrow \infty} \sum_{k=1}^n X_{t_k^*} (B_{t_k} - B_{t_{k-1}}), \quad (9.4)$$

for ever-finer partitions $0 = t_0 < t_1 < \cdots < t_{n-1} < t_n = t$, where $t_k^* \in [t_{k-1}, t_k]$. Unfortunately, the definition does not work. Unlike previous integrals, the choice of point t_k^* in the subinterval $[t_{k-1}, t_k]$ matters. The integral is not well-defined for arbitrary $t_k^* \in [t_{k-1}, t_k]$. Furthermore, the integral requires a precise definition of the meaning of the limit in Equation (9.4), as well as several regularity conditions for the process $(X_t)_{t \geq 0}$.

This brings us to the *Ito integral*, named after Kiyoshi Ito, a brilliant 20th century Japanese mathematician whose name is most closely associated with stochastic calculus. The Ito integral is based on taking each t_k^* to be the left endpoint¹ of the subinterval $[t_{k-1}, t_k]$. That is,

$$\int_0^t X_s dB_s = \lim_{n \rightarrow \infty} \sum_{k=1}^n X_{t_{k-1}} (B_{t_k} - B_{t_{k-1}}). \quad (9.5)$$

¹A different type of stochastic integral, called the *Stratonovich integral*, is obtained by choosing $t_k^* = (t_{k-1} + t_k)/2$ to be the midpoint of the subinterval $[t_{k-1}, t_k]$.

The Ito integral requires

1. $\int_0^t E(X_s^2) ds < \infty$.
2. X_t does not depend on the values $\{B_s : s > t\}$ and only on $\{B_s : s \leq t\}$. We say that X_t is *adapted* to Brownian motion.
3. The limit in Equation (9.5) is defined in the *mean-square* sense. A sequence of random variables $X_0, X_1, \dots$ is said to *converge to X in mean-square* if $\lim_{n \rightarrow \infty} E((X_n - X)^2) = 0$.

The Ito integral has many familiar properties, such as linearity. However, new rules of *stochastic calculus* will be needed for computations.

One of the most important properties of the Ito integral is that the process

$$\left(\int_0^t X_s dB_s \right)_{t \geq 0}$$

is a martingale with respect to Brownian motion.

The following properties of the Ito integral are summarized without proof.

Properties of the Ito Integral

The Ito integral

$$I_t = \int_0^t X_s dB_s$$

satisfies the following:

1. For processes $(X_t)_{t \geq 0}$ and $(Y_t)_{t \geq 0}$, and constants α, β ,

$$\int_0^t (\alpha X_s + \beta Y_s) dB_s = \alpha \int_0^t X_s dB_s + \beta \int_0^t Y_s dB_s.$$

2. For $0 < r < t$,

$$\int_0^t X_s dB_s = \int_0^r X_s dB_s + \int_r^t X_s dB_s.$$

- 3.

$$E(I_t) = 0.$$

- 4.

$$\text{Var}(I_t) = E \left(\left(\int_0^t X_s dB_s \right)^2 \right) = \int_0^t E(X_s^2) ds.$$

5. $(I_t)_{t \geq 0}$ is a martingale with respect to Brownian motion.

The Ito integral does not satisfy the usual integration-by-parts formula. Consider

$$\int_0^t B_s dB_s.$$

A formal application of integration by parts gives

$$\int_0^t B_s dB_s = B_t^2 - B_0^2 - \int_0^t B_s dB_s = B_t^2 - \int_0^t B_s dB_s,$$

which leads to $\int_0^t B_s dB_s = B_t^2/2$. However, this must be wrong as the Ito integral has mean 0, and thus $E\left(\int_0^t B_s dB_s\right) = 0$. However, $E(B_t^2/2) = t/2$.

To evaluate $\int_0^t B_s dB_s$, consider the approximating sum

$$\begin{aligned} & \sum_{k=1}^n B_{t_{k-1}} (B_{t_k} - B_{t_{k-1}}) \\ &= \sum_{k=1}^n \left(\frac{1}{2} (B_{t_k} + B_{t_{k-1}}) - \frac{1}{2} (B_{t_k} - B_{t_{k-1}}) \right) (B_{t_k} - B_{t_{k-1}}) \\ &= \frac{1}{2} \sum_{k=1}^n (B_{t_k}^2 - B_{t_{k-1}}^2) - \frac{1}{2} \sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2 \\ &= \frac{1}{2} B_t^2 - \frac{1}{2} \sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2. \end{aligned}$$

It can be shown that $\sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2$ converges to the constant t in mean-square, that is,

$$\lim_{n \rightarrow \infty} E \left(\left(\sum_{k=1}^n (B_{t_k} - B_{t_{k-1}})^2 - t \right)^2 \right) = 0.$$

This gives

$$\int_0^t B_s dB_s = \frac{1}{2} (B_t^2 - t),$$

which is a martingale. Recall that $B_t^2 - t$ is the quadratic martingale, shown in Example 8.19. Multiplying a martingale by a constant, does not change the martingale property.

ITO'S LEMMA

If one disqualifies the Pythagorean Theorem from contention, it is hard to think of a mathematical result which is better known and more widely applied in the world today

than “Ito’s Lemma.” This result holds the same position in stochastic analysis that Newton’s fundamental theorem holds in classical analysis. That is, it is the sine qua non of the subject.

–National Academy of Sciences

The most important result in stochastic calculus is Ito’s Lemma, which is the stochastic version of the chain rule. It has been called the fundamental theorem of stochastic calculus.

Ito’s Lemma

Let g be a real-valued function that is twice continuously differentiable. Then,

$$g(B_t) - g(B_0) = \int_0^t g'(B_s) dB_s + \frac{1}{2} \int_0^t g''(B_s) ds.$$

This is often written in shorthand differential form

$$dg(B_t) = g'(B_t)dB_t + \frac{1}{2}g''(B_t)dt.$$

■ **Example 9.4** Let $g(x) = x^2$. By Ito’s Lemma,

$$B_t^2 = \int_0^t 2B_s dB_s + \frac{1}{2} \int_0^t 2 ds = 2 \int_0^t B_s dB_s + t.$$

That is,

$$\int_0^t B_s dB_s = \frac{1}{2}B_t^2 - \frac{t}{2}.$$

■

■ **Example 9.5** Evaluate $d(\sin B_t)$.

Solution Let $g(x) = \sin x$. By Ito’s Lemma,

$$d(\sin B_t) = \cos B_t dB_t - \frac{1}{2} \sin B_t dt.$$

In integral form,

$$\sin B_t = \int_0^t \cos B_s dB_s - \frac{1}{2} \int_0^t \sin B_s ds.$$

■

■ **Example 9.6** Evaluate

$$\int_0^t B_s^2 dB_s \quad \text{and} \quad \int_0^t (B_s^2 - s) dB_s.$$

Solution

(i) Use Ito's Lemma with $g(x) = x^3$. This gives

$$B_t^3 = \int_0^t 3B_s^2 dB_s + \frac{1}{2} \int_0^t 6B_s ds.$$

Rearranging gives

$$\int_0^t B_s^2 dB_s = \frac{1}{3}B_t^3 - \int_0^t B_s ds.$$

(ii) By linearity of the Ito integral,

$$\begin{aligned} \int_0^t (B_s^2 - s) dB_s &= \int_0^t B_s^2 dB_s - \int_0^t s dB_s \\ &= \frac{1}{3}B_t^3 - \int_0^t B_s ds - \left(tB_t - \int_0^t B_s ds \right) \\ &= \frac{1}{3}B_t^3 - tB_t. \end{aligned}$$

The second equality is by integration by parts, which is valid for stochastic integrals with deterministic integrands.

Since Ito integrals are martingales, the process $\left(\frac{1}{3}B_t^3 - tB_t\right)_{t \geq 0}$ is a martingale. ■

Here is a heuristic derivation of Ito's Lemma. For a function g , consider its Taylor series expansion

$$g(t + dt) = g(t) + g'(t)dt + \frac{1}{2}g''(t)(dt)^2 + \dots.$$

Higher-order terms, starting with $(dt)^2$, are negligible. Hence,

$$dg(t) = g(t + dt) - g(t) = g'(t)dt.$$

For a given function h ,

$$g(h(t) + dh(t)) = g(h(t)) + g'(h(t))dh(t) + \frac{1}{2}g''(h(t))(dh(t))^2 + \dots.$$

Under suitable regularity conditions, the higher-order terms drop out, giving the usual chain rule $dg(h) = g'(h)dh$.

Replacing $h(t)$ with B_t , the Taylor series expansion is

$$g(B_t + dB_t) = g(B_t) + g'(B_t)dB_t + \frac{1}{2}g''(B_t)(dB_t)^2 + \frac{1}{6}g'''(B_t)(dB_t)^3 + \dots.$$

However, what is different for Brownian motion is that the term $(dB_t)^2$ is not negligible and cannot be eliminated. Intuitively, $dB_t = B_{t+dt} - B_t$ is a stochastic

element with the same distribution as B_{dt} , which is normally distributed with mean 0 and variance dt . Thus, B_{dt} takes values on the order of the standard deviation $\sqrt{dt}$. This gives $(dB_t)^2 \approx (\sqrt{dt})^2 = dt$. Thus, the $(dB_t)^2 = dt$ term is retained in the expansion.

Higher-order terms beyond the quadratic term are dropped from the expansion, as $(dB_t)^k \approx (\sqrt{dt})^k = (dt)^{k/2}$, which is negligible for $k > 2$. This leaves

$$dg(B_t) = g(B_t + dB_t) - g(B_t) = g'(B_t)dB_t + \frac{1}{2}g''(B_t)dt,$$

the differential form of Ito's Lemma.

Here are the heuristic stochastic calculus rules for working with stochastic differentials:

$$(dt)^2 = 0, \quad (dt)(dB_t) = 0, \quad (dB_t)^2 = dt.$$

An extended version of Ito's Lemma allows g to be a function of both t and B_t . The extended result can be motivated by considering a second-order Taylor series expansion of g .

Extension of Ito's Lemma

Let $g(t, x)$ be a real-valued function whose second-order partial derivatives are continuous. Then,

$$\begin{aligned} g(t, B_t) - g(0, B_0) &= \int_0^t \left(\frac{\partial}{\partial t} g(s, B_s) + \frac{1}{2} \frac{\partial^2}{\partial x^2} g(s, B_s) \right) ds \\ &\quad + \int_0^t \frac{\partial}{\partial x} g(s, B_s) dB_s. \end{aligned}$$

In shorthand differential form,

$$dg = \left(\frac{\partial g}{\partial t} + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} \right) dt + \frac{\partial g}{\partial x} dB_t.$$

We regret possible notational confusion in the statement of the lemma. It is common to use the letter t as the time variable, and thus t appears both as the upper limit of integration and as the dummy variable for the function g and its derivative. We trust the reader will safely navigate their way.

■ **Example 9.7** Evaluate $d(tB_t^2)$.

Solution Let $g(t, x) = tx^2$. Partial derivatives are

$$\frac{\partial g}{\partial t} = x^2, \quad \frac{\partial g}{\partial x} = 2tx, \quad \text{and} \quad \frac{\partial^2 g}{\partial x^2} = 2t.$$

By Ito's Lemma,

$$d(tB_t^2) = (B_t^2 + t) dt + 2tB_t dB_t.$$

Observe that the usual product rule would give the incorrect answer

$$d(tB_t^2) = B_t^2 dt + 2tB_t dB_t. \quad \blacksquare$$

Example 9.8 Use Ito's Lemma to evaluate $d(B_t^3)$ and $E(B_t^3)$.

Solution Let $g(t, x) = x^3$. By Ito's Lemma,

$$d(B_t^3) = 3B_t dt + 3B_t^2 dB_t,$$

and

$$B_t^3 = 3 \int_0^t B_s ds + 3 \int_0^t B_s^2 dB_s.$$

Taking expectations gives

$$E(B_t^3) = 3 \int_0^t E(B_s) ds + 3E\left(\int_0^t B_s^2 dB_s\right) = 3(0) + 0 = 0. \quad \blacksquare$$

9.3 STOCHASTIC DIFFERENTIAL EQUATIONS

To motivate the discussion, consider an exponential growth process, be it the spread of a disease, the population of a city, or the number of cells in an organism. Let X_t denotes the size of the population at time t . The deterministic exponential growth model is described by an ordinary differential equation

$$\frac{dX_t}{dt} = \alpha X_t, \text{ and } X_0 = x_0,$$

where x_0 is the initial size of the population, and α is the growth rate. The equation says that the population growth rate is proportional to the size of the population, where the constant of proportionality is α . A *solution* of the differential equation is a function X_t , which satisfies the equation. In this case, the solution is uniquely specified

$$X_t = x_0 e^{\alpha t}, \text{ for } t \geq 0.$$

The most common way to incorporate uncertainty into the model is to add a random error term, such as a multiple of white noise W_t , to the growth rate. This gives the *stochastic differential equation*

$$\frac{dX_t}{dt} = (\alpha + \beta W_t)X_t = \alpha X_t + \beta X_t \frac{dB_t}{dt},$$

or

$$dX_t = \alpha X_t dt + \beta X_t dB_t, \quad (9.6)$$

where α and β are parameters, and $X_0 = x_0$. Equation (9.6) is really a shorthand for the integral form

$$X_t - X_0 = \alpha \int_0^t X_s ds + \beta \int_0^t X_s dB_s. \quad (9.7)$$

A solution to the SDE is a stochastic process $(X_t)_{t \geq 0}$, which satisfies Equation (9.7).

For the stochastic exponential model, we show that geometric Brownian motion defined by

$$X_t = x_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)t + \beta B_t}, \text{ for } t \geq 0, \quad (9.8)$$

is a solution. Let

$$g(t, x) = x_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)t + \beta x}$$

with partial derivatives

$$\frac{\partial g}{\partial x} = \beta g, \quad \frac{\partial^2 g}{\partial x^2} = \beta^2 g, \quad \text{and} \quad \frac{\partial g}{\partial t} = \left(\alpha - \frac{\beta^2}{2}\right) g.$$

By the extended version of Ito's Lemma,

$$\begin{aligned} g(t, B_t) - g(0, B_0) &= x_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)t + \beta B_t} - x_0 \\ &= \left(\alpha - \frac{\beta^2}{2} + \frac{\beta^2}{2}\right) \int_0^t x_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)s + \beta B_s} ds \\ &\quad + \beta \int_0^t x_0 e^{\left(\alpha - \frac{\beta^2}{2}\right)s + \beta B_s} dB_s, \end{aligned}$$

which reduces to the solution

$$X_t - X_0 = \alpha \int_0^t X_s ds + \beta \int_0^t X_s dB_s.$$

Geometric Brownian motion can be thought of as the stochastic analog of the exponential growth function.

Differential equations are the meat and potatoes of applied mathematics. Stochastic differential equations are used in biology, climate science, engineering, economics, physics, ecology, chemistry, and public health.

 **Example 9.9 (Logistic equation)** Unfettered exponential growth is typically unrealistic for biological populations. The *logistic* model describes the growth of

a self-limiting population. The standard deterministic model is described by the ordinary differential equation

$$\frac{dP_t}{dt} = rP_t \left(1 - \frac{P_t}{K}\right),$$

where P_t denotes the population size at time t , r is the growth rate, and K is the *carrying capacity*, the maximum population size that the environment can sustain.

If $P_t \approx 0$, then $dP_t/dt \approx rP_t$, and the model exhibits near-exponential growth. On the contrary, if the population size is near carrying capacity and $P_t \approx K$, then $P_t/dt \approx 0$, and the population exhibits little growth.

The solution of the deterministic equation—obtained by separation of variables and partial fractions—is

$$P_t = \frac{KP_0}{P_0 + (K - P_0)e^{-rt}}, \text{ for } t \geq 0. \quad (9.9)$$

Observe that $P_t \rightarrow K$, as $t \rightarrow \infty$; that is, the population size tends to the carrying capacity.

A stochastic logistic equation is described by the SDE

$$dP_t = rP_t \left(1 - \frac{P_t}{K}\right) dt + \sigma P_t dB_t,$$

where $\sigma > 0$ is a parameter. Let $(X_t)_{t \geq 0}$ be the geometric Brownian motion process defined by

$$X_t = e^{\left(r - \frac{\sigma^2}{2}\right)t + \sigma B_t}.$$

It can be shown that the solution to the logistic SDE is

$$P_t = \frac{P_0 K X_t}{K + P_0 r \int_0^t X_s ds}.$$

When $\sigma = 0$, $X_t = e^{rt}$ and the solution reduces to Equation (9.9). See Figure 9.4 for sample paths of the stochastic logistic process. ■

■ **Example 9.10** Stochastic models are used in climatology to model long-term climate variability. These complex models are typically multidimensional and involve systems of SDEs. They are relevant for our understanding of global warming and climate change.

A simplified system that models the interaction between the atmosphere and the ocean's surface is described in Vallis (2010). Let T_t^A and T_t^S denote the atmosphere and sea surface temperatures, respectively, at time t . The system is

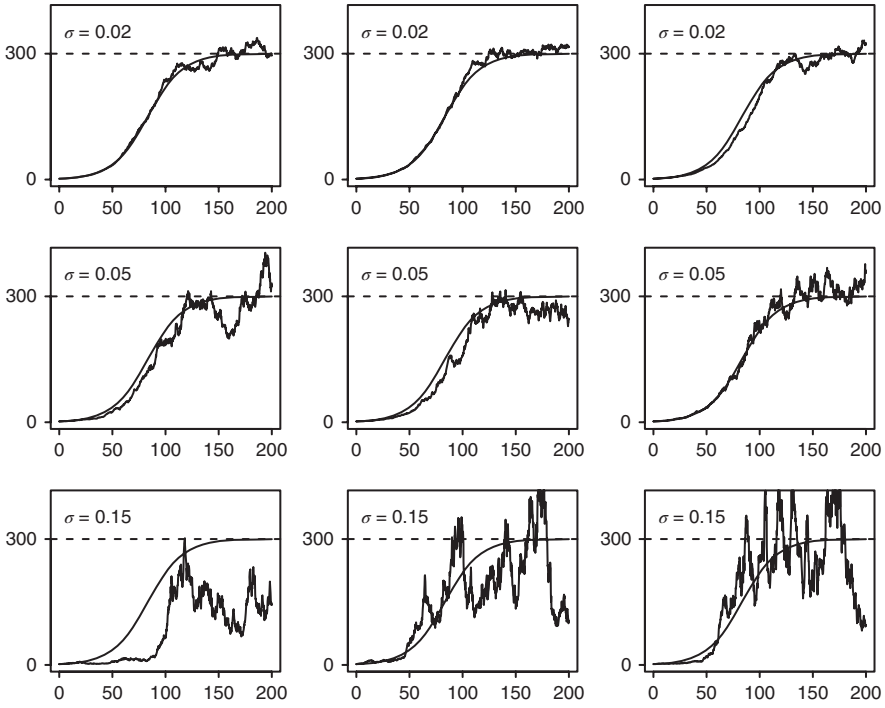


Figure 9.4 Sample paths for the logistic SDE, with $P_0 = 2$, $r = 0.06$, and $K = 300$. Smooth curve is the deterministic logistic function.

$$c_S \frac{dT^S}{dt} = aT^A - bT^S,$$

$$c_A \frac{dT^A}{dt} = cT^S - dT^A + \sigma B_t,$$

where c_A and c_S describe the heat capacity of the atmosphere and sea, respectively, and a, b, c, d , and σ are parameters. The model is based on Newton's law of cooling, by which the rate of heat loss of an object is proportional to the difference in temperature between the object and its surroundings. The Brownian motion term σB_t accounts for random fluctuations that affect the atmosphere.

Assuming the heat capacity of the ocean surface is much greater than that of the atmosphere, the model finds that rapid changes of atmospheric temperatures can affect long-term fluctuations in the ocean temperature, over possibly decades or centuries. The finding has significance in understanding how temperature changes at time scales greater than a year can occur in the earth's climate. ■

Ito's Lemma is an important tool for working with stochastic differential equations. The Lemma can be extended further to include a wide class of stochastic processes, which are solutions to SDEs of the form

$$dX_t = a(t, X_t)dt + b(t, X_t)dB_t, \quad (9.10)$$

where a and b are functions of t and X_t .

The integral form is

$$X_t - X_0 = \int_0^t a(s, X_s) ds + \int_0^t b(s, X_s) dB_s.$$

Such processes are called *diffusions* or *Ito processes*. A diffusion is a Markov process with continuous sample paths. The functions a and b are called, respectively, the *drift coefficient* and *diffusion coefficient*.

Standard Brownian motion is a diffusion with $a(t, x) = 0$ and $b(t, x) = 1$. The introductory example of this section shows that geometric Brownian motion is a diffusion with $a(t, x) = \alpha x$ and $b(t, x) = \beta x$, for parameters α, β .

Ito's Lemma for Diffusions

Let $g(t, x)$ be a real-valued function whose second-order partial derivatives are continuous. Let $(X_t)_{t \geq 0}$ be an Ito process as defined by Equation (9.10). Then

$$\begin{aligned} g(t, X_t) - g(0, X_0) &= \int_0^t \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x} \alpha(s, X_s) + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} \beta^2(s, X_s) \right) ds \\ &\quad + \int_0^t \left(\frac{\partial g}{\partial x} \beta(s, X_s) \right) dB_s. \end{aligned}$$

In shorthand differential form,

$$dg = \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial x} \alpha(t, X_t) + \frac{1}{2} \frac{\partial^2 g}{\partial x^2} \beta^2(t, X_t) \right) dt + \frac{\partial g}{\partial x} \beta(t, X_t) dB_t.$$

We showed in the introductory example that geometric Brownian motion is a solution to the SDE of Equation (9.6). However, we did not solve the equation directly. Rather, we offered a candidate process and then verified that it was in fact a solution.

In general, solving an SDE may be difficult. A closed-form solution might not exist, and numerical methods are often needed. However, for the stochastic exponential growth model, the SDE can be solved exactly with the help of Ito's Lemma for diffusions.

From Equation (9.6), divide through by X_t to obtain

$$\frac{dX_t}{X_t} = \alpha dt + \beta dB_t.$$

The left-hand side suggests the function dx/x , whose integral is $\ln x$. This suggests applying Ito's Lemma with $g(t, x) = \ln x$. Derivatives are

$$\frac{\partial g}{\partial t} = 0, \quad \frac{\partial g}{\partial x} = \frac{1}{x}, \quad \text{and} \quad \frac{\partial^2 g}{\partial x^2} = -\frac{1}{x^2}.$$

This gives

$$d \ln X_t = \left(\frac{1}{X_t} \alpha X_t - \frac{1}{2X_t^2} \beta^2 X_t^2 \right) dt + \frac{1}{X_t} \beta X_t dB_t = \left(\alpha - \frac{\beta^2}{2} \right) dt + \beta dB_t.$$

Integrating gives

$$\ln X_t - \ln x_0 = \left(\alpha - \frac{\beta^2}{2} \right) t + \beta B_t,$$

with solution

$$X_t = x_0 e^{\left(\alpha - \frac{\beta^2}{2} \right) t + \beta B_t}.$$

■ **Example 9.11 (Ornstein–Uhlenbeck process)** Mathematical Brownian motion is not necessarily the best model for physical Brownian motion. If B_t denotes the position of a particle, such as a pollen grain, at time t , then the particle's position is changing over time and it must have velocity. The velocity of the grain would be the derivative of the process, which does not exist for mathematical Brownian motion.

The *Ornstein–Uhlenbeck process*, called the Langevin equation in physics, arose as an attempt to model this velocity. In finance, it is known as the Vasicek model and has been used to model interest rates. The process is called *mean-reverting* as there is a tendency, over time, to reach an equilibrium position.

The SDE for the Ornstein–Uhlenbeck process is

$$dX_t = -r(X_t - \mu)dt + \sigma B_t,$$

where r, μ , and $\sigma > 0$ are parameters. The process is a diffusion with

$$a(t, x) = -r(x - \mu) \quad \text{and} \quad b(t, x) = \sigma.$$

If $\sigma = 0$, the equation reduces to an ordinary differential equation, which can be solved by separation of variables. From

$$\frac{dX_t}{X_t - \mu} = -r dt,$$

integrating gives

$$\ln(X_t - \mu) = -rt + C,$$

where $C = \ln(X_0 - \mu)$. This gives the deterministic solution

$$X_t = \mu + (X_0 - \mu)e^{-rt}.$$

Observe that $X_t \rightarrow \mu$, as $t \rightarrow \infty$.

The SDE can be solved using Ito's Lemma by letting $g(t, x) = e^{rt}x$, with partial derivatives

$$\frac{\partial g}{\partial t} = re^{rt}x, \quad \frac{\partial g}{\partial x} = e^{rt}, \quad \text{and} \quad \frac{\partial^2 g}{\partial x^2} = 0.$$

By Ito's Lemma,

$$\begin{aligned} d(e^{rt}X_t) &= (re^{rt}X_t - e^{rt}r(X_t - \mu))dt + e^{rt}\sigma dB_t \\ &= r\mu e^{rt}dt + e^{rt}\sigma dB_t. \end{aligned}$$

This gives

$$e^{rt}X_t - X_0 = r\mu \int_0^t e^{rs} ds + \sigma \int_0^t e^{rs} dB_s = \mu(e^{rt} - 1) + \sigma \int_0^t e^{rs} dB_s,$$

with solution

$$X_t = \mu + (X_0 - \mu)e^{-rt} + \sigma \int_0^t e^{-r(t-s)} dB_s.$$

See Figure 9.5 for realizations of the Ornstein–Uhlenbeck process.

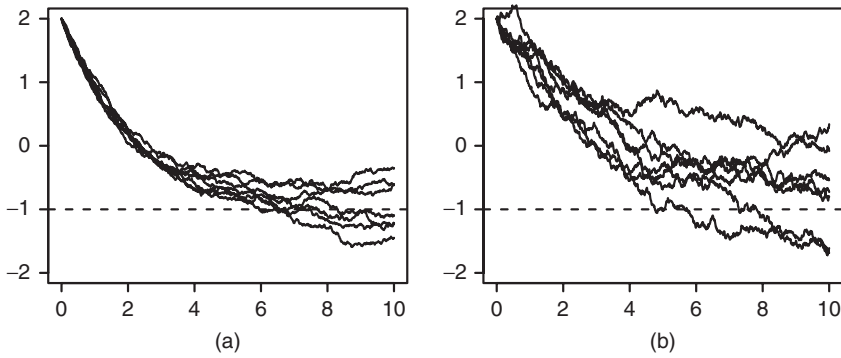


Figure 9.5 Realizations of the Ornstein–Uhlenbeck process with $X_0 = 2$ and $\mu = -1$.

(a) $r = 0.5$, $\sigma = 0.1$. (b) $r = 0.3$, $\sigma = 0.2$.

If X_0 is constant, then by Equation (9.2), X_t is normally distributed with

$$E(X_t) = \mu + (X_0 - \mu)e^{-rt}$$

and

$$\text{Var}(X_t) = \sigma^2 \int_0^t e^{-2r(t-s)} ds = \frac{\sigma^2}{2r} (1 - e^{-2rt}).$$

The limiting distribution, as $t \rightarrow \infty$, is normal with mean μ and variance $\sigma^2/2r$. ■

Numerical Approximation and the Euler–Maruyama Method

The differential form of a stochastic differential equation lends itself to an intuitive method for simulation. Given the SDE

$$dX_t = a(t, X_t)dt + b(t, X_t)dB_t,$$

the *Euler–Maruyama* method generates a discrete sequence $X_0, X_1, \dots, X_n$, which approximates the process X_t on an interval $[0, T]$. The method extends the popular Euler method for numerically solving deterministic differential equations.

Partition the interval $[0, T]$ into n equally spaced points

$$0 = t_0 < t_1 < \dots < t_{n-1} < t_n = T,$$

where $t_i = iT/n$, for $i = 0, 1, \dots, n$. The differential dt_i is approximated by $t_i - t_{i-1} = T/n$. The stochastic differential dB_{t_i} is approximated by $B_{t_i} - B_{t_{i-1}}$, which is normally distributed with mean 0 and variance $t_i - t_{i-1} = T/n$. Thus, dB_{t_i} can be approximated by $\sqrt{T/n}Z$, where Z is a standard normal random variable. Let

$$X_{i+1} = X_i + a(t_i, X_i)T/n + b(t_i, X_i)\sqrt{T/n}Z_i, \text{ for } i = 0, 1, \dots, n-1,$$

where $Z_0, Z_1, \dots, Z_{n-1}$ are independent standard normal random variables. The sequence $X_0, X_1, \dots, X_n$ is defined recursively and gives a discretized approximate sample path for $(X_t)_{0 \leq t \leq T}$.

■ **Example 9.12 (Ornstein–Uhlenbeck process)** To simulate the solution of the Ornstein–Uhlenbeck SDE

$$dX_t = -r(X_t - \mu)dt + \sigma dB_t, \text{ for } 0 \leq t \leq T,$$

let

$$X_{i+1} = X_i - r(X_i - \mu)T/n + \sigma\sqrt{T/n}Z_i, \text{ for } i = 0, 1, \dots, n-1.$$

With $n = 1000$, we generate the process with $X_0 = 2$, $\mu = -1$, $r = 0.5$, and $\sigma = 0.1$. Realizations are shown in Figure 9.5(a). ■

R : Ornstein–Uhlenbeck Simulation

```
# ornstein.R
> mu <- -1
> r <- 0.5
> sigma <- 0.1
> T <- 10
> n <- 1000
> xpath <- numeric(n+1)
> xpath[1] <- 2 # initial value
> for (i in 1:n) {
+   xpath[i+1] <- xpath[i] - r*(xpath[i] - mu)*T/n
+   + sigma*sqrt(T/n)*rnorm(1) }
> plot(seq(0,T,T/n), xpath, type="l")
```

To simulate the random variable X_T , for fixed T , it is not necessary to store past outcomes X_t , for $t < T$. To generate one outcome of X_T the code simplifies.

```
> x <- 2 # initial value
> for (i in 1:n) {
+ x <- x - r*(x-mu)*T/n + sigma*sqrt(T/n)*rnorm(1) }
> x
[1] -0.9404498
```

Here, we simulate the mean of X_{10} based on 10,000 trials.

```
> trials <- 10000
> simlist <- numeric(trials)
> for (k in 1: trials) {
> x <- 2
> for (i in 1:n) {
+ x <- x - r*(x-mu)*T/n + sigma*sqrt(T/n)*rnorm(1) }
> simlist[k] <- x }
> mean(simlist)
[1] -0.9978892
```

From Example 9.11, the exact mean is

$$E(X_{10}) = \mu + (X_0 - \mu)e^{-r(10)} = -1 + 3e^{-5} = -0.9798.$$

■ Example 9.13 (Random genetic drift) The SDE

$$dX_t = \sqrt{X_t(1 - X_t)} dB_t, \text{ for } 0 \leq t \leq 1$$

arises as a model for *random genetic drift*. It is a continuous version of the discrete-time Wright–Fisher Markov chain introduced in Example 2.6. The latter is a model for the evolution of a population of N genes consisting of two alleles A and a . In the discrete-time process, the number of A alleles is obtained by drawing from replacement from the gene population. Given i A alleles at time n , the number of A alleles at time $n + 1$ has a binomial distribution with parameters $2N$ and $p = i/2N$. The Markov chain is absorbing with absorbing states 0 and $2N$.

The discrete-time process extends to a continuous-time diffusion $(X_t)_{t \geq 0}$ by a suitable scaling of time and space, where X_t denotes the *proportion* of A alleles in the gene population at time t . The diffusion process is absorbing with absorbing states 0 and 1.

Solving the SDE exactly is beyond the scope of this book. However, numerical methods are used (i) to approximate the sample paths of the process on the time

interval $[0, 1.5]$ and (ii) to simulate the probability density function of X_t , for $t = 0.1, 0.2, 0.4, 1$. See Figures 9.6 and 9.7.²

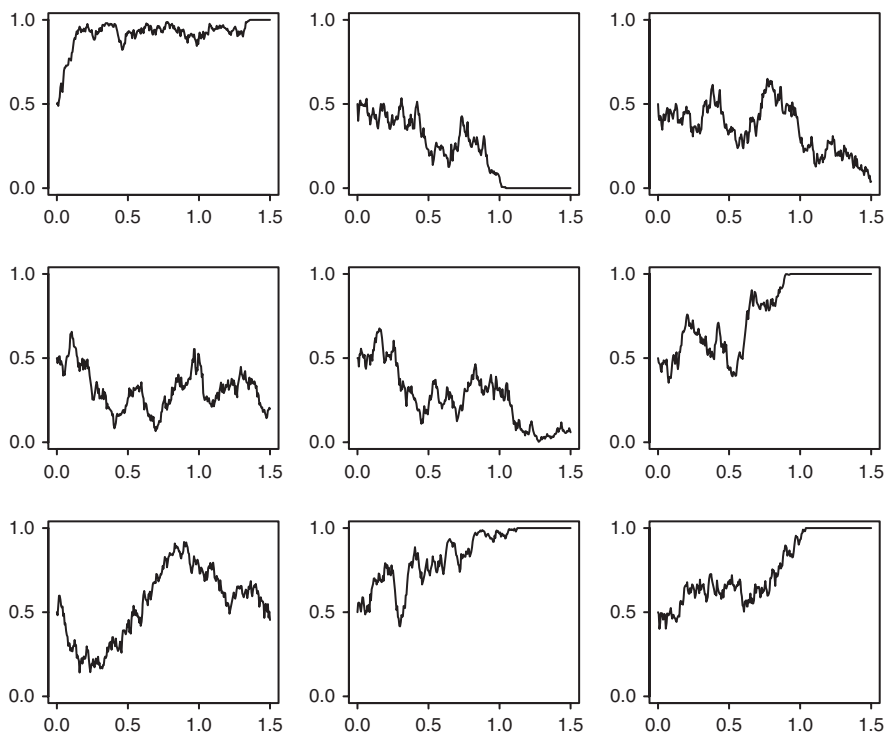


Figure 9.6 Sample paths for the solution of the random genetic drift SDE.

R: Euler–Maruyama Method for Simulating SDE

The following code generates the histograms in Figure 9.7.

```
# drift.R
> par(mfrow=c(2,2))
> times <- c( 0.1, 0.2,0.35,1)
> n = 100 # number of subintervals
> trials <- 10000
> for (k in 1:4) {
> t = times[k]
> simlist <- numeric(trials)
```

²In the Euler–Maruyama R code, to insure that the argument to the square foot function is non-negative, the absolute value of $x(1 - x)$ is taken. This gives an equivalent model to the original SDE.

```

> for (j in 1:trials) {
>   x <- 1/2 # initial state
>   for (i in 2:n) {
>     x <- x + sqrt(abs(x*(1-x)))*sqrt(t/n)*rnorm(1) }
>   simlist[j] <- x }
> hist(simlist,freq=F,main="", col="gray") }

```

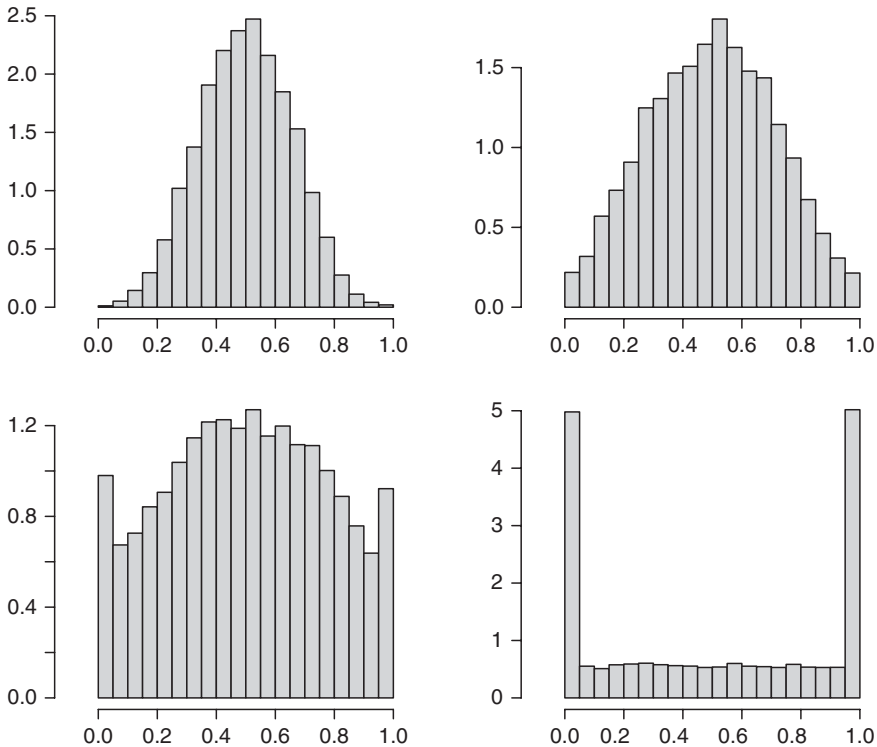


Figure 9.7 Simulating the distribution of X_t in the random genetic drift model, for $t = 0.1, 0.2, 0.35, 1.0$ (top-left to bottom-right).

Example 9.14 (Stochastic resonance) Stochastic resonance is a remarkable phenomenon whereby a signal, which is too weak to be detected, can be boosted by adding noise to the system. The idea is counter-intuitive, since we typically expect that noise (e.g., random error) makes signal detection more difficult. Yet the theory has found numerous applications over the past 25 years in biology, physics, and engineering, and has been demonstrated experimentally in the operation of ring lasers and in the neurons of crayfish.

The phenomenon was first introduced by Roberto Benzi in 1980 in the context of climate research, where it was proposed as a mechanism to explain how dramatic

climatic events such as the almost periodic occurrence of the ice ages might be caused by minute changes in the earth's orbit around the sun. The theory has prompted discussions of whether rapid climate change is a hallmark of human impact (e.g., noise in the system).

As explained in Benzi (2010), stochastic resonance can be observed by considering the SDE

$$dX_t = (X_t - X_t^3 + A \sin t)dt + \sigma dB_t.$$

Think of the sinusoidal term, called a *periodic forcing*, as representing a weak, external signal, with amplitude A . We are interested in studying the effect of the noise parameter σ on detection of the forcing signal.

The process is simulated using the Euler–Maruyama method.

R : Stochastic Resonance

```
# stochresonance.R
> T <- 100
> n <- 10000
> A <- 0.3
> sigma <- 0.2
> w <- 2*pi/40
> xpath <- numeric(n+1)
> xpath[1] <- 0
> for (i in 2:(n+1)) {
+   x[i] <- x[i-1] + (x[i-1] - x[i-1]^3 + A*sin(w*T*(i-1)/n))
+     *T/n + sigma*sqrt(T/n)*rnorm(1)
> plot(seq(0,T,T/n),x,type="l",ylim=c(-2.8,2.8),
+       xaxt="n",xlab="",ylab="",yaxt="n",lwd=0.5)
> axis(2,c(-1,0,1))
> axis(1,c(0,25,50,75,100))
> curve(A*sin(w*x),0,100,lty=2,add=TRUE)
```

The process $(X_t)_{t \geq 0}$ has two *stable* points, at ± 1 . For *small* σ (little noise), paths tend to stay near one of these values, although jumps may occur from one stable point to another. Three sample paths are shown in Figure 9.8 for $\sigma = 0.2$. The periodic forcing function (dashed curve) is not detectable. For this example, the amplitude of the sine function is $A = 0.3$, which is significantly smaller than the distance between the two stable points.

The effect of a relatively *large* random error term, with $\sigma = 2.0$, is apparent in Figure 9.9. The noise swamps any underlying structure. Again, the periodic forcing function is not detectable.

For Figure 9.10, an *optimal* value of σ is chosen at $\sigma = 0.8$. The *hidden* periodic forcing is now apparent. The added noise is sufficient for paths of the process to intersect with the range of the sine wave, which facilitates switching states. The system exhibits stochastic resonance.

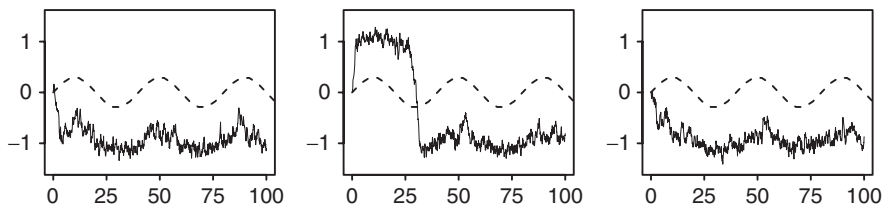


Figure 9.8

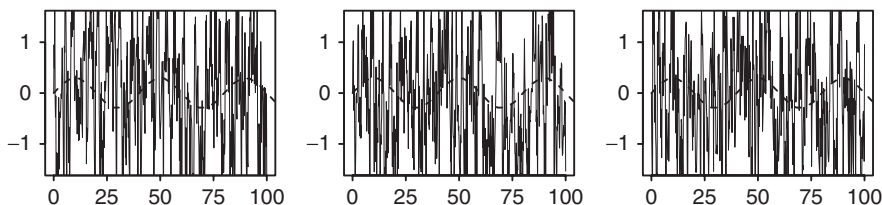


Figure 9.9

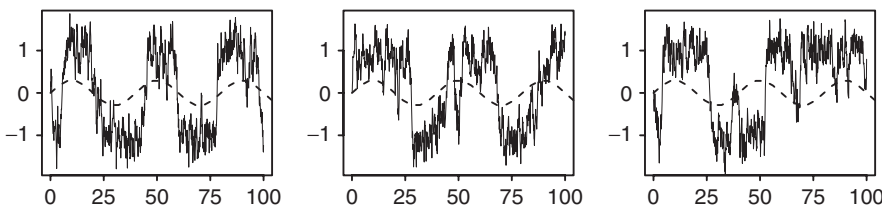


Figure 9.10

■

EXERCISES

- 9.1 Find the distribution of the stochastic integral $I_t = \int_0^t s B_s \, ds$.
- 9.2 Show that Brownian motion with drift coefficient μ and variance parameter σ^2 is a diffusion.
- 9.3 Find $E(B_t^4)$ by using Ito's Lemma to evaluate $d(B_t^4)$.

9.4 Use Ito's Lemma to show that

$$E(B_t^k) = \frac{k(k-1)}{2} \int_0^t E(B_s^{k-2}) ds, \text{ for } k \geq 2.$$

Use this result to find $E(B_t^k)$, for $k = 1, \dots, 8$.

9.5 Use the methods of Example 9.6 to derive a martingale that is a fourth-degree polynomial function of Brownian motion.

9.6 Consider the stochastic differential equation

$$dX_t = (1 - 2X_t)dt + 3 dB_t.$$

- (a) Use Ito's Lemma to find $d(e^{rt}X_t)$.
- (b) For suitable choice of r , simplify the drift coefficient in the resulting SDE. Solve the SDE and find the mean of X_t , and the asymptotic mean of the process.

9.7 Consider the SDE for the *square root process*

$$dX_t = dt + 2\sqrt{X_t} dB_t.$$

With $X_0 = x_0$, show that $X_t = (B_t + x_0)^2$ is a solution.

9.8 R : Show how to use the Euler–Maruyama method to simulate geometric Brownian motion started at $G_0 = 8$, with $\mu = 1$ and $\sigma^2 = 0.25$.

- (a) Generate a plot of a sample path on $[0, 2]$.
- (b) Simulate the mean and variance of G_2 . Compare with the theoretical mean and variance.

9.9 R Use the Euler–Maruyama method to simulate the square root process of Exercise 9.7 with $x_0 = 1$.

- (a) Estimate $E(X_3)$, $\text{Var}(X_3)$, and $P(X_3 < 5)$.
- (b) Using the fact that $X_t = (B_t + x_0)^2$ is a solution to the SDE, compare your simulations in (a) with the exact results.

9.10 R: The random drift model of Example 9.13 is an absorbing process with two absorbing states. Use the Euler–Maruyama method to estimate the expectation and standard deviation of the time until absorption.

9.11 R : The *Cox–Ingersoll–Ross (CIR) model*

$$dX_t = -r(X_t - \mu)dt + \sigma\sqrt{X_t}dB_t$$

has been used to describe the evolution of interest rates. The diffusion has the same drift coefficient as the Ornstein–Uhlenbeck process and is also mean-reverting. The CIR model has the advantage over the Ornstein–Uhlenbeck

process as a model for interest rates since, unlike the latter, the process is non-negative. However, unlike that process, the CIR model has no closed-form solution.

With $X_0 = 0$, $\mu = 1.25$, $r = 2$, and $\sigma = 0.2$, simulate the CIR process. Estimate the asymptotic mean and variance by taking $t = 100$. Demonstrate that the process is mean-reverting.

APPENDIX A

GETTING STARTED WITH R

There are many excellent R primers and tutorials on the web. To learn more about R, the best starting place is the homepage of the R Project for Statistical Computing at <http://www.r-project.org/>. Go there to download the software. The site contains links to books, manuals, demonstrations, and other resources. For this introduction, we assume that you have R up and running on your computer.

1. R as a calculator

When you bring up R the first window you see is the R console. You can type directly on the console, using R like a calculator.

```
> 1+1
[1] 2
> 2-2
[1] 0
> 3*3
[1] 9
> 4/4
[1] 1
> 5^5
[1] 3125
```

R has many built-in math functions. Their names are usually intuitive. R is case sensitive. All commands are in lowercase letters. The # symbol is for comments. Anything appearing after # is ignored by R.

```
> pi
[1] 3.141593
> cos(pi)
[1] -1
> exp(1)    #  $e^1$ 
[1] 2.718282
> abs(-6)   #  $|-6|$ 
[1] 6
> factorial(4)
[1] 24
> sqrt(9)
[1] 3
```

Exercises:

- 1.1. How many orderings are there of a standard 52-card deck?
- 1.2. An isosceles right triangle has two legs of length 5. Find the length of the hypotenuse.
- 1.3. Find the tangent of a 60-degree angle.
- 1.4. The *hyperbolic sine* is the function

$$h(x) = \frac{e^x - e^{-x}}{2}.$$

Evaluate the hyperbolic sine at $x = 1$. The R command for hyperbolic sine is `sinh(x)`. Verify that `sinh(1)` gives the same results.

2. Navigating the keyboard

Each command you type in R is stored in history. The up arrow (↑) key scrolls back along this history; the down arrow (↓) scrolls forward. These keystrokes are enormously helpful for navigating your R session.

Suppose you want to calculate $\log 8$ and you type

```
> Log(8)
Error: could not find function "Log"
```

You got an error because R commands start with lower case letters. Rather than retype the command, hit the up arrow key and fix the previously entered entry.

```
> log(8)
[1] 2.079442
```

You can use the <ESC> (escape) key to interrupt a running calculation or output. If R is expecting more input it will preface a line at the console with the + symbol. You can use escape to start anew. In the following, we forget the end parentheses. Repeatedly hitting the return key generated the + symbol as R is expecting the right parentheses to finish the command. By finally typing <ESC>, the command is voided.

```
> exp(1
+
+
+
+
+
>
```

3. Asking for help

Prefacing any R command with ? will produce a help file in another window.

```
> ?log
```

Reading the help file for the log function you learn that the function has the form `log(x, base = exp(1))`. The base argument defaults to `exp(1) = e`. That is, `log(x)` returns the natural logarithm of x . For logarithms in other bases give a second argument to `log`.

```
> log(8, 2)
[1] 3
> log(100, 10)
[1] 2
```

Exercises:

- 3.1. Read the help file for the `factorial` command to find the R command for computing the binomial coefficient $\binom{n}{k}$, also called “ n choose k .” Now compute the number of 5-card Poker hands in a standard 52-card deck $\binom{52}{5}$.
- 3.2. Use the up arrow key, and replace the 5 in the previous exercise with 13 to find the number of 13-card bridge hands in a standard 52-card deck.

4. Vectors

The real power of R lies in its ability to work with lists of numbers, called vectors. If a and b are integers, the command `a:b` creates a vector of integers from a to b .

```
> 1:10
[1] 1 2 3 4 5 6 7 8 9 10
```

```
> -3:5
[1] -3 -2 -1  0  1  2  3  4  5
```

For more control generating vectors, use the sequence command `seq`.

```
> seq(1,10)
[1]  1  2  3  4  5  6  7  8  9 10
> seq(1,20,2)
[1]  1  3  5  7  9 11 13 15 17 19
> seq(20,1,-4)
[1] 20 16 12  8  4
```

Create and manipulate vectors using the `c` concatenate command.

```
> c(2,3,5,7,11)
[1]  2  3  5  7 11
```

Assign a vector to a variable with the assignment operator `<-`. Here, the first five prime numbers are assigned to a variable called `primes`. The `primes` vector is then lengthened by concatenating three more primes to the original vector.

```
> primes <- c(2,3,5,7,11)
> primes
[1]  2  3  5  7 11
> primes <- c(primes,13,17,19)
> primes
[1]  2  3  5  7 11 13 17 19
```

Elements of vectors are indexed with brackets `[ ]`. Bracket arguments can be single integers or vectors.

```
> primes[4]
[1] 7
> primes[1:4]
[1] 2 3 5 7
> primes[c(1,4,5,10)]
[1]  2  7 11 NA
```

For the last command we asked for the first, fourth, fifth, and tenth element of `primes`. There is no tenth element so R returns an `NA`.

One often wants to find the elements of a vector that satisfy some property, such as those `primes` that are less than 10. This is done with the `which` command, which returns the indices of the vector that satisfy the given property.

```
> primes
[1] 2 3 5 7 11 13 17 19
> which(primes < 10)
[1] 1 2 3 4
> index <- which(primes < 10)
> primes[index]
[1] 2 3 5 7
```

Vectors can consist of numbers, characters, and even strings of characters.

```
> y <- c("Probability", "is", "very", "very",
        "cool")
> y
[1] "Probability" "is" "very" "very" "cool"
> y[c(1,2,5)]
[1] "Probability" "is" "cool"
```

When performing mathematical operations on vectors, the entire vector is treated as a single object.

```
> dog <- seq(0,30,4)
> dog
[1] 0 4 8 12 16 20 24 28
> dog+1
[1] 1 5 9 13 17 21 25 29
> dog*3
[1] 0 12 24 36 48 60 72 84
> 1/dog
[1] Inf 0.25000 0.12500 0.08333 0.06250 0.05000
[7] 0.041667 0.035714
> cat <- dog+1
> cat
[1] 1 5 9 13 17 21 25 29
> dog*cat
[1] 0 20 72 156 272 420 600 812
```

Notice that when a single number is added to a vector, that number gets added to each element of the vector. But when two vectors of the same length are added together, then corresponding elements are added. This applies to most binary operations.

Many mathematical functions can take vector arguments, returning vectors as output.

```
> factorial(1:7)
[1] 1 2 6 24 120 720 5040
```

```
> sqrt(seq(0,900,100)) #  $\sqrt{0}$ ,  $\sqrt{100}$ ,  $\sqrt{200}$ , ...,  $\sqrt{900}$ 
[1] 0.000 10.000 14.142 17.321 20.000 22.361
[7] 24.495 26.458 28.284 30.000
```

Here are some common, and intuitive, commands for working with vectors.

```
> x <- c(67.6, 68.7, 66.3, 66.2, 65.5, 70.2, 71.1)
> sum(x)
[1] 475.6
> mean(x)
[1] 67.943
> length(x)
[1] 7
> sort(x)
[1] 65.5 66.2 66.3 67.6 68.7 70.2 71.1
> sort(x,decreasing=T)
[1] 71.1 70.2 68.7 67.6 66.3 66.2 65.5
```

Exercises: Use vector operations:

- 4.1. Compute the squares of the first ten integers.
- 4.2. Compute the powers of 2 from 2^1 to 2^{20} .
- 4.3. For $n = 6$, use the `choose` command to compute the binomial coefficients $\binom{n}{k}$, for $k = 0, \dots, n$.
- 4.4. Let x be an n -element vector defined by $x_k = 2\pi k/n$, for $k = 1, \dots, n$. Find $(\cos x_1, \dots, \cos x_n)$ for $n = 5$. Repeat for $n = 13$, using the up arrow key rather than retyping the full command.
- 4.5. Compute the sum of the cubes of the first 100 integers.
- 4.6. Among the cubes of the first 100 integers, how many are greater than 10,000? Use `which` and `length`.

5. Generating random numbers

The `sample` command generates a random sample of a given size from the elements of a vector. The syntax is `sample(vec, size, replace=, prob=)`. Samples can be taken with or without replacement (the default is without). A probability distribution on the elements of `vec` can be specified. The default is that all elements of `vec` are equally likely.

```
> sample(1:10,1)
[1] 8
> sample(1:4,4)
[1] 3 1 4 2
```

```
> sample(c(-8,0,1,4,60),6,replace=T)
[1] 60 -8 1 1 4 60
```

To simulate ten rolls of a six-sided die, type

```
> sample(1:6,10,replace=T)
[1] 6 3 6 1 3 6 5 3 2 1
```

According to the Red Cross, the distribution of blood types in the United States is O: 44%, A: 42%, B: 10%, and AB: 4%. The following simulates a random sample of 10 people’s blood types.

```
> sample(c("O", "A", "B", "AB"), 8, replace=T,
        prob=c(0.44, 0.42, 0.10, 0.04))
[1] "A" "O" "O" "O" "A" "A" "A" "AB"
```

In a random sample of 50,000 people, how many have blood type B? Use the which command and the logical connective ==.

```
> samp <- sample(c("O", "A", "B", "AB"), 50000,
               replace=T, prob=c(0.44, 0.42, 0.10, 0.04))
> index <- which(samp=="B")
> length(index)
[1] 5042
```

Other logical connectives are given in Table A.1.

TABLE A.1 Logical Connectives			
==	equal to	&	and
!=	not equal to		or
>	greater than	!	not
>=	greater than or equal to		
<	less than		
<=	less than or equal to		

In the sample of 50,000 people, how many are either type B or type AB?

```
> length(which(samp=="AB" | samp=="B"))
[1] 7085
```

What proportion of people in the sample are not type O?

```
> length(which(samp != "O"))/50000
[1] 0.56388
```


Exercises:

- 5.1. Consider a probability distribution on $\{0, 2, 5, 9\}$ with respective probabilities 0.1, 0.2, 0.3, 0.4. Generate a random sample of size five.
- 5.2. For the above-mentioned distribution, generate a random sample of size one million and determine the proportion of sample values which are equal to 9.
- 5.3. Represent the cards of a standard 52-card deck with the numbers 1 to 52. Show how to generate a random five-card Poker hand from a standard 52-card deck. Let the numbers 1, 2, 3, and 4 represent the four aces in the deck. Write a command that will count the number of aces in a random Poker hand. Use `which` and `length`.
- 5.4. Generate 100,000 integers uniformly distributed between 1 and 20 and count the proportion of samples between 3 and 7.

6. Probability distributions

There are four commands for working with probability distributions in R. The commands take the root name of the probability distribution (see Table A.2) and prefix the root with `d`, `p`, `q`, or `r`. These give, respectively, continuous density or discrete probability mass function (`d`), cumulative distribution function (`p`), quantile (`q`), and random variable (`r`).

TABLE A.2 Probability Distributions in R

Distribution	Root	Distribution	Root
beta	beta	log-normal	lnorm
binomial	binom	multinomial	multinom
Cauchy	cauchy	negative binomial	nbinom
chi-squared	chisq	normal	norm
exponential	exp	Poisson	pois
F	f	Student's t	t
gamma	gamma	uniform	unif
geometric	geom	Weibull	weibull
hypergeometric	hyper		

Generate six random numbers uniformly distributed on $(0, 1)$.

```
> runif(6, 0, 1)
[1] 0.06300 0.44851 0.70231 0.20649 0.14377 0.74398
```

Generate 9 normally distributed variables with mean $\mu = 69$ and standard deviation $\sigma = 2.5$.

```
> rnorm(9, 69, 2.5)
[1] 69.548 69.931 68.923 71.153 71.779 68.975
[7] 67.429 65.621 70.148
```

Find $P(X \leq 2.5)$, where X has an exponential distribution with parameter $\lambda = 1$.

```
> pexp(2.5, 1)
[1] 0.91792
```

Suppose SAT scores are normally distributed with mean $\mu = 500$ and standard deviation $\sigma = 100$. Find the 95th quantile of the distribution of SAT scores.

```
> qnorm(0.95, 500, 100)
[1] 664.49
```

Find $P(X = 10)$, where X has a binomial distribution with parameters $n = 20$ and $p = 0.5$.

```
> dbinom(10, 20, 0.5)
[1] 0.1762
```

7. Plots and graphs

To graph functions and plot data, use `plot`. This workhorse command has enormous versatility. Here is a simple plot comparing two vectors.

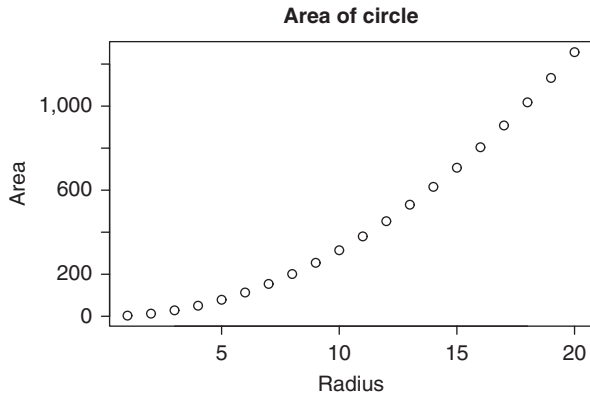
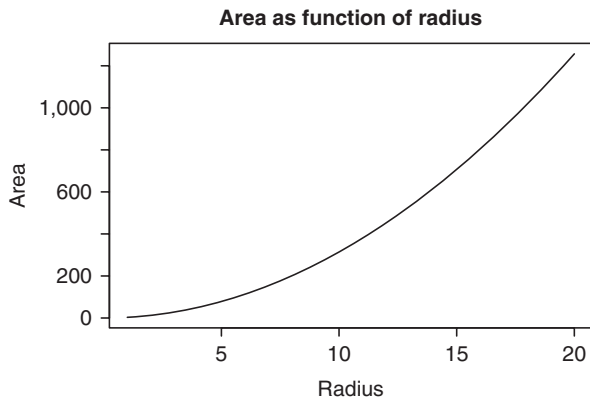
```
> radius <- 1:20
> area <- pi*radius^2
> plot(radius, area, main="Area as
function of radius")
```

To graph smooth functions, you can also use the `curve` command. See Figures A.1 and A.2. The syntax is `curve(expr, from=, to=)`, where `expr` is an expression that is a function of x .

```
> curve(pi*x^2, 1, 20, xlab="radius", ylab="area",
+ main="Area as function of radius")
```

For displaying data, a histogram is often used, obtained with the command `hist(vec)`. The default is a frequency histogram of counts. A density histogram has bars whose areas sum to one and is obtained using the `hist` command with parameter `freq=F`. A continuous density curve can be overlaid on the histogram by including the parameter `add=T`.

For example, male adult heights in the United States are normally distributed with mean 69 inches and standard deviation 2 inches. Here, we simulate 1,000

**Figure A.1****Figure A.2**

observations from such a distribution and plot the resulting data. The histogram is first graphed with counts, then with relative frequencies, and then overlaid with a normal density curve. See Figure A.3.

```
> heights <- rnorm(1000,69,2.0)
> hist(heights,main="Distribution of heights")
> hist(heights,main="Distribution of heights",
  freq=F)
> hist(heights,main="Distribution of heights",
  freq=F)
> curve(dnorm(x,69,2),60,80,add=T)
```

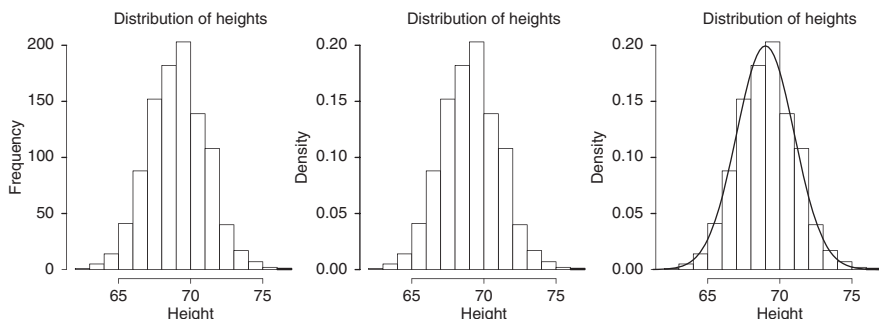


Figure A.3

8. Script files

When you have many R commands to manage, it is useful to keep your work in a script file. A script is a plain text file that contains R commands. The file can be saved and edited and you can execute the entire file or portions of it as you please. Many of the R examples in this book are contained in script files that you can download, execute, and modify.

Script files can be written in your favorite text editor. They can also be managed directly in your R environment.

In the Mac environment, click on **File: New Document** to bring up a new window for typing your R commands. To execute a portion of your code, highlight the text that you want to execute, and then hit **Command-Return**. The highlighted code will appear in the R console and be executed.

In the PC environment, click on **File: New Script** to bring up a new window for a script file. To execute a portion of your code, highlight the text. Under the task bar, press the **Run line or selection** button (third from the left) to execute your code.

Open the R script file **scriptsample.R**. The file contains two blocks of code. The first block contains commands for plotting the area of a circle as a function of radius for radii from 1 to 20. Highlight the three lines of R code in the first part of the file. Execute the code and you should see a plot similar to that in Figure A.1.

```
# scriptsample.R
### Area of circle
radius <- 1:20
area <- pi*radius^2
plot(radius,area, main="Area as function of radius")
```

```
### Coin flips
n <- 1000 # Number of coin flips
coinflips <- sample(0:1,n,replace=TRUE)
heads <- cumsum(coinflips)
prop <- heads/(1:n)
plot(1:n,prop,type="l",xlab="Number of coins",
     ylab="Running average",
     main="Running proportion of heads in 1000
           coin flips")
abline(h=0.5)
```

The second block of code simulates flipping 1,000 coins, with 1 representing heads and 0 representing tails. The running proportion of heads is plotted from 1 to 1,000. The vector `coinflips` contains the outcome of 1,000 flips. The `cumsum` command generates a cumulative sum and stores the resulting vector in `heads`. The k th element of `heads` gives the number of heads in the first k coin flips. The proportion of heads is then computed and stored in `prop`. Finally, the running proportions are plotted as in Figure A.4. We see that the proportion of heads appears to converge to $1/2$, an illustration of the law of large numbers.

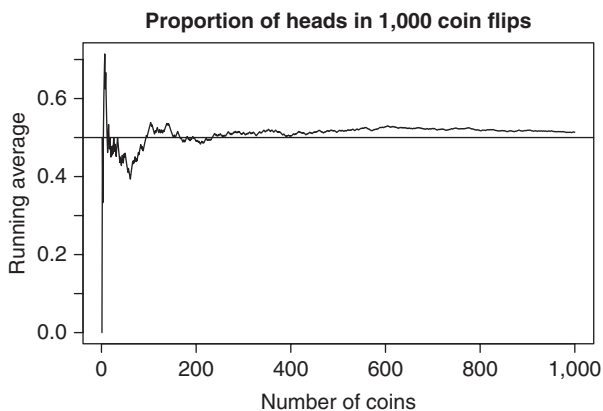


Figure A.4

Exercises:

- 8.1. Modify the **scriptsample.R** file to compute the volume of a sphere. Plot the volume as a function of radius for radii from 1 to 100.
- 8.2. Modify the **scriptsample.R** file so that the coin flips are biased and the probability of heads is 0.51. What happens to the resulting plot?

8.3. Given a list of numbers $x_1, \dots, x_n$, the *sample standard deviation* is

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2},$$

where $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ is the average of the x_i , obtained in R with the `mean` command. Write a command to compute the sample standard deviation for the integers from 1 to n . Use a script file. Compute the standard deviation for $n = 2, 10, 10000$. The R command for the standard deviation is `sd(x)`, where x is a vector. Compare the results of your command with that of the R command.

9. Functions

You can create your own functions in R. The syntax is

```
> name <- function(arg1, arg2, . . . ) expression
```

A function can have one, several, or no arguments. Here is a function to compute the area of a circle of given radius.

```
> area <- function(radius) pi*radius^2
```

The name of the function is `area`. It is a function of one variable.

```
> area(1)
[1] 3.141593
> area(7.5)
[1] 176.7146
```

The function `cone` computes the volume of a cone of height h and circular base of radius r .

```
> cone <- function(r,h) (1/3)*pi*r^2*h
> cone(1,1)
[1] 1.047198
> cone(3,10)
[1] 94.24778
```

If the function definition contains more than one line of code, enclose the code in curly braces `{}`. The function `allsums` takes a vector `vec` as its input and outputs the sum, sum of squares, and sum of cubes of the elements of `vec`.

```

> allsums <- function(vec) {
  s1 <- sum(vec)
  s2 <- sum(vec^2)
  s3 <- sum(vec^3)
  c(s1,s2,s3) }
> allsums(1:5)
[1] 15 55 225
> allsums(-5:5)
[1] 0 110 0

```

Exercises:

- 9.1. Write a function that takes as input the lengths of two sides of a right triangle and returns the length of the hypotenuse.
- 9.2. A tetrahedron die is a four-sided die with labels {1, 2, 3, 4}. Write a function with argument n that rolls n tetrahedron dice and computes their average value. Implement your function for $n = 1, 1000$, and $1,000,000$.

10. Other useful commands

```

> table(c(0,1,1,1,1,1,1,2,2,6,6,6,6,6,6,6))
# create a frequency table
0 1 2 6
1 5 2 7

> min(0:20) # minimum element
[1] 0

> max(0:20) # maximum element
[1] 20

> round(pi,3) # rounds to a given number of
decimal places
[1] 3.142

```

The `replicate` command is powerful and versatile. The syntax is `replicate(n,expr)`. The expression `expr` is repeated n times and output as a vector.

```

> replicate(6,2)
[1] 2 2 2 2 2 2

```

Choose 1,000 numbers uniformly distributed between 0 and 1 and find their mean. Then repeat the experiment five times.

```

> replicate(5,mean(runif(1000,0,1)))

```

```
[1] 0.51168 0.50591 0.48371 0.49162 0.50879
```

A Poisson distribution with parameter λ has mean $\mu = \lambda$ and standard deviation $\sigma = \sqrt{\lambda}$. To illustrate the central limit theorem we generate 80 observations $X_1, \dots, X_{80}$ from a Poisson distribution with parameter $\lambda = 4$ and compute the normalized sum

$$\frac{(X_1 + \dots + X_n)/n - \mu}{\sigma/\sqrt{n}} = \frac{(X_1 + \dots + X_{80})/80 - 4}{2/\sqrt{80}}.$$

The simulation is repeated 100,000 times and the resulting data are graphed as a histogram together with a standard normal density curve. See Figure A.5.

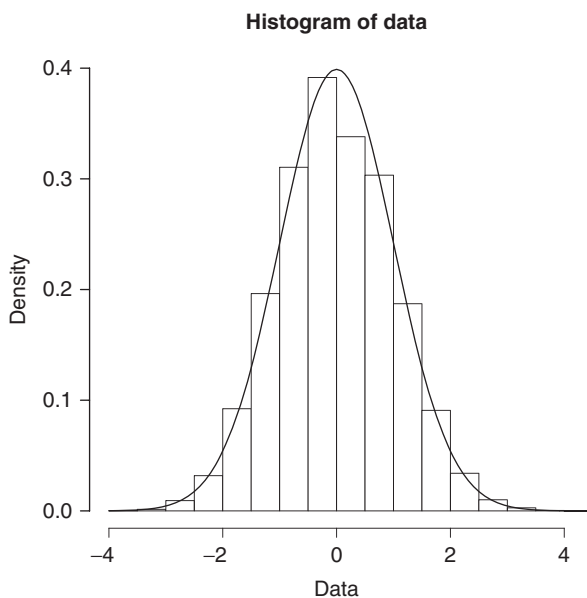


Figure A.5

```
> normsum <- function() (mean(rpois(80,4))-4)/
  (2/sqrt(80))
> normsum()
[1] -0.50312
> normsum()
[1] 1.062132
> data <- replicate(100000,normsum())
> hist(data,freq=F)
> curve(dnorm(x),-4,4,add=T)
```


11. Working with matrices

The `matrix` command is used to create matrices from vectors. Specify the number of rows or columns. The default is to fill the matrix by columns. To fill the entries of a matrix from left to right, top to bottom, add the parameter `byrow=T`.

```
> matrix(1:9,nrow=3)
      [,1] [,2] [,3]
[1,]     1     4     7
[2,]     2     5     8
[3,]     3     6     9
> matrix(1:9,nrow=3,byrow=T)
      [,1] [,2] [,3]
[1,]     1     2     3
[2,]     4     5     6
[3,]     7     8     9
```

If x is a scalar, then $x + A$ results in adding x to each component of A . If A and B are matrices, then $A + B$ adds entries componentwise. The same is true for all binary operators.

```
> A <- matrix(c(1,3,0,-4),ncol=2,byrow=T)
> B <- matrix(1:4,ncol=2,byrow=T)
> A
      [,1] [,2]
[1,]     1     3
[2,]     0    -4
> B
      [,1] [,2]
[1,]     1     2
[2,]     3     4
> 2+A
      [,1] [,2]
[1,]     3     5
[2,]     2    -2
> 3*B
      [,1] [,2]
[1,]     3     6
[2,]     9    12
> A+B
      [,1] [,2]
[1,]     2     5
[2,]     3     0
```

```

> A-B
      [,1] [,2]
[1,]    0    1
[2,]   -3   -8
> 2^B
      [,1] [,2]
[1,]    2    4
[2,]    8   16
> A/B
      [,1] [,2]
[1,]    1  1.5
[2,]    0 -1.0

```

For matrix multiplication, or matrix–vector multiplication use the matrix multiplication operator `%*%`.

```

> A %*% A
      [,1] [,2]
[1,]    1   -9
[2,]    0   16
> x <- c(2,1)
> A %*% x
      [,1]
[1,]    5
[2,]   -4
> y <- matrix(c(1,3),nrow=2)
y %*% x
      [,1] [,2]
[1,]    2    1
[2,]    6    3

```

Surprisingly, R does not have a primitive or built-in command for taking matrix powers. Here is a function that will take integer powers of a matrix.

```

matrixpower <- function(mat,k) {
  out <- mat
  for (i in 2:k) {
    out <- out %*% mat
  }
  out
}

> mat <-matrix(c(0.1,0.9,0.6,0.4),nrow=2,byrow=T)
> mat

```

```

      [,1] [,2]
[1,]  0.1  0.9
[2,]  0.6  0.4
> matrixpower(mat,2)
      [,1] [,2]
[1,]  0.55 0.45
[2,]  0.30 0.70

> matrixpower(mat,10)
      [,1] [,2]
[1,]  0.4005859 0.5994141
[2,]  0.3996094 0.6003906
> matrixpower(mat,50)
      [,1] [,2]
[1,]  0.4  0.6
[2,]  0.4  0.6

```

For A^T , the transpose of A , type `t(A)`.

```

> A <- matrix(c(3,1,-2,0,0,4),nrow=2,byrow=T)
> A
      [,1] [,2] [,3]
[1,]     3     1    -2
[2,]     0     0     4
> t(A)
      [,1] [,2]
[1,]     3     0
[2,]     1     0
[3,]    -2     4
> A %*% t(A)
      [,1] [,2]
[1,]    14    -8
[2,]    -8    16

```

To solve the linear system $A\mathbf{x} = \mathbf{b}$, type `solve(A,b)`. If A is invertible, typing `solve(A)` without the second argument returns the inverse A^{-1} .

```

> A <- matrix(c(2,0,1,1,-1,4,3,1,0),nrow=3,byrow=T)
> A
      [,1] [,2] [,3]
[1,]     2     0     1
[2,]     1    -1     4
[3,]     3     1     0
> b <- c(0,1,1)

```

```

> solve(A,b)
[1] -0.5  2.5  1.0
> A %*% c(-.5,2.5,1)
      [,1]
[1,]      0
[2,]      1
[3,]      1

> solve(A)
      [,1] [,2] [,3]
[1,]      1 -0.25 -0.25
[2,]     -3  0.75  1.75
[3,]     -1  0.50  0.50
> A %*% solve(A)
      [,1] [,2] [,3]
[1,]      1      0      0
[2,]      0      1      0
[3,]      0      0      1

```

For integer n , `diag(n)` gives the $n \times n$ identity matrix. If $\mathbf{x}$ is a vector, `diag(x)` returns a diagonal matrix whose diagonal elements are $\mathbf{x}$. If $\mathbf{A}$ is a matrix, `diag(A)` gives the diagonal elements of $\mathbf{A}$.

```

> diag(3)
      [,1] [,2] [,3]
[1,]      1      0      0
[2,]      0      1      0
[3,]      0      0      1
> A <- diag(c(1,4,0,2))
> A
      [,1] [,2] [,3] [,4]
[1,]      1      0      0      0
[2,]      0      4      0      0
[3,]      0      0      0      0
[4,]      0      0      0      2
> diag(A)
[1] 1 4 0 2

```

For an $n \times n$ matrix $\mathbf{A}$, the command `eigen(A)` returns the eigenvalues and eigenvectors of $\mathbf{A}$ in a two-component list. The first component of the list `eigen(A)$values` is a vector containing the n eigenvalues. The second component `eigen(A)$vectors` is an $n \times n$ matrix whose columns contain the corresponding eigenvectors. To illustrate, we diagonalize the matrix

$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. That is, we find an invertible matrix $\mathbf{S}$, and a diagonal matrix $\mathbf{D}$

such that $A = SDS^{-1}$. The diagonal elements of D are the eigenvalues of A . The columns of S are the corresponding eigenvectors.

```
> A <- matrix(replicate(9,1),nrow=3)
> A
      [,1] [,2] [,3]
[1,]     1     1     1
[2,]     1     1     1
[3,]     1     1     1
> eigen(A)$values
[1] 3 0 0
> eigen(A)$vectors
      [,1]      [,2]      [,3]
[1,] 0.57735027 0.70710678 0.40824829
[2,] 0.57735027 -0.70710678 0.40824829
[3,] 0.57735027 0.00000000 -0.81649658
> D <- diag(eigen(A)$values)
> D
      [,1] [,2] [,3]
[1,]     3     0     0
[2,]     0     0     0
[3,]     0     0     0
> S <- eigen(A)$vectors
> S
      [,1]      [,2]      [,3]
[1,] 0.57735027 0.70710678 0.40824829
[2,] 0.57735027 -0.70710678 0.40824829
[3,] 0.57735027 0.00000000 -0.81649658
> S %*% D %*% solve(S) # SDS-1 = A
      [,1] [,2] [,3]
[1,]     1     1     1
[2,]     1     1     1
[3,]     1     1     1
```

Exercises:

11.1. Let $A = \begin{pmatrix} 1 & 3 & -1 \\ 2 & 1 & 0 \\ 4 & -2 & 3 \end{pmatrix}$. Find the following:

- (i) A^3
- (ii) A^T
- (iii) A^{-1}
- (iv) $p(A)$, where $p(x) = x^5 - 3x^2 + 7$.

11.2. Construct a random 4×4 matrix each of whose elements is uniformly distributed on $(0, 1)$. Find the eigenvalues.

11.3. Solve the linear system

$$\begin{aligned} 2x + 3y - z &= 1, \\ x + y + z &= 2, \\ x + 2y - 3z &= 3. \end{aligned}$$

TABLE A.3 Summary of R Commands for Matrix Algebra

Command	Description
<code>matrix(vec)</code>	Creates a matrix from a vector
<code>A+B</code>	Matrix addition
<code>s*A</code>	Scalar multiplication
<code>A*B</code>	Elementwise multiplication
<code>A %*% B</code>	Matrix multiplication
<code>x %*% y</code>	If <i>x</i> and <i>y</i> are vectors, returns dot product
<code>solve(A)</code>	Inverse A^{-1}
<code>solve(A,b)</code>	Solves the linear system $Ax = b$.
<code>t(A)</code>	Transpose A^T
<code>det(A)</code>	Determinant of <i>A</i>
<code>diag(s)</code>	If <i>s</i> is a scalar, creates the $s \times s$ identity matrix
<code>diag(x)</code>	If <i>x</i> is a vector, creates diagonal matrix whose diagonal elements are <i>x</i>
<code>diag(A)</code>	If <i>A</i> is a matrix, returns the diagonal elements
<code>eigen(A)</code>	Eigenvalues and eigenvectors of <i>A</i> <code>eigen(A)\$values</code> is vector of eigenvalues <code>eigen(A)\$vectors</code> is matrix of eigenvectors

APPENDIX B

PROBABILITY REVIEW

This review of the basics of probability theory focuses on random variables and their distributions. Few results are proven and we refer the reader to a standard undergraduate probability textbook for more complete results. Conditional probability and conditional distribution are discussed in Chapter 1.

Probability begins with a *random experiment*, which is loosely defined as an experiment for which the outcome is uncertain. Given such an experiment, the *sample space* Ω is the set of all possible outcomes. Individual outcomes, that is, the elements of the sample space, are denoted by ω . An *event* A is a subset of the sample space. Say that A occurs if the outcome of the experiment is contained in A .

A probability P is a function that assigns to each event a number between 0 and 1 in such a way that the following conditions are satisfied:

1. $0 \leq P(A) \leq 1$, for all $A \subseteq \Omega$.
2. $P(\Omega) = 1$.
3. Given a sequence of disjoint events $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

We interpret $P(A)$ to mean *the probability that A occurs*.

A *random variable* is a real-valued function defined on a sample space. That is, the outcomes of a random variable are determined by a random experiment. For instance,

assume that three coins are flipped. Let X be the number of heads. Then, X is a random variable that takes values 0, 1, 2, or 3, depending on the outcome of the coin flips.

Write $P(X = x)$ for the probability that X takes the value x , and $P(X \leq x)$ for the probability that X takes a value less than or equal to x . More generally, for $R \subseteq \mathbb{R}$, write $P(X \in R)$ for the probability that X takes a value that is contained in R . The notation $\{X \in R\}$ is shorthand for $\{\omega : X(\omega) \in R\}$, which is the set of all outcomes ω with the property that $X(\omega)$ is contained in R .

The *distribution* of a random variable X describes the set of values of X and their corresponding probabilities.

The function $F(x) = P(X \leq x)$ is the *cumulative distribution function (cdf)* of X . The cdf takes values between 0 and 1 and is defined for all real numbers. The cdf gives complete probabilistic information about a random variable in the sense that knowing the cdf is equivalent to knowing the distribution of the random variable.

■ **Example B.1** For the random experiment of flipping a fair coin three times, let X denote the number of heads that occur. Letting H denote heads, T denote tails, and keeping track of the order of coin flips, the sample space is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

If all outcomes are equally likely, then each of the eight outcomes occurs with probability $1/8$. This gives

$$P(X = x) = \begin{cases} P(\{TTT\}) & = 1/8, & \text{if } x = 0, \\ P(\{HTT, THT, TTH\}) & = 3/8, & \text{if } x = 1, \\ P(\{HHT, HTH, THH\}) & = 3/8, & \text{if } x = 2, \\ P(\{HHH\}) & = 1/8, & \text{if } x = 3, \end{cases}$$

and cdf

$$F(x) = P(X \leq x) = \begin{cases} 0, & \text{if } x < 0, \\ 1/8, & \text{if } 0 \leq x < 1, \\ 4/8, & \text{if } 1 \leq x < 2, \\ 7/8, & \text{if } 2 \leq x < 3, \\ 1, & \text{if } x \geq 3. \end{cases}$$

■

B.1 DISCRETE RANDOM VARIABLES

A random variable that takes values in a finite or countably infinite set is called a *discrete random variable*. As a function of x , the function $P(X = x)$ is the *probability mass function (pmf)* of X . The pmf describes the distribution of a discrete random variable. For $R \subseteq \mathbb{R}$,

$$P(X \in R) = \sum_{x \in R} P(X = x).$$

The *expectation*, or mean, of a discrete random variable X is defined as

$$E(X) = \sum_x xP(X = x).$$

The expectation is a *weighted average* of the values of X , with weights given by the pmf. Intuitively, the expectation of X is the long-run average value of X over repeated trials.

If g is a function and X is a random variable, then $Y = g(X)$ is a *function of a random variable*, which itself is a random variable that takes the value $g(x)$ whenever X takes the value x . A useful formula for computing the expectation of a function of a random variable is

$$E(Y) = E(g(X)) = \sum_x g(x)P(X = x). \quad (\text{B.1})$$

The expectation is also computed as $E(Y) = \sum_y yP(Y = y)$, which requires knowledge of the distribution of Y .

■ **Example B.2** The radius R of a circle is a random variable that takes values 1, 2, 4, and 8 with respective probabilities 0.1, 0.2, 0.3, and 0.4. Find the expected area of the circle.

Solution Let Y be the area of the circle. The expected area is

$$\begin{aligned} E(Y) &= E(\pi R^2) \\ &= \sum_{r=1}^4 \pi r^2 P(R = r) \\ &= \pi (1(0.1) + 4(0.2) + 16(0.3) + 64(0.4)) \\ &= 31.3\pi. \end{aligned} \quad \blacksquare$$

The expectation of a linear function of a random variable is a linear function of the expectation. From Equation (B.1), for constants a and b ,

$$\begin{aligned} E(aX + b) &= \sum_x (ax + b)P(X = x) \\ &= a \sum_x xP(X = x) + b \sum_x P(X = x) \\ &= aE(X) + b, \end{aligned}$$

The *variance* of a random variable is a measure of variability or discrepancy from the mean. It is defined as

$$\text{Var}(X) = E((X - E(X))^2) = \sum_x (x - E(X))^2 P(X = x).$$

A computationally useful formula is $\text{Var}(X) = E(X^2) - (E(X))^2$. For constants a and b ,

$$\begin{aligned} \text{Var}(aX + b) &= E(((aX + b) - (aE(X) + b))^2) \\ &= E(a^2(X - E(X))^2) \\ &= a^2 \text{Var}(X). \end{aligned}$$

The *standard deviation* of a random variable is defined as $SD(X) = \sqrt{\text{Var}(X)}$.

B.2 JOINT DISTRIBUTION

The *joint probability mass function* of X and Y is $P(X = x, Y = y)$, which is a function of x and y . The *joint cumulative distribution function* is

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{i \leq x} \sum_{j \leq y} P(X = i, Y = j).$$

From the joint pmf of X and Y one can obtain the individual, or marginal, distributions of each random variable. For instance,

$$P(X = x) = P(X = x, -\infty < Y < \infty) = \sum_y P(X = x, Y = y).$$

Similarly, $P(Y = y) = \sum_x P(X = x, Y = y)$.

The *covariance* is a measure of linear association between two random variables. It is defined as

$$\text{Cov}(X, Y) = E((X - E(X))(Y - E(Y))) = E(XY) - E(X)E(Y).$$

The *correlation* between X and Y is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{SD(X)SD(Y)}.$$

The correlation satisfies $-1 \leq \text{Corr}(X, Y) \leq 1$ and is equal to ± 1 if one random variable is a linear function of the other.

If $g(x, y)$ is a function of two variables, and X and Y are random variables, then $g(X, Y)$ is a random variable whose expectation is

$$E(g(X, Y)) = \sum_x \sum_y g(x, y)P(X = x, Y = y).$$

In the case when $g(x, y) = x + y$,

$$\begin{aligned}
 E(X + Y) &= \sum_x \sum_y (x + y)P(X = x, Y = y) \\
 &= \sum_x x \sum_y P(X = x, Y = y) + \sum_y y \sum_x P(X = x, Y = y) \\
 &= \sum_x xP(X = x) + \sum_y yP(Y = y) \\
 &= E(X) + E(Y),
 \end{aligned}$$

which gives the important *linearity* property of expectation. For random variables $X_1, \dots, X_n$,

$$E(X_1 + \dots + X_n) = E(X_1) + \dots + E(X_n).$$

Example B.3 The solution of the classic *matching problem* is an elegant application of the linearity of expectation.

At the baseball stadium a group of n people wearing baseball hats all throw their hats into the air when their favorite player hits a home run. If the hats are mixed up at random when they fall to the ground, and each person picks up one hat, how many people, on average, will get their own hat back?

Solution Let $I_1, \dots, I_n$ be a sequence of random variables where

$$I_k = \begin{cases} 1, & \text{if the } k\text{th person gets their hat back,} \\ 0, & \text{otherwise,} \end{cases}$$

for $k = 1, \dots, n$. Then, $X = I_1 + \dots + I_n$ is the number of people who get their hat back. For each k ,

$$\begin{aligned}
 E(I_k) &= (1)P(\text{kth person gets their hat}) \\
 &\quad + (0)P(\text{kth person does not get their hat}) \\
 &= P(\text{kth person gets their hat}) \\
 &= \frac{1}{n},
 \end{aligned}$$

since there are n hats to choose from and exactly one belongs to the k th person. By linearity of expectation,

$$E(X) = E(I_1 + \dots + I_n) = E(I_1) + \dots + E(I_n) = \sum_{k=1}^n \frac{1}{n} = 1.$$

On average, one person gets their hat back. Remarkably, the solution does not depend on n . ■

For the variance of a sum of random variables,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

More generally,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j).$$

Events A and B are *independent* if $P(A \cap B) = P(A)P(B)$. Intuitively, events are independent if knowledge of whether or not one occurs has no influence on the probability of whether or not the other occurs.

We say that discrete random variables X and Y are *independent* if

$$P(X = x, Y = y) = P(X = x)P(Y = y) \text{ for all } x, y.$$

Equivalently,

$$P(X \in R, Y \in S) = P(X \in R)P(Y \in S) \text{ for all } R, S \subseteq \mathbb{R}.$$

If X and Y are independent random variables, then

$$\begin{aligned} E(XY) &= \sum_x \sum_y xyP(X = x, Y = y) \\ &= \sum_x \sum_y xyP(X = x)P(Y = y) \\ &= \sum_x xP(X = x) \sum_y yP(Y = y) = E(X)E(Y), \end{aligned}$$

and thus $\text{Cov}(X, Y) = 0$. Hence, for independent random variables, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Sequences of independent and identically distributed (i.i.d.) random variables are common models. For instance, an infinite sequence of fair coin flips can be modeled as an independent sequence $X_1, X_2, \dots$, where for each k , $X_k = 1$, if the k th flip is heads, and 0, if the k th flip is tails. The random variable $S_n = X_1 + \dots + X_n$ is the number of heads in the first n coin flips.

In statistics, one often models a simple random sample as an i.i.d. sequence of random variables from a common population.

B.3 CONTINUOUS RANDOM VARIABLES

A continuous random variable takes values in an uncountable set, most commonly $\mathbb{R}$, $(0, \infty)$ or (a, b) , with $a < b$. For continuous random variables $P(X = x) = 0$ for all x , and probabilities are computed by integrating the *probability density function*. The density function plays a role analogous to the pmf for discrete variables for computing probabilities.

A function f is a probability density function of X if

1. $f(x) \geq 0$, for all x
2. $\int_{-\infty}^{\infty} f(x) dx = 1$,
3. For all $R \subseteq \mathbb{R}$, $P(X \in R) = \int_R f(x) dx$.

The cumulative distribution function of X is

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt.$$

Differentiating with respect to x gives

$$\frac{d}{dx} F(x) = f(x).$$

That is, the density is the derivative of the cdf.

■ **Example B.4** Let X be a continuous random variable with density function

$$f(x) = cx^2, \text{ for } 0 < x < 3.$$

- (i) Find the constant c . (ii) Find $P(1 < X < 2)$. (iii) Find the density function of $Y = X^2$.

Solution

- (i) To find c , solve

$$1 = \int_{-\infty}^{\infty} f(x) dx = \int_0^3 cx^2 dx = 9c,$$

which gives $c = 1/9$.

- (ii) The desired probability is

$$P(1 < X < 2) = \int_1^2 f(x) dx = \int_1^2 \frac{x^2}{9} dx = \frac{7}{27}.$$

- (iii) To find the density of Y first find the cdf of Y and then differentiate. Since X takes values between 0 and 3, $Y = X^2$ takes values between 0 and 9. For $0 < y < 9$,

$$P(Y \leq y) = P(X^2 \leq y) = P(X \leq \sqrt{y}).$$

Taking the derivative with respect to y and applying the chain rule gives

$$f_Y(y) = \frac{d}{dy} P(X \leq \sqrt{y}) = \frac{1}{2\sqrt{y}} f_X(\sqrt{y}) = \frac{1}{2\sqrt{y}} \frac{y}{9} = \frac{\sqrt{y}}{18},$$

for $0 < y < 9$. ■

Formulas for expectation and variance are analogous to the discrete formulas with density function replacing pmf and integrals replacing sums. Thus, for a continuous random variable X with density function f ,

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx \quad \text{and} \quad \text{Var}(X) = \int_{-\infty}^{\infty} (x - E(X))^2 f(x) dx.$$

If g is a function, then $E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x) dx$.

For continuous random variables, the joint density of X and Y is the function $f(x, y)$ that satisfies

$$P((X, Y) \in R) = \iint_R f(x, y) dx dy, \quad \text{for all } R \subseteq \mathbb{R}^2.$$

The joint cdf of X and Y is the function

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds.$$

If X and Y are independent, then $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$ for all x and y . Equivalently, the joint density function factors into the product of the marginal densities. That is,

$$f(x, y) = f_X(x) f_Y(y), \quad \text{for all } x, y.$$

B.4 COMMON PROBABILITY DISTRIBUTIONS

Uniform (Discrete) Distribution

The simplest model for a random variable X taking values in a finite set is that all outcomes are equally likely. We say that X has a *uniform distribution*. Historically, probability began by considering equally likely outcomes, mostly in games of chance.

For a finite set R , let $|R|$ denote the number of elements of R . If X is uniformly distributed on $S = \{s_1, \dots, s_k\}$, then

$$P(X = s_i) = \frac{1}{k}, \quad \text{for } i = 1, \dots, k$$

and

$$P(X \in R) = \frac{|R|}{|S|} = \frac{|R|}{k}, \quad \text{for } R \subseteq S.$$

In the discrete uniform case, probability reduces to counting. The probability that X is contained in R is the number of elements of R divided by the number of elements of the sample space.

Mean and variance are

$$E(X) = \frac{s_1 + \cdots + s_k}{k} \quad \text{and} \quad \text{Var}(X) = \left(\frac{s_1^2 + \cdots + s_k^2}{k} \right) - \left(\frac{s_1 + \cdots + s_k}{k} \right)^2.$$

For the case $S = \{1, \dots, k\}$, this gives

$$E(X) = \frac{k+1}{2} \quad \text{and} \quad \text{Var}(X) = \frac{k^2-1}{12}.$$

Bernoulli Distribution

A *Bernoulli* random variable takes values 1 and 0, with probabilities p and $1-p$, respectively. It is common to refer to the dichotomous values of a Bernoulli variable as *success* and *failure*.

If X has a Bernoulli distribution with parameter $0 < p < 1$, then

$$E(X) = p \quad \text{and} \quad \text{Var}(X) = p(1-p).$$

Binomial Distribution

Assume that $X_1, \dots, X_n$ is an i.i.d. sequence of Bernoulli random variables with common parameter p , where each X_i represents success or failure on the i th trial. Let $X = X_1 + \cdots + X_n$. Then, X counts the number of successes in n trials, and has a *binomial distribution with parameters n and p* .

The pmf of the binomial distribution is derived by a counting argument using the fact that the event $\{X = k\}$ can be expressed as the set of all 0-1 sequences of length n with exactly k 1s. The number of such sequences is counted by the binomial coefficient $\binom{n}{k}$. It follows that

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad \text{for } k = 0, 1, \dots, n,$$

with

$$E(X) = np \quad \text{and} \quad \text{Var}(X) = np(1-p).$$

Example B.5 Whether or not tomato seeds germinate in Angel's garden is modeled as independent Bernoulli random variables with germination (success) probability $p = 0.8$. If 100 seeds are planted, find the probability that at least 75 germinate.

Solution Let X be the number of seeds that germinate. Then, X has a binomial distribution with parameters $n = 100$ and $p = 0.8$. The desired probability is

$$P(X \geq 75) = \sum_{k=75}^{100} \binom{100}{k} (0.8)^k (0.2)^{100-k} = 0.9125.$$

Since $P(X \geq 75) = 1 - P(X < 75) = 1 - P(X \leq 74)$, the probability is obtained in R by typing

```
> 1-pbinom(74,100,0.8)
[1] 0.9125246
```

■

Geometric Distribution

Given a sequence $X_1, X_2, \dots$ of i.i.d. Bernoulli variables with parameter p , with X_i representing success or failure on the i th trial, let N be the index of the first trial in which success occurs. That is $\{N = k\}$ if and only if $X_i = 0$ for all $i < k$ and $X_k = 1$. Then, N has a *geometric distribution with parameter p* and

$$P(N = k) = (1 - p)^{k-1}p, \text{ for } k = 1, 2, \dots,$$

with

$$E(N) = \frac{1}{p} \text{ and } \text{Var}(N) = \frac{1-p}{p^2}.$$

■ **Example B.6** In Texas hold 'em poker, players are initially dealt two cards. If the deck is reshuffled after each play, find the expected number of deals until a player gets at least one ace.

Solution The outcome of each deal is modeled as a Bernoulli variable with parameter

$$\begin{aligned} p &= P(\text{Player gets at least one ace}) \\ &= 1 - P(\text{Player gets no aces}) \\ &= 1 - \left(\frac{48}{52}\right) \left(\frac{47}{51}\right) = \frac{33}{221} = 0.1493. \end{aligned}$$

The number of deals required for a player to get at least one ace has a geometric distribution with parameter p . The expected number of required deals is $1/p = 221/33 = 6.697$. ■

Poisson Distribution

The Poisson distribution arises as a model for counts of independent events that occur in some fixed region of time or space. Examples include the number of traffic accidents along a stretch of highway, the number of births on a hospital ward, and the number of wrong numbers to your cell phone. The distribution is sometimes called the *law of rare events* and arises when the chance that some event occurs in a small interval of time or space is small.

The distribution depends on a parameter $\lambda > 0$, which can be interpreted as the *rate* of occurrence in a unit interval. A random variable X has a *Poisson distribution with parameter λ* , if

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \text{ for } k = 0, 1, \dots$$

The distribution has the property that mean and variance are both equal to λ . For the mean,

$$\begin{aligned} E(X) &= \sum_{k=0}^{\infty} k P(X = k) = \sum_{k=0}^{\infty} k \frac{e^{-\lambda} \lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-1)!} \\ &= e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = e^{-\lambda} \lambda \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} \lambda e^{\lambda} = \lambda. \end{aligned}$$

Example B.7 During the peak month of May, tornados hit Oklahoma at the rate of about 21.7 per month, according to the National Weather Service. Find the probability that there will be fewer than 15 tornados next May.

Solution If it is assumed that successive tornado hits are independent events, then the Poisson distribution is a reasonable model. Let X be the number of tornados which hit Oklahoma next May. Then,

$$P(X < 15) = P(X \leq 14) = \sum_{k=0}^{14} \frac{e^{-21.7} (21.7)^k}{k!} = 0.054.$$

In R, type

```
> ppois(14, 21.7)
[1] 0.05400056
```

The Poisson distribution is closely related to the binomial distribution and arises as a limiting distribution when the number of trials is large and the success probability is small. If X has a binomial distribution with large n and small p , then the distribution of X will be approximately equal to a Poisson distribution with parameter $\lambda = np$.

Poisson Approximation of Binomial Distribution

Consider the binomial distribution with parameters n and p_n . Assume that

$$\lim_{n \rightarrow \infty} p_n = 0 \text{ and } \lim_{n \rightarrow \infty} np_n = \lambda > 0.$$

Then, the binomial pmf converges to the pmf of a Poisson distribution with parameter λ . That is, for $k = 0, 1, \dots$,

$$\lim_{n \rightarrow \infty} \frac{n!}{k!(n-k)!} p_n^k (1-p_n)^{n-k} = \frac{e^{-\lambda} \lambda^k}{k!}.$$

Proof. Consider the binomial probability with $p = \lambda/n$, which gives

$$\begin{aligned} \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} &= \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{n(n-1) \cdots (n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{n^k \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{-k} \left(1 - \frac{\lambda}{n}\right)^n \\ &= \frac{\lambda^k}{k!} \left[\left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) \right] \left(1 - \frac{\lambda}{n}\right)^{-k} \left(1 - \frac{\lambda}{n}\right)^n. \end{aligned} \quad (\text{B.2})$$

Take the limit as $n \rightarrow \infty$, and consider the four factors on the right-hand side of Equation (B.2).

(i) Since λ and k are constants, $\lambda^k/k!$ stays unchanged in the limit.

(ii) For fixed k ,

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) = 1^{k-1} = 1.$$

(iii)

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^{-k} = 1^{-k} = 1.$$

(iv) Recall that the constant $e = 2.71827 \dots$ is defined as the limit

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

Make the substitution $1/x = -\lambda/n$, so that $n = -\lambda x$. This gives

$$\lim_{n \rightarrow \infty} \left(1 - \frac{\lambda}{n}\right)^n = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^{-\lambda x} = \left[\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x \right]^{-\lambda} = e^{-\lambda}.$$

Plugging in the four limits in Equation (B.2) gives the result. ■

■ **Example B.8** Mutations in DNA sequences occur from environmental factors and mistakes when a cell copies its DNA in preparation for cell division. The mutation rate per nucleotide of human DNA has been estimated at about 2.5×10^{-8} . There are about 3.3×10^9 nucleotide bases in the human DNA genome. Find the probability that exactly 80 DNA sites mutate in a random person's DNA.

Solution Let X be the number of mutations. If successive mutations are independent, then X has a binomial distribution with $n = 3.3 \times 10^9$ and $p = 2.5 \times 10^{-8}$. The distribution is approximated by a Poisson distribution with parameter

$$\lambda = np = (3.3 \times 10^9)(2.5 \times 10^{-8}) = 82.5.$$

The desired probability is

$$P(X = 80) \approx \frac{e^{-82.5}(82.5)^{80}}{80!} = 0.043.$$

Note that the exact probability using the binomial distribution is

$$P(X = 80) = \binom{3.3 \times 10^9}{80} (2.5 \times 10^{-8})^{80} (1 - 2.5 \times 10^{-8})^{3.3 \times 10^9 - 80}.$$

The approximate probability using the Poisson distribution can be compared with the exact probability in R. We display 12 significant digits, and see that the approximation is good to nine digits.

```
> options(digits=12)
> dpois(80, 82.5)
[1] 0.0428838140788
> dbinom(80, 3.3*10^9, 2.5*10^(-8))
[1] 0.042883814558
```

■

Multinomial Distribution

The multinomial distribution generalizes the binomial distribution. Consider a sequence of n i.i.d. random variables, where each variable takes one of k possible values. Assume that the i th value occurs with probability p_i , with $p_1 + \cdots + p_k = 1$. For $i = 1, \dots, k$, let X_i denote the number of times outcome i occurs. Then, $(X_1, \dots, X_k)$ has a *multinomial distribution with parameters $n, p_1, \dots, p_k$* . The joint pmf is

$$P(X_1 = x_1, \dots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \text{ for } x_1 + \cdots + x_k = n.$$

Marginally, each X_i has a binomial distribution with parameters n and p_i .

■ **Example B.9** According to the Red Cross, the distribution of blood types in the United States is as follows: O: 44%, A: 42%, B: 10%, and AB: 4%. In a sample of six people, find the probability that three are type O, two are A, one is B, and none are AB.

Solution Let X_O , X_A , X_B , and X_{AB} denote the number of people in the sample with the respective blood types. Then, (X_O, X_A, X_B, X_{AB}) has a multinomial distribution with parameters 6, 0.44, 0.42, 0.10, 0.04. The desired probability is

$$\begin{aligned} P(X_O = 3, X_A = 2, X_B = 1, X_{AB} = 0) \\ &= \frac{6!}{3!2!1!0!} (0.44)^3 (0.42)^2 (0.10)^1 (0.04)^0 \\ &= 0.09016. \end{aligned}$$

■

Uniform (Continuous) Distribution

For $a < b$, the uniform distribution on (a, b) is a continuous model for equally likely outcomes on a bounded interval. The density function is constant. A random variable X is *uniformly distributed on (a, b)* if the density of X is

$$f(x) = \frac{1}{b-a}, \text{ for } a < x < b.$$

Mean and variance are

$$E(X) = \frac{a+b}{2} \text{ and } \text{Var}(X) = \frac{(b-a)^2}{12}.$$

For $a < c < d < b$,

$$P(c < X < d) = \frac{d-c}{b-a} = \frac{\text{Length of } (c, d)}{\text{Length of } (a, b)}.$$

Exponential Distribution

The exponential distribution is a positive continuous distribution which often arises as a model for arrival, or waiting, times. Applications include the time when customers arrive at a queue, when electronic components fail, and when calls come in to a call center. The distribution depends on a parameter $\lambda > 0$, which has the interpretation of the arrival rate.

A random variable X has an *exponential distribution with parameter λ* , if the density function of X is

$$f(x) = \lambda e^{-\lambda x}, \text{ for } x > 0.$$

The cdf is

$$F(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}, \text{ for } x > 0.$$

Mean and variance are

$$E(X) = \frac{1}{\lambda} \text{ and } \text{Var}(X) = \frac{1}{\lambda^2}.$$

The exponential distribution plays a prominent role in probability models because of its *memoryless* property. A random variable X is memoryless if

$$P(X > s + t | X > s) = P(X > t), \text{ for all } s, t > 0.$$

The exponential distribution is the only continuous distribution which is memoryless.

Example B.10 The lifetime of an electronic component is modeled with an exponential distribution. If components fail on average after 1,200 hours, find the probability that the component lasts more than 1,300 hours.

Solution Let X be the time until the component fails. Since the parameter of an exponential distribution is the reciprocal of the mean, model X with an exponential distribution with parameter $\lambda = 1/1200$. The desired probability is

$$\begin{aligned} P(X > 1300) &= \int_{1300}^{\infty} f(x) dx \\ &= \int_{1300}^{\infty} \frac{1}{1200} e^{-x/1200} dx \\ &= e^{-1300/1200} = 0.3385. \end{aligned}$$

In R, type

```
> 1-pexp(1300, 1/1200)
[1] 0.3384654
```

■

Normal Distribution

The normal distribution plays a central role in statistics, is a common model for numerous natural and biological phenomenon, and arises as the limit for many random processes and distributions. It is also called the Gaussian distribution after Carl Friedrich Gauss who discovered its utility as a model for astronomical measurement errors. The distribution is parameterized by two numbers μ and σ^2 , which are the mean and variance, respectively, of the distribution.

A random variable X has a *normal distribution with parameters μ and σ^2* , if the density function of X is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \text{ for } -\infty < x < \infty.$$

The shape of the density is the famous *bell curve*. The density is symmetric about the line $x = \mu$. The inflection points, where the curvature of the density changes sign, occur one standard deviation from the mean, at the points $\mu \pm \sigma$.

The normal distribution has the property that a linear function of a normal random variable is normal. If X is normally distributed with mean μ and variance σ^2 , then $Y = aX + b$ is normally distributed with mean

$$E(Y) = E(aX + b) = aE(X) + b = a\mu + b$$

and variance

$$\text{Var}(Y) = \text{Var}(aX + b) = a^2 \text{Var}(X) = a^2 \sigma^2.$$

A common heuristic for working with the normal distribution is the *68–95–99.7 rule*, which says that for a normal distribution, the probability of being one, two, and three standard deviations from the mean is, respectively about 0.68, 0.95, and 0.997.

The *standard normal distribution* is a normal distribution with mean 0 and variance 1.

■ **Example B.11** Assume that babies' birth weights are normally distributed with mean 120 ounces and standard deviation 20 ounces. (i) Find the approximate probability that a random baby's birth weight is between 100 and 140 ounces. (ii) Find the exact probability that a baby's birth weight is less than 136 ounces. (iii) *Low birth weight* is defined as the 5th percentile of the birth weight distribution. At what weight is a baby's birth weight considered low?

Solution

- (i) Let X be a random baby's birth weight. The desired probability is $P(100 < X < 140)$. Observe that 100 is one standard deviation below the mean and 140 is one standard deviation above the mean. The event $\{100 < X < 140\}$ is the event that the birth weight is within one standard deviation of the mean. By the 68–95–99.7 rule, the probability is about 0.68.
- (ii) The desired probability is

$$P(X < 136) = \int_{-\infty}^{136} f(x) \, dx = \int_{-\infty}^{136} \frac{1}{\sqrt{2\pi(20^2)}} e^{-\frac{(x-120)^2}{2(20)^2}} \, dx.$$

There is no elementary closed form for the cdf of the normal distribution, and numerical methods are needed to solve integrals such as this. In R, type

```
> pnorm(136, 120, 20)
[1] 0.78814460
```

- (iii) The problem asks for the 5th percentile of the normal distribution. Equivalently, we seek the number q such that $P(X \leq q) = 0.05$. In R, use the quantile function.

```
> qnorm(0.05, 120, 20)
[1] 87.102927461
```

Babies with birth weights less than 87 ounces are considered to have low birth weight. ■

Bivariate Normal Distribution

The bivariate normal distribution for X and Y , and the multivariate normal distribution for $X_1, \dots, X_m$, are generalizations of the normal distribution to higher dimensions. The bivariate normal distribution is specified by five parameters: $\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho$. These are the means and variances of X and Y , and their correlation. If $\mu_X = \mu_Y = 0$ and $\sigma_X^2 = \sigma_Y^2 = 1$, this gives the bivariate standard normal distribution with joint density

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}},$$

for $-\infty < x, y < \infty$ and $-1 < \rho < 1$.

The marginal and conditional distributions of a bivariate normal distribution are normal. In particular, the conditional distribution of X given $Y = y$ is normal with mean ρy and variance $1 - \rho^2$. Similarly, the conditional distribution of Y given $X = x$ is normal with mean ρx and variance $1 - \rho^2$.

If X and Y have a bivariate normal distribution, then $aX + bY$ is normally distributed for all nonzero constants a and b .

If $\rho = 0$, we say that X and Y are *uncorrelated*. If normal random variables are uncorrelated, then they are independent. Note that this is not true for random variables in general. If two random variables X and Y are uncorrelated it does not necessarily mean that they are independent.

Gamma Distribution

The gamma distribution is a nonnegative continuous distribution which depends on two parameters. The distribution encompasses a large family of unimodal, skewed and symmetric distribution shapes and is a popular distribution for modeling positive continuous processes. It also arises naturally as the distribution of a sum of i.i.d. exponential random variables.

A random variable X has a *gamma distribution with parameters* $r > 0$ and $\lambda > 0$, if the density function of X is

$$f(x) = \frac{1}{\Gamma(r)} \lambda^r x^{r-1} e^{-\lambda x}, \text{ for } x > 0,$$

where $\Gamma(r)$ is the *gamma function*

$$\Gamma(r) = \int_0^\infty t^{r-1} e^{-t} dt.$$

If r is a positive integer, then $\Gamma(r) = (r-1)!$ For a gamma random variable,

$$E(X) = \frac{r}{\lambda} \text{ and } \text{Var}(X) = \frac{r}{\lambda^2}.$$

A sum of n independent exponential random variables with common parameter λ has a gamma distribution with parameters n and λ .

Example B.12 Bella is fishing and the time it takes to catch a fish is exponentially distributed with mean 20 minutes. Every time she catches a fish, she throws it back in the water, and continues fishing. Find the probability that she will catch five fish in the first hour.

Solution Let X_1 denote how long it takes for Bella to catch her first fish. For $k = 2, \dots, 5$, let X_k be the time from when she throws her $(k-1)$ th fish in the water to when she catches her k th fish. Then, $X_1, \dots, X_5$ are i.i.d. exponential random variables with parameter $\lambda = 1/20$, and $S = X_1 + \dots + X_5$ is the total time it takes for Bella to catch five fish. The random variable S has a gamma distribution with parameters $r = 5$ and $\lambda = 1/20$. The desired probability is

$$P(S \leq 60) = \int_0^{60} \frac{1}{\Gamma(5)} \left(\frac{1}{20}\right)^5 x^4 e^{-x/20} dx = 0.185. \quad \blacksquare$$

Beta Distribution

The beta distribution is a continuous distribution on $(0, 1)$ that depends on two parameters. It is a generalization of the uniform distribution. A random variable X has a *beta distribution with parameters* $a > 0$ and $b > 0$, if the density function of X is

$$f(x) = \frac{1}{B(a, b)} x^{a-1} (1-x)^{b-1}, \text{ for } 0 < x < 1,$$

where

$$B(a, b) = \int_0^1 x^{a-1} (1-x)^{b-1} dx = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

For a beta random variable,

$$E(X) = \frac{a}{a+b} \quad \text{and} \quad \text{Var}(X) = \frac{ab}{(a+b+1)(a+b)^2}.$$

The uniform distribution on $(0, 1)$ is obtained for $a = b = 1$.

B.5 LIMIT THEOREMS

The classic limit theorems of probability are concerned with sequences of i.i.d. random variables. If $X_1, X_2, \dots$ is such a sequence with common mean $\mu = E(X_1) < \infty$, let $S_n = X_1 + \dots + X_n$. The *law of large numbers* says that the sequence of averages S_n/n converges to μ , as $n \rightarrow \infty$. There are two versions of the law—weak and strong.

Theorem B.1. (Weak law of large numbers). *For any $\epsilon > 0$,*

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{S_n}{n} - \mu\right| < \epsilon\right) = 1.$$

While the weak law asserts that for large n , the average S_n/n is with high probability close to μ , it does not say that having come close to μ , the sequence of averages will always stay close to μ .

If a sequence of numbers $x_1, x_2, \dots$ converges to a limit x then eventually, for n sufficiently large, the terms $x_n, x_{n+1}, x_{n+2}, \dots$ will all be arbitrarily close to x . That is, for any $\epsilon > 0$, there is some index N such that $|x_n - x| \leq \epsilon$, for all $n \geq N$.

The strong law of large numbers asserts that with probability 1 the sequence of averages $S_1/1, S_2/2, S_3/3, \dots$ behaves precisely in this way.

Theorem B.2. (Strong law of large numbers).

$$P\left(\lim_{n \rightarrow \infty} \frac{S_n}{n} = \mu\right) = 1.$$

By the law of large numbers, $S_n/n - \mu$ converges to 0, as $n \rightarrow \infty$. The *central limit theorem* asserts that $(S_n/n - \mu)/(\sigma/\sqrt{n})$ converges to a normally distributed random variable. Remarkably, this is true for any i.i.d. sequence $X_1, X_2, \dots$ with finite mean and variance.

Theorem B.3. (Central limit theorem). For all t ,

$$\lim_{n \rightarrow \infty} P\left(\frac{S_n/n - \mu}{\sigma/\sqrt{n}} \leq t\right) = P(Z \leq t),$$

where Z has a standard normal distribution.

The central limit theorem gives that for large n , $X_1 + \cdots + X_n$ has an approximate normal distribution with mean $n\mu$ and variance $n\sigma^2$.

Example B.13 The number of accidents per week in a stretch of highway has a Poisson distribution with parameter $\lambda = 2$. If the number of accidents is independent from week to week, what is the probability that over a year's time there will be more than 100 accidents?

Solution Let $X_1, \dots, X_{52}$ be the number of accidents, respectively, during each week of the year. Then, $S_{52} = X_1 + \cdots + X_{52}$ is the total number of accidents in a year. The X_i are i.i.d. with common mean and variance $\lambda = 2$. By the central limit theorem,

$$\begin{aligned} P(S_{52} > 100) &= P\left(\frac{S_{52}/52 - \mu}{\sigma/\sqrt{52}} > \frac{100/52 - 2}{\sqrt{2}/\sqrt{52}}\right) \\ &\approx P(Z > -0.392) = 0.652. \end{aligned}$$

We can compare the central limit approximation with the exact result. The sum of independent Poisson random variables has a Poisson distribution, and $X_1 + \cdots + X_{52}$ has a Poisson distribution with parameter $52\lambda = 104$. The exact probability is

$$P(S_{52} > 100) = 1 - P(S_{52} \leq 100) = 1 - \sum_{k=0}^{100} \frac{e^{-104} 104^k}{k!} = 0.629. \quad \blacksquare$$

B.6 MOMENT-GENERATING FUNCTIONS

Moment-generating functions are remarkably versatile tools for proving results involving sums and limits of random variables. Let X be a random variable. The *moment-generating function (mgf)* of X is the function $m(t) = E(e^{tX})$, defined for all t for which the expectation exists.

The name *moment-generating* comes from the fact that the moments of X can be derived by taking successive derivatives of the mgf. In particular,

$$m'(t) = \frac{d}{dt} E(e^{tX}) = E\left(\frac{d}{dt} e^{tX}\right) = E(Xe^{tX}),$$

and $m'(0) = E(X)$. In general, the k th derivative of the mgf gives

$$m^{(k)}(0) = E(X^k), \text{ for } k = 1, 2, \dots$$

■ **Example B.14** Let X be a Bernoulli random variable with success parameter p . Find the mgf of X .

Solution The mgf of X is

$$m(t) = E(e^{tX}) = e^{t(1)}p + e^{t(0)}(1-p) = 1-p + pe^t. \quad \blacksquare$$

■ **Example B.15** Let X have a standard normal distribution. Find the mgf.

Solution The mgf is

$$\begin{aligned} m(t) &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x^2-2tx)/2} dx \\ &= e^{t^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2/2} dx = e^{t^2/2}, \end{aligned}$$

as the last integral gives the density of a normal distribution with mean t and variance 1 which integrates to 1. Check that $m'(0) = 0 = E(X)$ and $m''(0) = 1 = E(X^2)$. ■

Here are four key properties of the mgf.

Properties of Moment-Generating Functions

1. If X and Y are independent, then the mgf of $X + Y$ is the product of their respective mgfs. That is,

$$\begin{aligned} m_{X+Y}(t) &= E(e^{t(X+Y)}) = E(e^{tX}e^{tY}) = E(e^{tX})E(e^{tY}) \\ &= m_X(t)m_Y(t). \end{aligned}$$

2. For constant c ,

$$m_{cX}(t) = m_X(ct).$$

3. Moment-generating functions uniquely determine the underlying probability distribution. That is, if $m_X(t) = m_Y(t)$ for all t , then the distributions of X and Y are the same.

4. *Continuity Theorem:* Let $X_1, X_2, \dots$ be a sequence of random variables with corresponding mgfs $m_{X_1}, m_{X_2}, \dots$. Assume that X is a random variable such that for all t , $m_{X_n}(t) \rightarrow m_X(t)$, as $n \rightarrow \infty$. Then,

$$\lim_{n \rightarrow \infty} P(X_n \leq x) = P(X \leq x),$$

at each x for which $P(X \leq x)$ is continuous.

■ **Example B.16** Let X and Y be independent Poisson random variables with respective parameters λ_1 and λ_2 . Use moment-generating functions to show that $X + Y$ has a Poisson distribution with parameter $\lambda_1 + \lambda_2$.

Solution The mgf of a Poisson random variable with parameter λ is

$$m(t) = \sum_{k=0}^{\infty} e^{tk} \frac{e^{-\lambda} \lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

Hence, the mgf of $X + Y$ is

$$m_{X+Y}(t) = m_X(t)m_Y(t) = e^{\lambda_1(e^t-1)}e^{\lambda_2(e^t-1)} = e^{(\lambda_1+\lambda_2)(e^t-1)},$$

which is the mgf of a Poisson random variable with parameter $\lambda_1 + \lambda_2$. ■

APPENDIX C

SUMMARY OF COMMON PROBABILITY DISTRIBUTIONS

DISCRETE DISTRIBUTIONS	
Distribution	PMF, Expectation, Variance
Uniform (1, ... , n)	$P(X = k) = \frac{1}{n}, \quad k = 1, \dots, n$ $E(X) = \frac{n+1}{2} \quad Var(X) = \frac{n^2-1}{12}$
Binomial	$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$ $E(X) = np \quad Var(X) = np(1-p)$
Poisson	$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots$ $E(X) = \lambda \quad Var(X) = \lambda$
Geometric	$P(X = k) = (1-p)^{k-1} p, \quad k = 1, 2, \dots$ $E(X) = \frac{1}{p} \quad Var(X) = \frac{1-p}{p^2}$ $P(X > t) = (1-p)^t$
Negative binomial	$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r},$ $k = r, r+1, \dots, \quad r = 1, 2, \dots$ $E(X) = \frac{r}{p} \quad Var(X) = \frac{r(1-p)}{p^2}$
Hypergeometric	$P(X = k) = \frac{\binom{D}{k} \binom{N-D}{n-k}}{\binom{N}{n}}, k = 0, 1, \dots, n$ $E(X) = \frac{nD}{N} \quad Var(X) = \frac{nD(N-D)}{N^2} \left(1 - \frac{n-1}{N-1}\right)$

CONTINUOUS DISTRIBUTIONS	
Distribution	Density, CDF, Expectation, Variance
Uniform (a, b)	$f(x) = \frac{1}{b-a}, \quad a < x < b$ $F(x) = \frac{x-a}{b-a}, \quad a < x < b$ $E(X) = \frac{a+b}{2} \quad Var(X) = \frac{(b-a)^2}{12}$
Exponential	$f(x) = \lambda e^{-\lambda x}, \quad x > 0$ $F(x) = 1 - e^{-\lambda x}, \quad x > 0$ $E(X) = \frac{1}{\lambda} \quad Var(X) = \frac{1}{\lambda^2}$
Normal	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$ $E(X) = \mu \quad Var(X) = \sigma^2$
Gamma	$f(x) = \lambda e^{-\lambda x} \frac{(\lambda x)^{r-1}}{\Gamma(r)}, \quad x > 0,$ where $\Gamma(r) = \int_0^\infty t^{r-1} e^{-t} dt$ $E(X) = \frac{r}{\lambda} \quad Var(X) = \frac{r}{\lambda^2}$
Beta	$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 < x < 1,$ where $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$ $E(X) = \frac{\alpha}{\alpha+\beta} \quad Var(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$

APPENDIX D

MATRIX ALGEBRA REVIEW

D.1 BASIC OPERATIONS

A *matrix* is a rectangular array of elements. Rectangles are arranged in rows and columns. The general form of an $m \times n$ matrix (m rows and n columns) is

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The element a_{ij} is the entry in the i th row and j th column of A . An $n \times n$ matrix is said to be *square*.

A *row vector* is a $1 \times n$ matrix. A *column vector* is an $n \times 1$ matrix. Matrices are denoted with bold uppercase letters. Vectors are denoted with bold lowercase letters. The i th component of the vector $\mathbf{x}$ is denoted x_i . The Euclidean space $\mathbb{R}^n$ consists of all n -element vectors of real numbers.

Given n -element vectors $\mathbf{x}$ and $\mathbf{y}$, the *dot product*, or *inner product*, of $\mathbf{x}$ and $\mathbf{y}$ is the number

$$\mathbf{x} \cdot \mathbf{y} = x_1 y_1 + \cdots + x_n y_n = \sum_{i=1}^n x_i y_i.$$

A real number is called a *scalar*. If s is a scalar and $\mathbf{A}$ a matrix, then *scalar multiplication* is defined as the product $s\mathbf{A}$, which is the matrix obtained by multiplying each of the elements of $\mathbf{A}$ by s . For example,

$$(-3) \begin{pmatrix} 4 & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -12 & 0 \\ 3 & -3 \end{pmatrix}.$$

Matrix addition is the operation of adding two matrices of the same dimension. Corresponding elements are added. For example,

$$\begin{pmatrix} 3 & 1 & 0 \\ 2 & -4 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 1 & -2 \\ 0 & -4 & 5 \end{pmatrix} = \begin{pmatrix} 5 & 2 & -2 \\ 2 & -8 & 6 \end{pmatrix}.$$

Linear Combination

A *linear combination* of vectors $\mathbf{x}_1, \dots, \mathbf{x}_k$ is a vector of the form

$$s_1\mathbf{x}_1 + \dots + s_k\mathbf{x}_k,$$

where $s_1, \dots, s_k$ are scalars. The s_i are called the *coefficients* of the linear combination.

Observe that $\begin{pmatrix} 8 & -1 & -3 \end{pmatrix}$ is a linear combination of $\begin{pmatrix} 1 & 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 2 & -1 & 1 \end{pmatrix}$, and $\begin{pmatrix} 3 & 3 & 3 \end{pmatrix}$ since

$$\begin{pmatrix} 8 & -1 & -3 \end{pmatrix} = (2) \begin{pmatrix} 1 & 1 & 0 \end{pmatrix} + (3) \begin{pmatrix} 2 & -1 & 1 \end{pmatrix} + (0) \begin{pmatrix} 3 & 3 & 3 \end{pmatrix}.$$

The coefficients of this linear combination are 2, 3, and 0.

We define the matrix–vector product $\mathbf{A}\mathbf{x}$ of an $m \times n$ matrix $\mathbf{A}$ and an $n \times 1$ column vector $\mathbf{x}$. Write the columns of $\mathbf{A}$ as $\mathbf{a}_1, \dots, \mathbf{a}_n$. Then, $\mathbf{A}\mathbf{x}$ is defined as the $m \times 1$ column vector, which is the linear combination of the columns of $\mathbf{A}$ whose coefficients are the components of $\mathbf{x}$. That is,

$$\mathbf{A}\mathbf{x} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

The i th component of $\mathbf{A}\mathbf{x}$ is

$$(\mathbf{A}\mathbf{x})_i = \sum_{j=1}^n a_{ij}x_j, \text{ for } i = 1, \dots, m.$$

Equivalently, the i th component of $\mathbf{A}\mathbf{x}$ is the dot product of the i th row of $\mathbf{A}$ and the vector $\mathbf{x}$. For example,

$$\begin{pmatrix} 2 & 1 & 1 \\ 3 & -1 & 2 \\ 1 & 0 & -4 \\ 4 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2(1) + 1(1) + 1(-2) \\ 3(1) + (-1)(1) + 2(-2) \\ 1(1) + 0(1) + (-4)(-2) \\ 4(1) + 2(1) + 1(-2) \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 9 \\ 4 \end{pmatrix}$$

$$= (1) \begin{pmatrix} 2 \\ 3 \\ 1 \\ 4 \end{pmatrix} + (1) \begin{pmatrix} 1 \\ -1 \\ 0 \\ 2 \end{pmatrix} + (-2) \begin{pmatrix} 1 \\ 2 \\ -4 \\ 1 \end{pmatrix}.$$

D.2 LINEAR SYSTEM

A *linear system* of m equations in n unknowns is a collection of linear equations of the form

$$\begin{array}{ccccccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

Such a system can be written succinctly in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Linear systems can have no solutions, infinitely many solutions, or exactly one solution. For instance, the system

$$\begin{array}{l} 2x_1 + x_2 = 5 \\ 2x_1 + x_2 = 4 \end{array}$$

has no solutions. The system

$$\begin{array}{l} 2x_1 + x_2 = 5 \\ 4x_1 + 2x_2 = 10 \end{array}$$

has infinitely many solutions of the form

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a \\ 5 - 2a \end{pmatrix}, \quad \text{for all real } a.$$

And the system

$$\begin{aligned} 2x_1 + x_2 &= 5 \\ x_1 - x_2 &= -2 \end{aligned}$$

has the unique solution $\mathbf{x} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

D.3 MATRIX MULTIPLICATION

If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is an $n \times p$ matrix, then the matrix product $\mathbf{AB}$ is defined as the $m \times p$ matrix whose i th column is the matrix–vector product of $\mathbf{A}$ and the i th column of $\mathbf{B}$. Writing

$$\mathbf{B} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \\ | & | & \cdots & | \end{pmatrix}$$

gives

$$\mathbf{AB} = \mathbf{A} \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{Ab}_1 & \mathbf{Ab}_2 & \cdots & \mathbf{Ab}_p \\ | & | & \cdots & | \end{pmatrix}.$$

The ij th element of $\mathbf{AB}$ is

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

Equivalently, the ij th element of $\mathbf{AB}$ is the dot product of the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$. For example,

$$\begin{pmatrix} 0 & 1 & 4 \\ 1 & 2 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 1 & 0 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 4 & 3 \end{pmatrix}.$$

Matrix multiplication is not commutative. That is, $\mathbf{AB}$ does not necessarily equal $\mathbf{BA}$.

D.4 DIAGONAL, IDENTITY MATRIX, POLYNOMIALS

Given an $n \times n$ matrix $\mathbf{A}$, the entries $a_{11}, \dots, a_{nn}$ are called the diagonal elements of $\mathbf{A}$. An $n \times n$ matrix $\mathbf{A}$ is a *diagonal matrix* if $a_{ij} = 0$, for all $i \neq j$.

The $n \times n$ *identity matrix*, denoted $\mathbf{I}_n$, is the diagonal matrix all of whose diagonal elements are 1. The columns of the $n \times n$ identity matrix are called the *standard basis vectors of $\mathbb{R}^n$* , denoted $\mathbf{e}_1, \dots, \mathbf{e}_n$. That is, $\mathbf{e}_k$ is the n -element column vector of all 0s except for a 1 in the k th position.

If $\mathbf{A}$ is an $n \times n$ matrix, then $\mathbf{AI}_n = \mathbf{I}_n\mathbf{A} = \mathbf{A}$.

If A is a square matrix, then $AA = A^2$ is also a square matrix. Similarly, A^k is well-defined for all integer k . It follows that if $p(x)$ is a polynomial function and A is a square matrix, then $p(A)$ is well-defined. For instance, let $p(x) = x^3 - 5x + 6$ and $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Then,

$$p(A) = A^3 - 5A + 6I = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix} - (5) \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + (6) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -3 \\ -3 & 7 \end{pmatrix}.$$

D.5 TRANSPOSE

Given an $m \times n$ matrix A , the *transpose* A^T is the $n \times m$ matrix whose ij th element is the ji th element of A .

A matrix A is *symmetric* if $A = A^T$. That is, $a_{ij} = a_{ji}$ for all i, j . A symmetric matrix is necessarily square.

D.6 INVERTIBILITY

A square matrix A is *invertible* if there exists a matrix B such that $AB = BA = I$. The matrix B is denoted A^{-1} and is called *the inverse of A*.

Since $AA^{-1} = I$, it follows that the i th column of A^{-1} is the solution of the linear system $Ax = e_i$, where e_i is the i th standard basis vector.

The solution of a general linear system $Ax = b$ is unique if and only if A is invertible. In that case, the solution is $x = A^{-1}b$.

Properties of Inverse, Transpose

1. $(A^T)^T = A$
2. $(A^{-1})^{-1} = A$
3. $(A^T)^{-1} = (A^{-1})^T$
4. $(AB)^T = B^T A^T$
5. $(AB)^{-1} = B^{-1} A^{-1}$

D.7 BLOCK MATRICES

It is sometimes convenient to partition a matrix A into smaller *blocks*, such as

$$A = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ \hline 13 & 14 & 15 & 16 \end{array} \right) = \left(\begin{array}{c|c} B & C \\ \hline D & E \end{array} \right),$$

where

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 9 & 10 & 11 \end{pmatrix}, \quad C = \begin{pmatrix} 4 \\ 8 \\ 12 \end{pmatrix}, \quad D = (13 \ 14 \ 15), \quad \text{and} \quad E = (16).$$

Matrix operations on block matrices can be carried out by treating the blocks as matrix elements. Thus,

$$\begin{aligned} A^2 &= \left(\begin{array}{c|c} B & C \\ \hline D & E \end{array} \right) \left(\begin{array}{c|c} B & C \\ \hline D & E \end{array} \right) \\ &= \left(\begin{array}{c|c} B^2 + CD & BC + CE \\ \hline DB + ED & DC + E^2 \end{array} \right) \\ &= \left(\begin{array}{ccc|c} 90 & 100 & 110 & 120 \\ 202 & 228 & 254 & 280 \\ 314 & 356 & 398 & 440 \\ \hline 426 & 484 & 542 & 600 \end{array} \right). \end{aligned}$$

D.8 LINEAR INDEPENDENCE AND SPAN

Given a set of vectors, if at least one vector can be written as a linear combination of the others, then the vectors are called *linearly dependent*. If none of the vectors in the set can be written as a linear combination of the other vectors, then the vectors are called *linearly independent*.

The vectors $\left\{ \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 \\ -3 \\ 7 \end{pmatrix} \right\}$ are linearly dependent, as

$$\begin{pmatrix} 4 \\ -3 \\ 7 \end{pmatrix} = (2) \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + (-1) \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}.$$

The vectors $\{e_1, e_2, e_3\} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ are linearly independent.

The *span* of a set of vectors is the set of all linear combinations of those vectors.

The span of $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ is the set of all vectors of the form

$$a \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ b \\ 0 \end{pmatrix}, \quad \text{for scalars } a \text{ and } b.$$

Geometrically, this set is the x - y plane in $\mathbb{R}^3$.

D.9 BASIS

A *basis* for $\mathbb{R}^n$ is a set of n vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ in $\mathbb{R}^n$, which are linearly independent and span $\mathbb{R}^n$. The fact that $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ span $\mathbb{R}^n$ means that every vector in $\mathbb{R}^n$ can be written as a linear combination of the $\mathbf{b}_i$. Together with linear independence we obtain the following result.

Theorem D.1. *If $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ is a basis for $\mathbb{R}^n$, then every vector in $\mathbb{R}^n$ can be written uniquely as a linear combination of the $\mathbf{b}_i$.*

To obtain this unique representation for a given set of vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, let $\mathbf{B}$ be the square matrix obtained by making the $\mathbf{b}_i$ the columns of $\mathbf{B}$. That is,

$$\mathbf{B} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & \cdots & | \end{pmatrix}.$$

The matrix $\mathbf{B}$ is invertible. If $\mathbf{x} = s_1\mathbf{b}_1 + \cdots + s_n\mathbf{b}_n$ for some choice of $s_1, \dots, s_n$, then

$$\mathbf{x} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} s_1 \\ \vdots \\ s_n \end{pmatrix} = \mathbf{B} \begin{pmatrix} s_1 \\ \vdots \\ s_n \end{pmatrix}$$

and thus

$$\begin{pmatrix} s_1 \\ \vdots \\ s_n \end{pmatrix} = \mathbf{B}^{-1}\mathbf{x}.$$

D.10 VECTOR LENGTH

The *length* (also magnitude or norm) of a vector $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}.$$

A vector $\mathbf{x}$ has unit length if $\|\mathbf{x}\| = 1$. For any nonzero vector $\mathbf{v}$, the vector $\left(\frac{1}{\|\mathbf{v}\|}\right)\mathbf{v}$ has unit length, since

$$\left\| \left(\frac{1}{\|\mathbf{v}\|} \right) \mathbf{v} \right\| = \left(\frac{1}{\|\mathbf{v}\|} \right) \|\mathbf{v}\| = 1.$$

A most important inequality in linear algebra is the Cauchy–Schwarz inequality, which says that for any n -element vectors $\mathbf{x}$ and $\mathbf{y}$,

$$|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|.$$

For example, letting $\mathbf{x} = (1 \ \cdots \ 1)$ and $\mathbf{y} = (y_1 \ \cdots \ y_n)$, the inequality yields

$$|y_1 + \cdots + y_n| \leq \sqrt{n} \sqrt{y_1^2 + \cdots + y_n^2}.$$

Equivalently,

$$(y_1 + \cdots + y_n)^2 \leq n(y_1^2 + \cdots + y_n^2).$$

D.11 ORTHOGONALITY

Vectors $\mathbf{x}$ and $\mathbf{y}$ are said to be *orthogonal* if $\mathbf{x} \cdot \mathbf{y} = 0$. Geometrically, orthogonal vectors are perpendicular.

A set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is *orthonormal* if the vectors have unit length and are pairwise orthogonal. That is,

$$\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

A square matrix whose columns are orthonormal is called an *orthogonal matrix*. If $\mathbf{U}$ is an orthogonal matrix, then $\mathbf{U}$ is invertible and $\mathbf{U}^{-1} = \mathbf{U}^T$. The latter follows from the definition of orthonormal vectors since the ij th element of $\mathbf{U}^T \mathbf{U}$ is the dot product of the i th column of $\mathbf{U}$ and the j th column of $\mathbf{U}$.

The matrix

$$\mathbf{U} = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} & -\sqrt{2/3} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix}$$

is an orthogonal matrix as

$$\begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{6} & -\sqrt{2/3} \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{6} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{6} & 1/\sqrt{2} \\ 1/\sqrt{3} & -\sqrt{2/3} & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

D.12 EIGENVALUE, EIGENVECTOR

Let $\mathbf{A}$ be a square matrix. If there exists a scalar λ and nonzero (column) vector $\mathbf{x}$ such that $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$, we say that λ is an *eigenvalue* of $\mathbf{A}$ with corresponding *eigenvector* $\mathbf{x}$.

For example, let $\mathbf{A} = \begin{pmatrix} 3 & 0 \\ 1 & -4 \end{pmatrix}$. Observe that

$$\begin{pmatrix} 3 & 0 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -4 \end{pmatrix} = (-4) \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

which shows that $\lambda = -4$ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Further observe that

$$\begin{pmatrix} 3 & 0 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} 7 \\ 1 \end{pmatrix} = \begin{pmatrix} 21 \\ 3 \end{pmatrix} = (3) \begin{pmatrix} 7 \\ 1 \end{pmatrix},$$

which shows that $\lambda = 3$ is an eigenvalue of $\mathbf{A}$ with corresponding eigenvector $\begin{pmatrix} 7 \\ 1 \end{pmatrix}$.

One can show that $\lambda = -4$ and $\lambda = 3$ are the only eigenvalues of $\mathbf{A}$.

D.13 DIAGONALIZATION

A square matrix $\mathbf{A}$ is *diagonalizable* if there exists an invertible matrix $\mathbf{S}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{SDS}^{-1}$. If $\mathbf{A}$ is diagonalizable then the entries of $\mathbf{D}$ are the eigenvalues of $\mathbf{A}$ and the columns of $\mathbf{S}$ are corresponding eigenvectors.

For example, let $\mathbf{A} = \begin{pmatrix} 6 & -3 & 7 \\ 6 & -3 & 6 \\ 0 & 0 & -1 \end{pmatrix}$. The eigenvalues of $\mathbf{A}$ are $\lambda = 3, -1, 0$, with respective eigenvectors $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}$.

Let $\mathbf{S} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}$, with $\mathbf{S}^{-1} = \begin{pmatrix} 2 & -1 & 2 \\ 0 & 0 & 1 \\ -1 & 1 & -1 \end{pmatrix}$. This gives

$$\begin{aligned} \mathbf{SDS}^{-1} &= \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ 0 & 0 & 1 \\ -1 & 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 6 & -3 & 7 \\ 6 & -3 & 6 \\ 0 & 0 & -1 \end{pmatrix} = \mathbf{A}. \end{aligned}$$

That is, $\mathbf{A}$ is diagonalizable.

A sufficient condition for an $n \times n$ matrix to be diagonalizable is that there exists n distinct eigenvalues. The following theorem gives a necessary and sufficient condition for diagonalizability.

Theorem D.2. *An $n \times n$ matrix $\mathbf{A}$ is diagonalizable if and only if there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of $\mathbf{A}$.*

One advantage of diagonalizability is that it simplifies matrix products. If $\mathbf{A}$ is diagonalizable, then

$$\mathbf{A}^k = (\mathbf{S}\mathbf{D}\mathbf{S}^{-1})^k = \mathbf{S}\mathbf{D}^k\mathbf{S}^{-1}.$$

In the previous example, for $k \geq 1$,

$$\begin{aligned}\mathbf{A}^k &= \mathbf{S}\mathbf{D}^k\mathbf{S}^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 3^k & 0 & 0 \\ 0 & (-1)^k & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & -1 & 2 \\ 0 & 0 & 1 \\ -1 & 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} (2)3^k & -3^k & (2)3^k - (-1)^k \\ (2)3^k & -3^k & (2)3^k \\ 0 & 0 & (-1)^k \end{pmatrix}.\end{aligned}$$

A square matrix $\mathbf{A}$ is *orthogonally diagonalizable* if there exists an orthogonal matrix $\mathbf{U}$ and a diagonal matrix $\mathbf{D}$ such that $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^T$.

Theorem D.3 (*Spectral Theorem*). A matrix is orthogonally diagonalizable if and only if it is symmetric.

ANSWERS TO SELECTED ODD-NUMBERED EXERCISES

Solutions for Chapter 1

- 1.1** (b) X_t is the student's status at the end of year t .
 State space (discrete): $S = \{\text{Drop Out, Frosh, Sophomore, Junior, Senior, Graduate}\}$.
 Index set (discrete): $I = \{0, 1, 2, \dots\}$.
 (e) X_t is the arrival time of student t .
 State space (continuous): $[0, 60]$
 Index set (discrete): $\{1, 2, \dots, 30\}$.
 (f) X_t is the order of the deck of cards after t shuffles.
 State space (discrete): Set of all orderings of the deck ($52!$ elements).
 Index set (discrete): $\{0, 1, 2, \dots\}$

$$\begin{aligned}
 \mathbf{1.3} \quad \sum_{i=1}^k P(A|B_i \cap C)P(B_i|C) &= \sum_{i=1}^k \left(\frac{P(A \cap B_i \cap C)}{P(B_i \cap C)} \right) \left(\frac{P(B_i \cap C)}{P(C)} \right) \\
 &= \sum_{i=1}^k \frac{P(A \cap B_i \cap C)}{P(C)} = \frac{1}{P(C)} \sum_{i=1}^k P(A \cap B_i \cap C) \\
 &= \frac{P(A \cap C)}{P(C)} = P(A|C).
 \end{aligned}$$

- 1.5** (a) Uniform on $\{1, 2, 3, 4, 5, 6\}$.
 (b) Uniform on $\{2, 3, 4, 5, 6\}$.

- 1.7** Let X denote the time until the rat finds the cheese. Let 1, 2, and 3 denote each door, respectively. Then,

$$\begin{aligned} E(X) &= E(X|1)P(1) + E(X|2)P(2) + E(X|3)P(3) \\ &= (2 + E(X))\frac{1}{3} + (3 + E(X))\frac{1}{3} + (1)\frac{1}{3} \\ &= 2 + E(X)\frac{2}{3}. \end{aligned}$$

Thus, $E(X) = 6$ minutes.

- 1.11** Let x_k be the probability of reaching n when the gambler's fortune is k . As in Example 1.10.

$$x_k = x_{k+1}p + x_{k-1}q, \text{ for } 1 \leq k \leq n-1,$$

with $x_0 = 0$ and $x_n = 1$, which gives

$$x_{k+1} - x_k = (x_k - x_{k-1})\frac{p}{q}, \text{ for } 1 \leq k \leq n-1.$$

It follows that

$$x_k - x_{k-1} = \cdots = (x_1 - x_0)(p/q)^{k-1} = x_1(p/q)^{k-1}, \text{ for all } k.$$

This gives $x_k - x_1 = \sum_{i=2}^k x_1(p/q)^{i-1}$, or

$$x_k = \sum_{i=1}^k x_1(p/q)^{i-1} = x_1 \frac{1 - (p/q)^k}{1 - p/q}.$$

For $k = n$, this gives

$$1 = x_n = x_1 \frac{1 - (p/q)^n}{1 - p/q}.$$

Thus, $x_1 = (1 - p/q)/(1 - (p/q)^n)$, which gives

$$x_k = \frac{1 - (p/q)^k}{1 - (p/q)^n}, \text{ for } k = 0, \dots, n.$$

- 1.13** (a) $f_{Y|X}(y|x) = 2y/(1 - x^2)$, for $x < y < 1$.
 (b) The conditional distribution is uniform on $(0, y)$.

- 1.15** The area of the circle is π . The equation of the circle is $x^2 + y^2 = 1$. The joint density is

$$f(x, y) = \frac{1}{\pi}, \quad \text{for } -1 < x < 1, -\sqrt{1-x^2} < y < \sqrt{1-x^2}$$

Integrating out the y term gives the marginal density

$$f_X(x) = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy = \frac{2\sqrt{1-x^2}}{\pi}, \quad \text{for } -1 < x < 1.$$

The conditional density is

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{1/\pi}{2\sqrt{1-x^2}/\pi} = \frac{1}{2\sqrt{1-x^2}},$$

for $-1\sqrt{1-x^2} < y < \sqrt{1-x^2}$. The conditional distribution of Y given $X = x$ is uniform on $(-\sqrt{1-x^2}, \sqrt{1-x^2})$.

- 1.17** $E(X|X > 2) = 4.16525$.

1.21

$$\begin{aligned} \int_0^\infty P(T > t) dt &= \int_0^\infty \int_t^\infty f(s) ds dt = \int_0^\infty \int_0^s f(s) dt ds \\ &= \int_0^\infty sf(s) ds = E(T). \end{aligned}$$

- 1.23** (b) For $m > n$,

$$\begin{aligned} E(S_m|S_n) &= E(S_n + X_{n+1} + \cdots + X_m|S_n) \\ &= E(S_n|S_n) + E(X_{n+1} + \cdots + X_m|S_n) \\ &= S_n + \sum_{i=n+1}^m E(X_i|S_n) = S_n + \sum_{i=n+1}^m E(X_i) \\ &= S_n + (m-n)\mu. \end{aligned}$$

- 1.27** Let T be the total amount spent at the restaurant. Then,

$$E(T) = 200(15) = \$3000,$$

and

$$\text{Var}(T) = 9(200) + 15^2(40^2) = 361800, \quad SD(T) = \$601.50.$$

- 1.29** Yes.

Solutions for Chapter 2

- 2.1** (a) 0.6;
 (b) $P_{32}^2 = 0.27$;
 (c) $P_{31}\alpha_3/(\alpha P)_1 = (0.3)(0.5)/(0.17) = 15/17 = 0.882$;
 (d) $(0.182, 0.273, 0.545) \cdot (1, 2, 3) = 2.363$.

2.3 $P_{10}^3 = 0.517$.

- 2.5** (a)

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1/4 & 0 & 3/4 & 0 \\ 0 & 1/4 & 0 & 3/4 \\ 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

(b) $19/64$.

(c) 0.103 .

- 2.9**

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{pmatrix} 0 & 0 & 3/5 & 0 & 2/5 \\ 1/7 & 2/7 & 0 & 0 & 4/7 \\ 0 & 2/9 & 2/3 & 1/9 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 3/4 & 0 & 0 & 1/4 & 0 \end{pmatrix} \end{matrix}.$$

2.11 (b) $P_{0.5}^3 = 0.01327$.

2.25

Socializing	Traveling	Milling	Feeding	Resting
0.148	0.415	0.096	0.216	0.125

Solutions for Chapter 3

3.1 $\pi = \left(\frac{1}{5}, \frac{1}{3}, \frac{2}{5}, \frac{1}{15} \right).$

3.3 P and R are regular. Q is regular for $0 < p < 1$.

3.5 (a) All non-negative vectors of the form (a, a, b, c, a) , where $3a + b + c = 1$.

3.7 The transition matrix is doubly stochastic. The stationary distribution is uniform.

3.11 (a)
$$\pi_j = \begin{cases} 1/(2k), & \text{if } j = 0, k, \\ 1/k, & \text{if } j = 1, \dots, k-1. \end{cases}$$

(b) 2,000 steps.

- 3.13** Communication classes are $\{4\}$ (recurrent, absorbing); $\{1, 5\}$ (recurrent); $\{2, 3\}$ (transient). All states have period 1.

$$P = \begin{matrix} & \begin{matrix} 2 & 3 & 1 & 5 & 4 \end{matrix} \\ \begin{matrix} 2 \\ 3 \\ 1 \\ 5 \\ 4 \end{matrix} & \begin{pmatrix} 1/2 & 1/6 & 1/3 & 0 & 0 \\ 1/4 & 0 & 0 & 1/4 & 1/2 \\ 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

3.17 $P^n = \begin{pmatrix} 1/2^n & 1 - 1/2^n \\ 0 & 1 \end{pmatrix}$ and $\lim_{n \rightarrow \infty} P^n = \begin{pmatrix} 2 & +\infty \\ 0 & +\infty \end{pmatrix}.$

- 3.19** Let p_x be the expected time to hit d for the walk started in x . By symmetry, $p_b = p_e$ and $p_c = p_e$. Solve

$$p_a = \frac{1}{2}(1 + p_b) + \frac{1}{2}(1 + p_c),$$

$$p_b = \frac{1}{2}(1 + p_a) + \frac{1}{2}(1 + p_c),$$

$$p_c = \frac{1}{4}(1 + p_b) + \frac{1}{4} + \frac{1}{4}(1 + p_a) + \frac{1}{4}(1 + p_c).$$

This gives $p_a = 10$.

3.23 $\pi_i = 2(k+1-i)/(k(k+1)),$ for $i = 1, \dots, k$.

- 3.25** (a) For $k = 2$, $\pi = (1/6, 2/3, 1/6)$. For $k = 3$, $\pi = (1/20, 9/20, 9/20, 1/20)$.

- 3.29** Communication classes are: (i) $\{a\}$ transient; (ii) $\{e\}$ recurrent; (iii) $\{c, d\}$ transient; and (iv) $\{b, f, g\}$ recurrent. The latter class has period 2. All other states have period 1.

- 3.33** For all states i and j , and $m > 0$,

$$P_{ij}^{N+m} = \sum_k P_{ik}^m P_{kj}^N.$$

Since $P_{kj}^N > 0$ for all k , the only way the expression above could be zero is if $P_{ik}^m = 0$ for all k , which is not possible since P^m is a stochastic matrix whose rows sum to 1.

- 3.43** (a) The chain is ergodic for all $0 \leq p, q \leq 1$, except $p = q = 0$ and $p = q = 1$.
 (b) The chain is reversible for all $p = q$, with $0 < p < 1$.

3.47

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1/3 & 4/9 & 2/9 \\ 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix} \end{matrix}.$$

3.53 (a) The probability that A wins is $1/(2-p)$.

(c) $\alpha = 1270/6049 \approx 0.210$ and $\beta = 737/6049 \approx 0.122$. For the first method, A wins with probability 0.599. For the second method, A wins with probability 0.565.

3.61 Yes. T is a stopping time.

3.63

$$\pi = (0.325, 0.207, 0.304, 0.132, 0.030, .003, .0003).$$

```
3.67 > allsixes <- function() {
+ i <- 0
+ ct <- 0
+ while (ct < 5)
+ { x <- sample(1:6, 5-ct, replace=T)
+ sixes <- sum(x==6)
+ ct <- ct + sixes
+ i <- i+1 }
+ i
+ }
> sim <- replicate(10000, allsixes())
> mean(sim)
[1] 13.0873
```

Solutions for Chapter 4

4.1 $P_{0,j} = 1$, if $j = 0$, and 0, otherwise.

$$P_{1,j} = \begin{cases} a, & \text{if } j = 0, \\ b, & \text{if } j = 1, \\ c, & \text{if } j = 2. \end{cases}$$

For the second row of P ,

$$\begin{matrix} & 0 & 1 & 2 & 3 & 4 & 5 & \cdots \\ 2 & (a^2 & 2ab & 2ac + b^2 & 2bc & c^2 & 0 & \cdots) \end{matrix}$$

4.3 From Exercise 4.2, $G_X(s) = e^{\lambda(s-1)}$ and $G_Y(s) = e^{\mu(s-1)}$. Then,

$$G_{X+Y}(s) = G_X(s)G_Y(s) = e^{\lambda(s-1)}e^{\mu(s-1)} = e^{(\lambda+\mu)s-1}.$$

Thus, $X + Y$ has a Poisson distribution with parameter $\lambda + \mu$.

4.7 $G(s) = (1 - p) + ps^3$.

$$\mu = G'(1) = 3p.$$

$$\sigma^2 = G''(1) + G'(1) - G'(1)^2 = 6p + 3p - (3p)^2 = 9p(1 - p).$$

$$E(Z^4) = \mu^4 = (3p)^4 = 81p^4.$$

$$\text{Var}(Z^4) = 9p(1 - p)(3p)^3(81p^4 - 1)/(3p - 1).$$

4.11 $E(Z_n) = G'1_n(1) = G'(G_{n-1}(1))G'_{n-1}(1) = G'(1)G'_{n-1}(1)$
 $= \mu G'_{n-1}(1) = \mu E(Z_{n-1}).$

The result follows by induction.

4.15 Solve $s = 1/4 + s/4 + s^2/4 + s^3/4$. $e = 0.414$.

4.17 (a)

$$e = \begin{cases} (1 - p)/p, & \text{if } p > 1/2, \\ 1, & \text{if } p \leq 1/2. \end{cases}$$

(b)

$$P_{2i,2j} = \binom{2i}{2j} p^{2j} (1 - p)^{2i}.$$

4.21 (a) $\mu = c/(1 - p)^2$
 (c) 0.693, 0.803

4.23

$$G_Z(s) = \frac{1}{a_0}(G(s) - a_0).$$

4.29 (a)

```
> pgf <- function(s) {
+   0.8 + 0.1*s^4 + 0.1*s^9
+ }
> x <- 0.5 # initial value
> e <- pgf(x)
> for (i in 1:100) {
+   e <- pgf(e)
+ }
> e
[1] 0.9152025
```

(b) 0.101138

```

4.33 > branch <- function(n,lam) { ## Poisson
+   z <- c(1,rep(0,n))
+   for (i in 2:(n+1)) {
+     z[i] <- sum(rpois(z[i-1],lam))
+   }
+   return(z) }
+   # Assume extinction occurs by 50th generation
> n <- 10000
> simlist <- replicate(n, sum(branch(50,0.60)))
> mean(simlist)
[1] 2.5308
> var(simlist)
[1] 9.594811

```

Solutions for Chapter 5

5.1 Let

$$P = \begin{matrix} & \begin{matrix} \text{Truck} & \text{Car} \end{matrix} \\ \begin{matrix} \text{Truck} \\ \text{Car} \end{matrix} & \begin{pmatrix} 1/5 & 4/5 \\ 1/4 & 3/4 \end{pmatrix}, \end{matrix}$$

with stationary distribution $\pi = (5/21, 16/21)$. By the strong law of large numbers the toll collected is about

$$1000 \left(5 \left(\frac{5}{21} \right) + 1.5 \left(\frac{16}{21} \right) \right) = \$2333.33.$$

5.3 (a) Compute $(10000) \times \lambda P^2$, with $\lambda = (0.6, 0.3, 0.1)$. This gives Car: 3,645, Bus: 4,165, Bike: 2,190.

For long-term totals, find the stationary distribution and compute $(10000) \times \pi$ to get Car: 2083, Bus: 4583, Bike: 3333.

(b) Current: $271(0.6) + 101(0.3) + 21(0.1) = 195$ g. Long-term: $271(0.208) + 101(0.458) + 21(0.333) = 109.75$ g.

5.7 Assume that the chain is currently at state i . Let j be the proposal state, chosen uniformly on $\{0, 1, \dots, n\}$. Let $U \sim \text{Uniform}(0, 1)$. Accept j as the next state of the chain if

$$U < \frac{\binom{n}{j} p^j (1-p)^{n-j}}{\binom{n}{i} p^i (1-p)^{n-i}} = \frac{i!(n-i)!}{j!(n-j)!} \left(\frac{p}{1-p} \right)^{j-i}.$$

Otherwise, stay at state i .

5.15

$$P = \begin{matrix} & \begin{matrix} 123 & 132 & 213 & 231 & 312 & 321 \end{matrix} \\ \begin{matrix} 123 \\ 132 \\ 213 \\ 231 \\ 312 \\ 321 \end{matrix} & \begin{pmatrix} 1/2 & 1/8 & 1/8 & 1/8 & 1/8 & 0 \\ 1/8 & 1/2 & 1/8 & 0 & 1/8 & 1/8 \\ 1/8 & 1/8 & 1/2 & 1/8 & 0 & 1/8 \\ 1/8 & 0 & 1/8 & 1/2 & 1/8 & 1/8 \\ 1/8 & 1/8 & 0 & 1/8 & 1/2 & 1/8 \\ 0 & 1/8 & 1/8 & 1/8 & 1/8 & 1/2 \end{pmatrix} \end{matrix}.$$

```

5.19 > trials <- 20000
> n <- 50
> p <- 1/4
> sim <- numeric(trials)
> for (k in 1:trials) {
+   state <- 0
+   # run chain for 60 steps to be near stationarity
+   for (i in 1:60) {
+     y <- sample(0:n,1)
+     acc <- factorial(state)*factorial(n-state)/
+       (factorial(y)*factorial(n-y))
+     *(p/(1-p))^(y-state)
+     if (runif(1) < acc) state <- y
+   }
+   sim[k] <- if (state >= 10 & state <= 15) 1 else 0
+ }
> mean(sim) # estimate of P(10 <= X <= 15)
[1] 0.6712
# exact probability
> pbinom(15,n,p)-pbinom(9,n,p)
[1] 0.6732328

```

Solutions for Chapter 6

- 6.1** (a) 0.048;
 (b) 0.1898;
 (c) 0.297.
- 6.3** (a) 0.082;
 (b) 0.0257;
 (c) 0.01299.
- 6.7** (a) 1/2;
 (b) 1/4;
 (c) 1/6.

- 6.9** Let X be geometrically distributed with parameter p . The cumulative distribution function for X is

$$P(X \leq x) = \sum_{k=1}^x P(X = k) = \sum_{k=1}^x (1-p)^{k-1} p = p \frac{1 - (1-p)^x}{1 - (1-p)} = 1 - (1-p)^x.$$

This gives,

$$P(X > s+t | X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{(1-p)^{s+t}}{(1-p)^s} = (1-p)^t = P(X > t).$$

- 6.11** Let X be a memoryless, continuous random variable. Let $g(t) = P(X > t)$. By memorylessness,

$$P(X > t) = P(X > s+t | X > s) = \frac{P(X > s+t)}{P(X > s)}.$$

Thus, $g(s+t) = g(s)g(t)$. It follows that

$$g(t_1 + \cdots + t_n) = g(t_1) \cdots g(t_n).$$

Let $r = p/q = \sum_{i=1}^p (1/q)$. Then, $g(r) = (g(1/q))^p$. Also,

$$g(1) = g\left(\sum_{i=1}^q \frac{1}{q}\right) = g\left(\frac{1}{q}\right)^q,$$

or $g(1/q) = g(1)^{1/q}$. This gives

$$g(r) = g(1)^{p/q} = g(1)^r = e^{r \ln g(1)},$$

for all rational r . By continuity, for all $t > 0$, $g(t) = e^{-\lambda t}$, where

$$\lambda = -\ln g(1) = -\ln P(X > 1).$$

- 6.15** (a) 0.112.
 (b) 0.472.
 (c) 0.997.

6.17

$$E\left(\sum_{n=1}^{N_t} S_n^2\right) = \frac{\lambda t^3}{3}.$$

6.19 $E(T) = 88.74$.

6.25 The expected time of failure was 8:43 a.m. on the last day of the week.

- 6.27** (a) 0.747;
 (b) 0.632;
 (c) 0.20.

6.29 0.77.

6.35 $P(N_C = 0) = e^{(1-e^{-1})\pi} = 0.137.$

6.39 $P(N_1 = 1) = (e - 2)/e = 0.264.$

- 6.41** The goal scoring Poisson process has parameter $\lambda = 2.68/90$. Consider two independent thinned processes, each with parameter $p\lambda$, where $p = 1/2$. By conditioning on the number of goals scored in a 90-minute match, the desired probability is

$$\sum_{k=0}^{\infty} \left(\frac{e^{-90\lambda/2} (90\lambda/2)^k}{k!} \right)^2 = \sum_{k=0}^{\infty} \left(\frac{e^{-1.34} 1.34^k}{k!} \right)^2 = 0.259.$$

6.43 Mean and variance are 41.89.

Solutions for Chapter 7

7.3

$$Q = \begin{pmatrix} -a & a/2 & a/2 \\ b/2 & -b & b/2 \\ c/2 & c/2 & -c \end{pmatrix}.$$

$$\pi = \left(\frac{bc}{ac + bc + ab}, \frac{ac}{ac + bc + ab}, \frac{ab}{ac + bc + ab} \right).$$

7.7 (a)

$$P'_{11}(t) = -P_{11}(t) + 3P_{13}(t)$$

$$P'_{12}(t) = -2P_{12}(t) + P_{11}(t)$$

$$P'_{13}(t) = -3P_{13}(t) + 2P_{12}(t)$$

$$P'_{21}(t) = -P_{21}(t) + 3P_{23}(t)$$

$$P'_{22}(t) = -2P_{22}(t) + P_{21}(t)$$

$$P'_{23}(t) = -3P_{23}(t) + 2P_{22}(t)$$

$$P'_{31}(t) = -P_{31}(t) + 3P_{33}(t)$$

$$P'_{32}(t) = -2P_{32}(t) + P_{31}(t)$$

$$P'_{33}(t) = -3P_{33}(t) + 2P_{32}(t)$$

(b)

$$\begin{aligned}
 \mathbf{P}(t) &= \begin{pmatrix} -1 & 0 & 1 \\ -3 & 1 & 1 \\ 3 & 0 & 1 \end{pmatrix} \begin{pmatrix} e^{-4t} & 0 & 0 \\ 0 & e^{-2t} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1/4 & 0 & 1/4 \\ -3/2 & 1 & 1/2 \\ 3/4 & 0 & 1/4 \end{pmatrix} \\
 &= \frac{1}{4} \begin{pmatrix} 3 + e^{-4t} & 0 & 1 - e^{-4t} \\ 3 + 3e^{-4t} - 6e^{-2t} & 4e^{-2t} & 1 - 3e^{-4t} + 2e^{-2t} \\ 3 - 3e^{-4t} & 0 & 1 + 3e^{-4t} \end{pmatrix}
 \end{aligned}$$

7.11

$$\begin{aligned}
 \frac{d}{dt} e^{tA} &= \frac{d}{dt} \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n = \sum_{n=0}^{\infty} \frac{1}{n!} A^n \frac{d}{dt} t^n = \sum_{n=1}^{\infty} \frac{1}{n!} A^n n t^{n-1} \\
 &= A \sum_{n=1}^{\infty} \frac{t^{n-1}}{(n-1)!} A^{n-1} = A \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n = A e^{tA}.
 \end{aligned}$$

The second equality is done similarly.

7.13 Taking limits on both sides of $\mathbf{P}'(t) = \mathbf{P}(t)\mathbf{Q}$ gives that $\mathbf{0} = \boldsymbol{\pi}\mathbf{Q}$. This uses the fact that if a differentiable function $f(t)$ converges to a constant then the derivative $f'(t)$ converges to 0.

7.15 (a)

$$\mathbf{Q} = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 1 & 0 & 0 & -1 \end{pmatrix} \end{matrix}.$$

7.17 The population process is a Yule process. The distribution of X_8 , the size of the population at $t = 8$, is negative binomial, with mean and variance

$$E(X_8) = 651,019 \quad \text{and} \quad SD(X_8) = 325,509.$$

7.21 Make 4 an absorbing state. We have

$$(-V)^{-1} = \left(- \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 2 & -3 & 1 \\ 0 & 0 & 3 & -4 \end{pmatrix} \right)^{-1} = \begin{pmatrix} 10 & 9 & 4 & 1 \\ 9 & 9 & 4 & 1 \\ 8 & 8 & 4 & 1 \\ 6 & 6 & 3 & 1 \end{pmatrix},$$

with row sums (24, 23, 21, 16). The desired mean time is 23.

7.25 (a) $\boldsymbol{\psi} = (0.1, 0.3, 0.3, 0.3)$.

(b) We have that $(q_1, q_2, q_3, q_4) = (1, 1/2, 1/3, 1/4)$. The stationary distribution $\boldsymbol{\pi}$ is proportional to $(0.1, 0.3(2), 0.3(3), 0.3(4))$. This gives

$$\boldsymbol{\psi} = \frac{1}{2.8} (0.1, 0.6, 0.9, 1.2) = (0.036, 0.214, 0.321, 0.428).$$

7.27 (a) If the first dog has i fleas, then the number of fleas on the dog increases by one the first time that one of the $N - i$ fleas on the other dog jumps. The time of that jump is the minimum of $N - i$ independent exponential random variables with parameter λ . Similarly, the number of fleas on the first dog decreases by one when one of the i fleas on that dog first jumps.

(b) The local balance equations are $\pi_i(N - i)\lambda = \pi_{i+1}(i + 1)\lambda$. The equations are satisfied by the stationary distribution

$$\pi_k = \binom{N}{k} \left(\frac{1}{2}\right)^k, \text{ for } k = 0, 1, \dots, N,$$

which is a binomial distribution with parameters N and $p = 1/2$.

(c) 0.45 minutes.

7.29 (b) The embedded chain transition matrix, in canonical form, is

$$\tilde{\mathbf{P}} = \begin{matrix} & \begin{matrix} 1 & 5 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 5 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 2/3 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 3/5 & 0 & 2/5 & 0 \end{pmatrix} \end{matrix}.$$

By the discrete-time theory for absorbing Markov chains, write

$$\tilde{\mathbf{Q}} = \begin{pmatrix} 0 & 2/3 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 2/5 & 0 \end{pmatrix} \quad \text{and} \quad \tilde{\mathbf{R}} = \begin{pmatrix} 1/3 & 0 \\ 0 & 0 \\ 0 & 3/5 \end{pmatrix}.$$

The matrix of absorption probabilities is

$$(\mathbf{I} - \tilde{\mathbf{Q}})^{-1} \tilde{\mathbf{R}} = \begin{matrix} & \begin{matrix} 1 & 5 \end{matrix} \\ \begin{matrix} 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 4/7 & 3/7 \\ 5/14 & 9/14 \\ 1/7 & 6/7 \end{pmatrix} \end{matrix}.$$

The desired probability is $3/7$.

7.31 The process is an M/M/2 queue with $\lambda = 2$, $\mu = 3$, and $c = 2$. The desired probability is

$$\pi_0 = \left(1 + \frac{2}{3} + \frac{1}{3}\right)^{-1} = \frac{1}{2}$$

7.33 (a) The long-term expected number of customers in the queue L is the mean of a geometric distribution on $0, 1, 2, \dots$, with parameter $1 - \lambda/\mu$, which is $\lambda/(\mu - \lambda)$. If both λ and μ increase by a factor of k , this does not change the value of L .

(b) The expected waiting time is $W = L/\lambda$. The new waiting time is $L/(k\lambda) = W/k$.

- 7.37** (c) Choose N such that $P(Y > N) < 0.5 \times 10^{-3}$, where Y is a Poisson random variable with parameter $9 \times 0.8 = 7.2$. This gives $N = 17$.

Solutions for Chapter 8

- 8.3** (a) 0.013.

(b) $f_{X_2|X_1}(x|0) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, for $-\infty < x < \infty$.

- (c) 3.

- (d) X_1 .

- 8.5** (a) For the joint density of B_s and B_t , since

$$\{B_s = x, B_t = y\} = \{B_s = x, B_t - B_s = y - x\},$$

it follows that

$$f_{B_s, B_t}(x, y) = f_{B_s}(x) f_{B_t - B_s}(y - x) = \frac{1}{\sqrt{2\pi s}} e^{-x^2/2s} \frac{1}{\sqrt{2\pi(t-s)}} e^{-(y-x)^2/2(t-s)}.$$

- (b) $E(B_s|B_t = y) = sy/t$ and $\text{Var}(B_s|B_t = y) = s(t-s)/t$.

- 8.7** One checks that the reflection is a Gaussian process with continuous paths. Furthermore, the mean function is $E(-B_t) = 0$ and the covariance function is

$$E((-B_s)(-B_t)) = E(B_s B_t) = \min\{s, t\}$$

- 8.11** 0.11.

- 8.13** $E(X_s X_t) = st\mu^2 + \sigma^2 s$.

- 8.19** $E(M_t) = \sqrt{\frac{2t}{\pi}}$ and $\text{Var}(M_t) = (1 - 2/\pi)t$.

- 8.23** (a) $\arcsin(\sqrt{r/t}) / \arcsin(\sqrt{r/s})$.

- (b) $\sqrt{s}/\sqrt{t}$.

- 8.25** 0.1688.

- 8.27** Write $Z_{n+1} = \sum_{i=1}^{Z_n} X_i$. Then,

$$\begin{aligned} E\left(\frac{Z_{n+1}}{\mu^{n+1}} \mid Z_n, \dots, Z_0\right) &= \frac{1}{\mu^{n+1}} E\left(\sum_{i=1}^{Z_n} X_i \mid Z_n, \dots, Z_0\right) \\ &= \frac{1}{\mu^{n+1}} E\left(\sum_{i=1}^{Z_n} X_i \mid Z_n\right) \\ &= \frac{1}{\mu^{n+1}} Z_n \mu = \frac{Z_n}{\mu^n}. \end{aligned}$$

8.31 (a) $4/9$.

(b) 20.

8.35 $SD(T) = \sqrt{2/3}a^2$.

8.43 (a) Black–Scholes price is \$35.32.

(b) Price is increasing in each of the parameters, except strike price, which is decreasing.

(c) $\sigma^2 \approx 0.211$.

Solutions for Chapter 9

9.1 The distribution is normal, with

$$E\left(\int_0^t sB_s ds\right) = \int_0^t sE(B_s) ds = 0,$$

and

$$\begin{aligned} \text{Var}\left(\int_0^t sB_s ds\right) &= E\left(\left(\int_0^t sB_s ds\right)^2\right) = \int_{x=0}^t \int_{y=0}^t E(xB_x yB_y) dy dx \\ &= \int_{x=0}^t \int_{y=0}^x xyE(B_x B_y) dy dx + \int_{x=0}^t \int_{y=x}^t xyE(B_x B_y) dy dx \\ &= \int_{x=0}^t \int_{y=0}^x xy^2 dy dx + \int_{x=0}^t \int_{y=x}^t x^2 y dy dx \\ &= \int_{x=0}^t \frac{x^4}{3} dx + \int_{x=0}^t x^2 \left(\frac{t^2}{2} - \frac{x^2}{2}\right) dx \\ &= \frac{t^5}{15} + \frac{t^5}{6} - \frac{t^5}{10} = \frac{2t^5}{15}. \end{aligned}$$

9.3 By Ito's Lemma, with $g(t, x) = x^4$,

$$d(B_t^4) = 6B_t^2 dt + 4B_t^3 dB_t,$$

which gives

$$B_t^4 = 6 \int_0^t B_s^2 ds + 4 \int_0^t B_s^3 dB_s,$$

and

$$E(B_t^4) = 6 \int_0^t E(B_s^2) ds = 6 \int_0^t s ds = 3t^2.$$

9.5 The desired martingale is $B_t^4 - 6tB_t^2 + 3t^2$.

9.9 (b) $E(X_3) = 4$; $\text{Var}(X_3) = 30$; $P(X_3 < 5) = 0.7314$.

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INDEX

New York Times, 217

absorbing state, 120

Adams, Douglas, 1

Aldous, David, 361

animal tracking, 349

arcsine distribution, 341

Austen, Jane, 190

Babbage, Charles, 158

Bayer, David, 217

Bayes' rule, 14

Benzi, Roberto, 395

Bernoulli, Daniel, 72

Bernoulli–Laplace model of diffusion, 72, 149

Bienaymé, Irénée-Jules, 159

Bienaymé-Galton-Watson process, 159

birthday problem, 241

Black, Fisher, 359

Black–Scholes, 359

Bohr, Niels, 255

branching process, 158

extinction, 168

immigration, 178

linear fractional, 176

offspring distribution, 159

total progeny, 173, 461

Brown, Robert, 320

Brownian bridge, 347

Brownian motion, 8, 320

arcsine distribution, 341

first hitting time, 337

Gaussian process, 331

geometric, 352

integrated, 374

invariance principle, 328

Markov process, 336

martingale, 358

random walk, 326

reflection principle, 339

simulation, 324

standard, 321

strong Markov property, 337

transformations, 334

zeros, 341

card shuffling, 212

Carlyle, Thomas, 223

Ceres, 330

Cesaro's lemma, 79

Chapman–Kolmogorov, 55

checkerboard, 193

Chutes and Ladders, 119

co-occurrence matrix, 193

- coding function, 190
- coin tossing, 343
- Colton, Charles Caleb, 372
- complete spatial randomness, 250
- conditional distribution, 15
- conditional expectation, 18
- conditional variance, 31
- continuous-time Markov chain, 265
 - absorbing, 288
 - birth-and-death, 297
 - detailed balance, 295
 - embedded chain, 268
 - exponential alarm clock, 270
 - forward, backward equation, 275
 - fundamental matrix, 288
 - generator, 273
 - global balance, 293
 - holding time, 268
 - limiting distribution, 283
 - local balance, 295
 - Poisson subordination, 306
 - stationary distribution, 284
 - time reversibility, 294
 - transition rates, 270
 - tree theorem, 295
 - uniformization, 306
 - Yule process, 299
- counting process, 223
- coupling, 140
- coupling from the past, 205
- coupon collector's problem, 216
- Cox–Ingersoll–Ross model, 398
- cryptography, 190
- cutoff phenomenon, 217
- damping factor, 113
- Darwin, Charles, 192
- detailed balance equations, 114
- Diaconis, Persi, 42, 190, 217
- diffusion, 389
- DNA sequence, 280
 - evolutionary distance, 282
 - Felsenstein model, 288
 - Hasegawa, Kishino, Yao model, 314
 - Jukes–Cantor model, 280
 - Kimura model, 281
- Doebelin, Wolfgang, 140
- Donsker, Monroe, 327
- doubly stochastic matrix, 73, 144
- Ehrenfest model, 88, 110, 317, 459
- Einstein, Albert, 320
- embedding, 241
- empirical distribution function, 350
- Eugene Onegin, 43
- Euler–Maruyama method, 391
- expected return time, 103, 105, 135
- Fermat, Pierre, 8
- Fibonacci sequence, 220
- Fill, Jim, 361
- first hitting time, 134
- first passage time, 103
- first return time, 149
- first-step analysis, 125
- fundamental matrix, 125
- Galapagos Islands, 192
- Galton, Sir Francois, 158
- gambler's ruin, 7, 12, 44, 123
- Gauss, Carl Friedrich, 330
- Gaussian process, 330
- Geman, Donald, 197
- Geman, Stuart, 197
- Gibbs distribution, 202
- Gibbs sampler, 197
- Gibbs, Josiah, 197, 198
- Gilbert–Shannon–Reeds model, 217
- Goldfeather, Jack, 190
- graph
 - bipartite, 459
 - complete, 47
 - cycle, 47, 62
 - directed, 50
 - hypercube, 47
 - lollipop, 90
 - weighted, 50
- graphic
 - regular, 92
- greatest common divisor, 107
- Hastings, W.K., 187
- independent increments, 224, 321
- invariance principle, 328
- Ising model, 202
- Ito integral, 379
- Ito process, 389
- Ito's lemma, 381
 - diffusion, 389
- Ito, Kiyoshi, 379
- James, William, 265
- Kac, Mark, 88
- Kakutani, Shizuo, 101
- Knuth, Donald, 181
- Kolmogorov–Smirnov statistic, 349

- Laplace, Pierre-Simon, xvii
- law of large numbers, 182
- law of total expectation, 22
- law of total probability, 11
- law of total variance, 32
- length-biased sampling, 256
- limit theorem
 - coupling proof, 140
 - ergodic Markov chains, 109, 139
 - finite irreducible Markov chains, 103, 135
 - linear algebra proof, 142
 - regular Markov chains, 83, 135
- Little's formula, 302
- little-oh notation, 234
- logistic equation, 386
- lognormal distribution, 353
- Lotka, Alfred, 172
- Love Letter, 172
- Lucretius, 320
- M/M/ ∞ queue, 316
- M/M/1 queue, 301
- M/M/c queue, 304
- Markov chain
 - absorbing, 119, 126
 - aperiodic, 109
 - birth-and-death, 48, 118
 - canonical decomposition, 101
 - closed class, 101
 - communication, 94
 - definition, 41
 - eigenvalue, 92
 - eigenvector, 92
 - ergodic, 109, 139
 - first-step analysis, 105
 - irreducible, 95, 103
 - limiting distribution, 76
 - periodic, 109
 - periodicity, 106
 - recurrent state, 97
 - null recurrent , 104, 137
 - positive recurrent , 104, 137
 - regular, 83, 135
 - simulation, 65
 - stationary distribution, 80
 - time reversal, 153
 - time reversibility, 114
 - time-homogenous, 41
 - transient state, 97
- Markov chain Monte Carlo
 - eigenvalue connection, 210
 - rate of convergence, 210
- Markov process, 336
- Markov, Andrei Andreyevich, 40, 69
- martingale, 356
 - optional stopping theorem, 361
 - quadratic, 358
- matrix exponential, 277
- mean-reverting process, 390
- memorylessness, 229
- Mendeleev, Dmitri, 76
- Merton, Robert, 359
- Metropolis, Nicholas, 187
- Metropolis–Hastings algorithm, 187
- Monte Carlo simulation, 9
- move-ahead-1, 72
- move-to-front, 49, 72
- multistate Markov model, 289
- multivariate normal distribution, 330
- nearest-neighbor distance, 251
- option, 355
- optional stopping theorem, 361
- order statistics, 244
- Ornstein–Uhlenbeck process, 390, 392
- Pólya's Theorem, 101
- Pólya, George, 101
- PageRank, 3, 112, 150
- parlaying bets, 178
- Pascal, Blaise, 8
- patterns, 131
- perfect sampling, 205
 - monotonicity, 208
- period, 107
- periodic forcing, 396
- periodicity, 106
- Perron–Frobenius theorem, 143
- phase transition, 203
- Poincaré, Henri, 89
- Poisson approximation of binomial distribution, 431
- Poisson process, 223
 - arrival times, 227
 - compound, 263
 - doubly stochastic, 263
 - gamma distribution, 232
 - interarrival times, 227
 - mixed, 263
 - nonhomogeneous, 253
 - power law, 255
 - rate, 225
 - simulation, 248
 - spatial, 249
 - superposition, 240
 - thinned process, 239
 - translation, 226
 - uniform distribution, 245

- Poisson subordination, 306
- polygraphs, 14
- positive matrix, 82, 143
- power-law distribution, 189
- present value, 262
- probability generating function, 164
- Project Gutenberg, 190
- Propp, Jim, 205
- Propp–Wilson algorithm, 205
- Pushkin, Alexander, 43
- queueing theory, 301
- R Project for Statistical Computing, 400
- R scripts
 - absorption, 290
 - adjacent, 186
 - bbridge, 348
 - bivariatenormal, 199
 - bm, 325
 - bmhitting, 365
 - branching, 161, 171
 - buswaiting, 230
 - cancerstudy, 65
 - coinflips, 345
 - darwin, 195
 - decode, 191
 - drift, 394
 - forest, 157
 - gamblersruin, 13
 - globalbalance, 293
 - gym, 80
 - ising, 205
 - kstest, 352
 - matrixexp, 278
 - maxssrw, 330
 - ornstein, 392
 - pagerank, 113
 - pattern, 132
 - perfect, 208
 - poissonsimsim, 248
 - powerlaw, 190
 - Psubordination, 309
 - ReedFrost, 10
 - returntime, 106
 - scriptsample, 410
 - snakes, 119, 127
 - spatialPoisson, 251
 - stochresonance, 396
 - trivariate, 201
 - waitingparadox, 256
- random genetic drift, 393
- random sum of random variables, 28, 160
- random transpositions, 191
- random walk, 7, 326
 - $\mathbb{Z}$, 99, 114
 - $\mathbb{Z}^2$, 100, 459
 - $\mathbb{Z}^3$, 148
- absorbing boundary, 45
- bias, 152
- continuous time, 299
- cycle, 53, 62, 115
- graph, 4, 46, 89, 90, 115, 185
- partially reflecting boundary, 118
- reflecting boundary, 71, 146, 152
- simple, symmetric, 328, 357
- webgraph, 112
- weighted graph, 71, 90, 116, 129, 185
- recurrence, 96, 98
- Reed–Frost model, 5
- reflection principle, 339
- regeneration, 133
- regular matrix, 82, 139
- rejection method, 184
- reliability, 255
- Riemann–Stieltjes integral, 374
- Russell, Bertrand, 321
- Scholes, Myron, 359
- second largest eigenvalue, 212
- SIR model, 5
- size-biased sampling, 256
- Snakes and Ladders, 119
- spectral representation formula, 212
- spectral theorem, 454
- square root process, 398
- stationarity, 81
- stationary increments, 224, 321
- Stirling’s approximation, 100
- stochastic differential equation, 385
- stochastic matrix, 42
 - doubly stochastic, 73, 144
- stochastic process, 6
 - definition, 6
- stochastic resonance, 395
- stokhazesthai, 1
- stopping time, 133
- Stratonovich integral, 379
- strong Markov property, 97, 133, 136
- strong stationary time, 214
- substitution cipher, 190
- Tennyson, Alfred, 158
- time reversibility, 114
- top-to-random shuffle, 214
- total variation distance, 212
- transience, 96, 98

transposition scheme, 72
tree, 295

Ulam, Stanislaw, 10

Vasicek model, 390

Wald's equation, 370

Watson, Henry William, 159

Weierstrass, Karl, 334

white noise, 378

Wiener measure, 326

Wiener, Norbert, 321, 326

Wilson, David, 205

worm, 172

Wright–Fisher model, 45

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